

1. Introduction: Why Derivatives in Microeconomics?

In economics, we rarely care about the "total" amount in isolation. We care about **change**.

- **The Marginal Concept:** Whenever you hear the word "Marginal" (Marginal Cost, Marginal Utility, Marginal Revenue), you are talking about a **derivative**.
- **The Slope:** A derivative tells us the slope of a function at a specific point. It answers the question: *"If I change my input (Quantity) by a tiny amount, how much does my output (Profit/Cost) change?"*

2. Core Rules for the "Micro" Toolkit

You don't need to be a mathematician to master Micro, but you must master these three rules:

A. The Power Rule (The Bread and Butter)

If your function is $f(x) = ax^n$, then the derivative $f'(x)$ is:

$$f'(x) = n \cdot ax^{n-1}$$

- **In plain English:** Bring the exponent down to multiply, then subtract 1 from the exponent.

Handwritten notes for the Power Rule:

- $x^0 = 1$
- $x^1 = x$
- $x^2 = x^2$
- $x^3 = x^3$
- $x^4 = x^4$
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- $x^{100} = x^{100}$

B. The Constant Rule

If $f(x) = c$ (where c is a number like 100), then $f'(x) = 0$.

- **Economic Logic:** Fixed costs do not change when you produce one more unit, so their "marginal" effect is zero.

C. The Sum/Difference Rule

The derivative of a sum is the sum of the derivatives.

- If $TC = 50 + 2Q + Q^2$, we derive each part separately.

$$d(q) = 2 + 2Q$$

3. Partial Derivatives: Handling Multiple Goods

In Micro, utility often depends on two goods: X and Y. We use **Partial Derivatives** to see how utility changes when we change *only one* variable, holding the other constant.

Example: $U(x, y) = x^2 y^3$

$$d(x) = 2xy^3 \quad d(y) = 3x^2y^2$$

- To find Marginal Utility of x (dU/dx): Treat y as a constant number.
- **Result:** $2x y^3$

4. Economic Applications

We use derivatives to find the **Optimum** (Maximum or Minimum).

- **Profit Maximization:** Set the derivative of the Profit function to zero ($d\pi = 0$). This is where $MR = MC$.

Economic Term	Mathematical Operation
Marginal Cost (MC)	Derivative of Total Cost (TC)
Marginal Revenue (MR)	Derivative of Total Revenue (TR)
Marginal Utility (MU)	Partial Derivative of Utility (dU/dx)
Profit Max	Set $\pi' = 0$

5. Class Exercises

Exercise 1: Marginal Cost (MC)

A firm has the following Total Cost function:

$$TC(Q) = 150 + 10Q + 2Q^2$$

1. Find the Marginal Cost (MC) function.
2. Calculate the MC when the firm produces 5 units.

Solution:

$$d(Q) = 10 + 4Q \quad Q = 5 \Rightarrow MC = 10 + 4 \times 5$$

$$MC = 30$$

Exercise 2: Profit Maximization

A firm faces a Total Revenue of $TR = 100Q - Q^2$ and a Total Cost of $TC = 20Q$.

1. Find the Profit function $f(PR) = TR - TC$.
2. Find the derivative of the Profit function.
3. Calculate the Quantity (Q) that maximizes profit.

Solution:

$$PF = (100Q - Q^2) - (20Q) = 80Q - Q^2$$

$$d(PF) = 80 - 2Q$$

$$80 - 2Q = 0 \Rightarrow Q = 40$$

Exercise 3: Marginal Utility (Partial Derivatives)

A consumer has the following Utility function:

$$U(x, y) = 10x^{0.5} y^{0.2}$$

1. Find the Marginal Utility of x (MU_x).
2. Find the Marginal Utility of y (MU_y).

Solution:

$$MU_x = \frac{\partial U}{\partial x} \Rightarrow \frac{\partial}{\partial x} (10x^{0.5} y^{0.2}) = 10 \cdot 0.5 x^{-0.5} y^{0.2} = 5 \frac{1}{x^{0.5}} y^{0.2} = \frac{5x^{0.2} y^{0.2}}{x^{0.5}}$$

$$MU_y = \frac{\partial U}{\partial y} \Rightarrow \frac{\partial}{\partial y} (10x^{0.5} y^{0.2}) = 10x^{0.5} \cdot 0.2 y^{-0.8} = 2x^{0.5} \frac{1}{y^{0.8}} = \frac{2x^{0.5}}{y^{0.8}}$$

Exercise 4: Revenue Maximization

A monopolist faces a demand curve given by $P = 200 - 4Q$.

1. Find the Total Revenue function (TR = P * Q).
2. Find the Marginal Revenue (MR) function.
3. Determine the quantity Q that maximizes Total Revenue.

$$TR = 200Q - 4Q^2$$

$$\frac{\partial TR}{\partial Q} = 200 - 8Q$$

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Exercise. Utility, Income Effect and Substitution Effect

We have a consumer with the following Utility Function:

- $U(x,y) = x^2 y^3$
- Income $M=200€$
- $P_x = 4€, P_y = 6€$

- A- Calculate the amount of Good X and Y that maximizes Utility. Draw the Graph.
- B- If the price of Good X changes to $P_x=1$. Calculate the new equilibrium and draw the graph.
- C- Calculate the Income Effect (IE) and Substitution Effect (SE) using the Slutsky Method. Draw the Graph.
- D- Calculate the Income Effect (IE) and Substitution Effect (SE) using the Hicks Method. Draw the Graph.

Supply and Demand Exercise

In the sustainable mobility market, the annual demand for "Premium" electric scooters is given by the function:

$$Q_d = 240,000 - 400P$$

On the other hand, the supply from local manufacturers corresponds to the following function:

$$Q_s = -60,000 + 600P$$

Where Q represents the number of units and P the price in euros per unit.

- a- Calculate the market equilibrium price and quantity.
- b- The government, to finance infrastructure for bike lanes, imposes a tax of €50 per unit sold, which producers must pay. Calculate the new equilibrium.
- c- Determine the tax incidence: how much of the tax is actually paid by the consumer and how much by the producer?
- d- Calculate the income of the Government.



MICROECONOMICS



UTC2 CONSUMER BEHAVIOUR

UTC 2 CONSUMER BEHAVIOUR

- ❑ INTRODUCTION
- ❑ CONSUMER PREFERENCES
- ❑ INDIFFERENCE CURVES AND MARGINAL RATE OF SUBSTITUTION
- ❑ UTILITY
- ❑ BUDGET CONSTRAINTS
- ❑ CONSUMER CHOICE

INTRODUCTION

- In this lesson, we will study what we can find behind the demand curve.

Behind each individual demand curve, we can find the individual choices of each consumer.

- How do consumers behave?

They act rationally, aiming to maximize their satisfaction derived from consumption, considering the constraints imposed by their incomes.

- How does each consumer allocate their income for the purchase of different goods and services?

PRINCIPLE OF RATIONALITY: Economic agents choose the best alternative among all the feasible alternatives, according to their:

- ✓ Set of options they face and their preferences.
- ✓ Budget constraints (prices and income).
- ✓ The marginal utility of product/service.

CONSUMER PREFERENCES

Consumer preference theory is a theory that explains how consumers make decisions. It is based on the idea that consumers are rational and will choose the product or service they believe will satisfy their needs.

Consumer preference is a term that refers to consumers' choices to maximize their satisfaction.

- The consumer must choose a combination of goods (market basket or bundle of goods/**consumption bundles**).
- The consumer will choose the combination that produces **the most utility** (satisfaction).

CONSUMER PREFERENCES

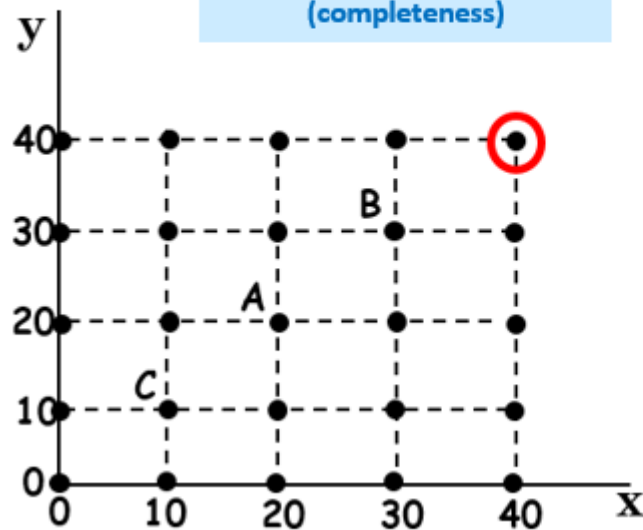
The theory of **consumer preferences** is based on three main **assumptions**:

- ❑ **Completeness:** This assumption implies that consumers can rank all combinations of goods available to them. In other words, given any two bundles of goods A and B, consumers can determine whether they prefer A to B, B to A, or are indifferent between the two.
- ❑ **Transitivity:** The transitivity assumption asserts that if a consumer prefers Bundle A to Bundle B, and Bundle B to Bundle C, then the consumer will also prefer Bundle A to Bundle C. This assumption is essential for rational consumer decision-making.
- ❑ **Non-satiation:** Otherwise known as the "**more is better**" assumption, non-satiation implies that given two bundles of goods, consumers will prefer the bundle offering more of at least one good and no less of other goods.

CONSUMER PREFERENCES

An order of preferences must be established over the 25 available alternatives
(**completeness**)

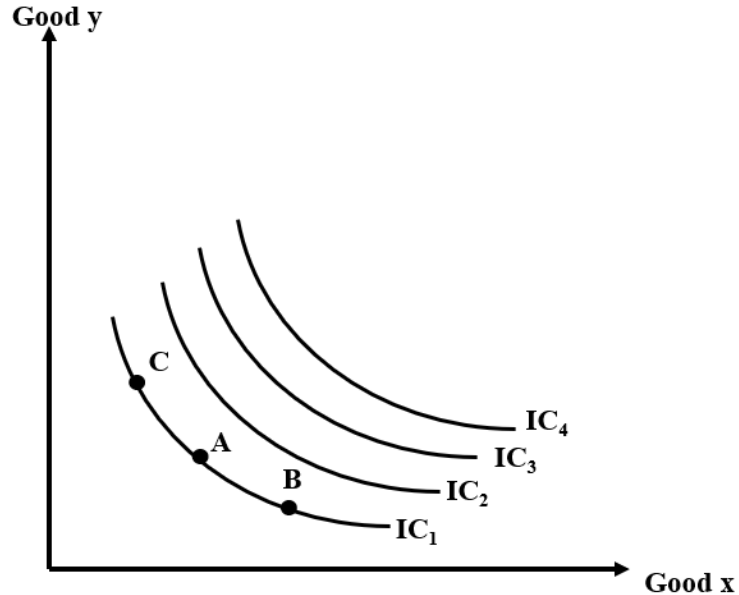
These 25 combinations (x,y) must be ordered consistently
(**consistency**)



That means that the transitivity property has to be fulfilled:
If B is preferred to A and A is **preferred to** C, then B is **preferred to** C.
(**transitivity**)

Finally, you must choose the best combination, that is the most preferred (the consumer prefers to have more than less)

INDIFFERENCE CURVES



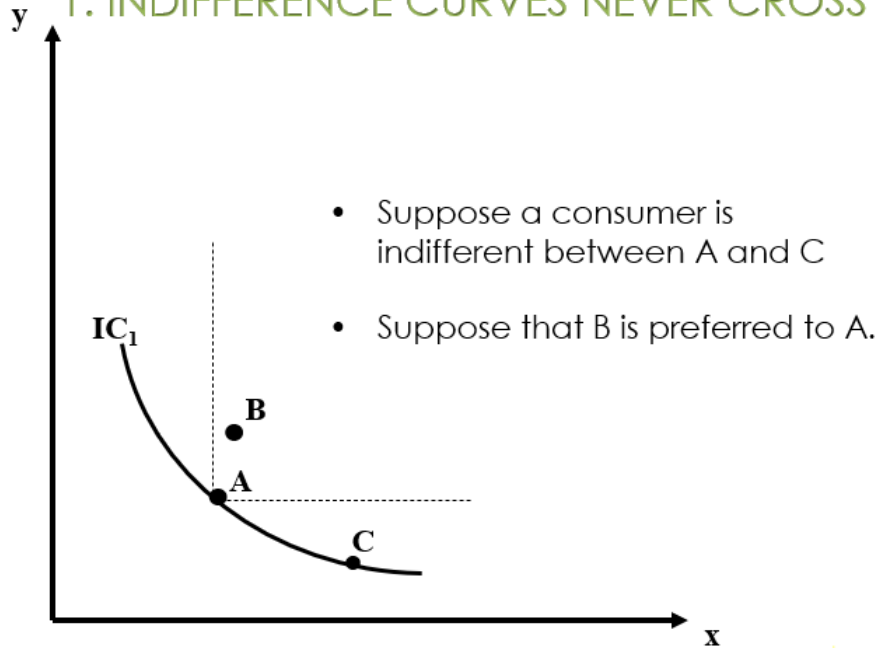
- **Consumer choice** depends on **preferences** regarding the two goods.
- An indifference curve is a **graphical representation of consumer preferences**.
- It shows **the various bundles of consumption** that make the consumer **equally happy**.
- Indifference curve analysis is simplified by assuming that the consumer spends all his/her money on 2 products.

Definition: An **Indifference Curve** is the set of all baskets for which the consumer is indifferent

Definition: An **Indifference Map** illustrates the set of indifference curves for a particular consumer

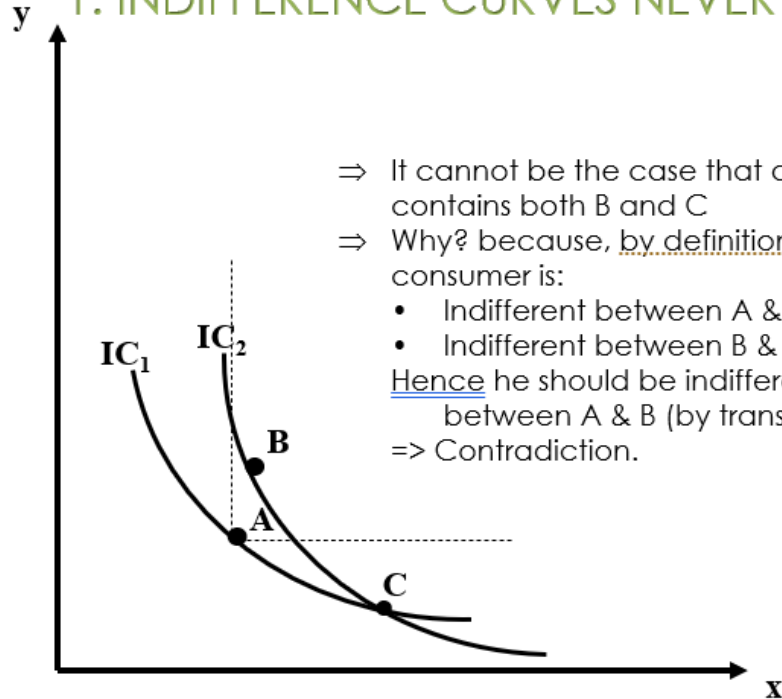
PROPERTIES OF INDIFFERENCE CURVES

1. INDIFFERENCE CURVES NEVER CROSS



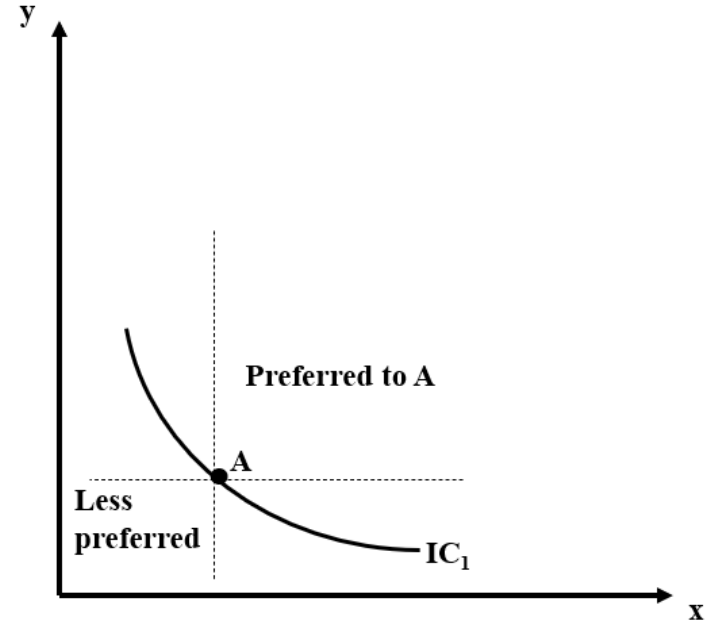
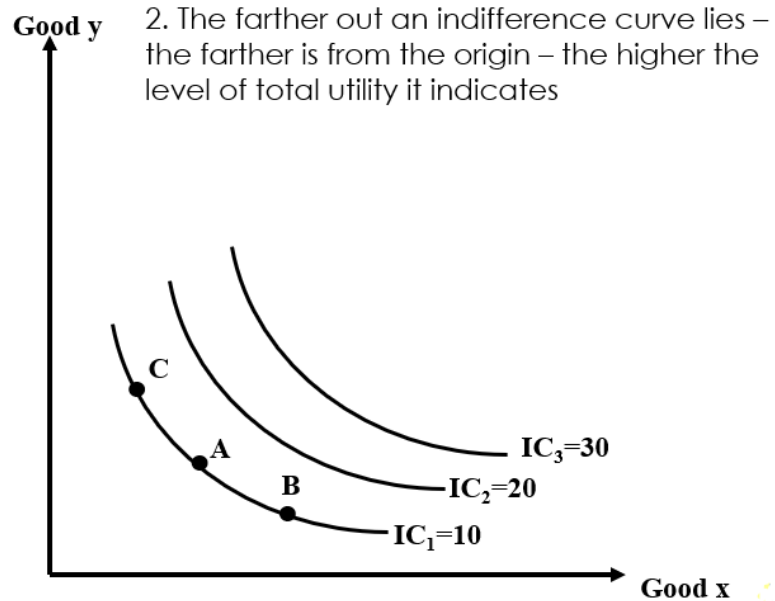
PROPERTIES OF INDIFFERENCE CURVES

1. INDIFFERENCE CURVES NEVER CROSS



- ⇒ It cannot be the case that an IC contains both B and C
- ⇒ Why? because, by definition of IC the consumer is:
- Indifferent between A & C
 - Indifferent between B & C
- Hence he should be indifferent between A & B (by transitivity).
⇒ Contradiction.

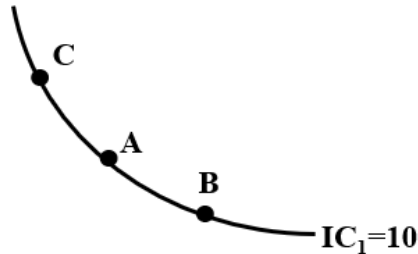
PROPERTIES OF INDIFFERENCE CURVES



PROPERTIES OF INDIFFERENCE CURVES

Good y

3. The indifference curves slope downward: more is better but if A, B and C are on the same indifference curve, an increase in X must be compensated by a decrease in Y

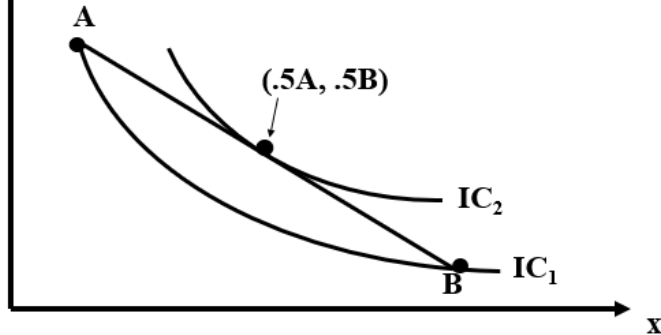


Good x

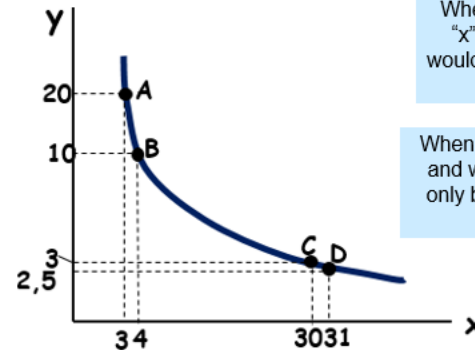


PROPERTIES OF INDIFFERENCE CURVES

4. Indifference curves have a convex shape: i) steeper slope in A, flatter slope in B; ii) average bundles are preferred to the extremes



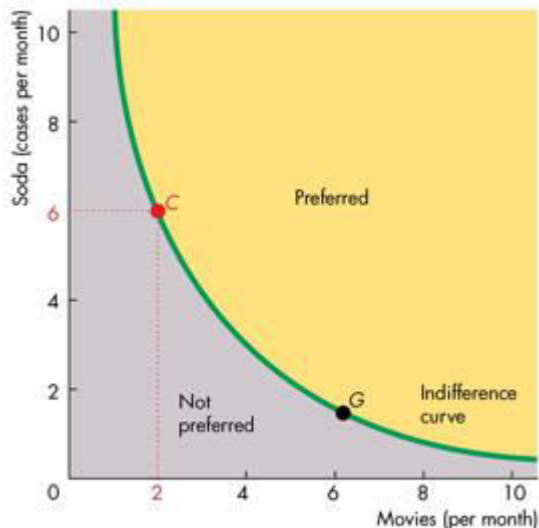
(3) Curves are **strictly convex (bowed inwards)**: a good is **less valued** as we **have more** of it.



When the consumer has 3 units of "x" and wants to get a fourth, he would be willing to give up 10 units of "y".

When the consumer has 30 units of "x" and wants to get one more, he would only be willing to give up half a unit of "y".

PROPERTIES OF INDIFFERENCE CURVES



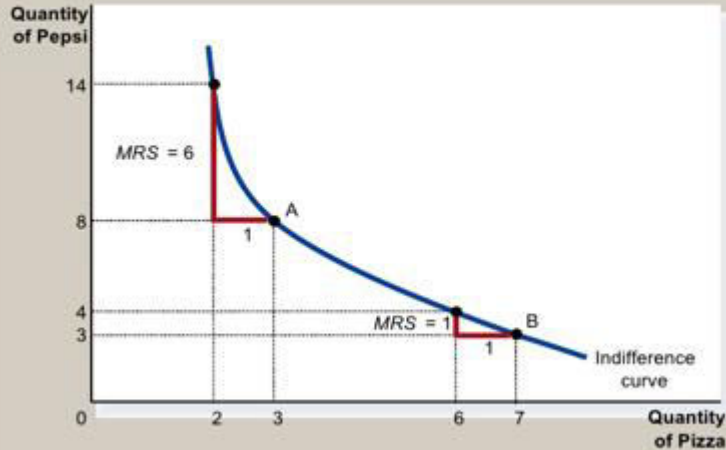
(a) An indifference curve

- The figure on the left illustrates Lisa's indifference curve.
- We can represent Lisa's preference as a map.
- An indifference curve is a line that shows combinations of goods among which a consumer is indifferent.
- **At point C**, Lisa sees 2 movies and drinks 6 cases of soda a month.
- Lisa can sort all possible combinations of goods into three groups: **preferred to C**, **not preferred to C**, and **just as good as C**.
- An indifference curve joins all those points that Lisa says are just as good as C.
- G is such a point. **Lisa is indifferent between C and G (C and G PROVIDE THE SAME UTILITY.)**
- Lisa **prefers** any point on the indifference curve to any point below the indifference curve.
- Lisa **prefers** any point above the indifference curve to any point on the curve.

AN INDEFERENCE CURVE BRINGS TOGETHER ALL THE COMBINATIONS OF GOODS (X, Y) THAT PROVIDE THE SAME LEVEL OF UTILITY TO A PERSON. WE WILL DESIGNATE IT WITH THE LETTER “U”

INDIFFERENCE CURVES and MARGINAL RATE OF SUBSTITUTION: MRS

Figure 4 Bowed Indifference Curves



We will often find it useful to refer to the slope of an indifference curve at a particular point.

The slope of an indifference curve (at each of its points) is known as **the marginal rate of substitution (MRS)**.

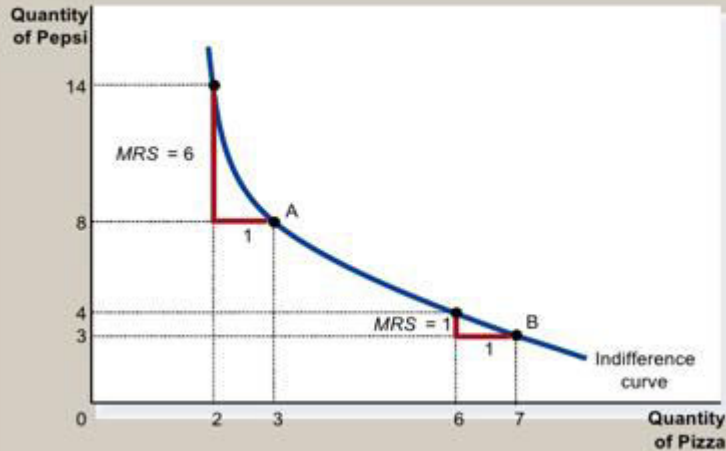
The **MRS** measures the **rate at which the consumer is willing to substitute one good for the other – while staying inside the same indifference curve.**

MRS = change in Y/change in X
(MRS = dY/dX)

$$MRS = \frac{\Delta Y}{\Delta X}$$

INDIFFERENCE CURVES and MARGINAL RATE OF SUBSTITUTION: MRS

Figure 4 Bowed Indifference Curves



An indifference curve exhibits a **diminishing marginal rate of substitution**:

1. The more of good x you have, the more you are willing to give up to get a little of good y.
2. The indifference curves
 - Get flatter as we move out along the horizontal axis
 - Get steeper as we move up along the vertical axis.

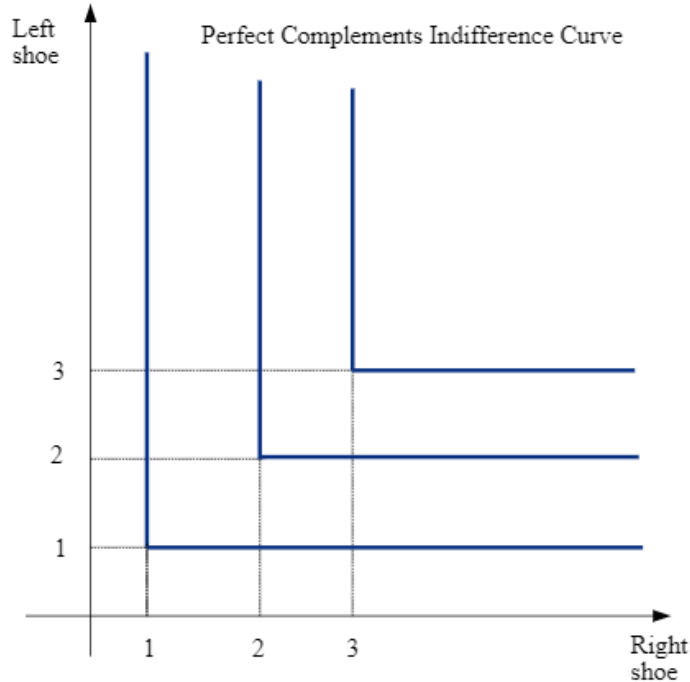
INDIFFERENCE CURVES: PARTICULAR CASES 1



- **IC** for goods which are **PERFECT SUBSTITUTES**
- Two goods are perfect substitutes if the consumer is willing to substitute one good for the other at a constant rate.
- Thus, the indifference curves are straight lines with a slope of -1

$$MRS_{x,y} = \frac{A}{B} (\text{constant MRS})$$

INDIFFERENCE CURVES: PARTICULAR CASES 2



- IC for goods which are **PERFECT COMPLEMENTS**
- Perfect complements are goods that are always consumed together in fixed proportions. In some sense, the goods “complement” each other.
- Thus, the indifference curves are L-shaped.

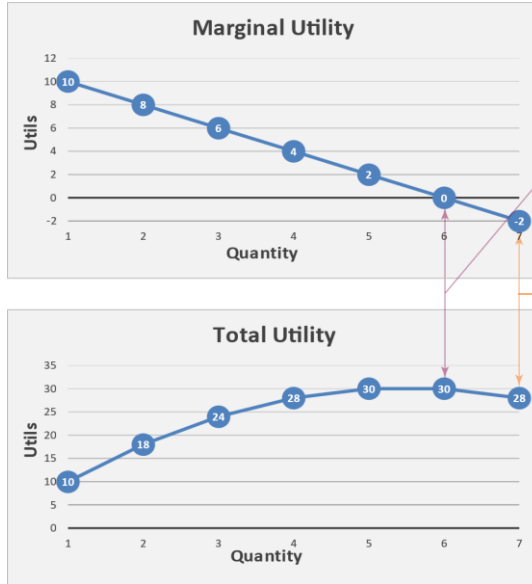
$$MRS_{x,y} = 0 \text{ or } \infty \text{ or undefined}$$

UTILITY FUNCTION

- The **utility** is a numerical score representing the utility (satisfaction) a consumer obtains from a given consumption basket.
- **Util**: a hypothetical unit measuring satisfaction.
- **Utility function**: Formula that assigns a utility level to a consumer basket.
- **Total utility (TU)**: the total satisfaction from a given level of consumption.
- **Marginal utility (MU)**: the change in satisfaction (the extra utility) received from consuming one additional unit of a good or service.

TOTAL AND MARGINAL UTILITY

The law of diminishing marginal utility states that the marginal utility from an additional unit of consumption declines as the quantity of consumed goods increases.



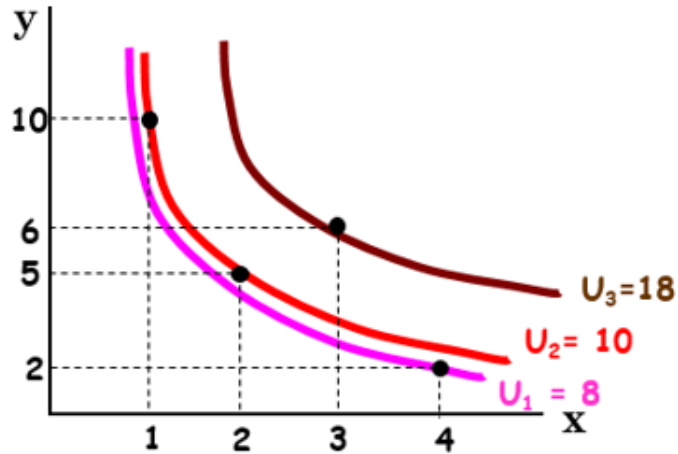
For a single item, total utility is maximized when marginal utility = 0.

Total utility declines when marginal utility becomes negative.

Quantity	Marginal Utility	Total Utility
1	10	10
2	8	18
3	6	24
4	4	28
5	2	30
6	0	30
7	-2	28

UTILITY FUNCTION

With the utility function we try to find a mathematical function that represents the order of consumer preferences.



Example:

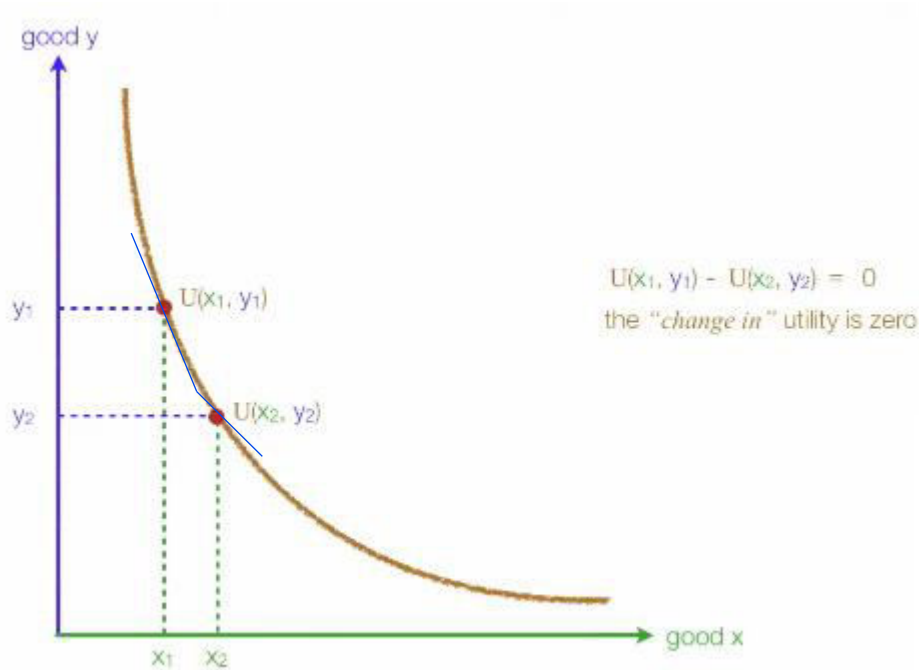
$$U = X \cdot Y$$

Example:

Red IC: we know that a consumer is indifferent between consuming the combination $(1, 10)$ and the combination $(2, 5)$.

Furthermore, these combinations are more preferred than combination $(4, 2)$ – purple IC, but less than combination $(3, 6)$ – brown IC.

UTILITY



- The **slope of the tangent line** is the **MRS**
- The slope is not constant, and **it changes** as we move down the IC.
- The slope of the tangent line **is the derivative of the Indifference Curve**

The MRS can be expressed as a function of **marginal profits**:

$$MRS = -\frac{dY}{dX} = \frac{U'_x}{U'_y}$$

UTILITY

Definition: The **marginal utility** of good x is the additional utility that the consumer gets from consuming a little more of x

$$MU_x = \frac{dU}{dx}$$

- It is the derivative of the utility function with respect to x.
- It assumes that the consumption of all other goods in the consumer's basket remains constant.

Definition: The **marginal utility** of good y is the additional utility that the consumer gets from consuming a little more of y

$$MU_y = \frac{dU}{dy}$$

- It is the derivative of the utility function with respect to y.
- It assumes that the consumption of all other goods in the consumer's basket remains constant.

$$MU_x / MU_y = - dy/dx$$

$$MU_x = dU/dX \text{ (a derivative of the utility function with respect to x)}$$

$$MU_y = dU/dY \text{ (a derivative of the utility function with respect to y)}$$

$$\text{Since } MRS_{x,y} = - \frac{dy}{dx}$$

The slope of the indifference curve is:

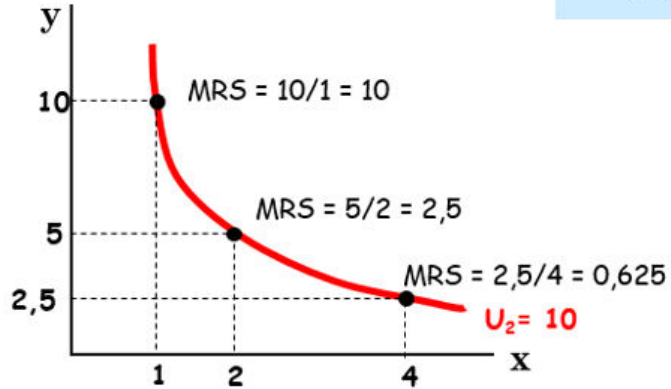
$$MRS_{x,y} = \frac{MU_x}{MU_y} = - \frac{dy}{dx}$$

The MRS can be expressed as a function of **marginal profits**:

$$MRS = - \frac{dY}{dX} = \frac{U'_x}{U'_y}$$

UTILITY

$$U = X \cdot Y$$



To find the slope (MRS) at each point we have to take the partial derivative of the function $U = xy$
Let's do it for point B.

$$MRS = MU_x / MU_y$$

$$MU_x = dU/dX$$

$$dU/dX = y \quad (5)$$

$$MU_y = dU/dY$$

$$dU/dY = x \quad (2)$$

$$MRS = 5/2$$

UTILITY: CONNECTION BETWEEN MARGINAL UTILITY & MARGINAL RATE OF SUBSTITUTION

The Marginal Rate of Substitution looks at the balance in changes of good X and good Y required for the consumer to be indifferent between his/her consumption bundles before and after trade. But what does indifference mean? It means that utility for both bundles is exactly equal. Therefore, $dU = 0$

There is some (negative) change in utility resulting from giving up a little bit of good Y which equals: $MU_y dy$

Similarly, there is some (positive) change in utility from getting a little more of good X, which equals: $MU_x dx$

Since we want to be indifferent before and after the trade, it must be that the sum of these changes equals zero. Our equation would thus look like this:

$$dU = MU_x \cdot dx + MU_y \cdot Dy = 0$$

UTILITY: CONNECTION BETWEEN MARGINAL UTILITY & MARGINAL RATE OF SUBSTITUTION

$$dU = MU_x \cdot dx + MU_y \cdot dy = 0$$

$$MU_x \cdot dx = - MU_y \cdot dy/dx$$

$$MU_x = - (MU_y \cdot dy)/dx \quad /: MU_y$$

$$MU_x/MU_y = - dy/dx$$

$$\text{Since } MRS_{x,y} = - \frac{dy}{dx}$$

The slope of the indifference curve is:

$$MRS_{x,y} = \frac{MU_x}{MU_y}$$



UTC 2 “A” EXERCISES (CANVAS): 1,2,3,4,5,6

BUDGET CONSTRAINTS

- Preferences do not fully explain consumer behaviour.
- The consumer's **budget constraint** limits the capacity to consume: the consumer is conditioned by income (I) and prices (Px, Py).

Budget constraint:

$$P_x \cdot X + P_y \cdot Y \leq I$$

- **Budget set:** Set of affordable baskets (combination of goods) at the given prices and income.

BUDGET CONSTRAINTS

Question:

A. *Pepito receives 1000€ as monthly income. He only consumes two goods: books and movies. Each movie has a price of 5€ and each book has a price of 20€. Indicate the formula for Pepito's budget constraint.*

B. *Represent graphically Pepito's budget constraint.*

BUDGET CONSTRAINTS

Consumption possibilities are **limited** by the available income: the consumer can only purchase a **limited** number of units of good x and good y.

The **BUDGET CONSTRAINT LINE** indicates all the combinations (x,y) for which the expenditure incurred by the consumer **is equal to the available income**.

$$I = P_x \cdot X + P_y \cdot Y$$

BUDGET CONSTRAINTS

- The **slope of the budget constraint** tells us how much we must give up of one good if we want to increase the other good by one unit.
- Let's assume we start from a consumption basket (x, y) , and the consumer wants to increase the consumption of good x (Δx) while keeping y constant. For the budget constraint to still hold, the consumption of the good y must change. Let Δy denote the variation in the quantity consumed of good y .
 - On the one hand, we need to satisfy the initial budget constraint:

$$M = p_x x + p_y y$$

- and on the other hand, the new budget constraint:

$$M = p_x (x + \Delta x) + p_y (y + \Delta y)$$

- By subtracting both expressions, we get:

$$0 = p_x \Delta x + p_y \Delta y$$

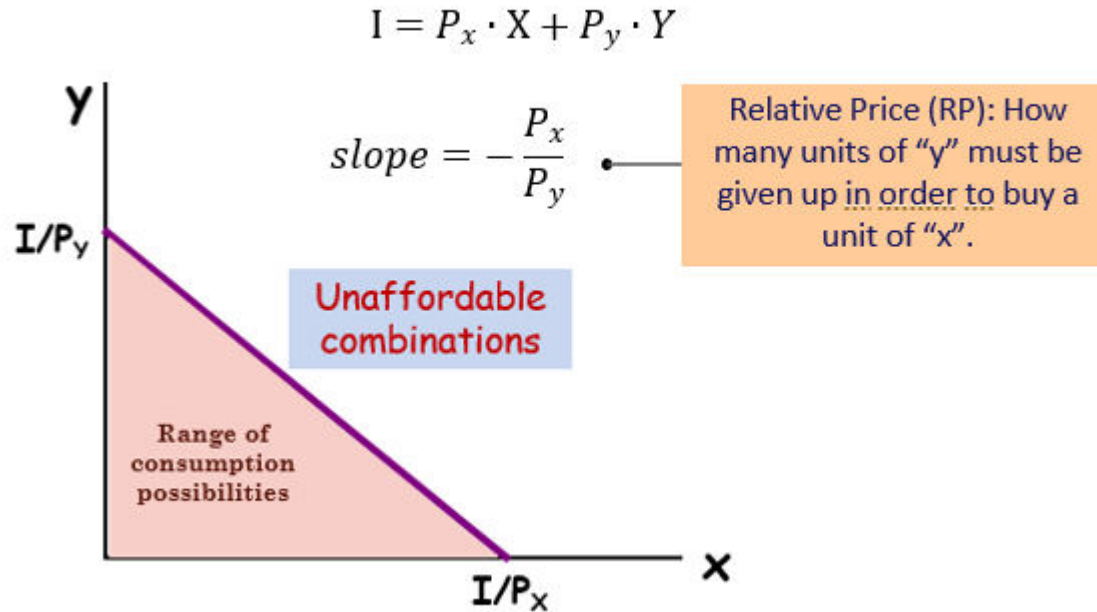


$$\frac{\Delta y}{\Delta x} = -\frac{p_x}{p_y}$$

THE SLOPE OF THE BUDGET CONSTRAINT/ LINE

The opportunity cost of one good in terms of the other good

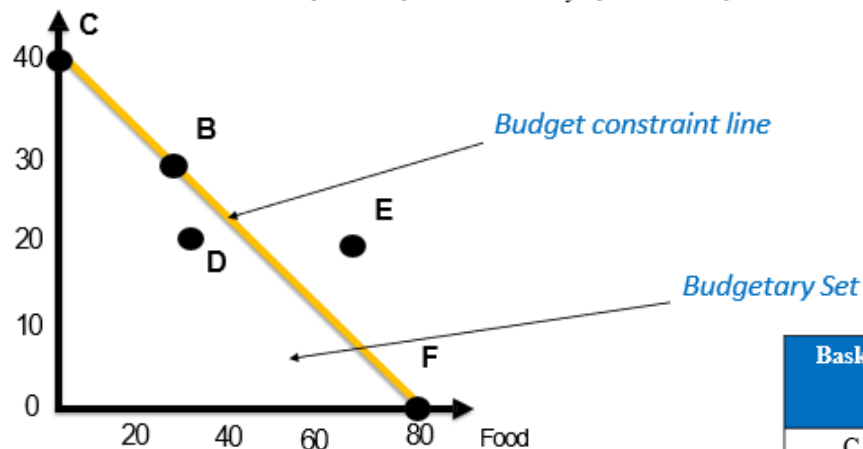
BUDGET CONSTRAINTS



BUDGET CONSTRAINTS

Budgetary Set

Clothes $P_x(\text{food}) = 1\text{€}$ $P_y(\text{clothes}) = 2\text{€}$ $I = 80\text{€}$

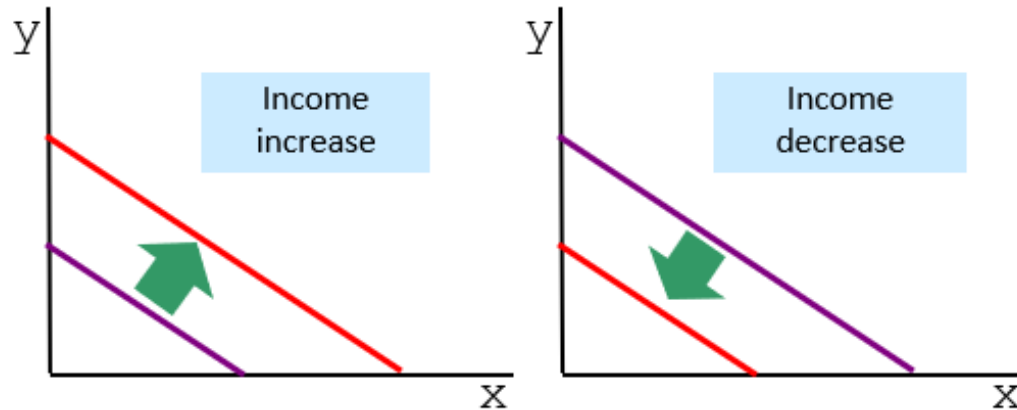


Basket	Food (x) ($P_x = 1\text{€}$)	Clothes (Y) ($P_y = 2\text{€}$)	Total expenditure ($P_x X + P_y Y$)
C	0	40	80€
B	20	30	80€
D	30	20	70€
E	60	20	100€
F	80	0	80€

BUDGET CONSTRAINTS

Changes in the budget line

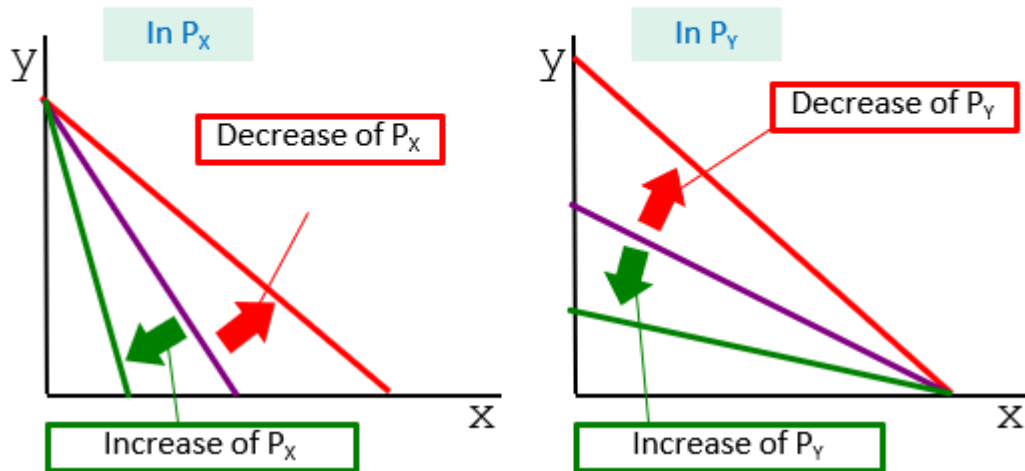
(1) Due to variations in income



BUDGET CONSTRAINTS

Changes in the budget line

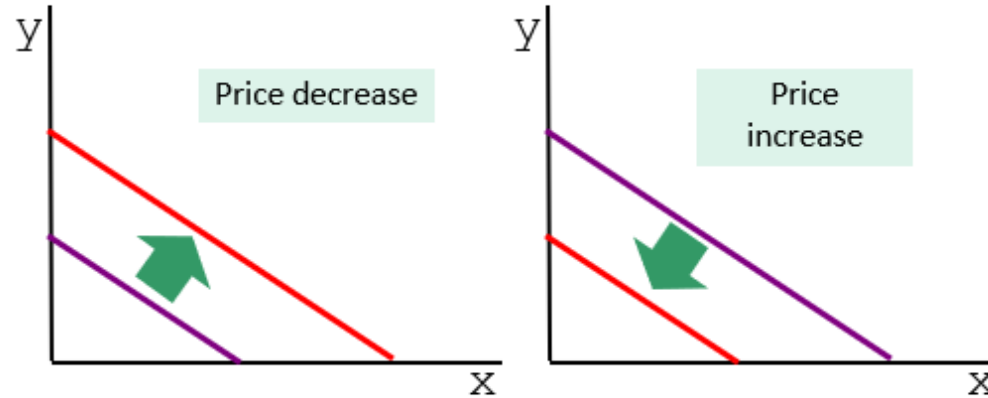
(2) Due to variations in one of the prices



BUDGET CONSTRAINTS

Changes in the budget line

(3) Equal and simultaneous variation in the two prices.



BUDGET CONSTRAINTS

Question:

With the same information that we had about Pepito, now we know that the price of movies is now 4€ and the price of books 25€. Represent graphically the changes in Pepito's budget constraint.

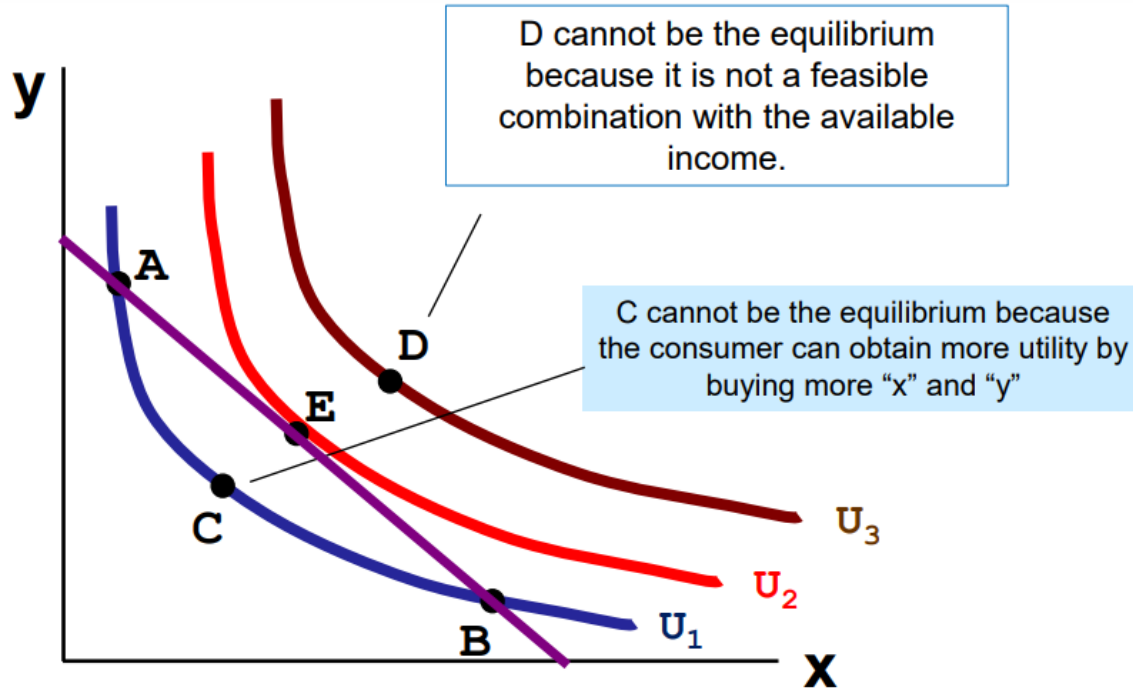
CONSUMER CHOICE

Utility Maximization and Consumer Equilibrium

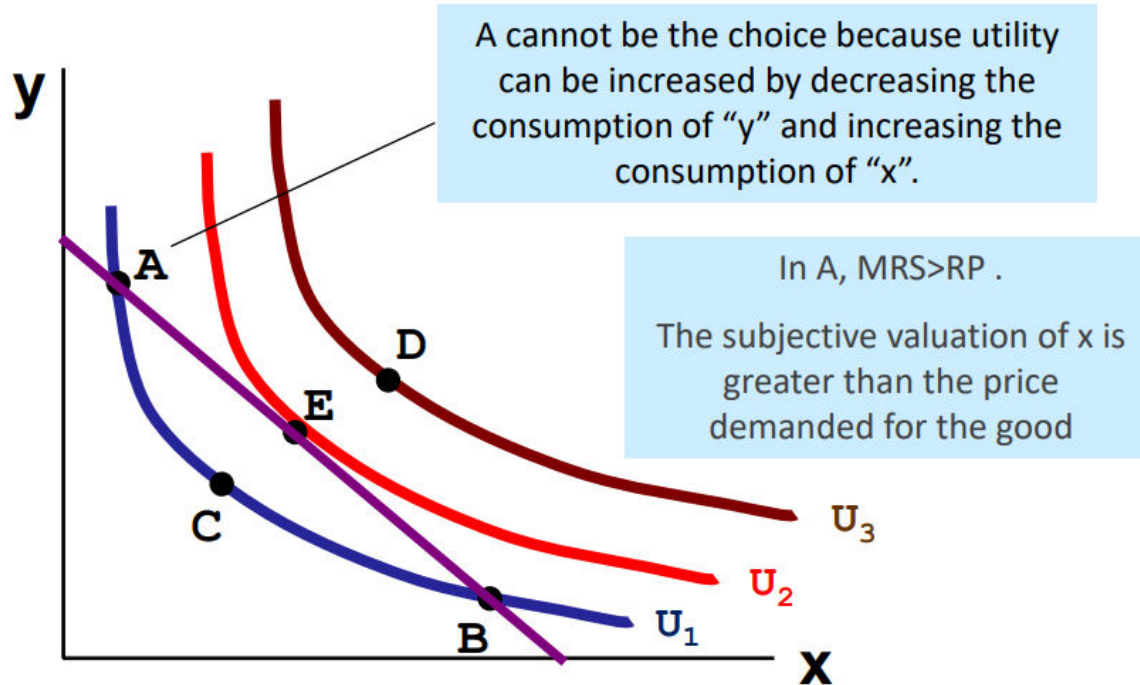
The consumer tries to find the **most preferred combination (x,y)** within the range of consumption possibilities.

The result will be that the preferred combination (the **EQUILIBRIUM**) will be obtained at the **point of tangency** between an indifference curve and the budget constraint line.

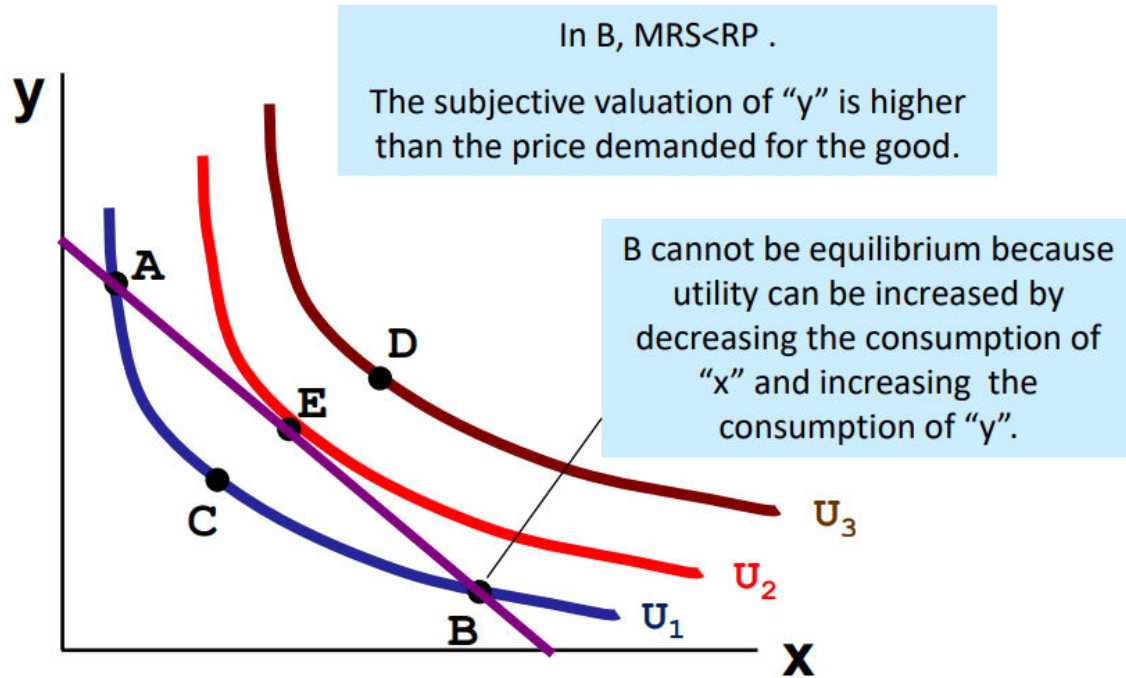
CONSUMER CHOICE



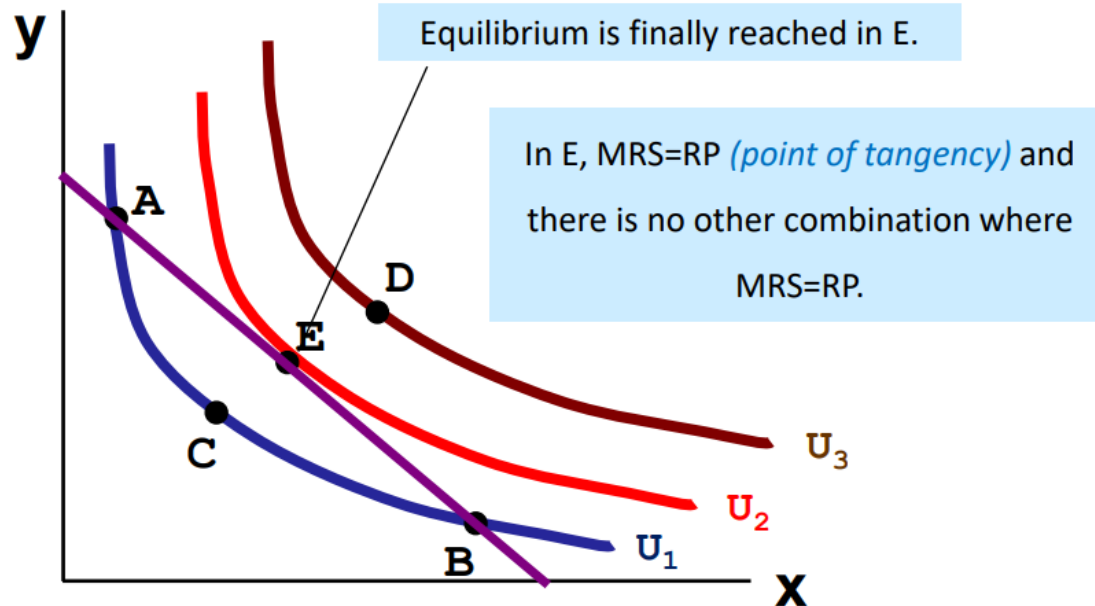
CONSUMER CHOICE



CONSUMER CHOICE

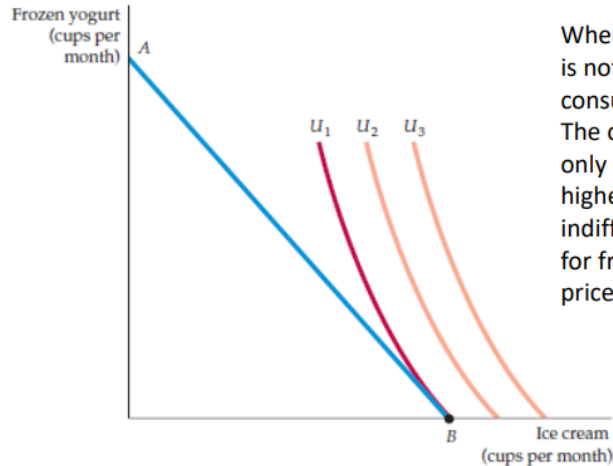


CONSUMER CHOICE



CONSUMER CHOICE

Corner Solution: Situation in which the marginal rate of substitution of one good for another in a chosen market basket is not equal to the slope of the budget line.

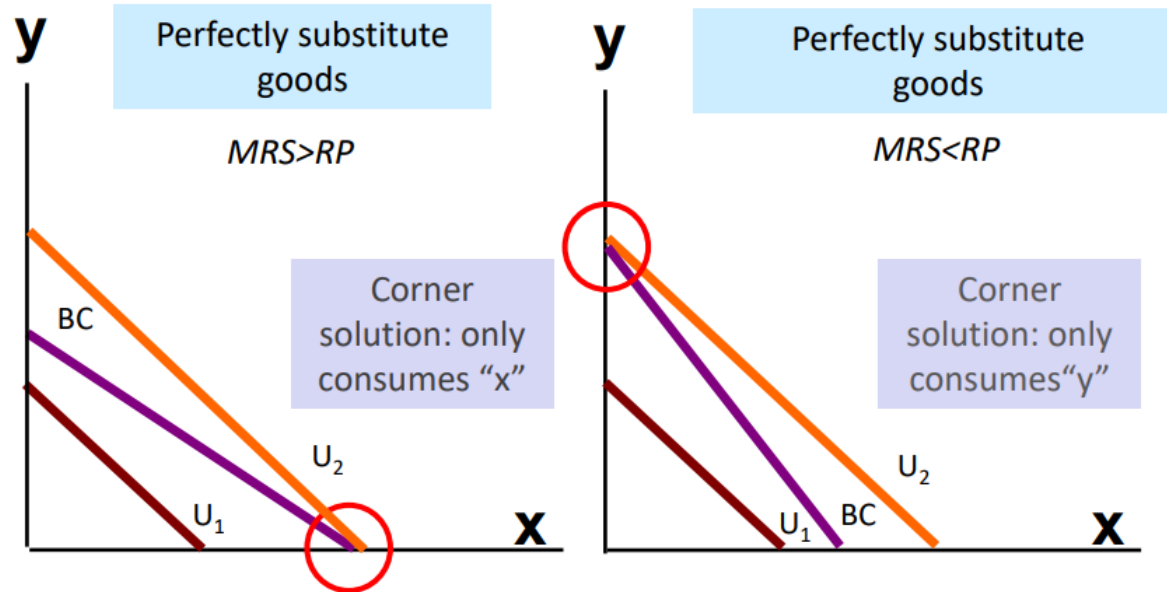


When the consumer's marginal rate of substitution is not equal to the price ratio for all levels of consumption, a corner solution arises. The consumer maximizes satisfaction by consuming only one of the two goods. Given budget line AB , the highest level of satisfaction is achieved at B on indifference curve U_1 , where the MRS (of ice cream for frozen yogurt) is greater than the ratio of the price of ice cream to the price of frozen yogurt.

SOURCE: Pindyck, R. S., Rubinfeld, D. L., & Mehta, P. L. (1995). *Microeconomics* (Vol. 4). Englewood Cliffs, NJ: Prentice Hall.

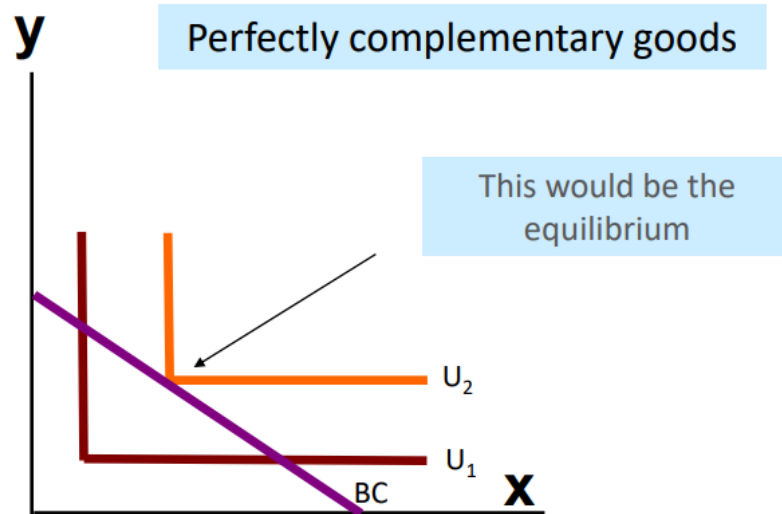
CONSUMER CHOICE

Particular cases in obtaining equilibrium.



CONSUMER CHOICE

Particular cases in obtaining equilibrium.



UTC 2 “A” EXERCISES (CANVAS): 7,8,9,10,11



THANK YOU!





MICROECONOMICS



UTC1 INTRODUCTION TO MICROECONOMICS

UTC1 INTRODUCTION TO MICROECONOMICS

- ❑ Microeconomics and Macroeconomics.
- ❑ Remembering the Supply and Demand Model.
- ❑ Elasticities.
- ❑ Consumer and Producer Surplus.

UTC1 INTRODUCTION TO MICROECONOMICS

☐ **Microeconomics and Macroeconomics.**

☐ Remembering the Supply and Demand Model.

☐ Elasticities.

☐ Consumer and Producer Surplus.

What is Economics?

Economics is a study of:

- how people use their time and material resources to improve their own well-being
- how our economic decisions have a broader impact that extends beyond the individual
- of the effect that government policies have on our economic decisions/our economic activity

As **Alfred Marshall** put it, economics is “a study of mankind in the ordinary business of life”.

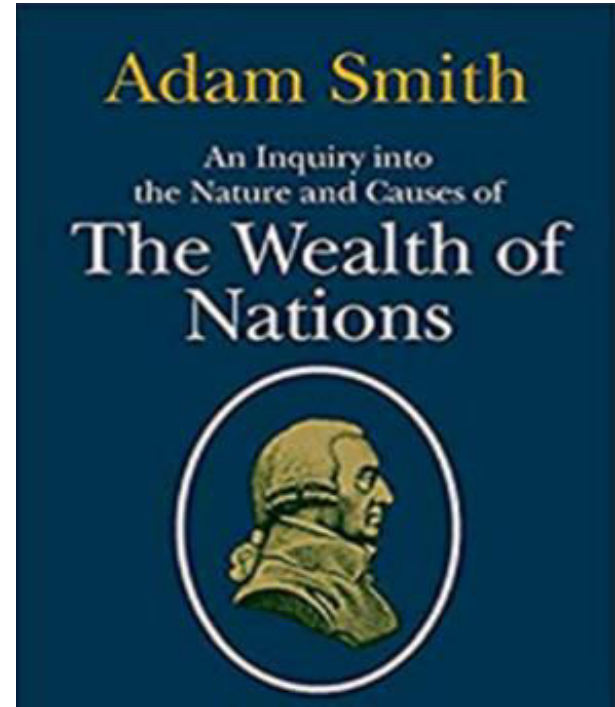
What is Economics?

Economics is a **social science** that studies the production, distribution and consumption of goods and services. It studies how individuals, businesses, governments, and nations make choices about how to allocate resources to satisfy their wants and needs.

- **Economics examines human behaviour that is motivated by self-interest and is concerned with how we use our time and material resources to make ourselves better off.**
- **Economics is the study of how society manages its scarce resources.**

What is Economics?

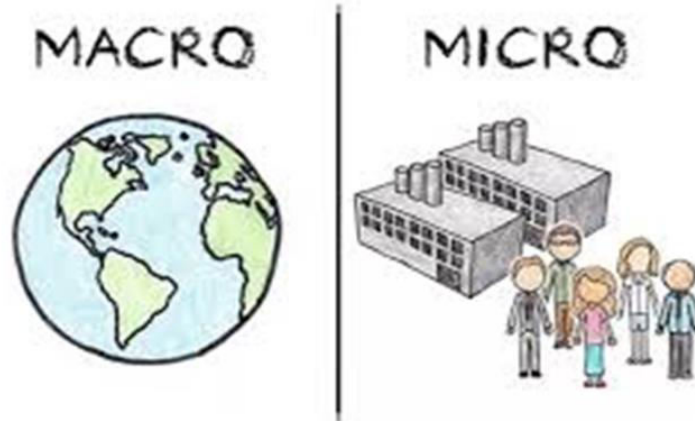
Adam Smith's 1776 book "An Inquiry into the Nature and Causes of the Wealth of Nations" is regarded as the beginning of economics as a discipline.



Microeconomics vs Macroeconomics

The study of economics can be subcategorized into two disciplines:

- Microeconomics
- Macroeconomics



Microeconomics vs Macroeconomics

Microeconomics is the study of how individuals make decisions and how these decisions interact. Individual agents may include: households, firms, buyers, and sellers.

Microeconomics includes concepts such as:

- ✓ supply and demand,
- ✓ price elasticity,
- ✓ utility,
- ✓ market structures, etc.



Microeconomics vs Macroeconomics

Macroeconomics is the study of the performance and structure of the whole economy rather than individual markets.

It includes concepts such as:

- ✓ inflation,
- ✓ international trade,
- ✓ unemployment,
- ✓ and national consumption, etc.



Microeconomics vs Macroeconomics



Microeconomics vs Macroeconomics

What type of issue is it? Micro vs Macro.

- A consumer's decision to purchase a laptop (i.e., what makes a consumer prefer one model to the other).
- Business decisions to introduce a new mobile phone (i.e., what resources will be required to produce the product).
- Explain the nature and causes of unemployment in a country.
- Business decisions about the cost of producing a new model of a car (i.e., what types of costs will be incurred).
- How can inflation be controlled?

Microeconomics vs Macroeconomics

Question:

Identify if the following news correspond to Microeconomics or Macroeconomics

CORONAVIRUS IMPACT



Spain's economy shrank 11% in 2020, in biggest drop since Civil War

ANTONIO MAQUEDA | Madrid | 29-01-2021 - 11:48 CET

Output grew 0.4% in the fourth quarter but it was not enough to compensate for the damage caused by the coronavirus pandemic

Microeconomics vs Macroeconomics

Question:

Identify if the following news correspond to Microeconomics or Macroeconomics

BREXIT



In Spain, old Franco law hampers post-Brexit house sales to Britons

RAFA BURGOS | Alicante | 25-01-2021 - 08:39 CET

A 1975 measure that still endures will now affect UK property buyers as non-EU residents in Spain, according to estate agents in the south of Alicante province

Microeconomics vs Macroeconomics

Question:

Identify if the following news correspond to Microeconomics or Macroeconomics

BANKING SECTOR



BBVA-Sabadell merger talks cap decade of forced consolidation for Spanish banks

ECONOMY & BUSINESS



PRISA refinances its debt until 2025 and agrees to sell Santillana Spain for €465m

Scarcity and choice

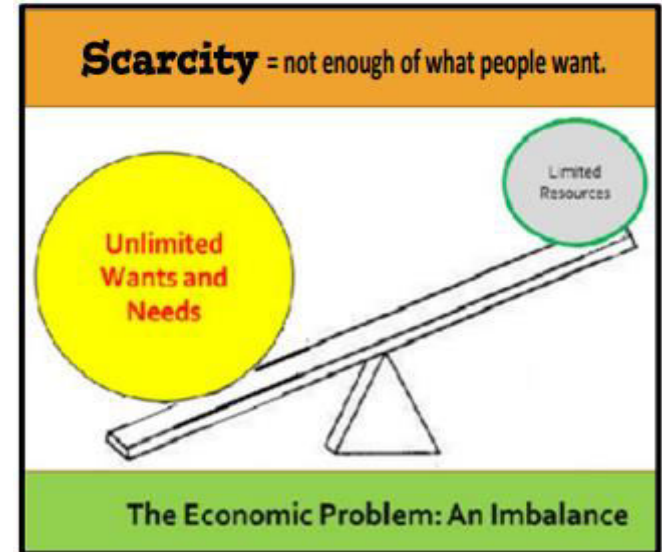
An economy has:

- limited resources at any moment
- unlimited human wants

Therefore, because of **scarcity**, choices have to be made.

“There ain't no such thing as a free lunch”

The issue of scarcity and choice lies at the heart of economics.



Scarcity and choice

- **Scarcity** means that society has limited resources and therefore cannot produce all the goods and services people wish to have. Not enough of the resources are available to satisfy everyone's wants and needs.
- Because resources are scarce people must make **choices**.
- Making decisions requires **comparing the costs and benefits of alternative courses of action**. The real **cost of your decision** is called **OPPORTUNITY COST** - the **value of your next best alternative choice**.

Productive resources (factors of production)



Land



Labour



Capital



Enterprise

In every society, people require goods and services which will enable them to survive and prosper.

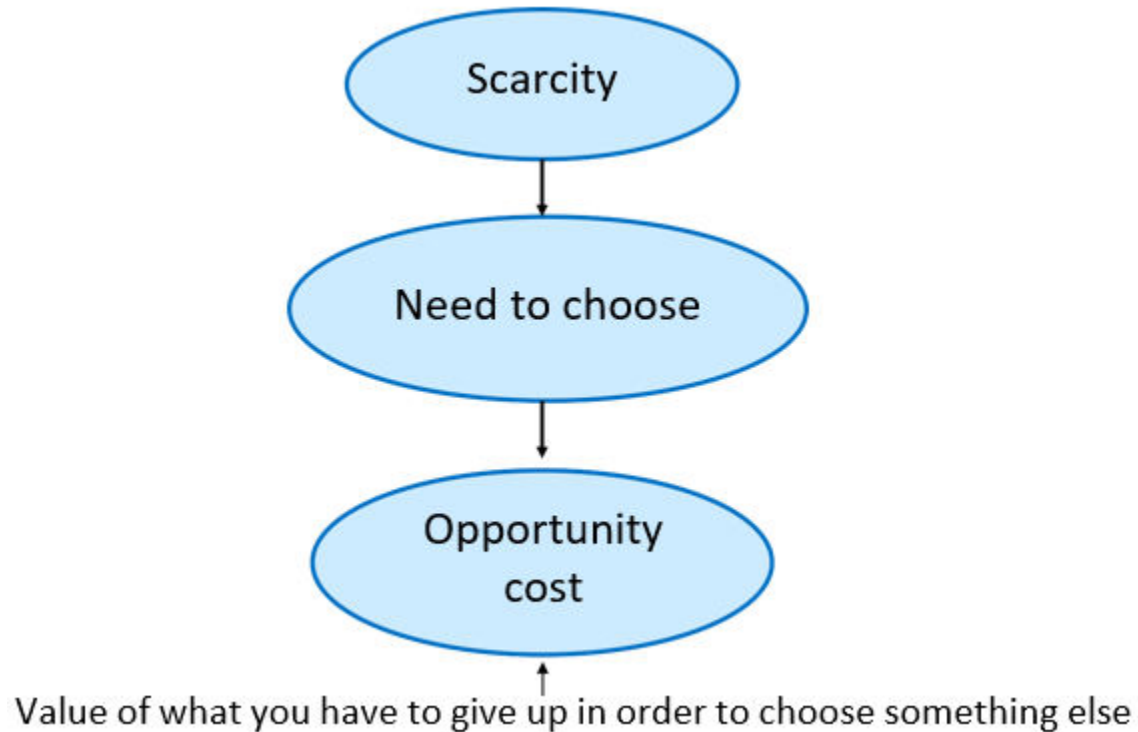
Every society is endowed with resources which can be used to produce the goods and services people require.

A **resource** is anything that can be used to produce something else.

The factors of production are rewarded for their contribution to the production of goods and services:

- the reward for land is **rent**,
- the reward for labour it is **wage**,
- the reward for capital it is **interest**,
- the reward for the entrepreneurship is called **profit**.

Scarcity and choice



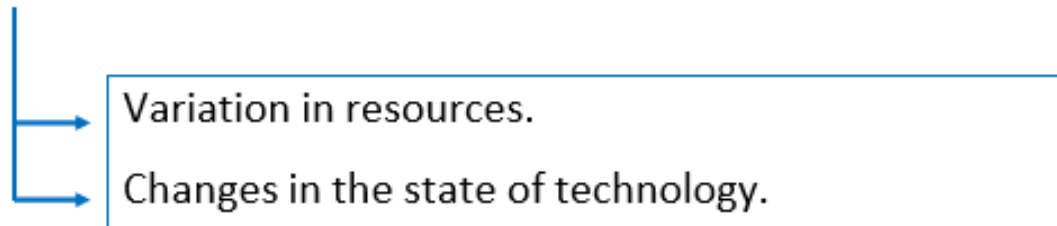
The Production Possibilities Frontier (PPF)

Reflects all the options available to society and the need to choose among them.

The PPF illustrates several essential aspects:

1. Resources are scarce.
2. There is an opportunity cost.
3. Given the resources, the economies can reach a potential production.

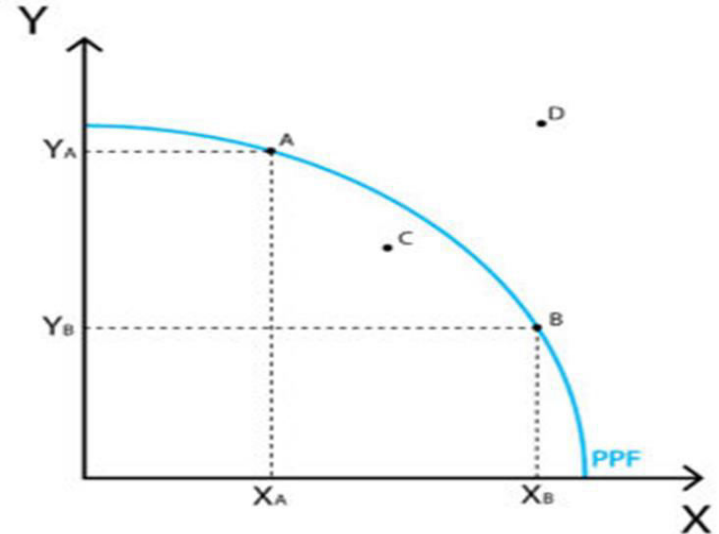
If the potential production of an economy is modified, the PPF shifts.



The Production Possibility Frontier

The production possibility frontier (PPF)

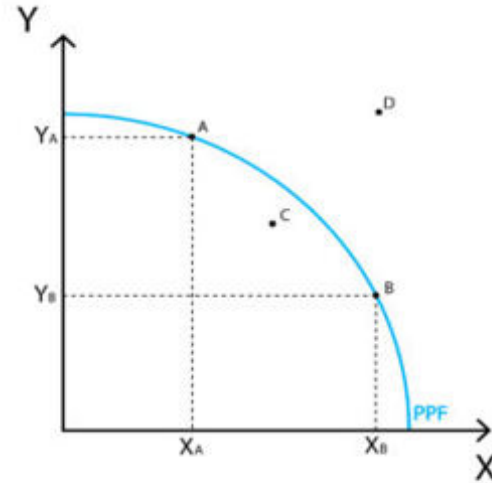
- ✓ Illustrates the trade-offs faced by an economy that produces only two goods.
- ✓ It shows the maximum quantity of one good that can be produced for any given quantity produced of the other.
- ✓ This is a simple economic model which is used to show how a society chooses to allocate scarce resources between competing uses.



What to Know about the PPF

PPF shows the **combination of output**, in this case, good X and good Y, that the economy can possibly produce.

- ✓ The economy can produce any **combination on or inside the frontier**.
- ✓ Points **on the PPF (A and B)** are **feasible and efficient in production**.
- ✓ Points **inside the frontier (C)** represent an **inefficient outcome**. The economy is producing less than it could from the available resources.
- ✓ Points **outside the frontier (D)** are **not feasible** given the economy's resources.



What to Know about the PPF

The Production Possibilities Frontier illustrates some of the key economic concepts:

- Efficiency
- Scarcity
- Trade-offs and choice
- Opportunity cost

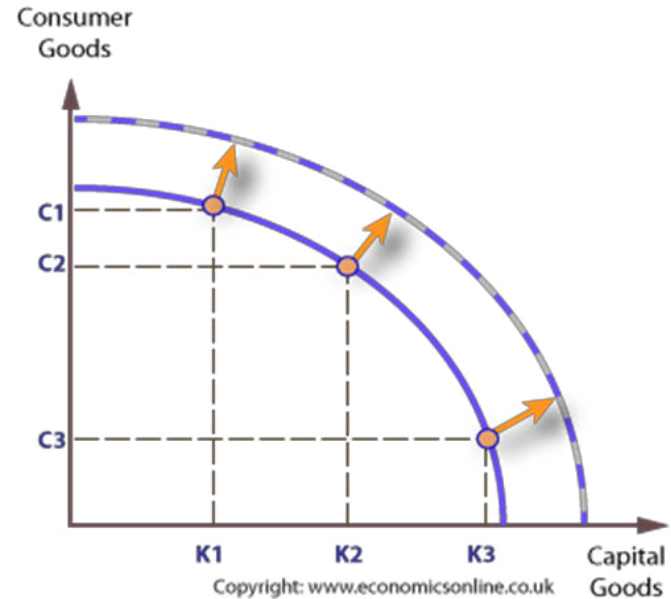
EFFICIENCY	SCARCITY	TRADE-OFFS	OPPORTUNITY COST
With the resources it has, the economy can produce at any point on the PPF , which is feasible and efficient in production (for example point A or point B). Point C represents an inefficient outcome. The economy is producing less than it could from the available resources. Any point outside the curve like point D is unattainable due to insufficient resources.	Because resources are scarce , not every outcome is feasible. No matter how resources are allocated between the two industries, the economy cannot produce the amount of goods X and Y represented by point D . Given the technology available for manufacturing goods X and Y, the economy does not have enough of the factors of production to support that level of output. Production at point D is not feasible .	Once we have reached an efficient point on the frontier, the only way of producing more of one good is to produce less of another. When the economy moves from point A to B, its production of X increases at the expense of producing less Y.	It shows the opportunity cost of one good measured in terms of the other good . It is the specific number of good Y I have to give up, to produce one unit of good X.

What causes an outward shift in the PPF?

Economic growth (the growing ability of the economy to produce goods and services) results in an outward shift of the Production Possibilities Frontier because production possibilities are expanded.

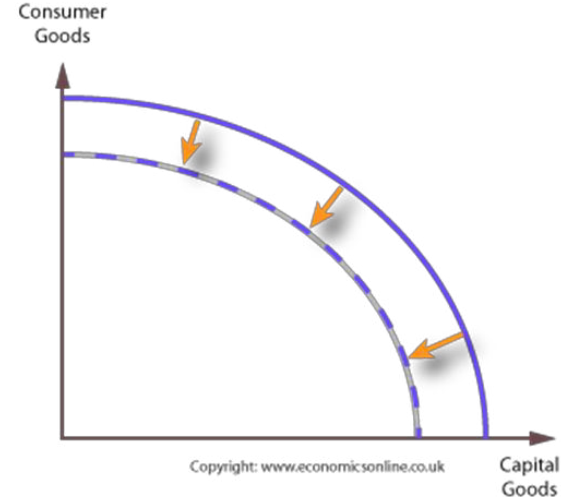
There are two sources of economic growth:

- An **increase in the quantity or quality of factors of production** (the resources to produce goods and services).
- Progress in technology** (the technical means for the production of goods and services).



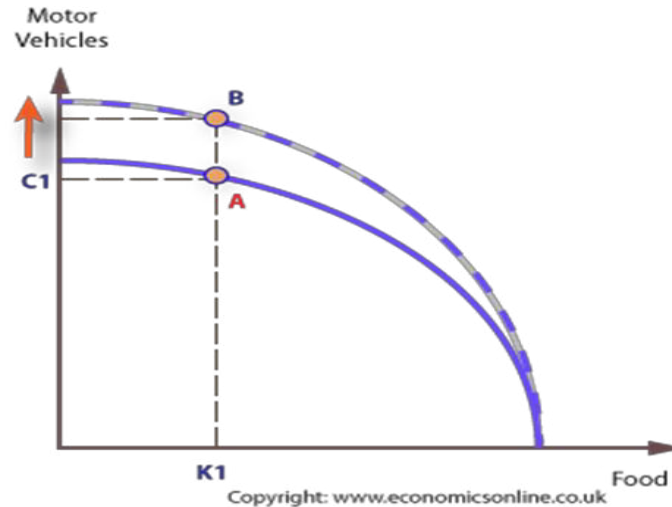
What causes an inward shift in the PPF?

A PPF will shift **inwards** when an economy has suffered a loss or exhaustion of some of its scarce resources. This reduces an economy's productive potential.



What causes an asymmetric shift in the PPF?

An economy can grow because of an increase in productivity in one sector of the economy - this is called **asymmetric growth**.

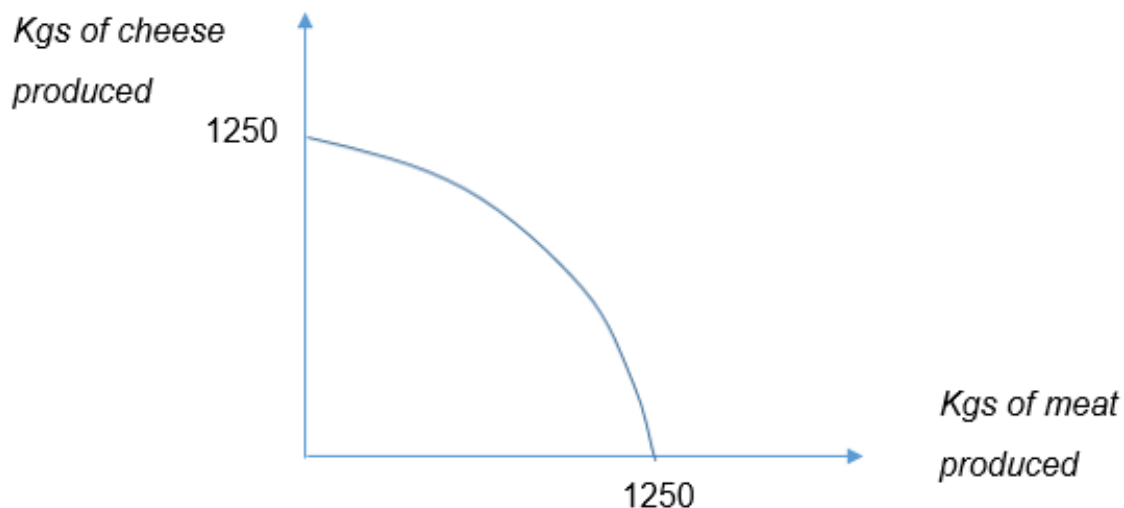




Question:

In the following PPF show two points in which the economy is being efficient, two in which it is being inefficient and two that currently are not possible to reach.

REMEMBER



Decide how the PPF would shift?

1. An economy employs new technology
2. Erosion of infrastructure
3. An economy employs new production methods
4. An economy increases its labour force
5. Factor mobility from one industry to the other
6. An economy discovers new raw materials
7. Resources run out
8. An economy employs a division of labour, allowing specialisation
9. Natural disaster
10. An improvement in technology applied to industry Y, such as motor vehicles, but not to X such as food production
11. Failure to invest

UTC1 EXERCISES (CANVAS): 1 and 2

UTC1 INTRODUCTION TO MICROECONOMICS

- ❑ Microeconomics and Macroeconomics.
- ❑ **Remembering the Supply and Demand Model.**
- ❑ Elasticities.
- ❑ Consumer and Producer Surplus.

UTC1 INTRODUCTION TO MICROECONOMICS

A model is a simple theoretical description that captures the essential elements about how the economy works.

And what is a market? It is an organization or system where exchange agreements for various goods or services are produced between buyers and sellers.

<i>Agents participating in the market</i>	<i>Goal (it guides its behavior)</i>
<i>Households or families</i>	<i>Maximize welfare</i>
<i>Companies / Firms</i>	<i>Maximize profit</i>
<i>Public organizations</i>	<i>Maximize social welfare</i>

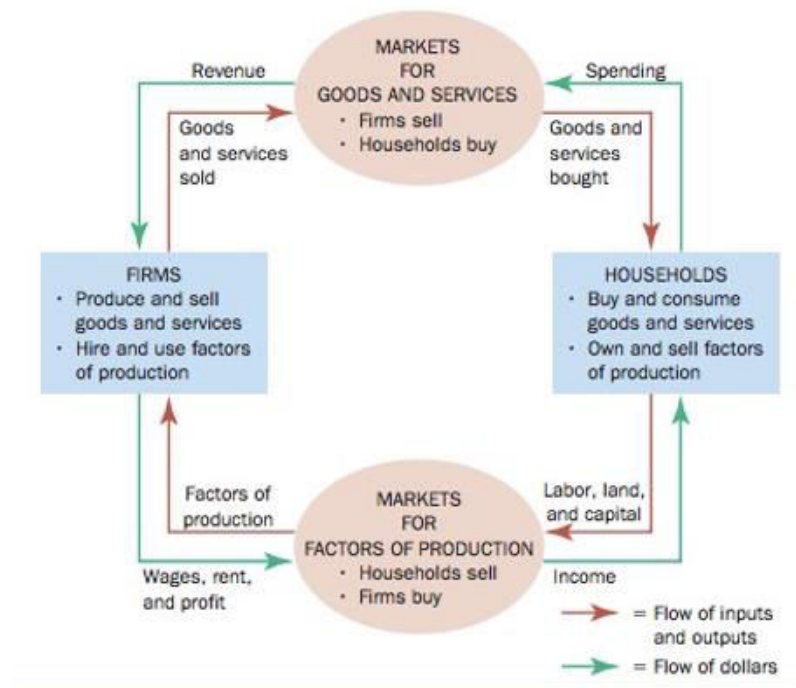
UTC1 INTRODUCTION TO MICROECONOMICS

We differ between two basic markets:

- *Factors of production market.*
- *Goods and services market.*

	<i>Goods and services market</i>	<i>Factors of production market</i>
<i>Households</i>	<i>CLAIMANTS</i>	<i>PROVIDERS</i>
<i>Companies / Firms</i>	<i>PROVIDERS</i>	<i>CLAIMANTS</i>

UTC1 INTRODUCTION TO MICROECONOMICS



Demand Curve

Demand function:

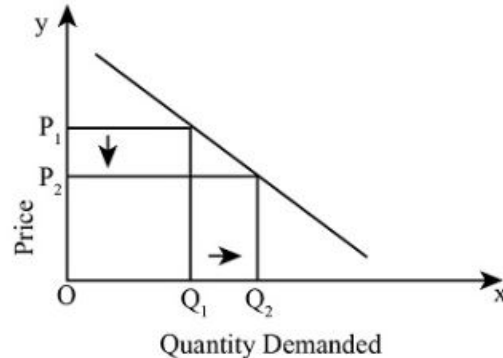
$$Q_{dx} = f(P_x, P_y, \dots P_z, I, T, E, Nb)$$

Ceteris Paribus

P_x : price of good x

The **law of demand** says (**ceteris paribus** = other things being equal, remaining the same): as the price of a good rises, the quantity demanded declines and vice versa.

$$Q_{dx} = f(P_x)$$



- The **buyers**, as a group, determine the demand for the product or service.
- The behaviour of buyers is described by **the law of demand**.

P_y, P_z : price of related products (substitutes, complements)

I : Income (for normal goods, for inferior goods)

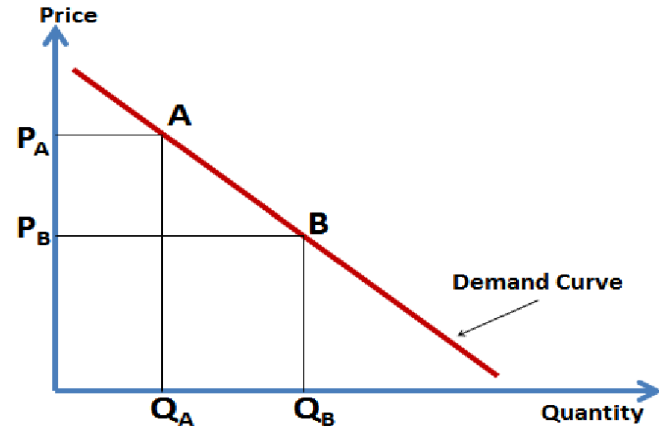
T : Tastes and Preferences for that good (impact of marketing efforts)

E : Expectation of future price

Nb : Number of buyers

Demand Curve

The quantity demanded: is the actual amount of a good or service consumers are willing to buy at some specific price (it is associated with a particular point on the demand curve).

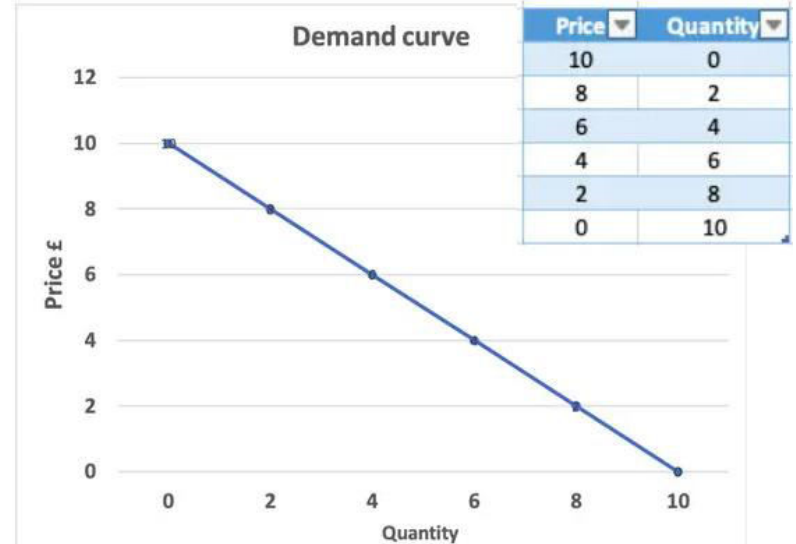


Demand: is not a particular quantity, but **a full description of the quantity** of a good or service buyers would like to purchase at each and every price (it is the set of all points on the demand curve that show how price and quantity demanded are associated) and is illustrated by the demand curve.

Demand Curve

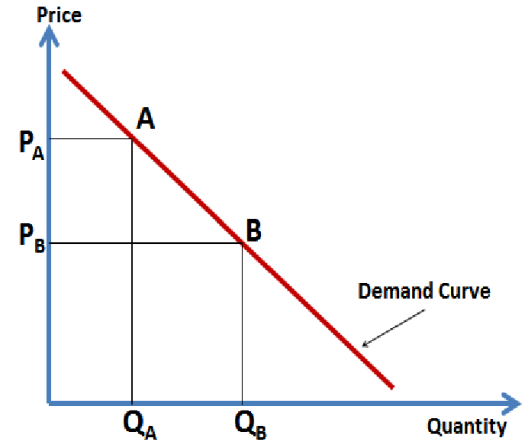
A demand curve: is a graphical representation of the relationship between the price of a good or service and the quantity demanded.

A demand schedule: a table that shows the relationship between the price of a good or service and the quantity demanded (how much of a good or service consumers will want to buy at different prices).



Demand Curve

Graph 1 shows a **negative** or **inverse relationship** between the **price** of a good or service and the **quantity demanded** of that good or service, *ceteris paribus*.



On the **vertical axis** (independent variable) we **plot the price of a good or service** and on the **horizontal axis** (dependent variable) the **quantity demanded of that good or service**. We consider a linear demand curve as it is simpler to analyse than a non-linear curve.

Shifts of the Demand Curve

Change in the quantity demanded vs change in demand:

- **Change in quantity demanded:** a change in the quantity of a good that people plan to buy that results from a change in the price of the good.
- **Change in demand (shifts of the demand curve):** a change in the quantity that people plan to buy when any influence other than the price of the good changes.

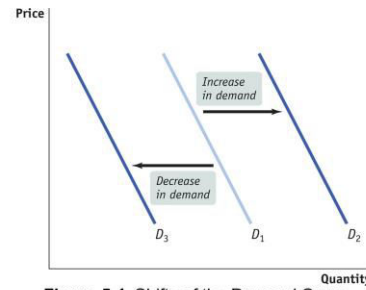
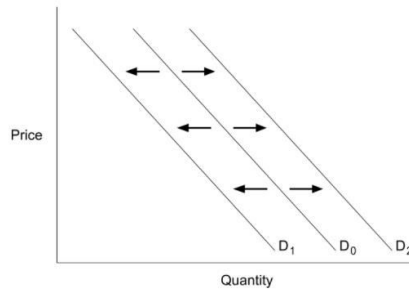
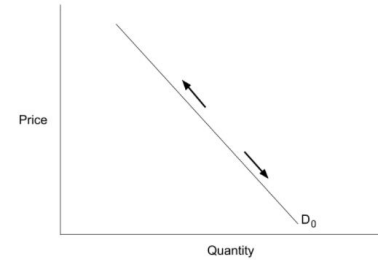


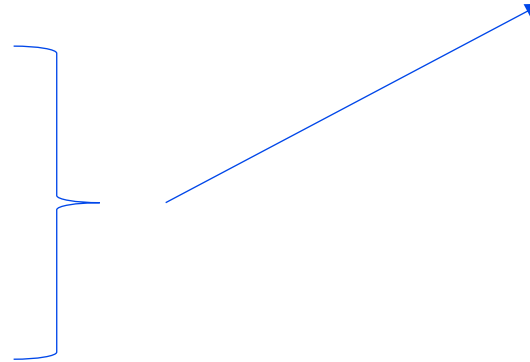
Figure 5.4 Shifts of the Demand Curve
Ray and Anderson: Krugman's Economics for AP, First Edition
Copyright © 2010 W. H. Freeman & Co.

Shifts of the Demand Curve

There are five **principal factors** that shift the demand curve for a good or service:

- Changes in the prices of related goods or services
- Changes in income
- Changes in tastes
- Changes in expectations
- Changes in the number of consumers

$$Q_{dx} = f(P_x, P_y, \dots P_z, I, T, E, N_b)$$



Shifts of the Demand Curve

Changes in the prices of related goods or services

Two goods are **substitutes** (substitute: a good that can be consumed in place of another good e.g., muffins and doughnuts).

When the price of a substitute rises, demand for the original good increases, shifting the demand curve to the right.

When the price of a substitute falls, demand for the original good decreases, shifting the demand curve to the left.

Substitutes	
Price of a substitute ↗	Demand ↗
Price of a substitute ↘	Demand ↘

Shifts of the Demand Curve

Changes in the prices of related goods or services

Two goods are **complements** (complement: a good that is consumed with another good e.g., computers and software).

When the price of a complement falls, demand for the original good increases, shifting the demand curve to the right.

When the price of a complement rises, demand for the original good decreases, shifting the demand curve to the left.

Complements	
Price of a complement $\nearrow$	Demand $\searrow$
Price of a complement $\searrow$	Demand $\nearrow$

Shifts of the Demand Curve

Changes in income

Most goods are **normal goods**, the demand for them increases when consumer income rises.

When income rises, the demand for a normal good increases, shifting the demand curve to the right.

When income falls, the demand for a normal good decreases, shifting the demand curve to the left.

Normal Goods	
Income ↗	Demand ↗
Income ↘	Demand ↘

Shifts of the Demand Curve

Changes in income

Goods for which demand decreases when income rises are known as **inferior goods** (less desirable goods than more expensive alternatives e.g. a bus ride vs a taxi ride).

When income falls, the demand for an inferior good increases, shifting the demand curve to the right.

When income rises, the demand for an inferior good decreases, shifting the demand curve to the left.

Inferior Goods	
Income ↗	Demand ↘
Income ↘	Demand ↗

Shifts of the Demand Curve

Changes in tastes

People have certain preferences, or tastes, that determine what they choose to consume, and these tastes can change. Preferences change when:

- people become better informed
- new goods become available

When tastes change in favour of a good, more people want to buy it at any given price, so the demand for the good increases, shifting the demand curve to the right.

When tastes change against a good, fewer people want to buy it at any given price, so the demand for the good decreases, shifting the demand curve to the left.

Tastes and Preferences	
Popularity ↗	Demand ↗
Popularity ↘	Demand ↘

Shifts of the Demand Curve

Changes in expectations

When **the price is expected to rise in the future**, the demand for the good increases today, **shifting the demand curve to the right**.

When **the price is expected to fall in the future**, the demand for the good decreases today, **shifting the demand curve to the left**.

Future Expectations	
Expectations of Future Prices ↗	Demand ↗
Expectations of Future Prices ↘	Demand ↘

Shifts of the Demand Curve

Changes in the number of consumers

The market demand curve shows how the combined quantity demanded by all costumers depends on the market price of the good. The market demand curve is the horizontal sum of the individual demand curves of all consumers in the market.

An individual demand curve shows the relationship between quantity demanded and price for an individual consumer.

When the **number of consumers rises**, the **market demand for the good increases**, shifting the demand curve to the right.
When the **number of consumers falls**, the **market demand for the good decreases**, shifting the demand curve to the left.

Number of Buyers in the Market

Number of Customers ↗	Demand ↗
Number of Customers ↘	Demand ↘



GO TO CANVAS:
**INTERACTIVE PRACTICE: change in
demand vs change in quantity
demanded**

Supply Curve

Supply function:

$$Q_{sx} = f(P_x, PF_1, \dots, PF_n, P_y, \dots, P_z, T, E, N_p)$$

Ceteris Paribus

P_x : price of good x

The **law of supply** says (**ceteris paribus** = other things being equal, remaining the same): as the price of a good rises, the quantity supplied of that good increases and vice versa.

The law of supply: $Q_{sx} = f(P_x)$

- The **sellers**, as a group, determine the supply of the product or service.
- The behaviour of sellers is described by **the law of supply**.

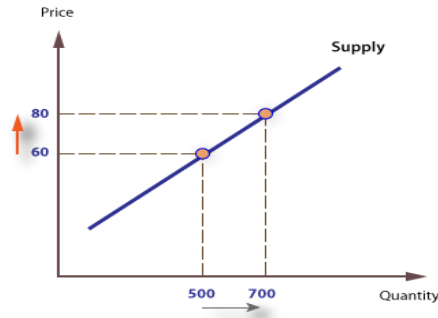
$PF_1, \dots, PF_n$: the price of factors of production/inputs

$P_y, \dots, P_z$: prices of related goods or services in production

T : Technology

E : Expectation of future price

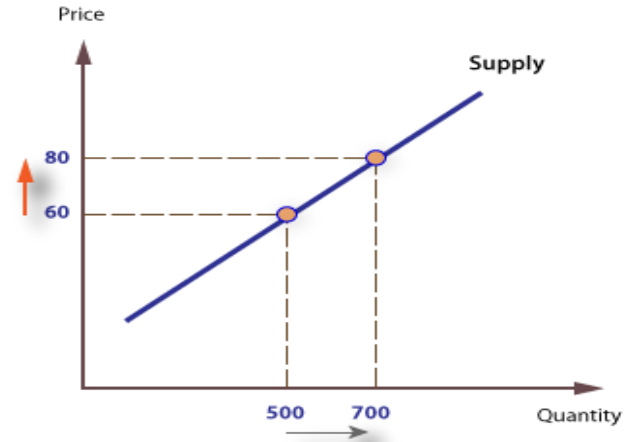
N_p : Number of producers/sellers



Supply Curve

The quantity supplied: is the actual amount of a good or service, people are willing to sell at some specific price.

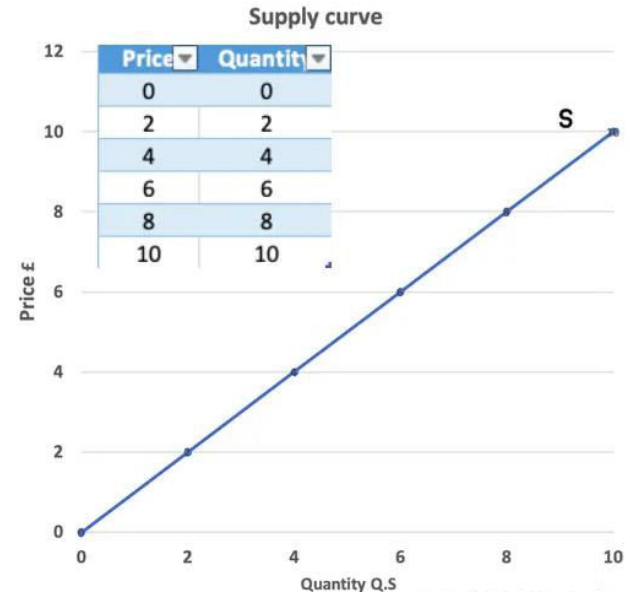
Supply: is not a particular quantity but a complete description of the quantity that sellers want to sell at each possible price.



Supply Curve

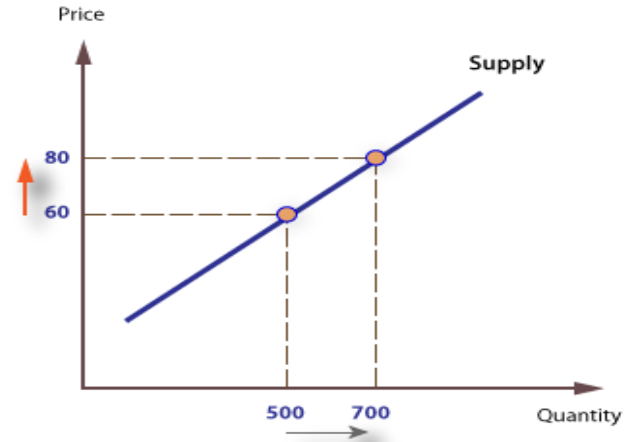
A supply curve: shows the relationship between the quantity supplied and price.

A supply schedule: a table which shows how much of a good or service would be supplied at different prices.



Supply Curve

Graph 2 shows a **positive** or **direct** relationship between the **price** of a good or service and the **quantity supplied** of that good or service, ceteris paribus.



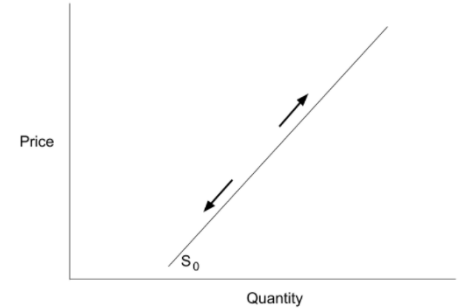
On the **vertical axis** (independent variable) we **plot the price of a good or service** and on the **horizontal axis** (dependent variable) the **quantity supplied of that good or service**. We consider a linear supply curve as it is simpler to analyse than a non-linear curve.

Shifts of the Supply Curve

Change in the quantity supplied vs change in supply:

- **Change in quantity supplied:**

a change in the quantity of a good that suppliers plan to sell that results from a change in the price of the good.



- **Change in supply (shifts of the supply curve):** a change in the quantity that suppliers plan to sell when any influence on selling plans other than the price of the good changes.

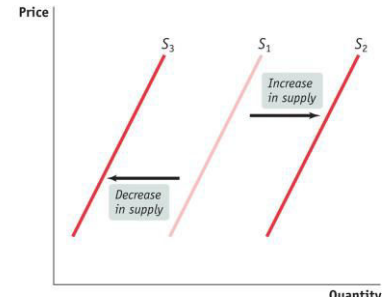
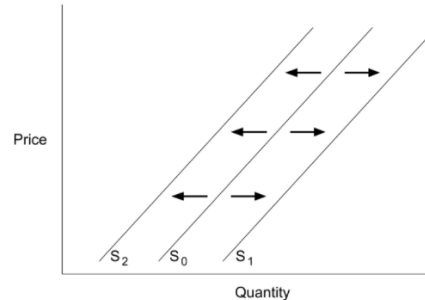


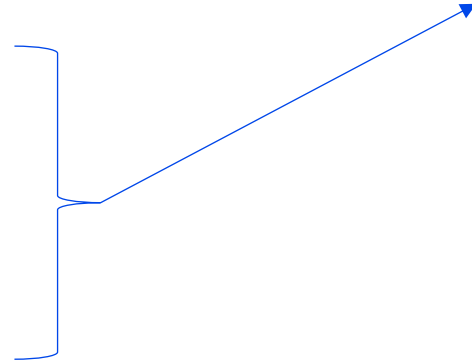
Figure 6.4 Shifts of the Supply Curve
Ray and Anderson: Krugman's Economics for AP, First Edition
Copyright © 2011 by Worth Publishers

Shifts of the Supply Curve

There are five **principal factors** that shift the supply curve for a good or service:

$$Q_{sx} = f(P_x, PF_1, \dots PF_n, P_y, \dots P_z, T, E, N_p)$$

- Changes in input prices
- Changes in the prices of related goods or services (in production)
- Changes in technology
- Changes in expectations
- Changes in the number of producers/sellers



Shifts of the Supply Curve

Changes in input prices

An input is any good or service that is used to produce another good or service.

When the **price of an input** (used to produce a good) **rises**, the **supply** of the good **decreases**, shifting the supply curve to the **left**.

When the **price of an input** (used to produce a good) **falls**, the **supply** of the good **increases**, shifting the supply curve to the **right**.

Changes in Input prices			
Price of Input	↗	Supply	↘
Price of Input	↘	Supply	↗

Shifts of the Supply Curve

Changes in the prices of related goods or services

A single producer often produces a mix of goods from the same resources rather than a single product.

- **Substitute in production:** a good that can be produced in place of another good. For example, a truck and an SUV in an auto factory or gasoline and heating oil (produced from the same raw material: crude oil in an oil refinery).

When the **price of a substitute in production falls**, the **supply** of the **original good increases**, shifting the supply curve for the original good to the **right**.

When the **price of a substitute in production rises**, the **supply** of the **original good decreases**, shifting the supply curve for the original good to the **left**.

Substitute in production			
Price of Substitute	↗	Supply	↘
Price of Substitute	↘	Supply	↗

Shifts of the Supply Curve

Changes in the prices of related goods or services

- **Complement in production:** a good that is produced along with another good. For example, producers of natural gas often find that natural gas wells also produce oil as a by-product of extraction.

When the **price of a complement in production rises**, the **supply** of the **original good increases**, shifting the supply curve for the original good to the **right**.

When the **price of a complement in production falls**, the **supply** of the **original good decreases**, shifting the supply curve to the **left**.

Complement in production			
Price of Complement	↗	Supply	↗
Price of Complement	↘	Supply	↘

Shifts of the Supply Curve

Changes in technology

When the **technology** used to produce the good **improves**, **supply** of the good **increases**, shifting the supply curve to the **right**.

When the best **technology** used to produce the good is **no longer available**, **supply** of the good **decreases**, shifting the supply curve to the **left**.

Technology			
Technology	↗	Supply	↗
Technology	↘	Supply	↘

Shifts of the Supply Curve

Changes in expectations

When the **price is expected to fall in the future**, the **supply** of the good **increases today**, shifting the supply curve to the **right**.

When the **price is expected to rise in the future**, the **supply** of the good **decreases today**, shifting the supply curve to the **left**.

Future Expectations			
Expectations of Future Prices	↗	Supply	↘
Expectations of Future Prices	↘	Supply	↗

Shifts of the Supply Curve

Changes in the number of producers

When the **number of producers rises**, the market **supply** of the good **increases**, shifting the supply curve to the **right**.

When the **number of producers falls**, the market **supply** of the good **decreases**, shifting the supply curve to the **left**.

Number of Producers in the Market			
Number of Producers	↗	Supply	↗
Number of Producers	↘	Supply	↘

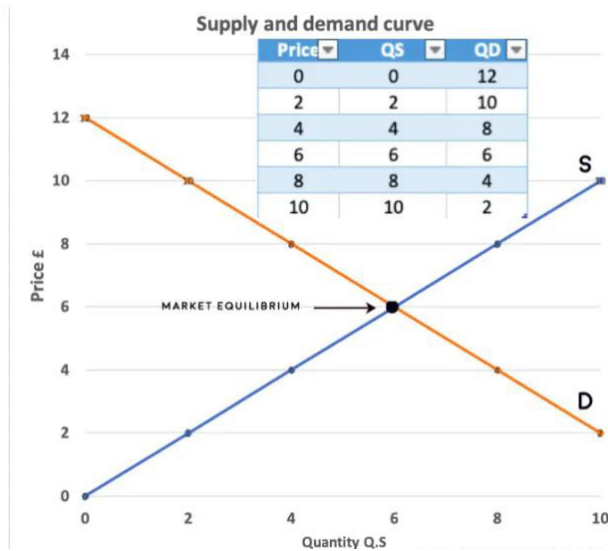


GO TO CANVAS:
**INTERACTIVE PRACTICE: change in
supply vs change in quantity supplied**

Market Equilibrium

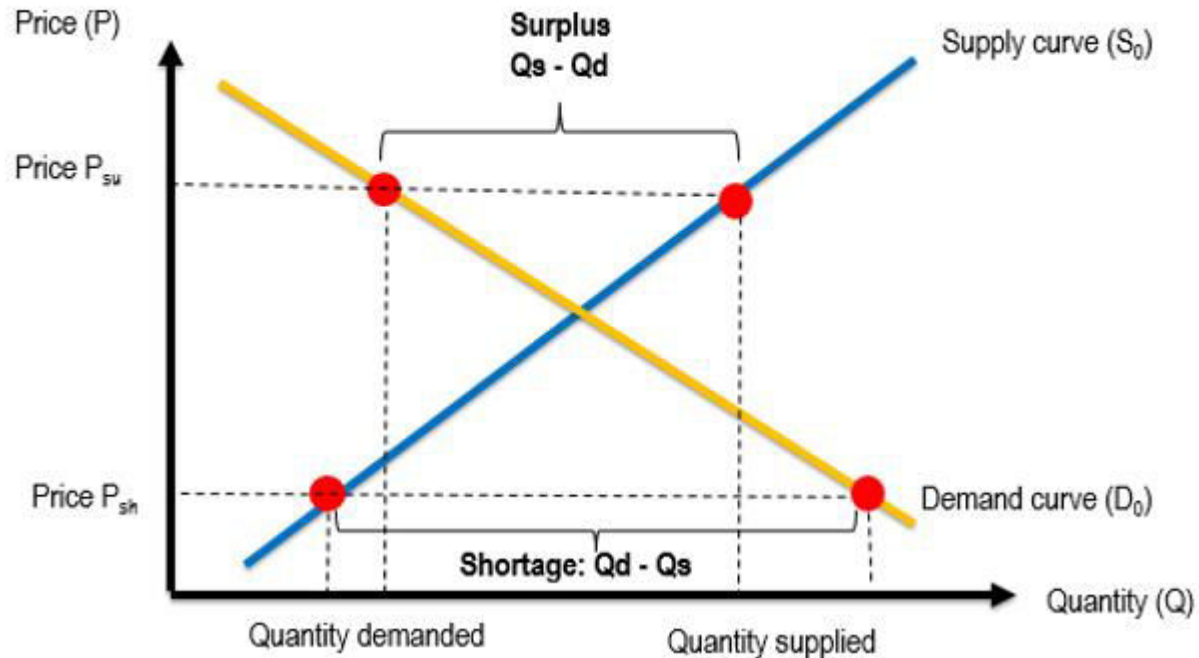
The market equilibrium includes the **equilibrium price** and **equilibrium quantity**

Market equilibrium



Market Equilibrium

Adjustments towards equilibrium





Question

Complete the following table with the effects of each of the shifts of supply and demand on the equilibrium price and the equilibrium quantity:

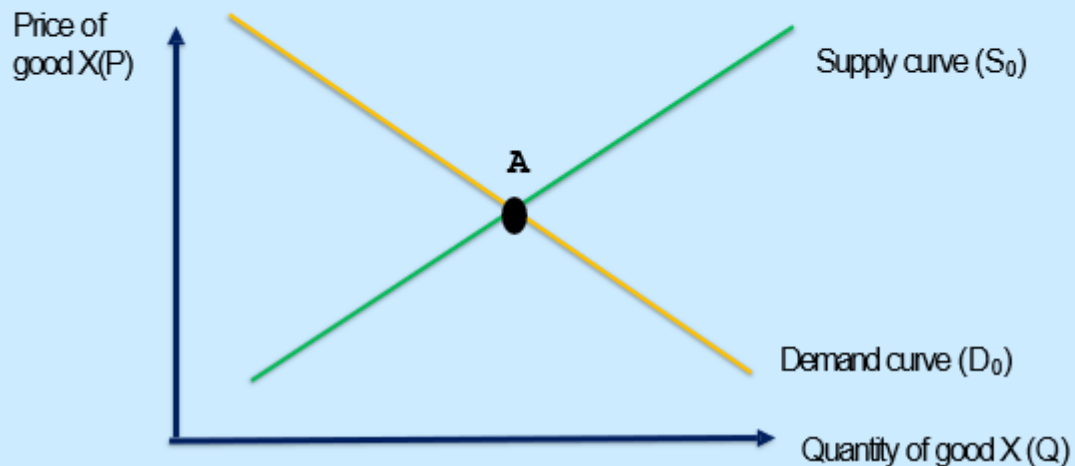
	<i>Equilibrium price</i>	<i>Equilibrium quantity</i>
<i>Increase in demand</i>		
<i>Decrease in demand</i>		
<i>Increase in supply</i>		
<i>Decrease in supply</i>		



REMEMBER

Question

What would happen to the market equilibrium if the price of good X increases?



Question

Study what changes could occur if there is an increase in supply that occurs at the same time as a decrease in demand.

Question

Study what changes could occur if there is an increase in supply that occurs at the same time as a decrease in demand.

UTC1 EXERCISES (CANVAS): 3,4,5,6



THANK YOU!





MICROECONOMICS



UTC1 INTRODUCTION TO MICROECONOMICS

UTC1 INTRODUCTION TO MICROECONOMICS

- ❑ Microeconomics and Macroeconomics.
- ❑ Remembering the Supply and Demand Model.
- ❑ **Elasticities.**
- ❑ Consumer and Producer Surplus.

The Price Elasticity of Demand and Supply

- What is price elasticity?
- How do you calculate price elasticity?
- What are the different types of price elasticity?

PRICE ELASTICITY – it measures how sensitive/responsive the quantity demanded and supplied of your product are to changes in price.

The Price Elasticity of Demand

The law of demand states that a fall in the price of a good raises the quantity demanded.

The price elasticity of demand measures how much the quantity demanded responds to a change in price.

- Demand for a good (a demand curve) is said to be **elastic** if the **quantity demanded responds substantially to changes in the price**.
- Demand for a good (a demand curve) is said to be **inelastic** if the **quantity demanded responds only slightly to changes in the price**.



The Price Elasticity of Demand

Basic formula: - the basic formula (as you move along the curve and drop the minus sign).

$$\text{Price Elasticity of Demand} = \frac{\% \text{ Change in the Quantity Demanded } (\Delta Q)}{\% \text{ Change in the Price } (\Delta P)}$$

Example:

- If the price of oil increases by 10% and over a period of several years, the quantity demanded falls by 5%, then the long run elasticity of demand for oil is:

$$\frac{-5\%}{10\%} = -0.5$$

or 0.5 (in absolute terms)

Midpoint formula - in practice, percent changes are best measured using the midpoint method, in which the percent changes are calculated using the average of starting and final values.

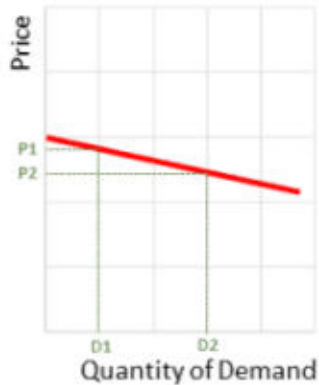
$$\text{Price elasticity of demand} = \frac{\left(\frac{Q_2 - Q_1}{\frac{Q_1 + Q_2}{2}} \right)}{\left(\frac{P_2 - P_1}{\frac{P_1 + P_2}{2}} \right)}$$

Example: At the initial price of \$10, the quantity demanded is 100. When the price rises to \$20, the quantity demanded falls to 90. What is the elasticity?

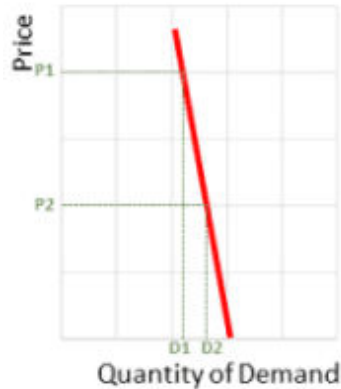
$$\frac{\frac{90 - 100}{(90 + 100)/2}}{\frac{\$20 - \$10}{(\$20 + \$10)/2}} = \frac{\frac{-10}{95}}{\frac{\$10}{\$15}} = \frac{-.105}{.666} \cong -.158$$

The Price Elasticity of Demand

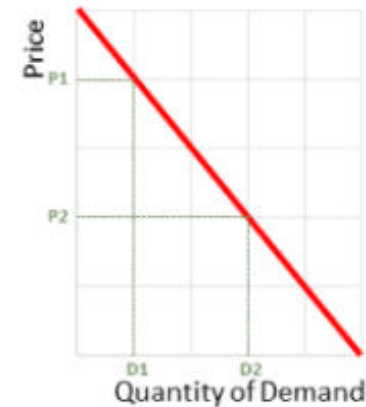
Demand is **elastic**:
when **price elasticity** > 1
The quantity demanded
moves proportionately
more than the price.



Demand is **inelastic**:
when **price elasticity** < 1
The quantity demanded
moves proportionately
less than the price.



Demand has **unit elasticity**:
when **price elasticity** $= 1$
The percentage change in
quantity demanded equals the
percentage change in price.



The Price Elasticity of Demand

If ...	Then ...	And It Is Called ...
% change in quantity > % change in price	$\frac{\% \text{ change in quantity}}{\% \text{ change in price}} > 1$	Elastic
% change in quantity = % change in price	$\frac{\% \text{ change in quantity}}{\% \text{ change in price}} = 1$	Unitary
% change in quantity < % change in price	$\frac{\% \text{ change in quantity}}{\% \text{ change in price}} < 1$	Inelastic

Examples:

Product category	Price elasticity
Restaurant Meals	2.3
Peanut Butter	1.7
Beef	1.6
Toothpaste	1
Rice	0.5
Cigarettes	0.4
Eggs	0.1

| PED | > 1 Elastic demand

| PED | = 1 Unit elastic demand

| PED | < 1 Inelastic demand

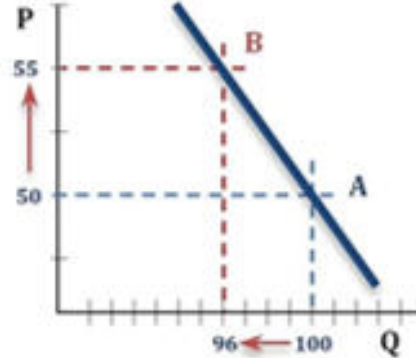
The Price Elasticity of Demand



INELASTIC DEMAND - Demand is inelastic when price elasticity < 1

It means that the **quantity demanded moves** proportionately **less** than the **price**.

GOOD Z		
PRICE P	QUANTITY DEMANDED Q	
50	100	initial situation
↓	↓	
55	96	final situation
5	-4	change
10%	-4%	%change $((nv-ov)/ov) \times 100$
	0.4	$\% \Delta Q / \% \Delta P$



This type of elasticity usually occurs in essential goods (necessities) (milk, bread, etc.) and goods that have very few substitutes (sugar, petrol, etc.), since consumers, with a modification in their price, will reduce the quantity demanded very little.

$|PED| < 1$ Inelastic demand

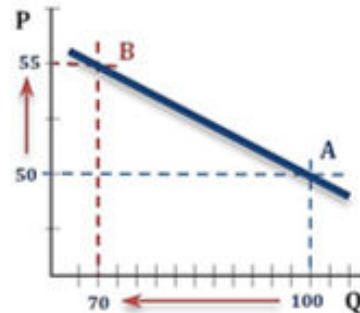
The Price Elasticity of Demand



ELASTIC DEMAND - Demand is elastic when price elasticity > 1

It means that the **quantity demanded moves** proportionately **more** than the **price**.

GOODY		
PRICE P	QUANTITY DEMANDED Q	
50	100	initial situation
↓	↓	
55	70	final situation
5	-30	change
10%	-30%	%change $((nv-ov)/ov) \times 100$
	3	% ΔQ /% ΔP



It usually manifests itself in value-added products, as they are very sensitive to price variations, for example luxuries. It is also true for products with many close substitutes and low brand loyalty.

$|PED| > 1$ Elastic demand

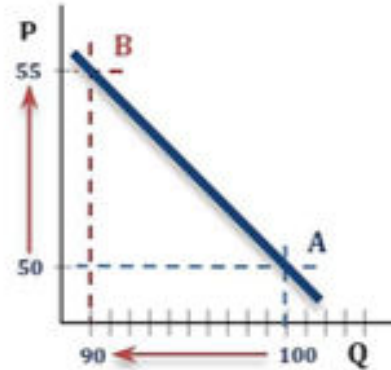
The Price Elasticity of Demand

UNIT ELASTIC DEMAND – Demand is unit elastic when price elasticity=1

It means that the percentage **change** in **quantity demanded equals** the percentage **change in price**.
the percentage

GOOD X		
PRICE P	QUANTITY DEMANDED Q	
50	100	initial situation
↓	↓	
55	90	final situation
5	-10	change
10%	-10%	%change ((nv-ov)/ov) x100
	1	%ΔQ/%ΔP

$|PED| = 1$ demand is unit elastic

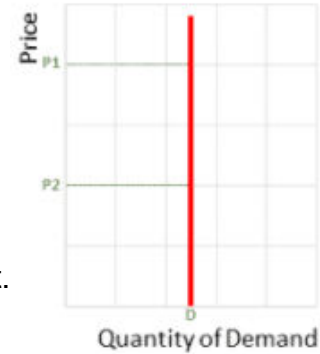


In this case, an increase in price by 1% leads to exactly 1% drop in volume.

The Price Elasticity of Demand: EXTREME CASES

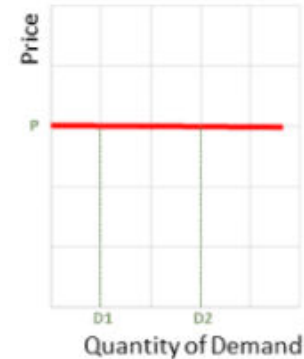
Demand is **perfectly inelastic**: when **price elasticity = 0** the demand curve is vertical, regardless of the price the quantity demanded stays the same.

Perfectly Inelastic Demand means that there is no change in the quantity of the product demanded when the price changes. This means that the supplier can charge whatever price they want and people will still be willing to buy that product.



Demand is **perfectly elastic**: when the **price elasticity of demand approaches infinity** and the demand curve becomes horizontal. **PED = ∞**

Perfectly elastic demand is when **the demand for the product is entirely dependent on the price** of the product. This means that if any producer increases his price by even a minimal amount, his demand will disappear. Customers will then switch to a different producer or supplier.



The Price Elasticity of Demand

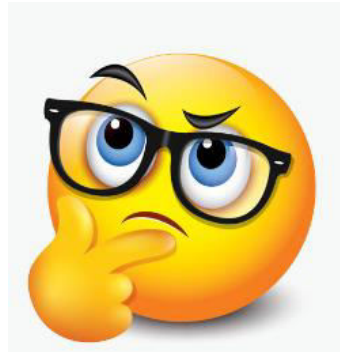
TYPE	PERFECTLY INELASTIC	INELASTIC	UNIT ELASTIC	ELASTIC	PERFECTLY ELASTIC
$\% \Delta Q$ vs $\% \Delta P$	$\% \Delta Q = 0$	$\% \Delta Q < \% \Delta P$	$\% \Delta Q = \% \Delta P$	$\% \Delta Q > \% \Delta P$	$\% \Delta P = 0$
how responsive the quantity demanded is to a change in a price	$\% \Delta Q$ - does not change	$\% \Delta Q$ is smaller than $\% \Delta P$	$\% \Delta Q$ equals $\% \Delta P$	$\% \Delta Q$ is greater than $\% \Delta P$	The quantity demanded varies while keeping the price constant.
E_p	$E_p = 0$	$E_p < 1$	$E_p = 1$	$E_p > 1$	$E_p = \infty$

DEMAND CURVE



The Price Elasticity of Demand

What influences the price elasticity of demand?



The Price Elasticity of Demand

1. The **availability of substitutes** strongly influences the sensitivity of quantity demanded to changes in price
 - For goods with **many** substitutes, switching brands when prices change is EASY, so demand is **elastic**
 - For goods with **fewer** substitutes, consumers find it HARD to adjust quantity demanded much when prices change... so demand is **inelastic**

Summary of Determinants of Elasticity of Demand

Less Elastic	More Elastic
Fewer Substitutes	More Substitutes
Short Run (less time)	Long Run (more time)
Necessities	Luxuries
Small Part of Budget	Large Part of Budget

2. The **time horizon** influences the elasticity of demand for a good
 - Immediately following a price increase, consumers may **not be able to alter their consumption patterns** (making demand **inelastic**)
 - Over time, however, consumers can adjust their behavior by finding substitutes (making demand **elastic**)

3. The **classification of the good** influences the elasticity of demand for a good
 - The **broader the classification**, the less likely consumers will be to finding a substitute (making demand **inelastic**)
 - The **narrower the classification**, the more likely the consumer will be to find substitutes (making demand **elastic**)
 - E.g. the demand for "food" (a broad classification) is less elastic than for the demand for "lettuce" (narrow classification)

4. The **nature of the good** to the consumer can also affect the elasticity of demand
 - For **necessities**, consumers do not change quantity demanded much when the price changes (making demand **inelastic**)
 - For **luxuries**, consumers will alter their behavior when the price rises (making demand **elastic**)

5. The **size of the purchase** (relative to the consumer's budget) influences the elasticity of demand
 - Consumers are **less** concerned about price changes **when the good feels cheap** (making demand **inelastic**)
 - Consumers become much **more** concerned about price changes **when the good feels expensive** (making demand **elastic**)

The Price Elasticity of Demand – PRACTICE

1. Which of the following two goods is more likely to be inelastically demanded?

- a. Demand for tangerines
- b. Demand for fruit

2. Which of the following two goods is more likely to be inelastically demanded?

- a. Demand for beef next month
- b. Demand for beef over the next decade

3. Which of the following two goods is more likely to be inelastically demanded?

- a. Demand for Exxon gasoline at the corner of 7th and Grand
- b. Demand for gasoline in the entire city

4. Which of the following two goods is more likely to be inelastically demanded?

- a. Demand for insulin
- b. Demand for vitamins

The Price Elasticity of Demand - PRACTICE

5. After a public information campaign highlighting that bacteria and other organisms cause and spread disease, will the demand curve for soap be more elastic or more inelastic?

- a. More elastic
- b. More inelastic

6. After the invention of nuclear power plants, will the demand curve for coal power plants be more elastic or more inelastic?

- a. More elastic
- b. More inelastic

7. After more employers allow employees to telecommute, will the demand curve for cars be more elastic or more inelastic?

- a. More elastic
- b. More inelastic

8. After an economic boom, will the demand curve for TVs be more elastic or more inelastic?

- a. More elastic
- b. More inelastic

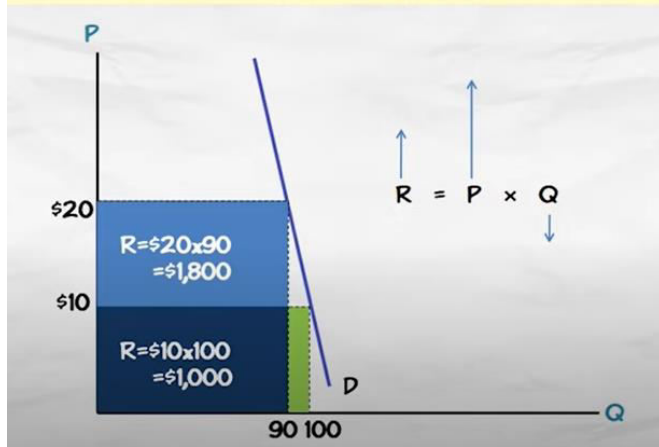
9. During the Middle Ages, the African city of Taghaza quarried salt in 200-pound blocks to be sent to the salt market in Timbuktu, in present-day Mali. Travelers report that Taghazans used salt instead of wood to construct buildings.

Compared with other towns without big salt mines, was the demand for wood more elastic or more inelastic in Taghaza?

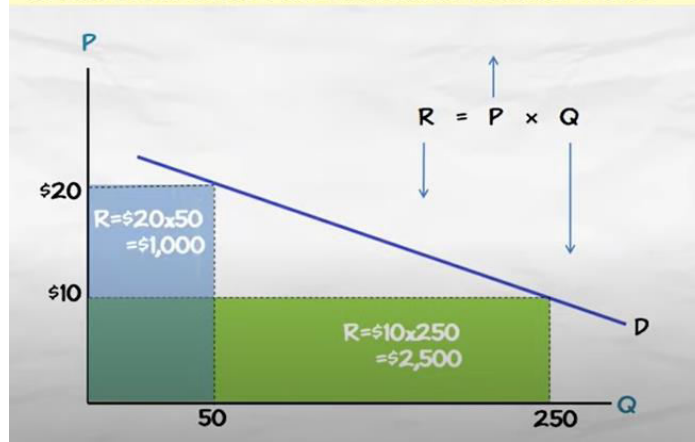
- a. More elastic
- b. More inelastic

The Price Elasticity of Demand

Revenues Rise as Price Rises if Demand is Inelastic



Revenues Fall as Price Rises with Elastic Demand Curves



Summary of Relationship between Elasticity of Demand and Revenues

Absolute Value of Elasticity	Name	Price and Revenue
$ E_D < 1$	Inelastic	P and R Move Together
$ E_D > 1$	Elastic	P and R Move Opposite
$ E_D = 1$	Unit Elastic	P Moves but R Stays the Same

The Price Elasticity of Demand - PRACTICE

The elasticity of demand for eggs has been estimated to be 0.1. If egg producers raised their prices by 10 percent, what will happen to their total revenue?

- a) It will increase
- b) It will decrease
- c) It won't change



The Price Elasticity of Demand - PRACTICE

1. If the elasticity of demand for college textbooks is -0.1 , and the price of textbooks increases by 20%, how much will the quantity demanded change, and in what direction?

- a. The quantity demanded increases by 2%
- b. The quantity demanded decreases by 20%
- c. The quantity demanded decreases by 2%
- d. The quantity demanded remains the same

2. If the elasticity of demand for spring break packages to Cancun is -5 , and if you notice that this year in Cancun the quantity of packages demanded increased by 10%, then what happened to the price of Cancun vacation packages?

- a. The price fell by 10 percent
- b. The price fell by 2 percent
- c. The price increased by 2 percent
- d. The price remained the same

3. The long-run elasticity of oil demand has been estimated at -0.5 . Does a 10% rise in oil prices increase or decrease total revenues to the oil producers?

- a. Increase
- b. Decrease

4. The elasticity of demand is 0.2 . Is the demand curve relatively steep or flat? Will a fall in price raise total revenue or lower it? Note: we present the elasticity in terms of its absolute value.

- a. Relatively steep; raise total revenue
- b. Relatively flat; raise total revenue
- c. Relatively steep; lower total revenue
- d. Relatively flat; lower total revenue

5. The elasticity of demand is 2.0 . Is the demand curve relatively steep or flat? Will a fall in price raise total revenue or lower it? Note: we present the elasticity in terms of its absolute value.

- a. Relatively steep; raise total revenue
- b. Relatively flat; raise total revenue
- c. Relatively steep; lower total revenue
- d. Relatively flat; lower total revenue

The Price Elasticity of Supply

The law of supply states that higher prices raise the quantity supplied.

The **price elasticity of supply** measures how much the quantity supplied responds to changes in the price.

- Supply of a good is said to be **elastic** if the **quantity supplied responds substantially to changes in the price**.
- Supply is said to be **inelastic** if the **quantity supplied responds only slightly to changes in the price**.



The Price Elasticity of Supply

Price elasticity of supply - the basic formula

$$\text{Price Elasticity of Supply} = \frac{\% \text{ Change in the Quantity Supplied } (\Delta Q)}{\% \text{ Change in the Price } (\Delta P)}$$

The midpoint formula: in practice, percent changes are best measured using the midpoint method, in which the percent changes are calculated using the average of starting and final values.

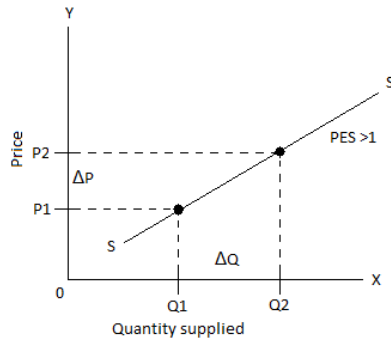
Example: At the initial price of \$10, the quantity supplied is 100. When the price rises to \$20, the quantity supplied is 110.

$$\frac{\frac{\Delta Q}{\text{Avg. } Q}}{\frac{\Delta P}{\text{Avg. } P}} = \frac{\frac{110 - 100}{(110 + 100)/2}}{\frac{\$20 - \$10}{(\$20 + \$10)/2}} = \frac{\frac{10}{105}}{\frac{\$10}{\$15}} = \frac{.095}{.666} \cong .143$$

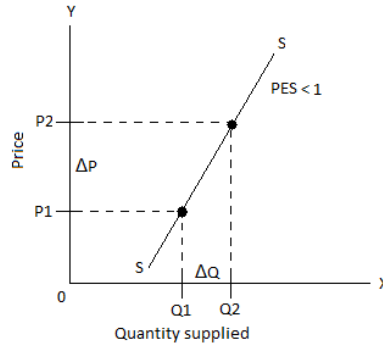
$$E_s = \frac{\% \Delta Q_s}{\% \Delta P} = \frac{\frac{Q_2 - Q_1}{(Q_2 + Q_1)/2}}{\frac{P_2 - P_1}{(P_2 + P_1)/2}}$$

The Price Elasticity of Supply

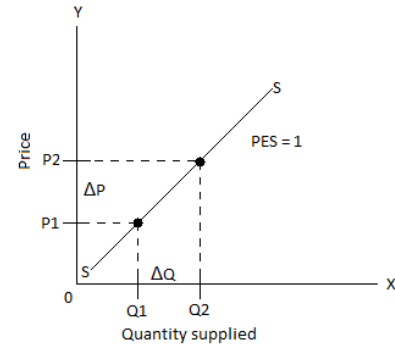
Supply is elastic:
when **price elasticity** > 1
The quantity supplied moves
proportionately more than the price.



Supply is inelastic:
when **price elasticity** < 1
The quantity supplied moves
proportionately less than the price.



Supply has unit elasticity:
when **price elasticity** $= 1$
The percentage change in
quantity supplied equals the
percentage change in price.



The Price Elasticity of Supply

- If the $E_s < 1$, the supply curve is inelastic
- If the $E_s > 1$, the supply curve is elastic
- If the $E_s = 1$, the supply curve is unit elastic

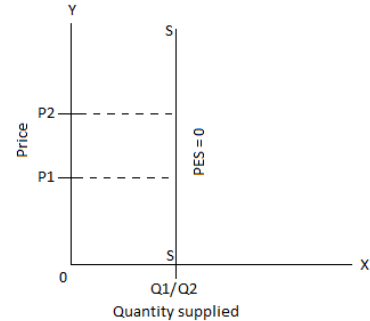
The Elasticity of Supply: EXTREME CASES

Supply is **perfectly inelastic**: when **price elasticity = 0**

The supply curve is vertical, regardless of the price, the quantity supplied stays the same.

Price has not effect on quantity supplied.

(example: Picasso's paintings)

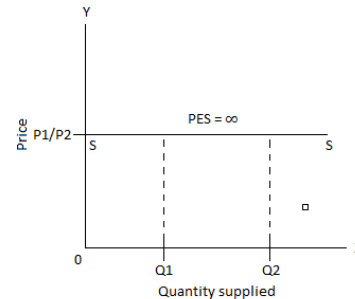


Supply is **perfectly elastic**: this occurs when **the price elasticity of supply approaches infinity** and

the supply curve is horizontal at some specific price.

If the price falls below that level, the quantity supplied falls to 0.

If the price rises above that level, the quantity supplied is extremely large.



The Price Elasticity of Supply

1. The main determinant of the elasticity of supply is how quickly **per-unit costs** increase with an increase in production
 - If increased production requires much higher costs, then the supply curve will be **inelastic**
 - If production can increase with constant costs then the supply curve will be **elastic**

2. The **time horizon** influences the elasticity of supply for a good
 - **Immediately** following a price increase, producers can **expand output only using their current capacity** (making supply **inelastic**)
 - **Over time**, however, producers can expand their capacity (making supply **elastic**)

Summary of Determinants of Elasticity of Supply

Less Elastic	More Elastic
Difficult to Increase Production at Constant Unit Cost	Easy to Increase Production at Constant Unit Cost
Short Run	Long Run
Large Share of Market for Inputs	Small Share of Market for Inputs
Global Supply	Local Supply

3. The elasticity of supply for a good also depends on whether it is a small or big demander in its input markets, i.e. the industry's **share of the demand for its inputs**
 - Supply is **elastic** when the industry is a small demander in its input markets because supply can be expanded **without causing a big increase in the demand for the industry's inputs**
 - Supply is **inelastic** when the industry is a big demander in its input markets

4. The **geographic scope** of the market determines the elasticity of supply for a good
 - The **narrower** the scope of the market of a good, the **more elastic** its supply
 - The **wider** the scope of the market of a good, the **less elastic** its supply

The Price Elasticity of Supply: PRACTICE

1. Which of the two goods is more likely to be elastically supplied?

- a. Supply of apples over the next growing season
- b. Supply of apples over the next decade

2. Which of the two goods is more likely to be elastically supplied?

- a. Supply of construction workers in Binghamton, NY
- b. Supply of construction workers in New York State

3. Which of the two goods is more likely to be elastically supplied?

- a. Supply of breakfast cereal
- b. Supply of food

4. Which of the two goods is more likely to be elastically supplied?

- a. Supply of gold
- b. Supply of computers

5. If a new process for manufacturing diamonds is created, will the supply curve for diamonds become more elastic or more inelastic?

- a. More elastic
- b. More inelastic

6. If pesticides and fertilizers are banned, will the supply curve for food become more elastic or more inelastic?

- a. More elastic
- b. More inelastic

7. If a larger share of oil output is required to make plastic, will the supply curve for plastic become more elastic or more inelastic?

- a. More elastic
- b. More inelastic



THANK YOU!





MICROECONOMICS



UTC1 INTRODUCTION TO MICROECONOMICS

UTC1 INTRODUCTION TO MICROECONOMICS

- ☐ Microeconomics and Macroeconomics.
- ☐ Remembering the Supply and Demand Model.
- ☐ Elasticities.
- ☐ **Consumer and Producer Surplus.**

Consumer and Producer Surplus

In practice, we want to find a possible **measure for the gains** that consumers and sellers obtain from **trading at the equilibrium price**.

Consumer Surplus

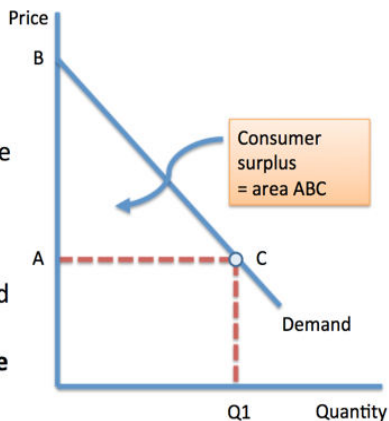
For the **customers**, this **measure of trade gain** is called **consumer surplus**.

- ❖ For a single consumer, the **individual consumer surplus** is the **difference** between the **maximum price** (reservation price) that she/he is willing to pay for a given amount of a good or service and the **price he or she actually pays**.
- ❖ A **consumer's willingness to pay** for a good or service is the maximum price at which they would buy that good or service.
- ❖ **Total consumer surplus** is the sum of the individual consumer surpluses of all buyers of a good or service in a market.
- ❖ The **consumer surplus** is measured by the **area below the market demand curve** and **above the equilibrium price**.

What is Consumer Surplus?

Consumer surplus is a measure of the economic welfare that people gain from purchasing and then consuming goods and services

- **Consumer surplus** is the difference between the total amount that consumers are willing and able to pay for a good or service (shown by the demand curve) and the total amount that they actually do pay (i.e. the market price).
- Consumer surplus is indicated by the area **under the demand curve** and **above the market price**.



FORMULA: $0.5 \times (\text{BASE} \times \text{HEIGHT})$

Producer Surplus

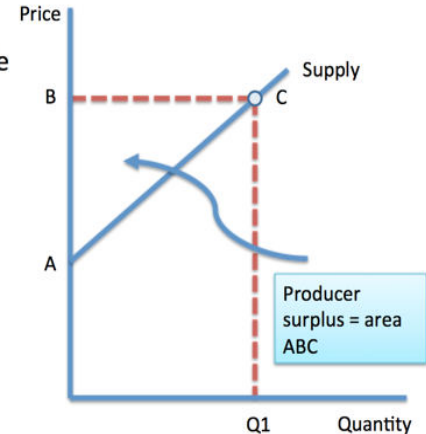
For the **producers/sellers**, the **measure of trade gain** obtained from selling a good or service at the equilibrium price is called **producer surplus**.

- ❖ **Individual producer surplus** is the net gain to an individual seller from selling a good.
- ❖ It is equal to the **difference** between **the price received** and the **seller's cost** (the minimum price at which the seller is willing to sell).
- ❖ **Cost**: the value of everything a seller must give up to produce a good.
- ❖ **The producer surplus for sellers** is the amount that sellers benefit from by selling at a market price that is higher than the minimum price they would be willing to sell for.
- ❖ **Total producer surplus** is the sum of the individual producer surpluses of all the sellers of a good in a market.
- ❖ **The producer surplus** is measured by the area **above the market supply curve** and **below the equilibrium price**.

What is Producer Surplus?

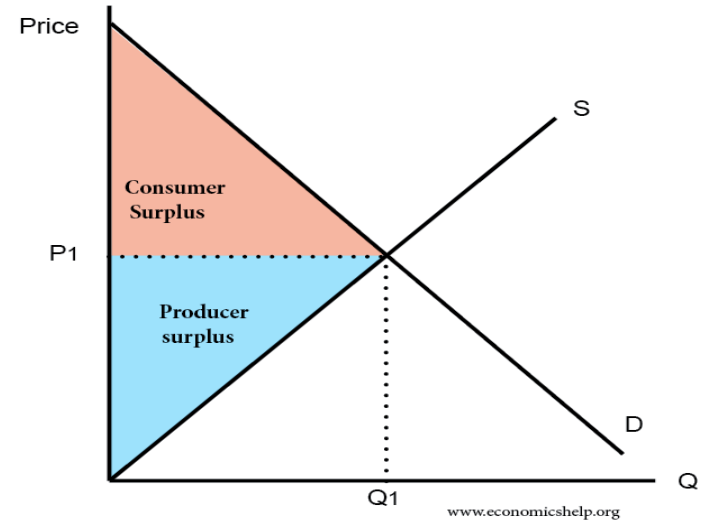
Producer surplus is a measure of producer welfare

- Producer surplus is the difference between what producers are willing and able to supply a good for and the price they actually receive
- Producer surplus shown by area above the supply curve and below the market price
- Higher prices provide an incentive to supply more to the market (profit motive)



The Sum of the Consumer and Producer Surplus in a Market

- ❖ The **sum of the consumer and producer surplus in a market** is a measure of the **economic surplus** that the participants obtain by trading in the market.
- ❖ It is **total surplus**, the **total benefit to society from the production and consumption** of the good.



Deadweight loss

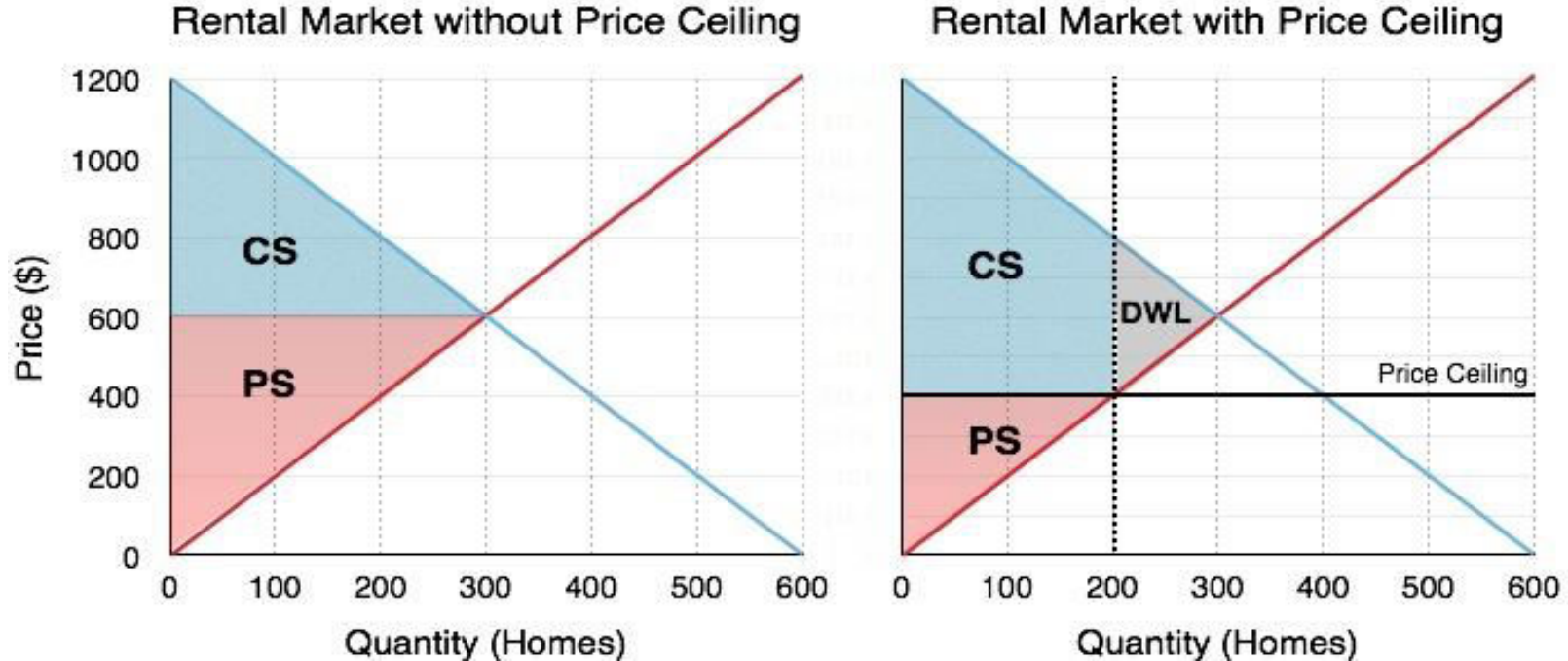
As long as $P_{\text{consumers are willing to pay}} > P_{\text{sellers are willing to accept}}$
there are mutually profitable trades that can be made

But when mutually profitable gains from trade are **not fully exploited** we **have lost consumer and producer surplus**.

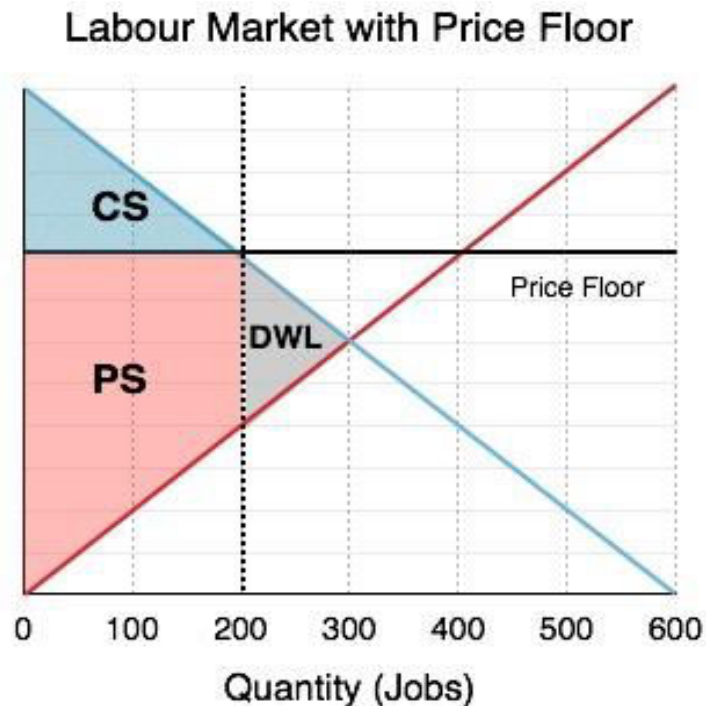
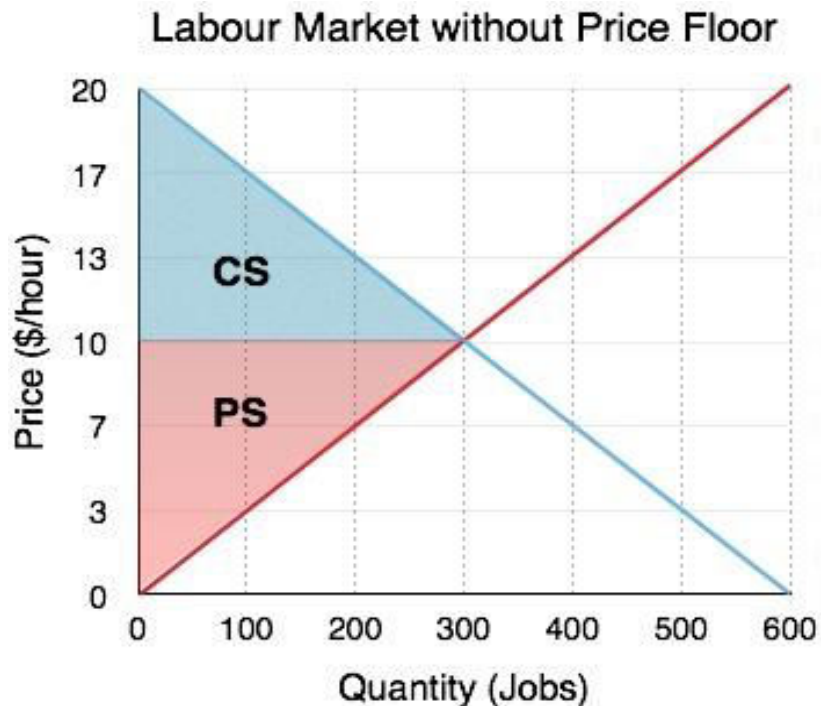
A **deadweight loss** is a **loss in total surplus** (social surplus) created by **market inefficiency**, which happens when **supply and demand are out of equilibrium**.

It represents the lost efficiency that occurs when the market doesn't reach its equilibrium.

Deadweight loss created by price ceiling



Deadweight loss created by price floor





UTC1 EXERCISES (CANVAS): 7



THANK YOU!



Page 1

LESSON 4

SLIDE N-44 Question:

$$X = K^{1/2} \cdot L^{1/2}$$

$$TC = 32L + 15K$$

$$CT = 50.000 \text{ €}$$

• Expansion Path

$$\text{Expansion Path} \Rightarrow MRTS = \frac{P_L}{P_K}$$

$$MRTS = \frac{dL}{dK} = \frac{\frac{1}{2} K^{1/2} L^{-1/2}}{\frac{1}{2} K^{-1/2} L^{1/2}} = K \cdot L^{-1} = \underline{\underline{\frac{K}{L}}}$$

$$\frac{K}{L} = \frac{32}{15} \Rightarrow \underline{\underline{K = \frac{32L}{15}}} \Rightarrow \text{Expansion Path}$$

Page 2

• Combination of labor and Capital

$50.000 = 32L + 15K$ We substitute K in Isocost

$$50.000 = 32L + 15\left(\frac{32L}{15}\right) \Rightarrow 50.000 = 64L$$

$$\boxed{L = 781'25 \quad K = 1666'6 \quad Q = 1140'9}$$

EXERCISE 3

① $X = 2L + K \quad K = 40$

$$\text{Av Prod } L = \frac{X}{L} = \frac{2L + 40}{L} = 2 + \frac{40}{L}$$

$$\text{Maj Prod } L = dL = 2$$

② $X = 10L^2 - KL^3 \quad K = 1$

$$\text{Av. Prod } L = \frac{X}{L} = \frac{10L^2 - L^3}{L} = 10L - L^2$$

Page 3

$$C) X = 48Lk - 3L^2 - 2K^2 \quad K=10$$

$$x = 480L - 3L^2 - 200$$

$$\text{Av. Prod } L = \frac{X}{L} = \frac{480L - 3L^2 - 200}{L} = 480 - 3L - \frac{200}{L}$$

EXERCISE 4

$$A) X = L^2 + L - L^3$$

$$\text{Av. Prod } L = \frac{X}{L} = \frac{L^2 + L - L^3}{L} = L + 1 - L^2$$

$$\text{Marg Prod } L = dL = 2L + 1 - 3L^2$$

$$\text{Technical Optimum} \Rightarrow \text{Av. Prod } L = \text{Marg. Prod. } L$$

$$L + 1 - L^2 = 2L + 1 - 3L^2 \Rightarrow 2L^2 = L \Rightarrow \frac{2L^2}{L} = \frac{L}{L} \Rightarrow$$

$$2L = 1 \quad \boxed{L = \frac{1}{2}}$$

$$\left[x = \left(\frac{1}{2}\right)^2 + \frac{1}{2} - \left(\frac{1}{2}\right)^3 = 0.625 \right]$$

We will not calculate Technical Maximum.

(B) $x = 20L^2 + 14L - 2L^3$

Av. Prod L = $\frac{x}{L} = \frac{20L^2 + 14L - 2L^3}{L} = 20L + 14 - 2L^2$

Marg. Prod L = $\frac{dx}{dL} = 40L + 14 - 6L^2$

Technical Optimum $\Rightarrow$ Av. Prod L = Marg. Prod L

$$20L + 14 - 2L^2 = 40L + 14 - 6L^2$$

$$4L^2 = 20L \Rightarrow \frac{4L^2}{L} = \frac{20L}{L} \Rightarrow 4L = 20$$

$$\boxed{\boxed{L=5}}$$

$$x = 20 \cdot 5^2 + 14 \cdot 5 - 2 \cdot 5^3$$

$$\boxed{\boxed{x=320}}$$

ALUMNO / A

~~Nicolás Clement Perro~~

CURSO

ASIGNATURA

FECHA

Page 5

EXERCISE 6

$$X = K L^{1/2}$$

$$P_L = 20 \quad P_K = 5 \quad TC = 100.000$$

(A) Expansion Path $\Rightarrow MRTS = \frac{P_L}{P_K}$

$$MRTS = \frac{dL}{dK} = \frac{1/2 K L^{-1/2}}{L^{1/2}} = \frac{K/2 L^{1/2}}{L^{1/2}} = \frac{K}{2L}$$

$$\frac{K}{2L} = \frac{20}{5} \Rightarrow \frac{K}{2L} = 4 \quad \boxed{K = 8L \Rightarrow \text{Expansion Path}}$$

(B)

$$100.000 = 20L + 5K$$

$$100.000 = 20L + 5(8L) \Rightarrow 100.000 = 20L + 40L$$

$$L = 1.666'66$$

$$K = 8 \times 1.666'66 = 13.333'36$$

③

$$X = KL^{1/2}$$

$$X = (13.333\overline{36}) \cdot (1.666\overline{66})^{1/2}$$

$$X = 544.338\overline{4}$$

Exercise 1.

We have a high-end electronic component market, with the following functions:

$$Q_d = 20.000 - 100P$$

$$Q_s = -4.000 + 200P$$

- A- Calculate the **equilibrium** price and quantity. **Draw the graph.**
- B- Calculate **Consumer Surplus, Producer Surplus and Total Society Benefit.**
- C- If the Government sets a Maximum Price **$P_{max}=50$** in this market. Calculate new Q_s , Q_d and the excess of Supply/Demand.
- D- Calculate the Deadweight Loss of this market

Exercise 2.

We have a market for Staple Grain, with the goal of protecting national farmers. In this market we have the following functions:

$$Q_d = 50.000 - 200P$$

$$Q_s = 10.000 + 200P$$

- A- Calculate the **equilibrium** price and quantity. **Draw the graph.**
- B- Calculate **Consumer Surplus, Producer Surplus and Total Society Benefit.**
- C- If the Government sets a Minimum Price **$P_{min}=140$** in this market. Calculate new Q_s , Q_d and the excess of Supply/Demand.
- D- Calculate the Deadweight Loss of this market

LESSON 4

EXERCISES

EXERCISE 1

Indicate whether the following statements are true or false. Justify your answer.

- A. The main objective of companies is to maximize their total revenue.
- B. Companies must take into account the accounting profit when making decisions because this profit is the one that really reflects the companies' costs.
- C. The opportunity cost reflects the profitability that we could obtain if we made the decision to invest in our own company.
- D. The production function shows the relationship between the combination of production factors and the quantity produced, the most efficient possible with the available resources.

EXERCISE 1

- E. A change in the combination of factors will imply an increase in the quantity produced, moving therefore to an isoquant further from the origin.
- F. For companies, it is positive to have a low average productivity
- G. Marginal productivity shows the effect that an increase in one unit of a production factor would have on the total quantity produced.

EXERCISE 2

According to the data in the table, calculate the marginal and average product.

Units of labour (L)	Total production (Q)
0	0
2	2000
4	3000
6	3500
8	3800
10	3900

EXERCISE 3

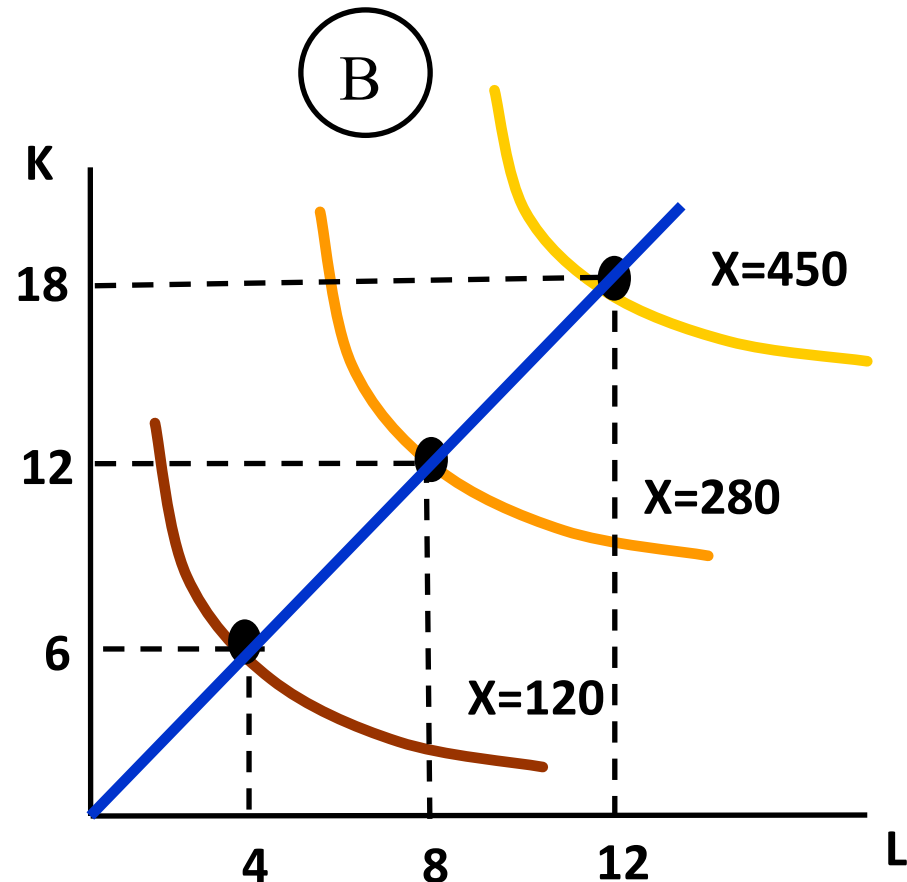
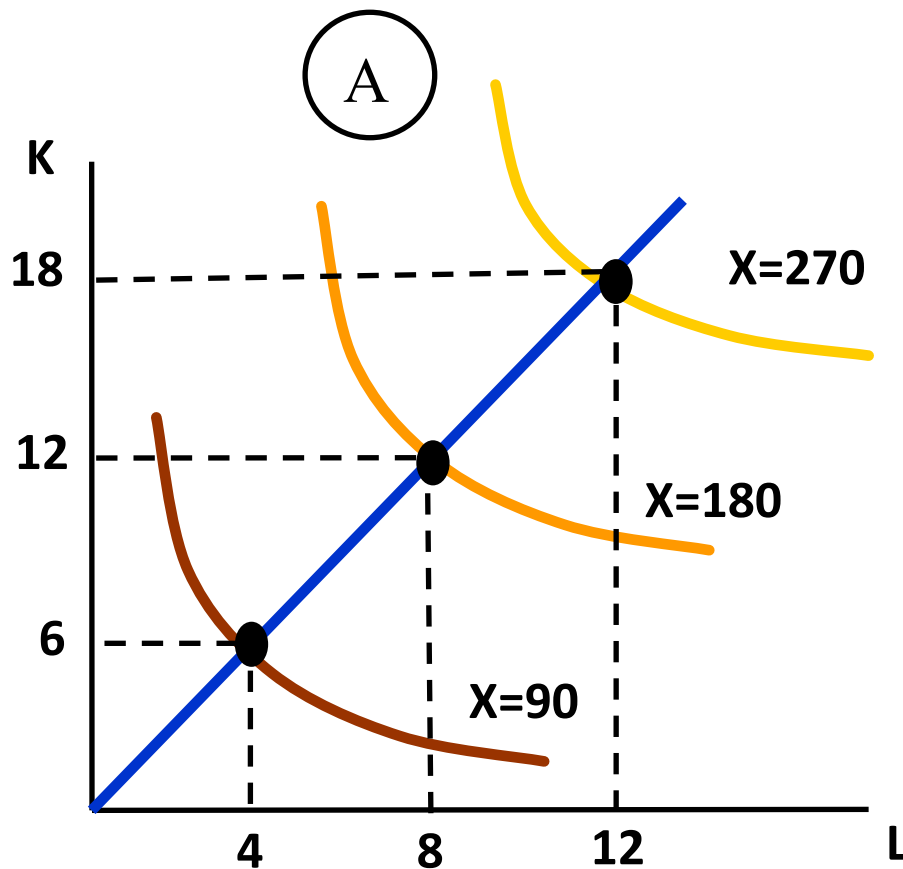
- A. Given the production function $X = 2L + K$, calculate the average product and marginal product function of labour if 20 units of capital are used.
- B. Given the production function $X = 10L^2 - KL^3$, calculate the average product and marginal product of labour if $K=1$.
- C. Given the production function $X = 48LK - 3L^2 - 2K^2$, calculate the average product and marginal product of labour if $K=10$.

EXERCISE 4

- A. Given the production function $X = L^2 + L - L^3$, calculate the average product, marginal product, technical optimum and technical maximum.
- B. Given the production function $X = 20L^2 + 14L - 2L^3$, calculate the average product, marginal product, technical optimum and technical maximum.

EXERCISE 5

According to the following graphs identify the type of economies of scale of the firm.



EXERCISE 6

The production function of a firm is the following:

$$X = KL^{1/2}$$

The price of capital (u) is 5€ and the price of labour (w) is 20€.

The firm's total cost is 100000 euros.

- A. What would be the expansion path in the long run?**
- B. At the point of tangency between the isocost and the isoquant, how many units of K and L will the firm need?**
- C. What would be the company's output?**

EXERCISE 7

A firm has the following costs:

Q	FC	VC
0	3000	0
10	3000	1000
20	3000	2500
30	3000	4500
40	3000	7000
50	3000	10000
60	3000	13500
70	3000	17500

- A. Calculate the marginal cost, average fixed cost, average variable cost and average total cost.
- B. Graph the marginal cost and the average total cost. Explain the graph.

EXERCISE 8

Indicate whether the following statements are true or false. Justify your answer.

- A. The law of diminishing marginal returns proves that the greater the number of workers, the greater the production
- B. Companies have the possibility of modifying all production factors in the short run, since it only depends on company's choice
- C. The isocost line will show us the total cost that the company will incur from the combination of factors used and their prices
- D. The optimal combination of production factors that the company should choose is the one that results when $MRTS = RP$ of inputs

EXERCISE 8

- E. When the marginal product is increasing, the marginal cost is also increasing since companies are producing more and using a greater amount of production factors to achieve it.
- F. The economies of scale show us that as the size of the company increases, the average total cost also increases.

LESSON 4: FIRM BEHAVIOR. PRODUCTION AND COSTS.

- 4.1. *Introduction*
- 4.2 Production function
- 4.3 Average and marginal product
- 4.4 The law of diminishing marginal returns
- 4.5 Long-run production. Returns to scale.
- 4.6 Expansion path
- 4.7 Marginal cost, average cost and total cost.
- 4.8 Long-run costs and economies of scale.

LESSON 4. FIRM BEHAVIOR. PRODUCTION AND COSTS.

The objective of the company is to maximize its profit:

$$P = \text{TR} - \text{TC}$$

TOTAL REVENUE

TOTAL COSTS

$$\text{TR} = P \cdot Q$$

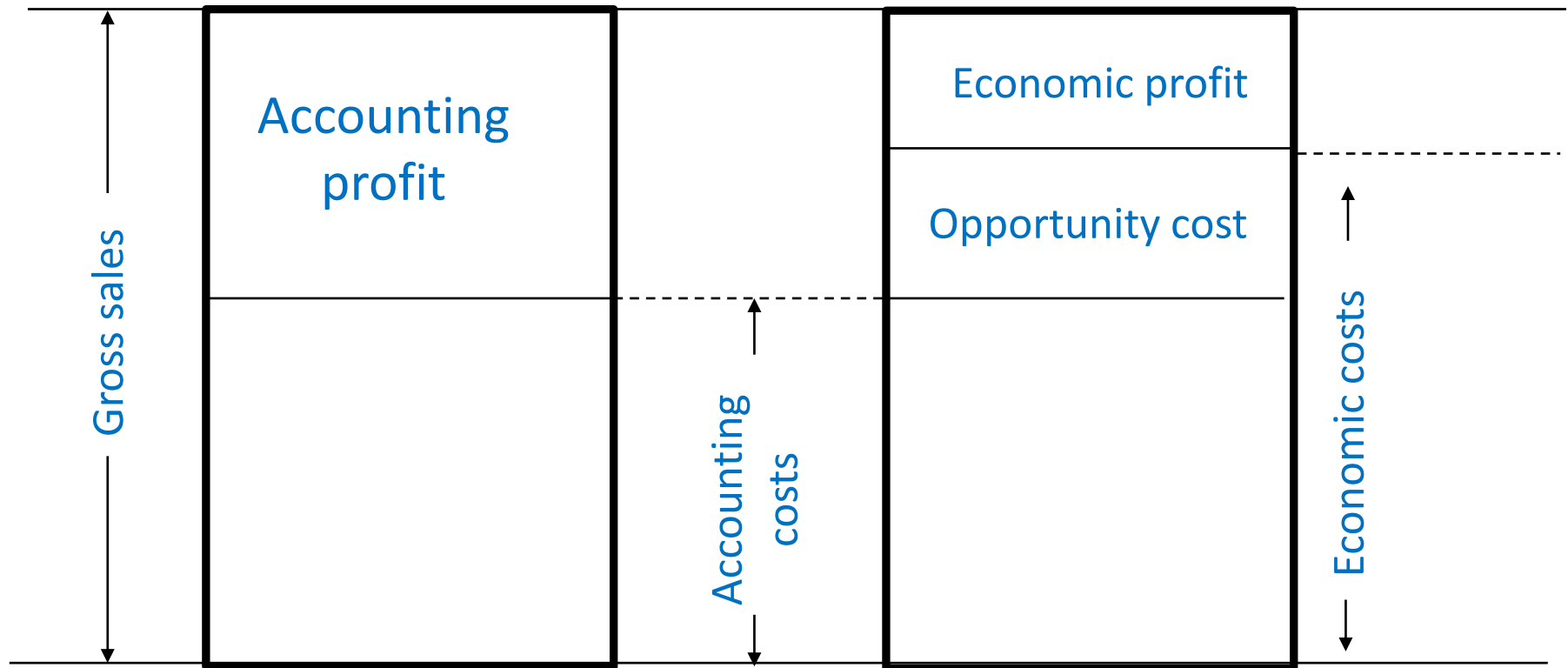
P and Q are determined
by the market
equilibrium

How is production done
and how much does it
cost to produce

Determinants of the
firm's supply curve

LESSON 4. FIRM BEHAVIOR. PRODUCTION AND COSTS.

- The costs are valued according to the *opportunity cost*.
- The profits are *economic*, not accounting profits



LESSON 4. FIRM BEHAVIOR. PRODUCTION AND COSTS.

For example:

An entrepreneur has 1 million euros to invest.

He/She has two possibilities:

- Invest in his/her company.
- Invest in financial securities with a 15% profitability per year (150.000€).

Finally he/she decides to invest in the company.

Let's suppose that there are three possibilities:

- That the accounting profit at the end of the year is 100,000€
- That the accounting profit at the end of the year is 150.000€
- That the accounting profit at the end of the year is 200.000€

LESSON 4. FIRM BEHAVIOR. PRODUCTION AND COSTS.

What **economic profit** would he/she obtain in each situation?

- If the accounting profit (AP) is 100000€, the AP is lower than the opportunity cost, therefore...
 $\Rightarrow$ If $AP < OC$, there will be losses
- If the AP = 150.000€, the opportunity cost and the accounting profit are the same:
 $\Rightarrow$ If $AP = OC$, there will be zero profit
- If the AP = 200.000€, the AP is higher than the opportunity cost, therefore...
 $\Rightarrow$ If $AP > OC$, there will be extraordinary profits

LESSON 4: FIRM BEHAVIOR. PRODUCTION AND COSTS.

- 4.1. Introduction
- 4.2 *Production function*
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LESSON 4. FIRM BEHAVIOR. PRODUCTION AND COSTS.

A certain quantity of good X (X_0) can be produced using different combinations of *production factors* (capital, labor and land) as it has already been studied.

However, for simplicity, we will only use *capital (K) and labor (L)*.

A *PRODUCTION FUNCTION* shows the relationship between a production level (X) and the quantities of production factors (K and L) used:

$$X = f(K, L)$$

X represents the maximum production that can be achieved when the production factors, K and L, are used as efficiently as possible.

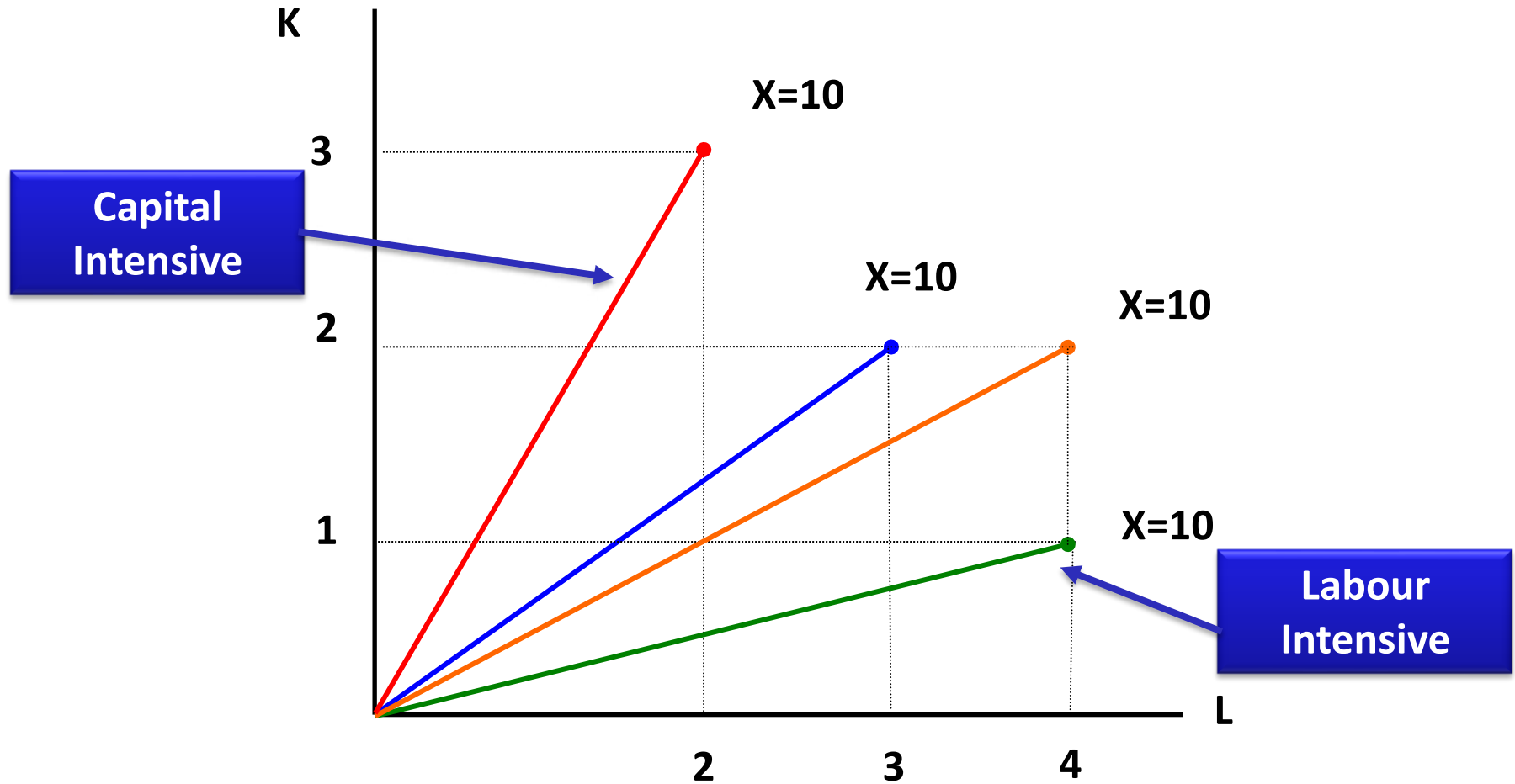
LESSON 4. FIRM BEHAVIOR. PRODUCTION AND COSTS.

For example, let's suppose that we want to produce 10 units of good X.

This can be done by combining different amounts of K and L:

X=10	K=2	L=3
X=10	K=3	L=2
X=10	K=1	L=4
X=10	K=2	L=4

LESSON 4. FIRM BEHAVIOR. PRODUCTION AND COSTS.



LESSON 4. FIRM BEHAVIOR. PRODUCTION AND COSTS.

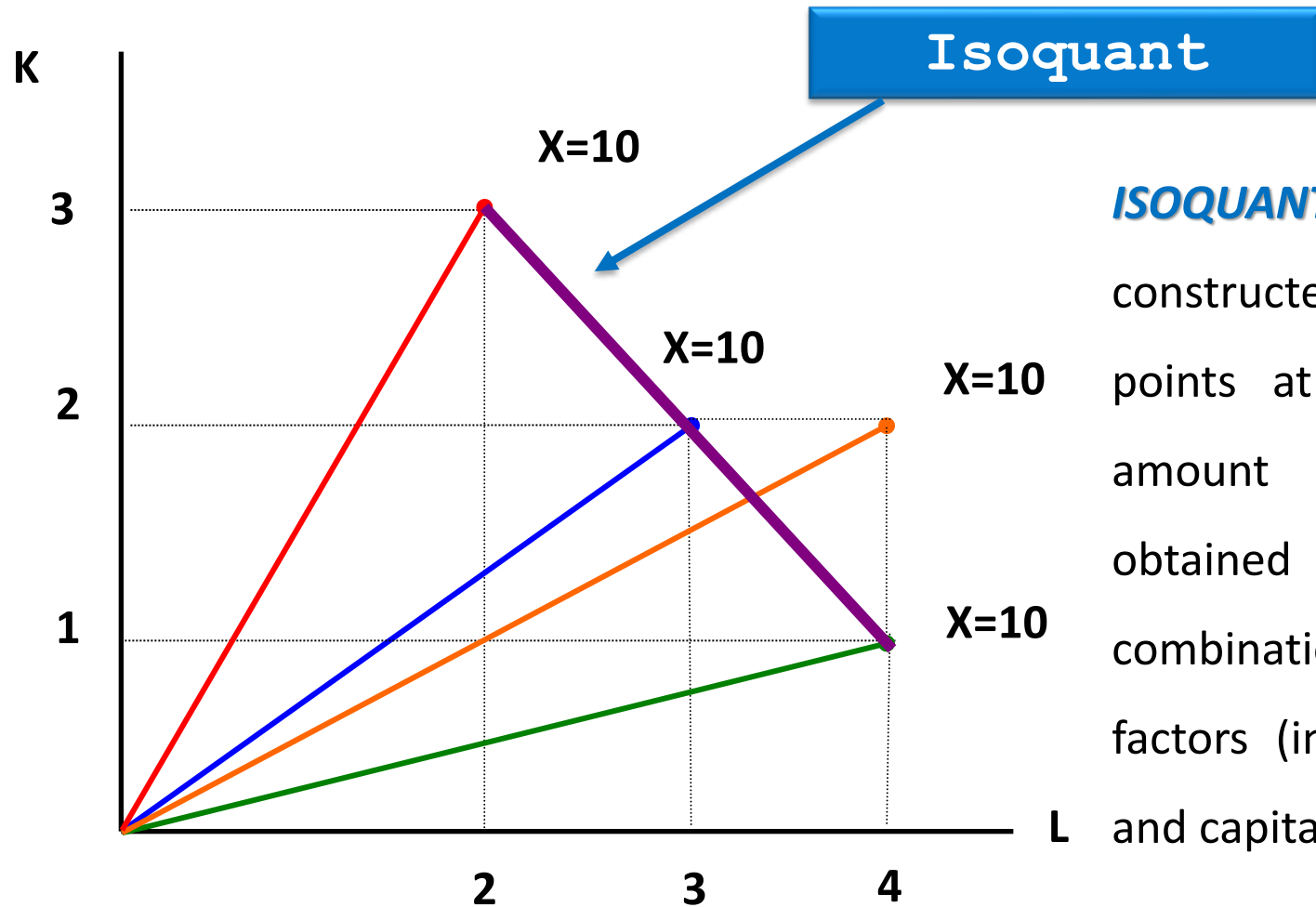
- All the techniques that are *feasible* form the **PRODUCTION SET**.
- The **PRODUCTION FUNCTION** shows us the set of techniques that in addition to being feasible are *efficient*.

QUESTION

Using the previous graph identify at least one technique that is feasible but not efficient.

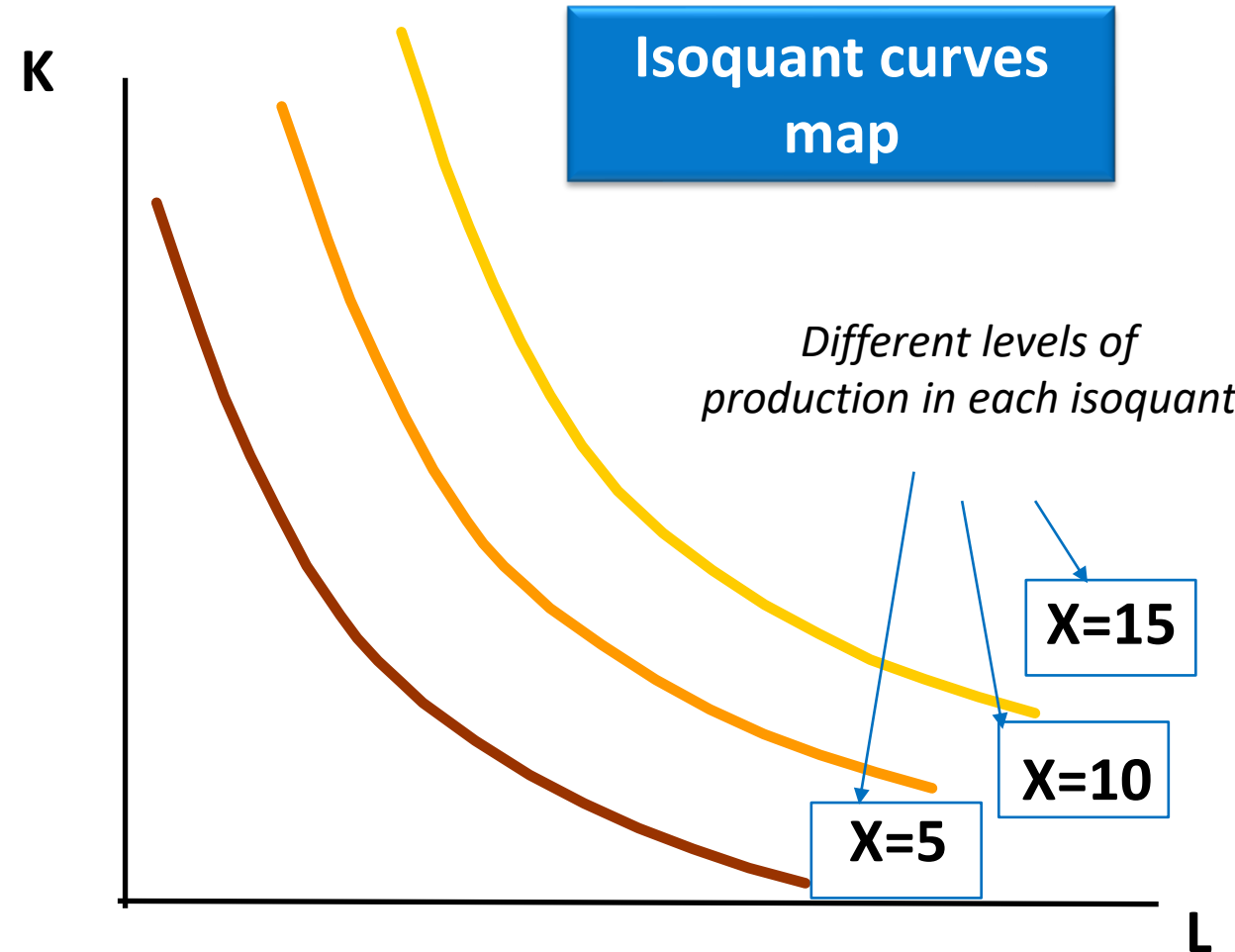
Think of another possible feasible but not efficient combination of production factors.

LESSON 4. FIRM BEHAVIOR. PRODUCTION AND COSTS.



ISOQUANTS are curves constructed by joining the points at which the same amount of production is obtained through various combinations of production factors (in this case, labour and capital).

LESSON 4. FIRM BEHAVIOR. PRODUCTION AND COSTS.



A *higher isoquant* indicates a greater amount of production and vice versa.

In each isoquant there is *an equivalence in productivity*; that is, the amount of the factor that substitutes and the substituted factor are equally productive; that is, *they reach the same level of production*

LESSON 4. FIRM BEHAVIOR. PRODUCTION AND COSTS.

- If the substitution is expressed per unit of one of the factors in relation to the other, then this substitution rate is called *marginal rate of technical substitution (MRTS)*. Therefore, the MRTS measures the amount of a factor that the firm must give up in order to increase the amount of the other factor by one unit remaining in the same isoquant.
- For example, the MRTS shows the amount of capital that a unit of labor can replace while maintaining the same level of production, that is, staying on the same isoquant (ceteris paribus).

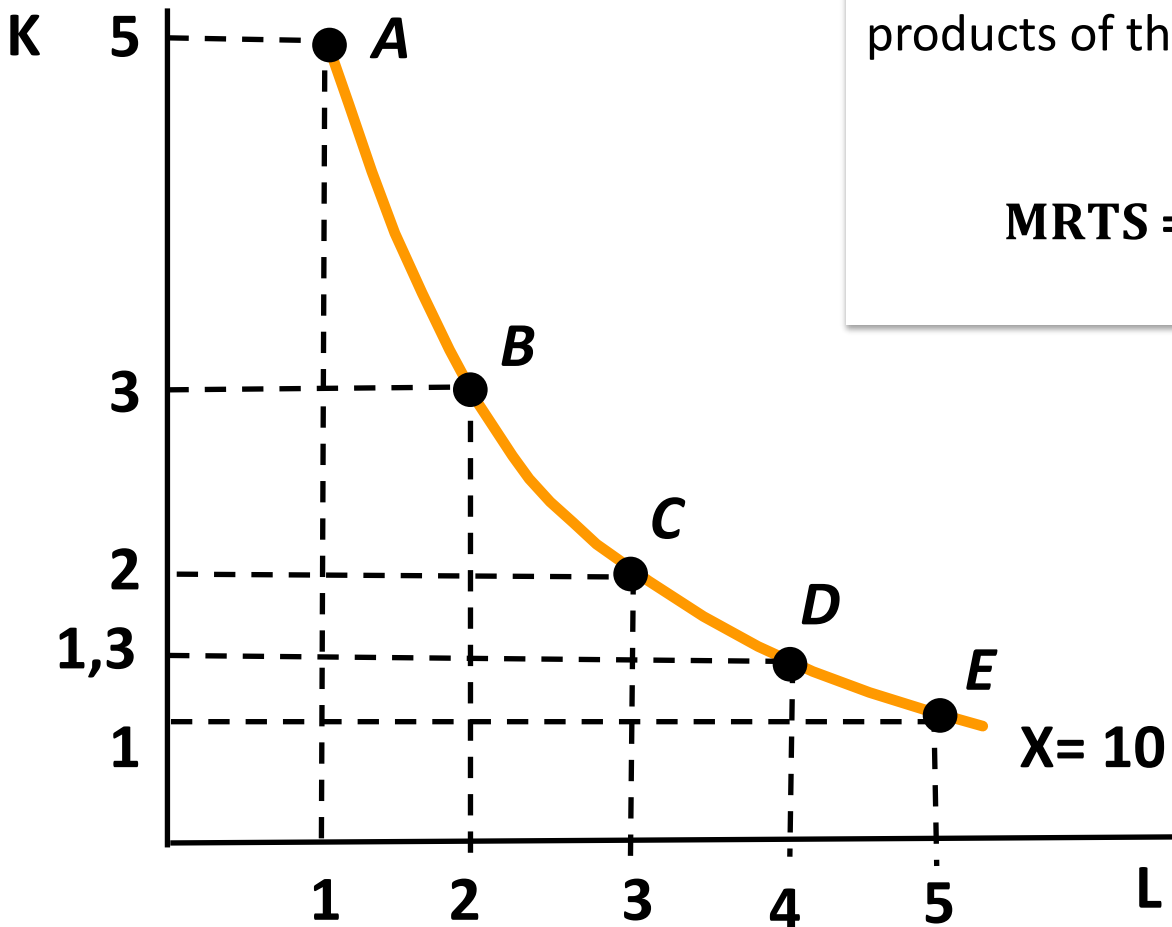
LESSON 4. FIRM BEHAVIOR. PRODUCTION AND COSTS.

The MRTS is the ratio of the marginal products of the production factors

$$\text{MRTS} = - \frac{\Delta K}{\Delta L} = \frac{\text{MPL}}{\text{MPK}} = \frac{\partial X / \partial L}{\partial X / \partial K}$$

QUESTION

What is the MRTS of moving from point A to point B?



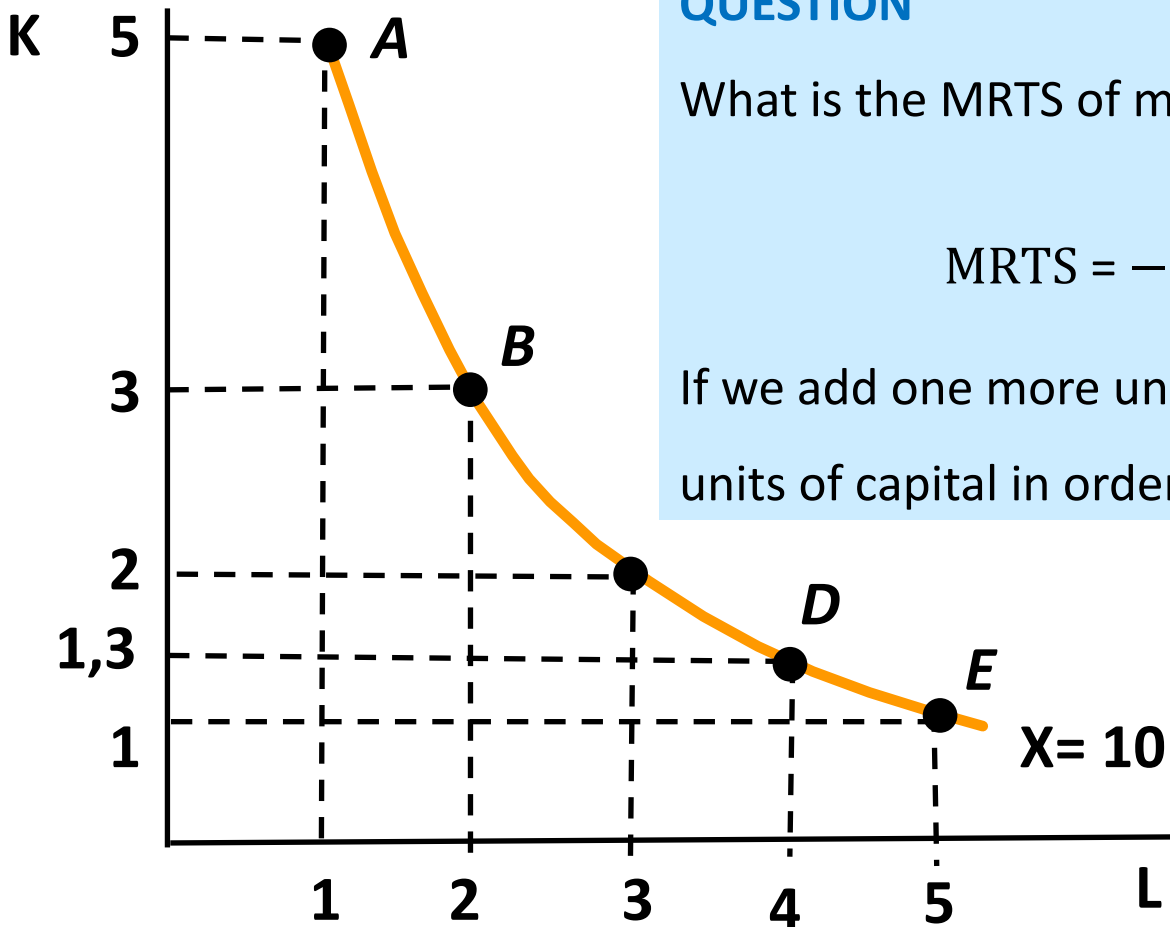
LESSON 4. FIRM BEHAVIOR. PRODUCTION AND COSTS.

QUESTION

What is the MRTS of moving from point A to point B?

$$\text{MRTS} = -\frac{\Delta K}{\Delta L} = -\frac{(3 - 5)}{(2 - 1)} = 2$$

If we add one more unit of labour we have to give up 2 units of capital in order to keep producing 10 units



QUESTION

The production function of a firm is the following:

$$X = K^{1/2} L^{1/2}$$

Calculate the marginal rate of technical substitution for this function.

LESSON 4: FIRM BEHAVIOR. PRODUCTION AND COSTS.

- 4.1. Introduction
- 4.2 Production function
- 4.3 *Average and marginal product*
- 4.4 The law of diminishing marginal returns
- 4.5 Long-run production. Returns to scale.
- 4.6 Expansion path
- 4.7 Marginal cost, average cost and total cost.
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LESSON 4. FIRM BEHAVIOR. PRODUCTION AND COSTS.

- The total product (quantity) obtained through the production function is known as the total productivity of the company.
- The output (quantity) per unit of a particular input (production factor) is also known as **average product**.

$$\text{Average Product (AP)} = \frac{\text{Output}}{\text{Input}} = \frac{X}{L}$$

 *Total output (quantity)*

 *Quantity of input
(labour or capital)*

- From a marginal perspective, the additional output produced as an input increases by one unit is known as **marginal product**.

LESSON 4. FIRM BEHAVIOR. PRODUCTION AND COSTS.

$$\text{Marginal Product (MP)} = \frac{\text{Change in Output}}{\text{Change in Input}} = \frac{\Delta X}{\Delta L} \rightarrow \text{Marginal Product of Labour}$$

Therefore, we can study the marginal product and the average product of any input.

All average and marginal magnitudes maintain the following relationship: *when the average magnitude increases the marginal magnitude is greater and when the average magnitude decreases the marginal magnitude is lower.*

Example with labour as input:

$$AP_L = \frac{Q_X}{Q_L} = \frac{X}{L} = \frac{q}{L}$$

$$MP_L = \frac{dQ_X}{dL} = \frac{dx}{dL}$$

LESSON 4. FIRM BEHAVIOR. PRODUCTION AND COSTS.

L	K ₀	X	AP _L	MP _L
0	10	0	---	---
1	10	10	10	10
2	10	30	15	20
3	10	60	20	30
4	10	80	20	20
5	10	95	19	15
6	10	108	18	13
7	10	112	16	4
8	10	112	14	0
9	10	108	12	-4

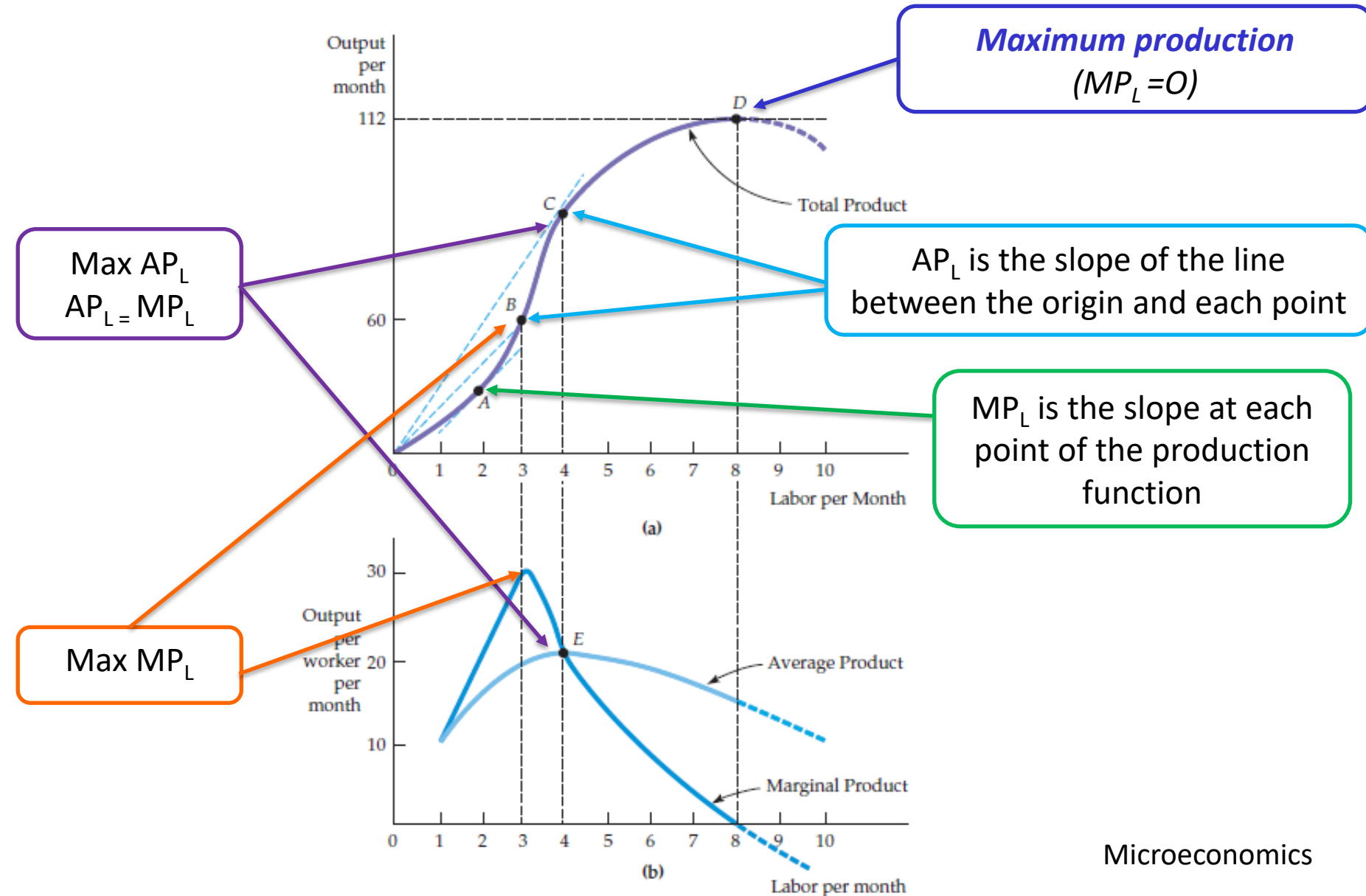
$$AP_L = \frac{X}{L} = \frac{60}{3} = 20$$

$$MP_L = \frac{\Delta X}{\Delta L}$$

$$MP_L = \frac{108 - 95}{6 - 5}$$

$$MP_L = 13$$

LESSON 4. FIRM BEHAVIOR. PRODUCTION AND COSTS.



QUESTION

The production function of a firm is the following:

$$X = K^{1/2} L^{1/2}$$

Calculate the average product for labour and capital if $K = 16$ and $L = 25$

LESSON 4: FIRM BEHAVIOR. PRODUCTION AND COSTS.

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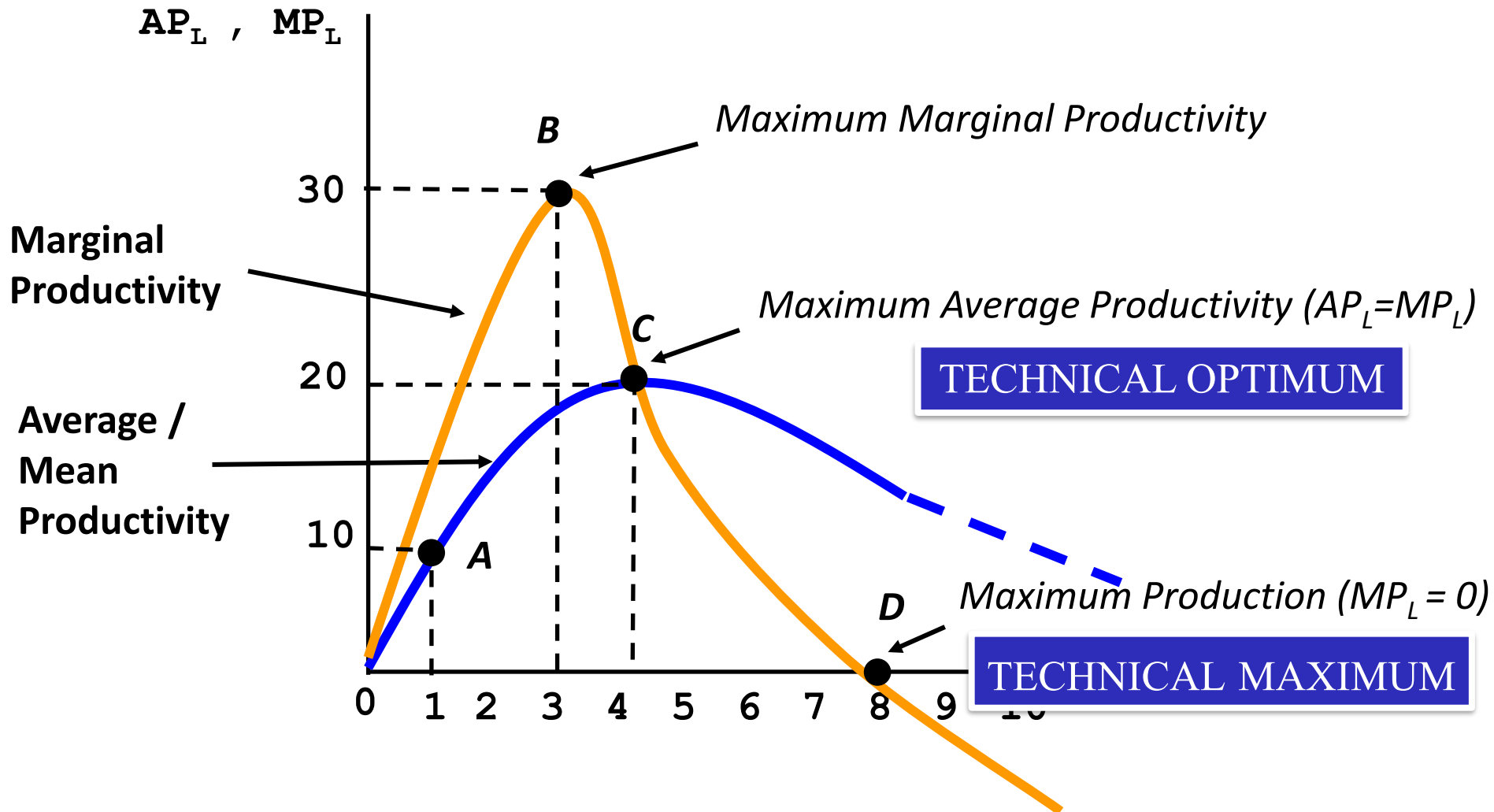
LESSON 4. FIRM BEHAVIOR. PRODUCTION AND COSTS.

Successive increases in one variable production factor (for example, labour) added to a fixed amount of other production factors (capital, land) will cause, when the time comes, that the amount of additional production attributed to each new worker in the company decreases.



There comes a time when marginal productivity starts decreasing

LESSON 4. FIRM BEHAVIOR. PRODUCTION AND COSTS.



LESSON 4. FIRM BEHAVIOR. PRODUCTION AND COSTS.

The firm is efficient since it reaches the maximum output per unit of labour. From this point on, the average productivity per worker begins to fall, and therefore the marginal productivity falls even more.

Here the firm reaches its technical maximum. Even if you want to, by increasing the number of units of work the firm can no longer increase the quantity produced.

Q	L	AP_L	MP_L
10	1	10	10
30	2	15	20
60	3	20	30
80	4	20	20
95	5	19	15
105	6	17.5	10
110	7	15.7	5
110	8	13.7	0
105	9	11.7	-5

QUESTION

The production function of a firm is the following:

$$X = 10L^2 + 100L - L^3$$

- a) Calculate the average product and marginal product
- b) Calculate the technical optimum
- c) Calculate the technical maximum

LESSON 4: FIRM BEHAVIOR. PRODUCTION AND COSTS.

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LESSON 4. FIRM BEHAVIOR. PRODUCTION AND COSTS.

- **LONG RUN:** Both production factors are variable; any amount of K and L can be used.

The long run is the minimum period of time needed to change the amounts of all factors.

- **SHORT RUN:** Period of time during which it is not possible to modify at least one of the factors used in a production process.

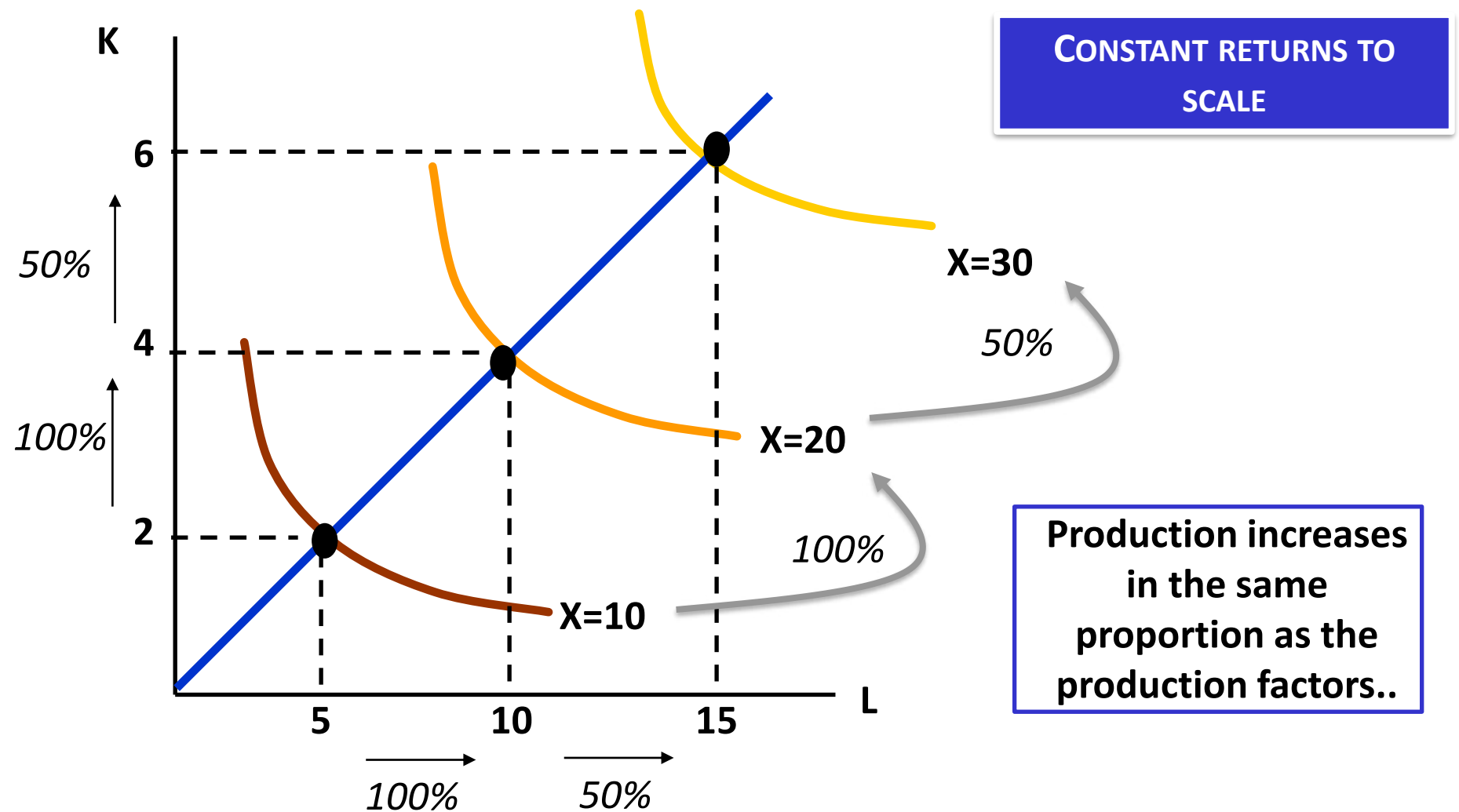
One of the production factors (at least) remains fixed. Any amount of the variable factors can be changed, but not the fixed one.

RETURNS TO SCALE try to measure how much does production increase when the two production factors vary simultaneously and in the same proportion.

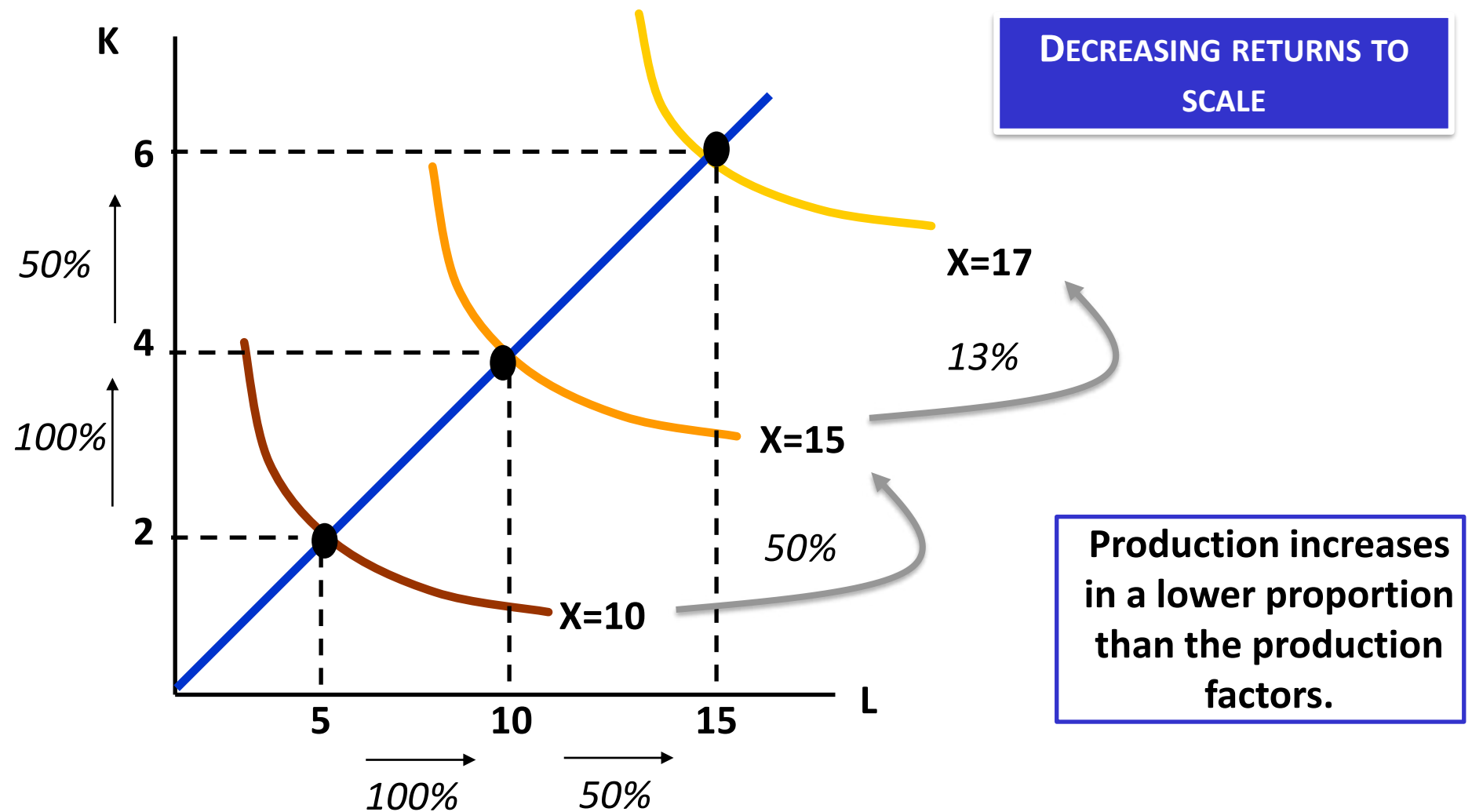
There are three possibilities:

- *Constant returns to scale*
- *Increasing returns to scale*
- *Decreasing returns to scale*

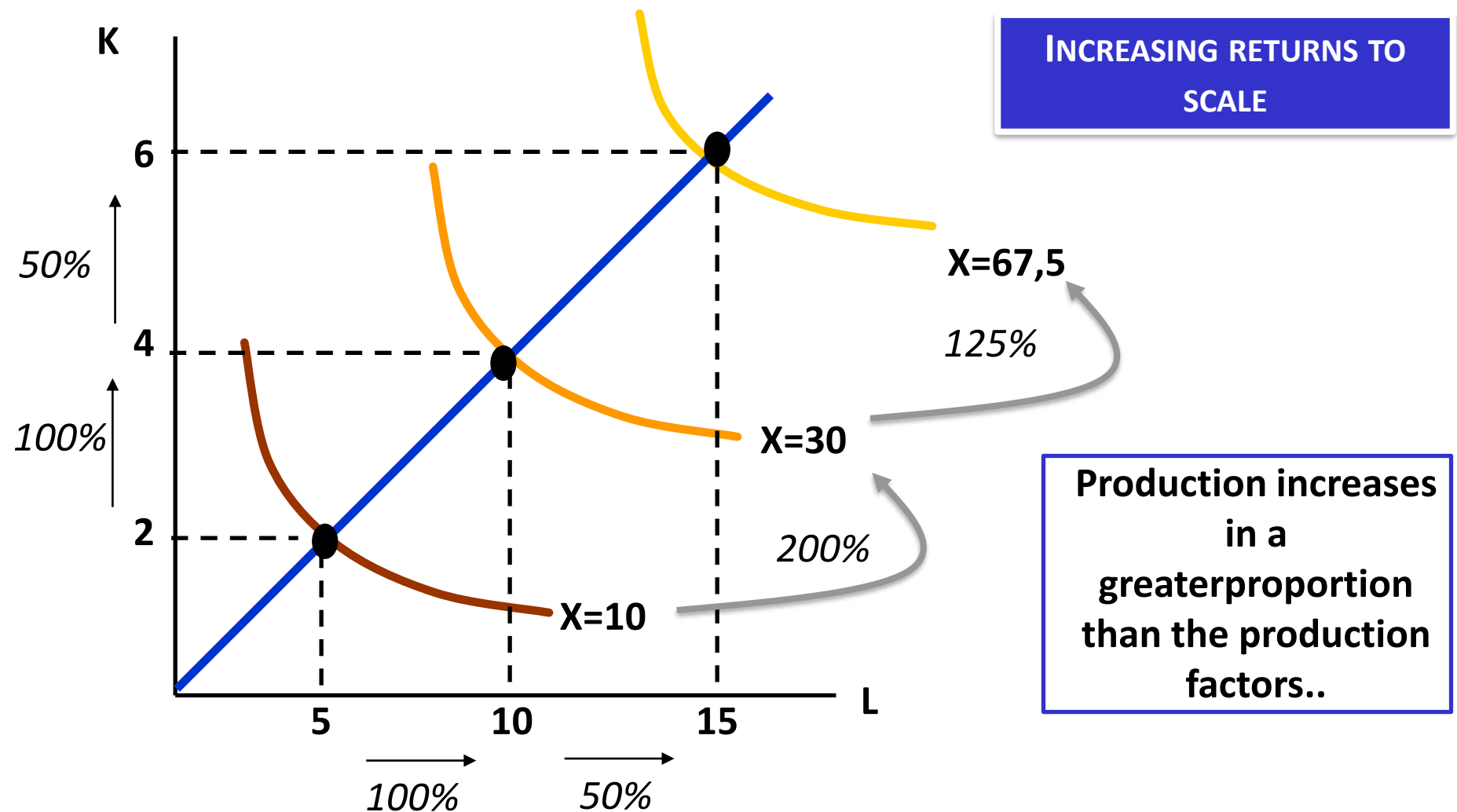
LESSON 4. FIRM BEHAVIOR. PRODUCTION AND COSTS.



LESSON 4. FIRM BEHAVIOR. PRODUCTION AND COSTS.



LESSON 4. FIRM BEHAVIOR. PRODUCTION AND COSTS.



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PRODUCTION SET:
ALL FEASIBLE TECHNIQUES



Production function: *only technically efficient ones*

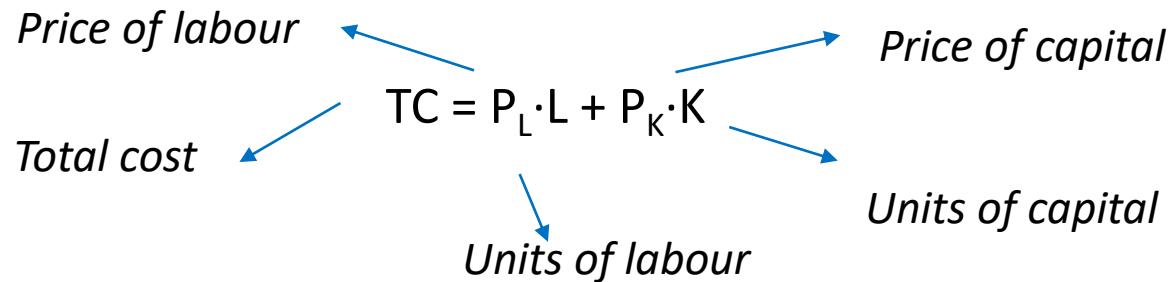


There is only a single economically efficient technique

If we link the different efficient production methods (technically and economically) the **expansion path** is obtained.

LESSON 4. FIRM BEHAVIOR. PRODUCTION AND COSTS.

To figure out the optimum we need to use isocost lines. An **ISOCOST** is a set of combinations of production factors (K, L) that have the same cost level:



Two questions:

- These are economic costs (*opportunity cost is included*), not accounting costs.
- It is more accurate to use the *cost of use of capital*, not the price of capital.

So:

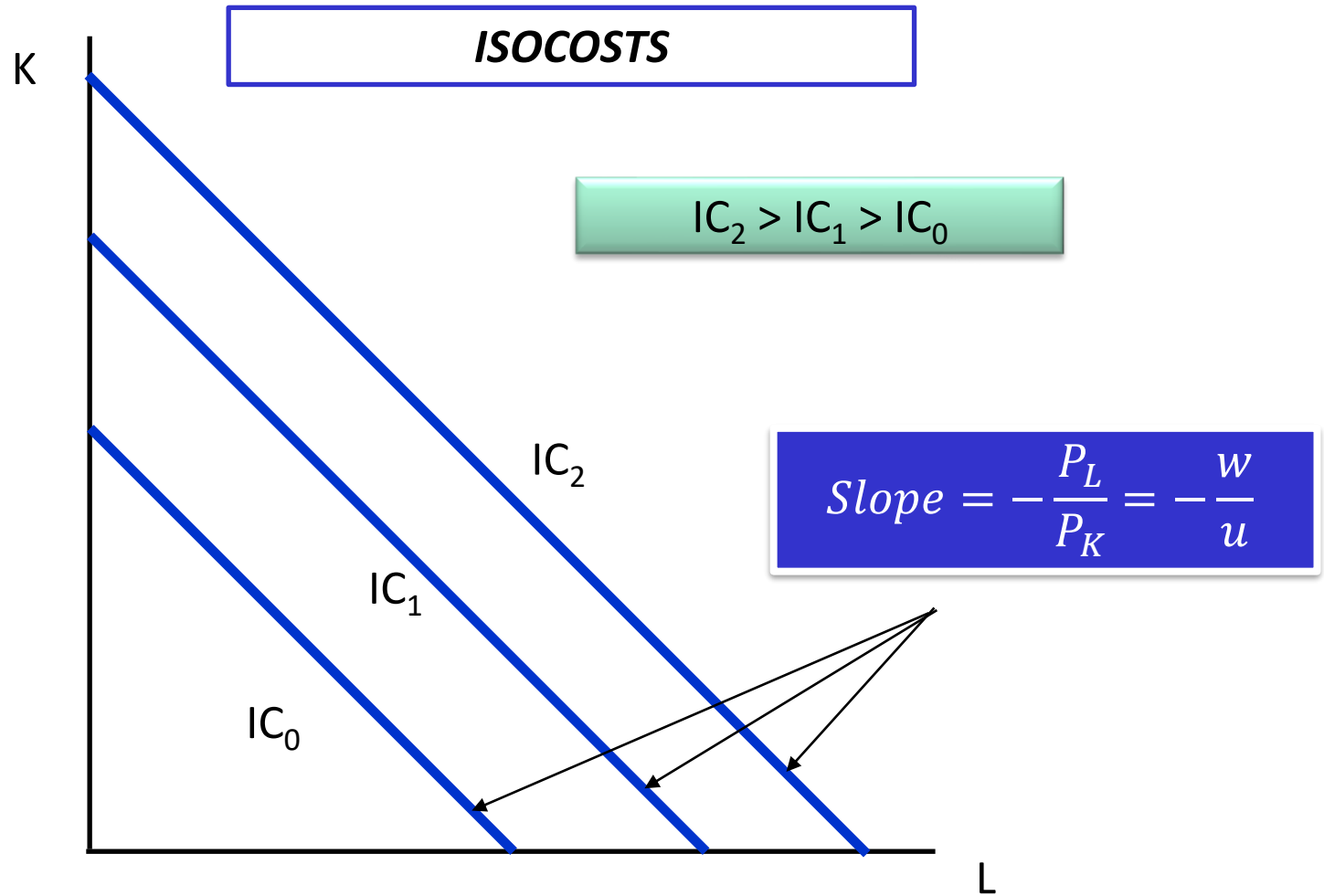
Diagram illustrating the revised total cost equation:

$$TC = w \cdot L + u \cdot K$$

Arrows point from the equation to the following labels:

- Wages (pointing to w)
- Cost of use of capital (pointing to u)

LESSON 4. FIRM BEHAVIOR. PRODUCTION AND COSTS.



QUESTION

A firm uses two production factors: capital and labour. The cost of use of capital is 15€/day and the daily wage is 32€. The total cost that the firm is going to spend daily is 650€.

- What would be the daily isocost for this firm?
- What would be the value of the slope of that isocost?

LESSON 4. FIRM BEHAVIOR. PRODUCTION AND COSTS.

The firm tries to produce a *certain amount of X* (X_1) incurring the *lowest possible costs*:

$$\text{Min } \boxed{TC = wL + uK} \longrightarrow \text{Isocost}$$

$$\boxed{X_1 = f(K, L)} \longrightarrow \text{Production function} \longrightarrow \text{Isoquants}$$

Producer's equilibrium will occur when two conditions are met:

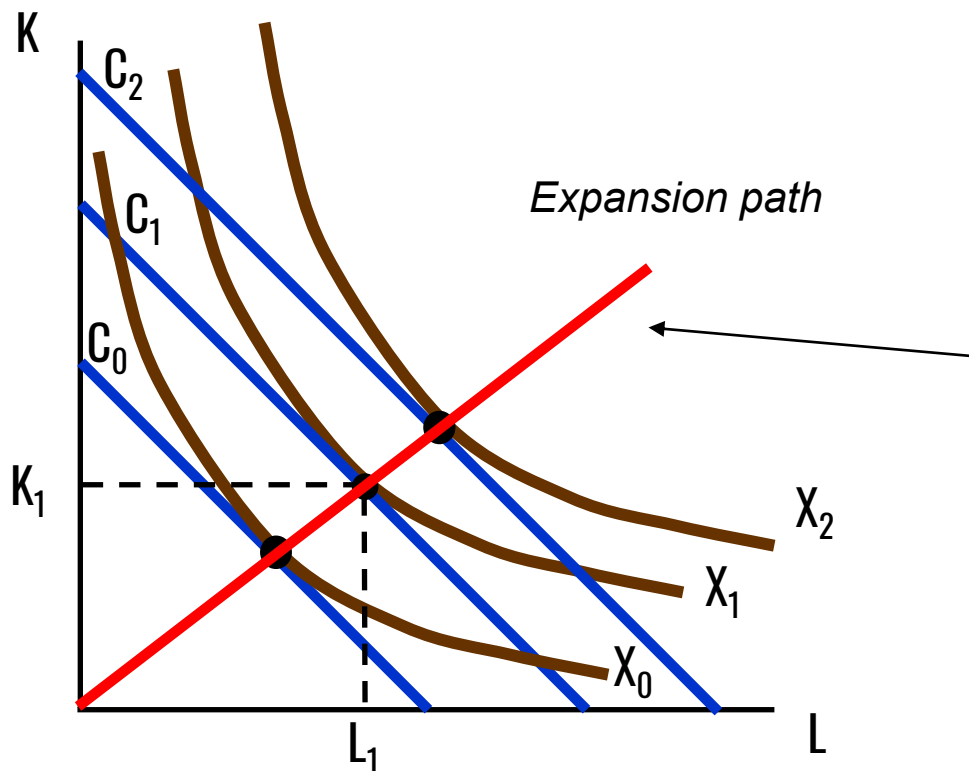
- (1) That the representative isoquant curve of X_1 is *tangent* to an isocost line, and
- (2) That the choice belongs to the production function.

The *economically efficient* technique is the combination of production factors (K, L) that results from the solution of the following system of equations:

$$(1) \text{ MRTS} = w/u$$

$$(2) X_1 = f(K, L)$$

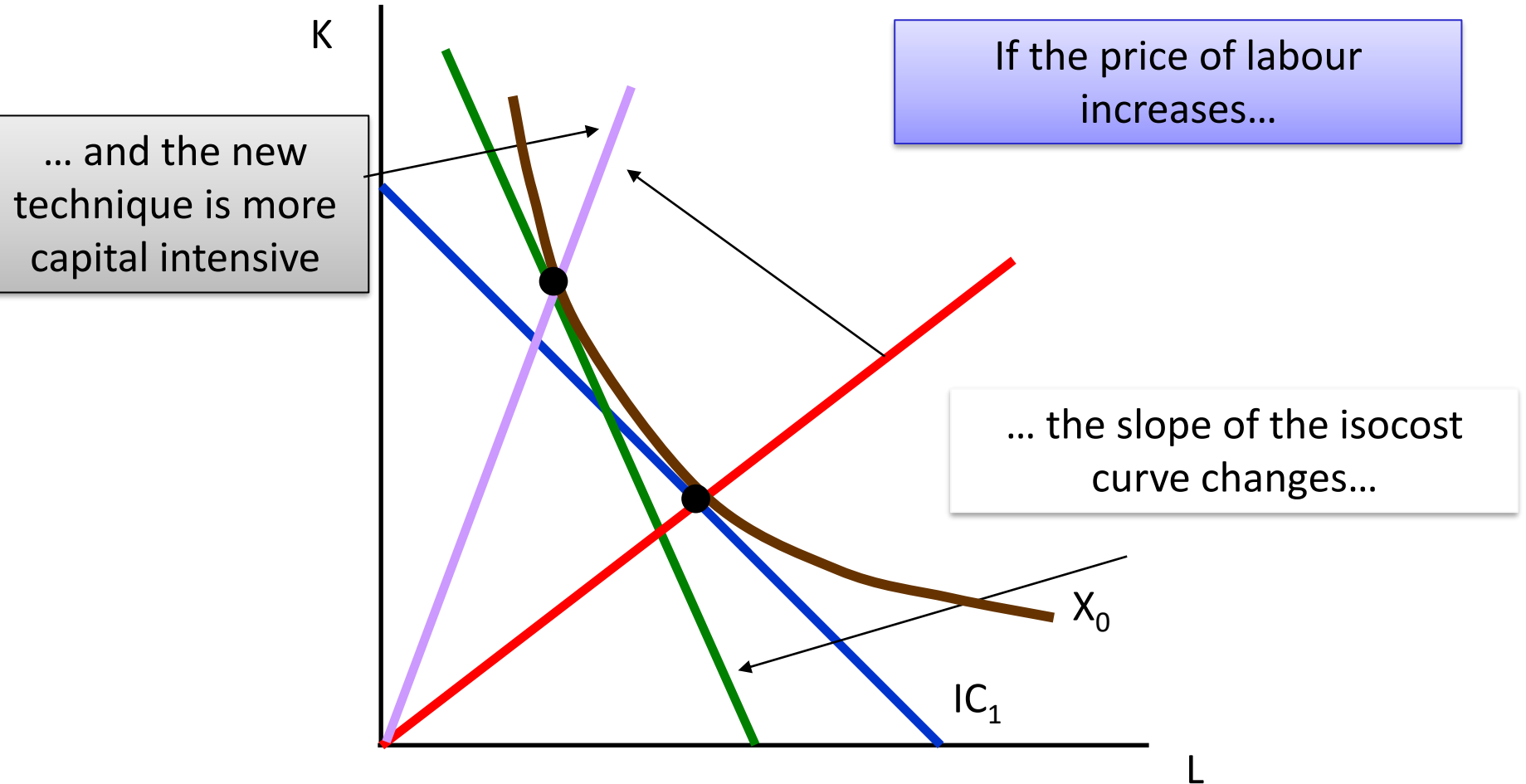
LESSON 4. FIRM BEHAVIOR. PRODUCTION AND COSTS.



In all expansion path points
 $MRTS = w/u$

The firm's **long-run expansion path** shows the least-cost combinations of factors of production (capital and labor) associated with each level of output in the long run.

LESSON 4. FIRM BEHAVIOR. PRODUCTION AND COSTS.

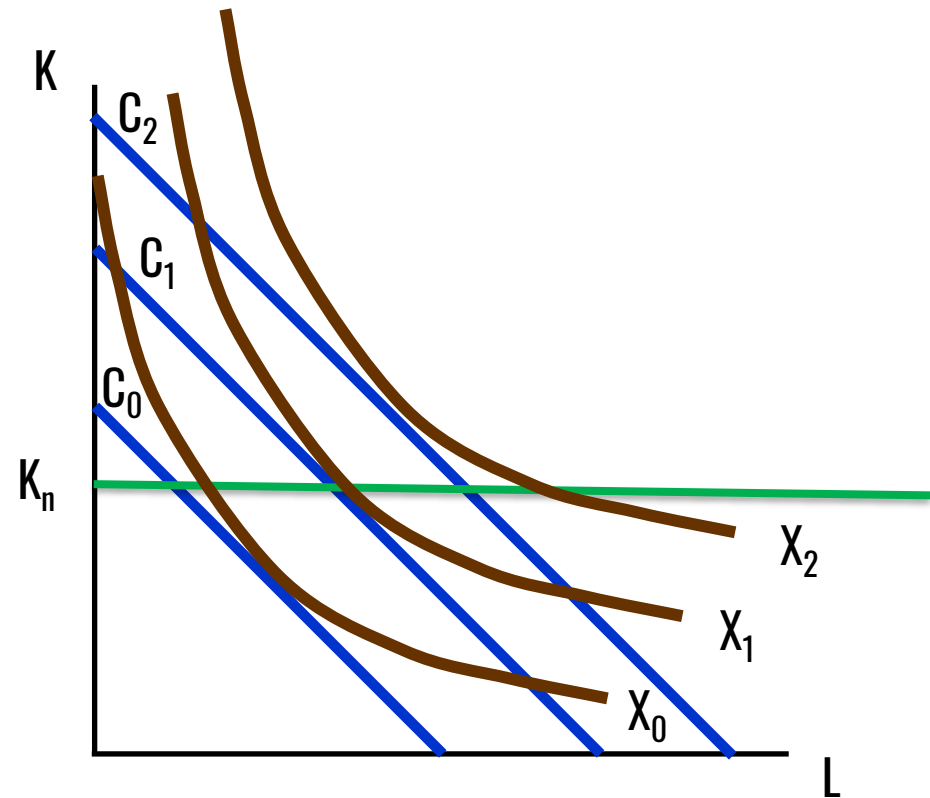


LESSON 4. FIRM BEHAVIOR. PRODUCTION AND COSTS.

And what about the short-run expansion path?

In the short-run there is rigidity in the expansion path since one of the factors of production is fixed (capital).

Therefore, the expansion path in the short run is a straight line given by the established level of capital.



QUESTION

The production function and isocost of a firm are the following:

$$X = K^{1/2}L^{1/2} \quad TC = 32L + 15K$$

- Get the expansion path for this firm.
- What combination of labor and capital would maximize production if the total cost of the company is 50000€?

LESSON 4: FIRM BEHAVIOR. PRODUCTION AND COSTS.

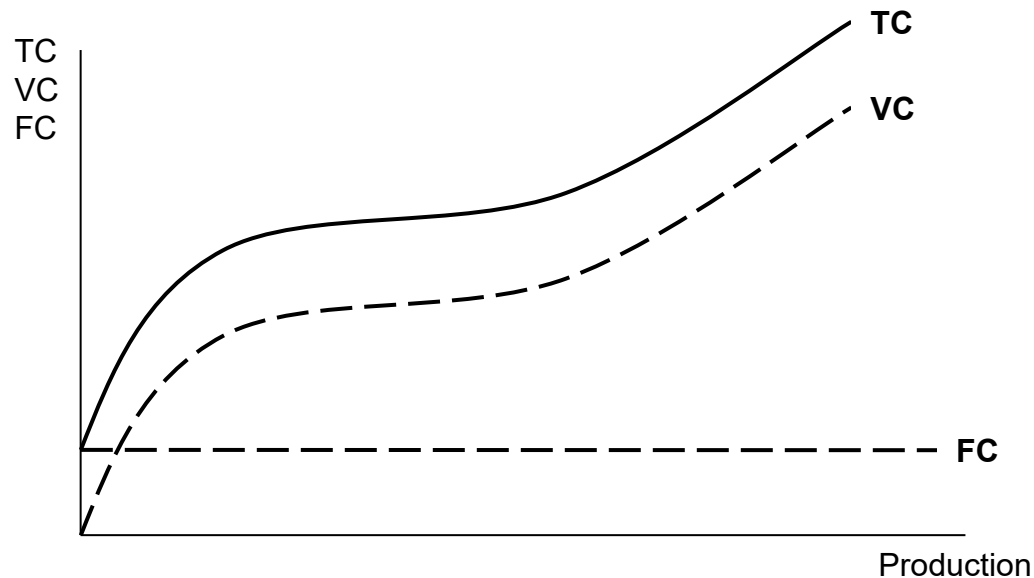
- 4.1. Introduction
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LESSON 4. FIRM BEHAVIOR. PRODUCTION AND COSTS.

$$\text{TOTAL COSTS (TC)} = \text{VARIABLE COSTS (VC)} + \text{FIXED COSTS (FC)}$$

The magnitude varies with
the level of production

They are not affected by variations in
the quantity produced



LESSON 4. FIRM BEHAVIOR. PRODUCTION AND COSTS.

- **AVERAGE COSTS** are costs per unit of product

$$ATC = \frac{TC}{Q}$$

If we distinguish between variable and fixed costs:

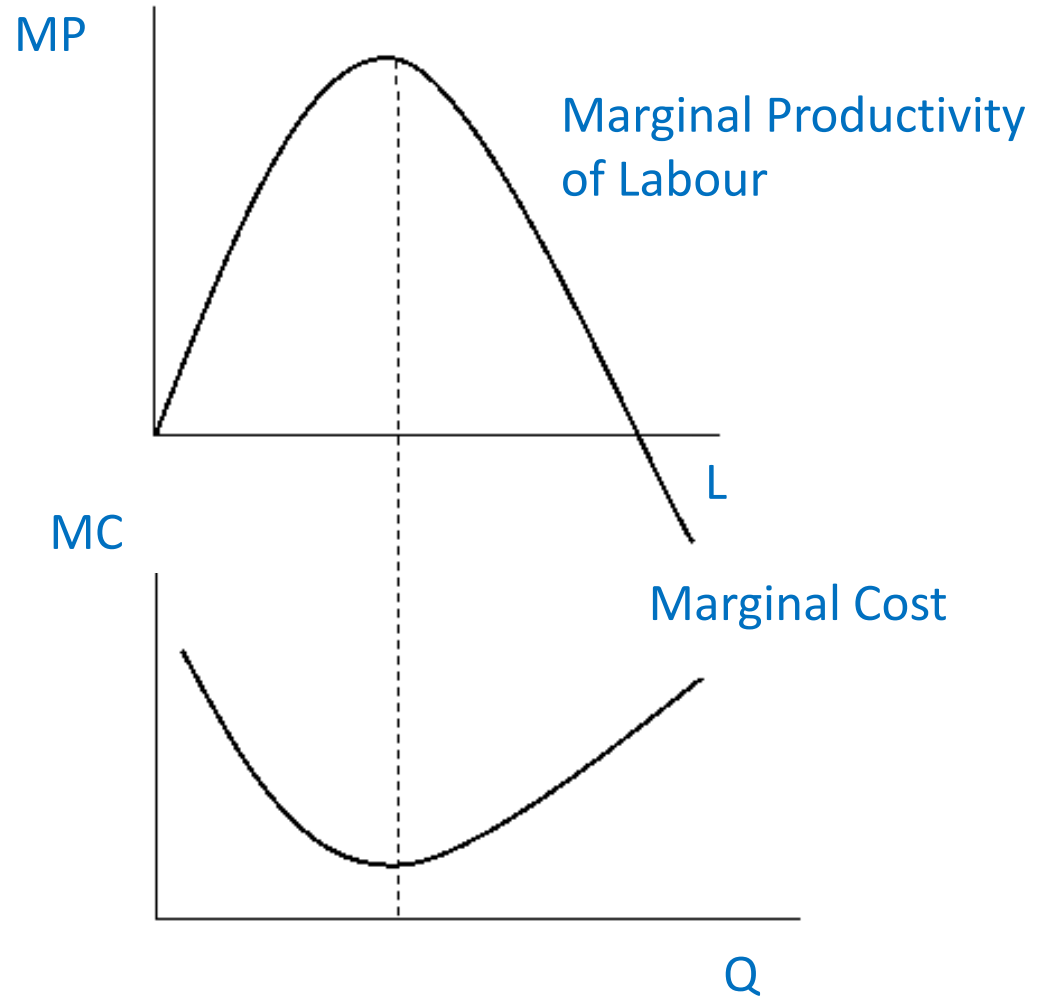
$$ATC = \frac{TC}{Q} = \frac{VC + FC}{Q} = \underbrace{\frac{VC}{Q}} + \underbrace{\frac{FC}{Q}}$$

$$ATC = AVC + AFC$$

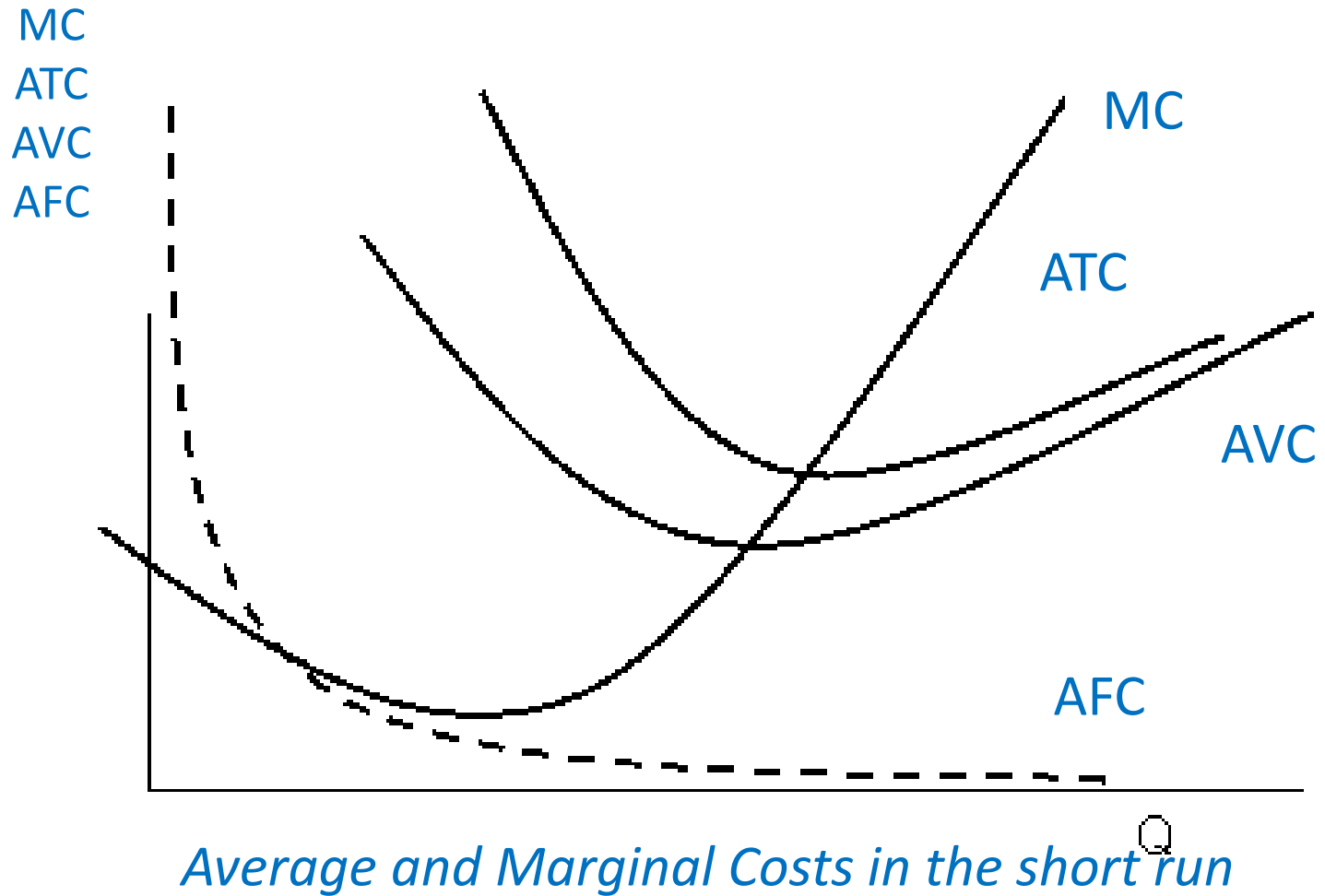
- **MARGINAL COST** is the increase in costs derived from increasing production by one additional unit

LESSON 4. FIRM BEHAVIOR. PRODUCTION AND COSTS.

We now link
production with
short-run costs



LESSON 4. FIRM BEHAVIOR. PRODUCTION AND COSTS.

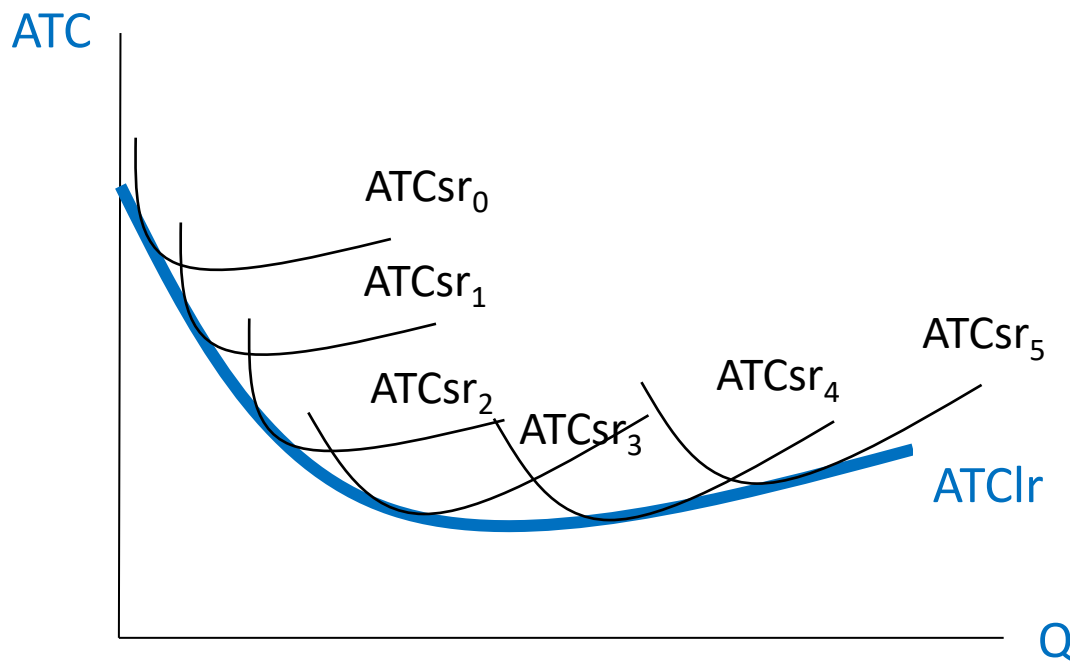


LESSON 4: FIRM BEHAVIOR. PRODUCTION AND COSTS.

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LESSON 4. FIRM BEHAVIOR. PRODUCTION AND COSTS.

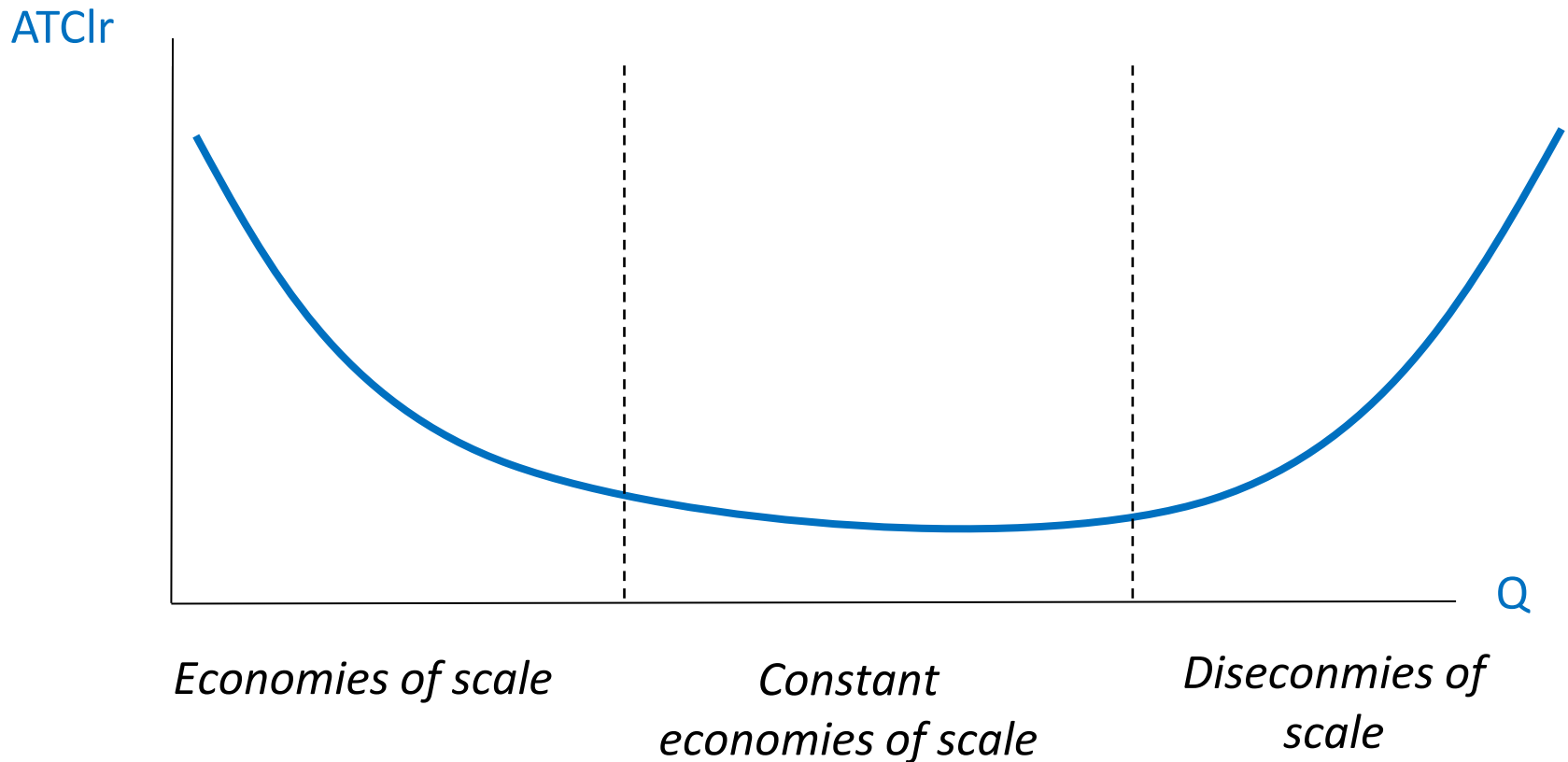
If we consider many production facilities (factory) sizes, the long-run cost curve would be:



In the long-run average total cost curve (**ATC_{lr}**) we consider that K is variable and the company can modify the size of the production facility.

LESSON 4. FIRM BEHAVIOR. PRODUCTION AND COSTS.

- The long-run average total cost curve is U-shaped.



LESSON 4. FIRM BEHAVIOR. PRODUCTION AND COSTS.

- In some economic sectors such as construction, decreasing returns to scale appear early
- Internal economies versus external economies
- Economies of scale (Internal economies)
- External economies
 - Agglomeration economies: Advantages arising from concentrating production in a specific area
 - Localization economies: Advantages arising from the localization together of firms from the same industry
 - Urbanization economies: Advantages arising from location near the economic activity
 - Economies of scope: Advantages arising from producing one product that lead to a reduction in the cost of production of another product.



MICROECONOMICS

TUC - 5: EXERCISES

EXERCISE 1

A company that is part of a Perfect Competition market has the following cost function:

$$TC = 2Q^3 - 72Q^2 + 964Q + 10580$$

Market Price is €1450

- A. Calculate how many units will produce the company to maximize profit.
- B. Calculate Total Profit.

EXERCISE 2

A company that is part of a Perfect Competition market has the following cost function:

$$TC = 2Q^3 - 75Q^2 + 1000Q + 361$$

The Supply and Demand functions on this market are the following:

$$Q_d = 10000 - 10P$$

$$Q_s = 20P - 2000$$

- A. Calculate how many units will produce the company to maximize profit.
- B. Calculate Total Profit.

EXERCISE 3

A company that is part of a Perfect Competition market has the following cost function:

$$TC = Q^3 - 20Q^2 + 150Q + 2500$$

$$VC = Q^3 - 20Q^2 + 150Q$$

The Supply and Demand functions on this market are the following:

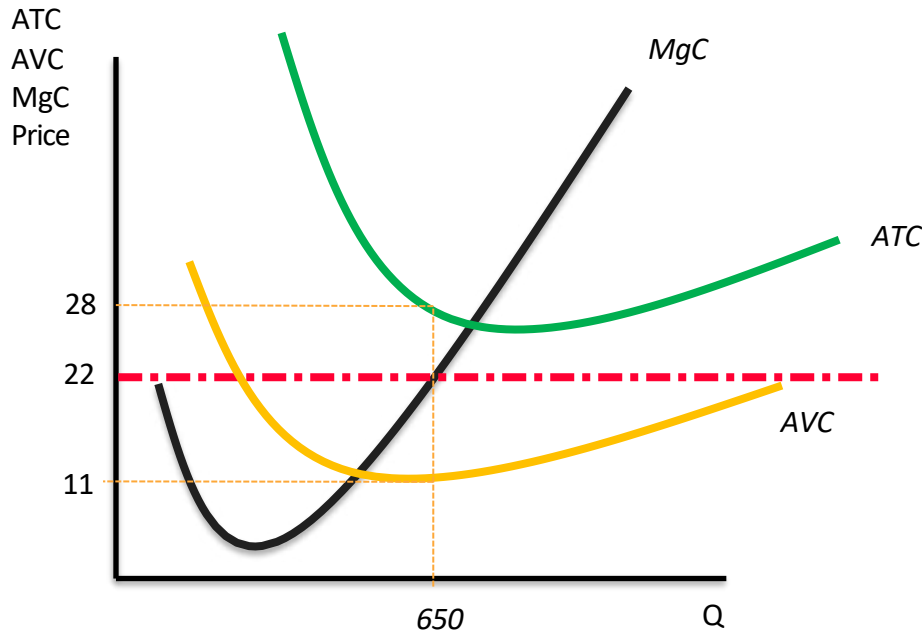
$$Q_d = 4500 - 10P$$

$$Q_s = 20P - 1500$$

- A. Calculate how many units will produce the company to maximize profit. Calculate Total Profit.
- B. Is the Company having a Profit? If not, what is its situation? Should the company keep producing? Or better close the production line?

EXERCISE 4

Calculate Total Cost, Total Income, Fix Cost, Variable Cost and Profit for this company. ¿Should they continue producing?



EXERCISE 5

Please indicate if the sentences are true or false:

- A. In perfect competition, the homogeneity of products means that the price is the same for all firms.
- B. In the long run, when the price is above the average total cost, firms will have extraordinary profits.
- C. In the long run, since firms will produce the quantity at which marginal revenue equals marginal cost, their revenue will be zero.
- D. Producer surplus is equal to consumer surplus in perfect competition, leading to a situation of economic market efficiency.
- E. In the long run, firms will always sell at the same price, with no possibility of it being affected.

EXERCISE 5

- F. When costs are decreasing, large extraordinary profits are generated, and there is a massive influx of companies, causing the price to fall below the initial price in the long run.
- G. In the long run, the extraordinary profits generated by increasing demand attracts more companies to the market, thus reducing the quantity supplied and increasing the price.
- H. Firms participating in a perfectly competitive market can change prices in the long run; they cease to be price-takers.

LESSON 5: ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

- 5.1. *Market structure*
- 5.2 Main characteristics of perfect competition
- 5.3 Profit maximization in firms
- 5.4 The decision to produce and the short-run market supply curve.
- 5.5 The efficiency of the competitive equilibrium in the short-run.
- 5.6 The long-run supply curve of the firm.
- 5.7 Competitive equilibrium. The functioning of competitive markets

LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

There are several market structures, the main ones are:



	Perfect competition	Monopolistic competition	Oligopoly	Monopoly
Number of firms	Many	Many	Low	One
Variety of goods	None	Some	Some	None
Control over prices	None	Little	Some	Complete
Barriers to entry and exit	None	Low	High	Complete

LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

Another way to figure out different market structures:

		Number of buyers		
		One	Some	Many
Number of sellers	One	Bilateral monopoly	Partial monopoly	Monopoly
	Some	Partial monopsony	Bilateral oligopoly	Oligopoly
	Many	Monopsony	Oligopsony	Perfect and monopolistic competition

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LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

For a market to be of perfect competition, the following conditions must be met:

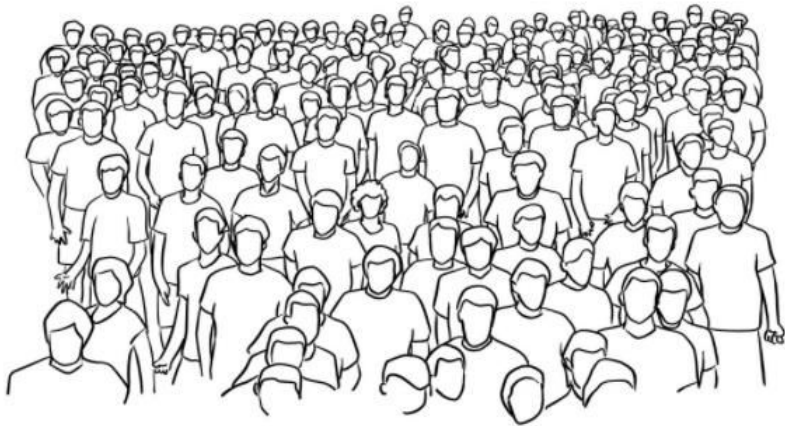
1. There must be a high number of buyers and sellers.
2. The product offered by the different companies in the market is homogeneous.
3. There is perfect information
4. Free mobility of resources: non-existence of entry and exit barriers for new companies in the market.



Firms are price-takers

LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

HIGH NUMBER OF BUYERS AND SELLERS



Existence of a high number of companies so that their individual decisions do not influence the price of the product. The companies are **price-takers**, they consider given the price of the product.

LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

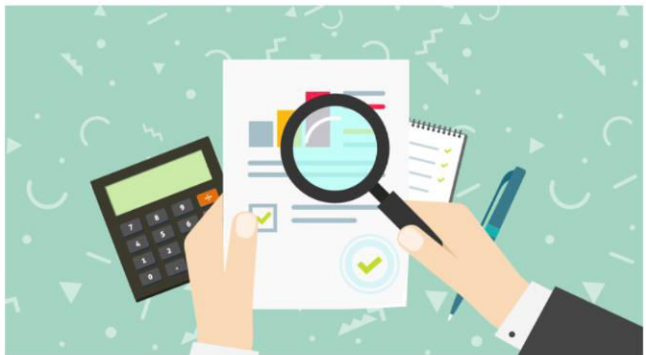
THE PRODUCT IS HOMOGENEOUS



All companies in an industry produce an identical product.

There are no brands or advertising.

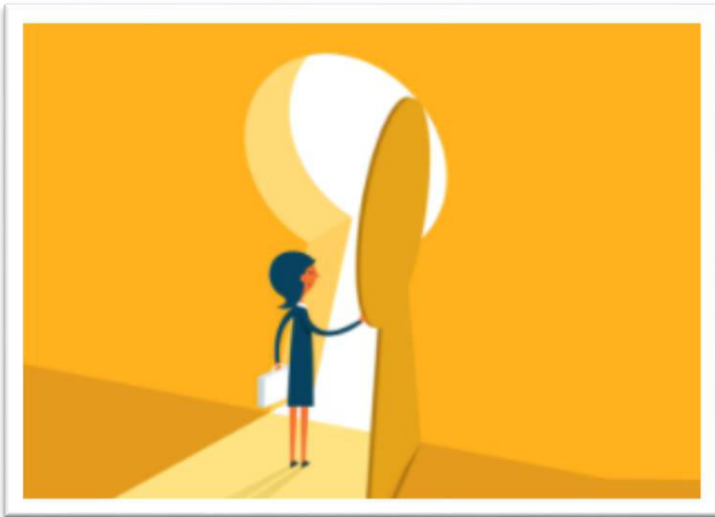
'TRANSPARENT' MARKET



All buyers and sellers in a market have perfect information about its conditions at no cost.

LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

NON-EXISTENCE OF ENTRY AND EXIT BARRIERS



Companies can easily enter or exit an industry: there are no patents, licenses, R&D expenses, significant plant investments, or legal restrictions.

LESSON 5: ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

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LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

The working hypothesis that guides the behavior of the firm is profit maximization.

$$P = TR - TC$$

Where: $P \rightarrow$ Profit

$TR \rightarrow$ Total revenue

$TC \rightarrow$ Total cost

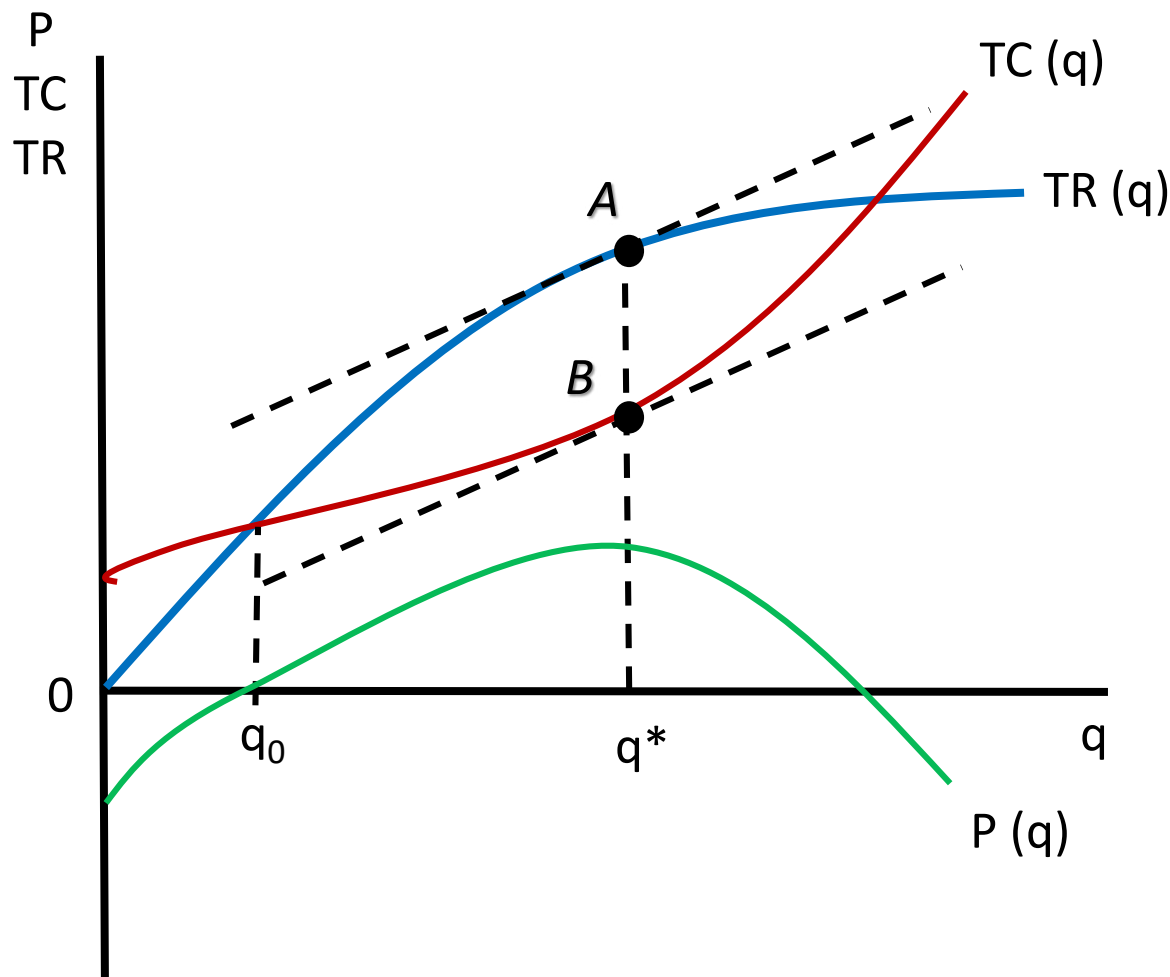


Are companies profit maximizers?

Large companies can behave as maximizers of sales, the company's market value, dividends, ...

However, we will maintain as a hypothesis that companies behave as profit maximizers.

LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.



In the short run, the firm will choose that level of production for which the difference between total revenue and total cost maximizes.

LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

Analytically, for any firm:

$$\text{Max: } P = TR - TC$$

$$\frac{\partial P}{\partial q} = 0 \rightarrow \frac{\partial TR}{\partial q} - \frac{\partial TC}{\partial q} = 0 \rightarrow \boxed{\frac{\partial TR}{\partial q}} = \boxed{\frac{\partial TC}{\partial q}}$$

Marginal Revenue (MR) points to $\frac{\partial TR}{\partial q}$

Marginal Cost (MC) points to $\frac{\partial TC}{\partial q}$

$$\left\{ \begin{array}{l} \text{Marginal Revenue (MR)} = \frac{\partial TR}{\partial q} \\ \text{Marginal Cost (MC)} = \frac{\partial TC}{\partial q} \end{array} \right.$$

LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

Therefore, when the firm maximizes profits, marginal revenue equals marginal cost:

$$\textit{Maximize Profit} \rightarrow MR = MC$$

Being marginal revenue (cost) the variation that total revenue (total cost) experiences when the level of production is increased by one unit.

LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

The marginal revenue for a competitive firm

Therefore total revenue equals the price (p) of the product multiplied by the number of units sold (q).

$$TR = p \cdot q$$

And since a competitive firm is a price-taker, price is constant ($p = \text{const}$).

For a *competitive firm* the marginal revenue will be:

$$MR = \frac{\partial TR}{\partial q} = \frac{\partial (p \cdot q)}{q} = p$$

LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

Therefore, since the profit maximization rule for any company is:

$$MR = MC$$

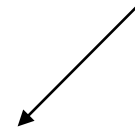
And since in a competitive market

$$MR = p$$

The specific profit maximization rule for a competitive firm is:

$$\text{Max Profit} \rightarrow p = MC = AR$$

*Average
Revenue*



LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

Exercise

A firm has the following total cost function:

$$TC = 250Q^2 + 300Q + 125$$

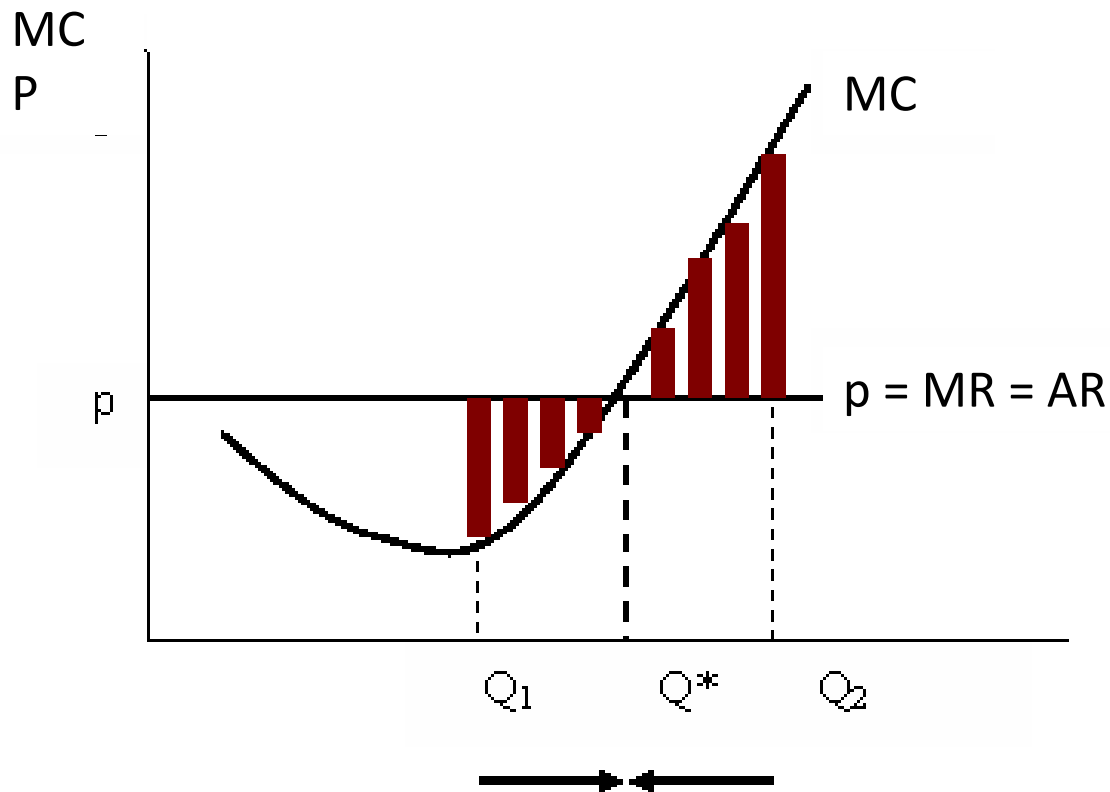
- A. Calculate the marginal cost for this firm

- B. When will this firm maximize profit?

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LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.



Since the profit maximization rule for a competitive firm is:

$$p = MC$$

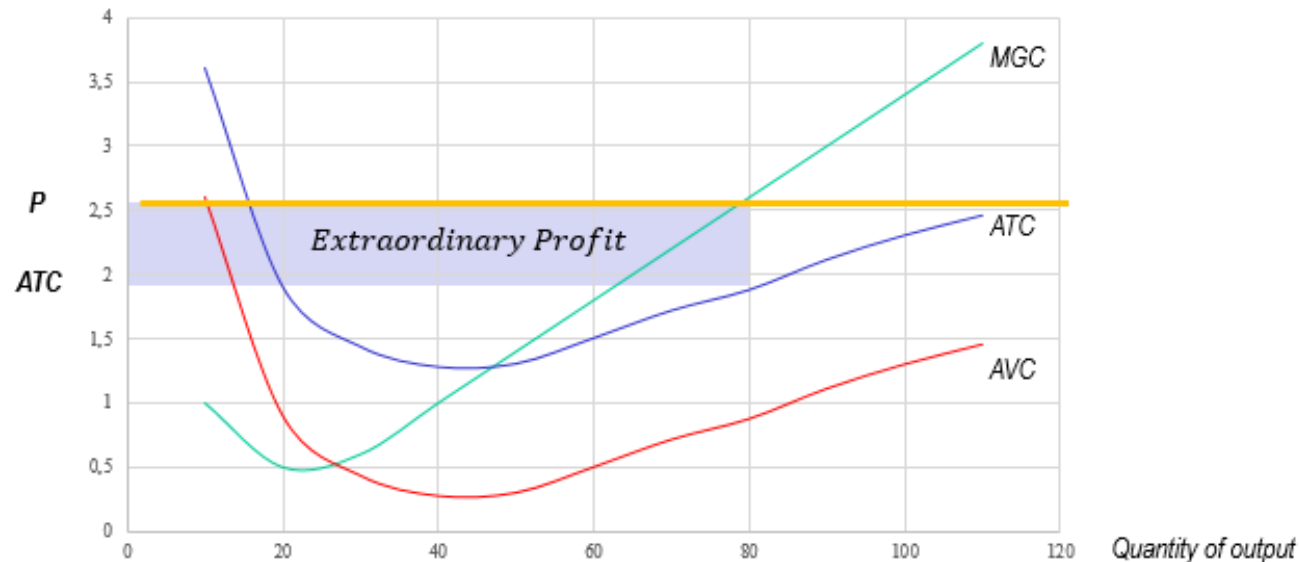
For a market price 'p', the firm will decide to produce the quantity Q^* .

LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

In the short run, a firm can achieve: extraordinary profit, normal profit (zero), or losses

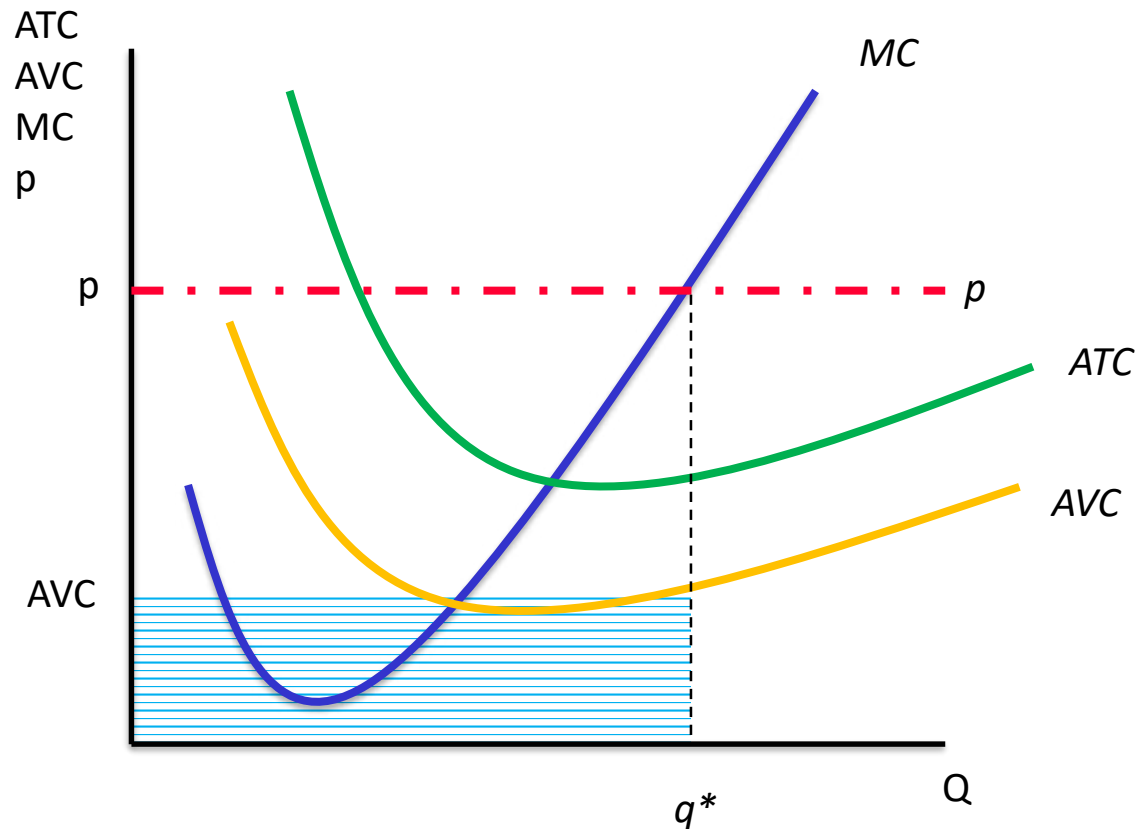
And, in the latter case, the firm must decide to continue producing or to shut down

Costs and revenue



LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

In the following graph:

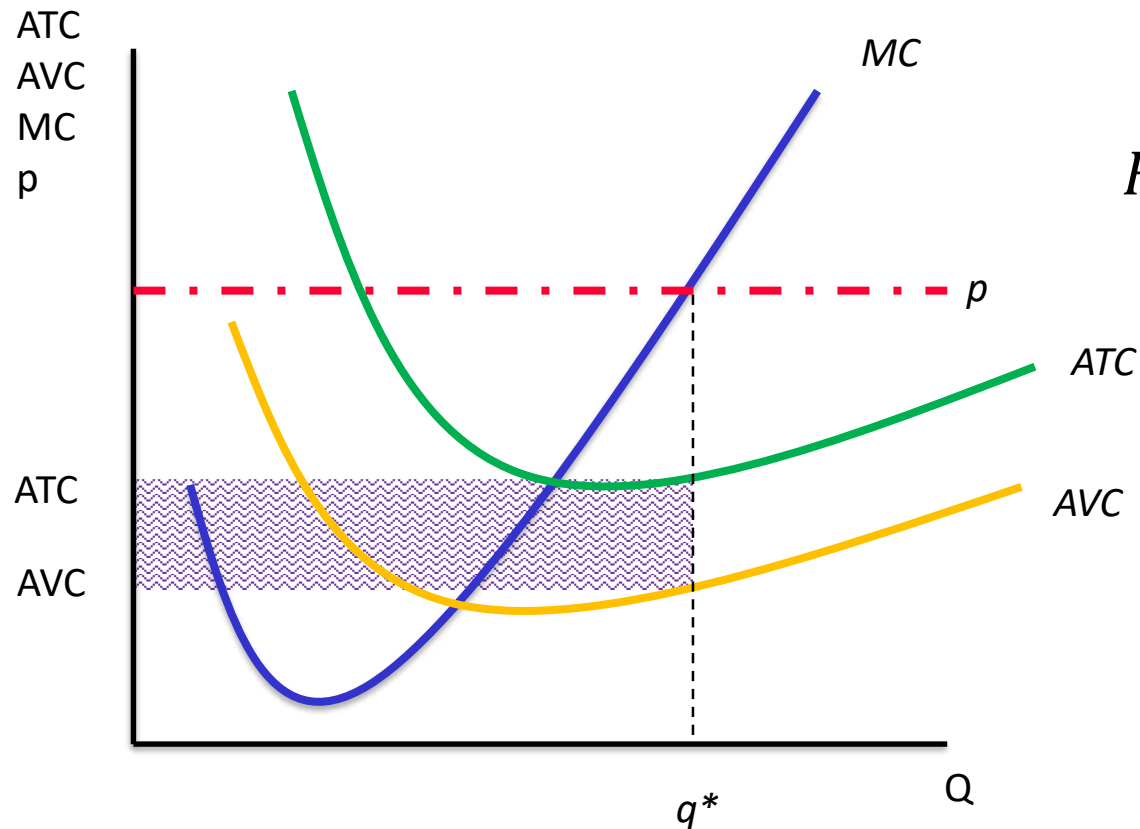


Variable Costs

$$VC = AVC \cdot q^*$$

LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

In the following graph:

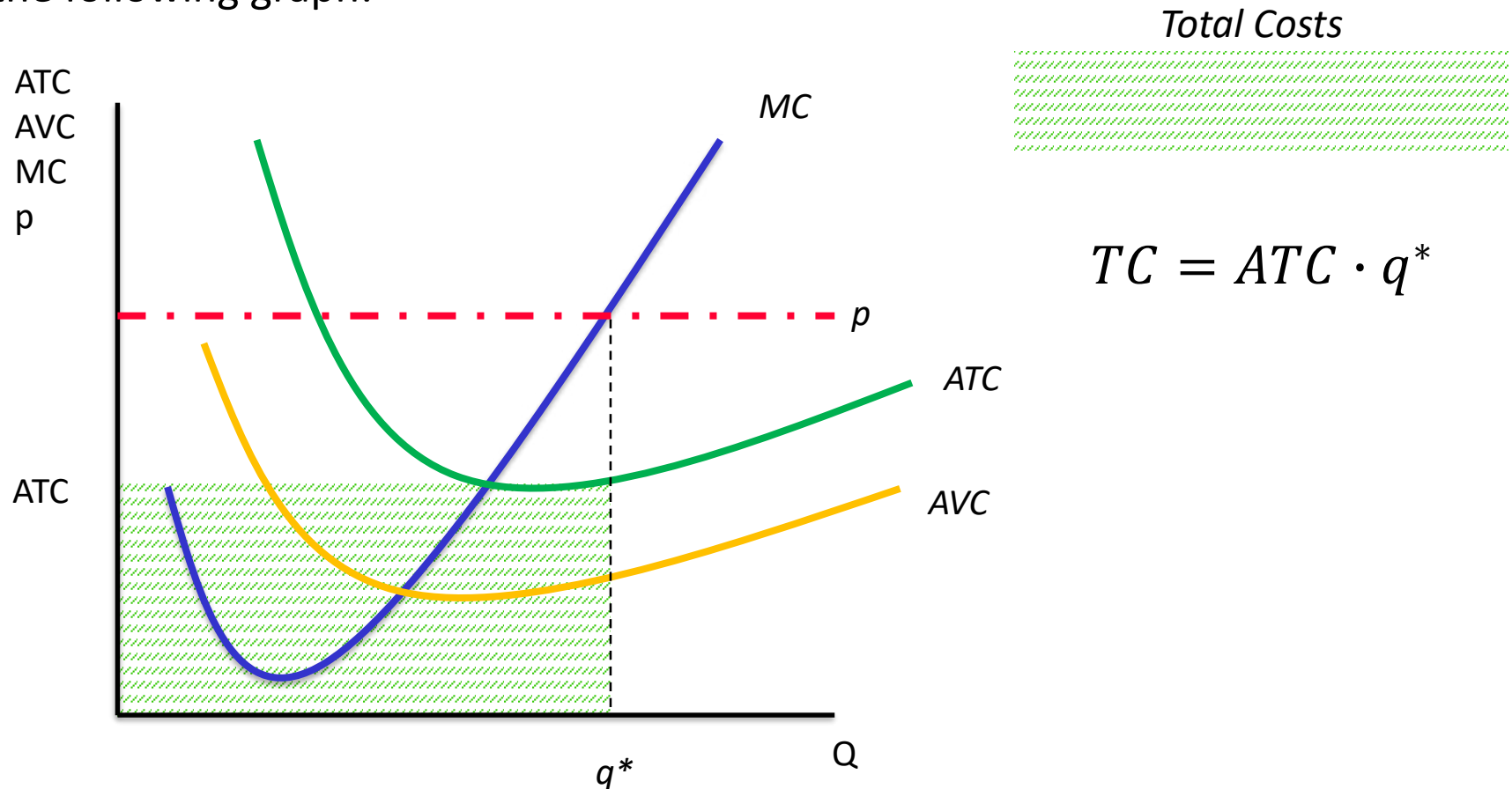


Fixed Costs

$$FC = \underbrace{(ATC - AVC)}_{AFC} \cdot q^*$$

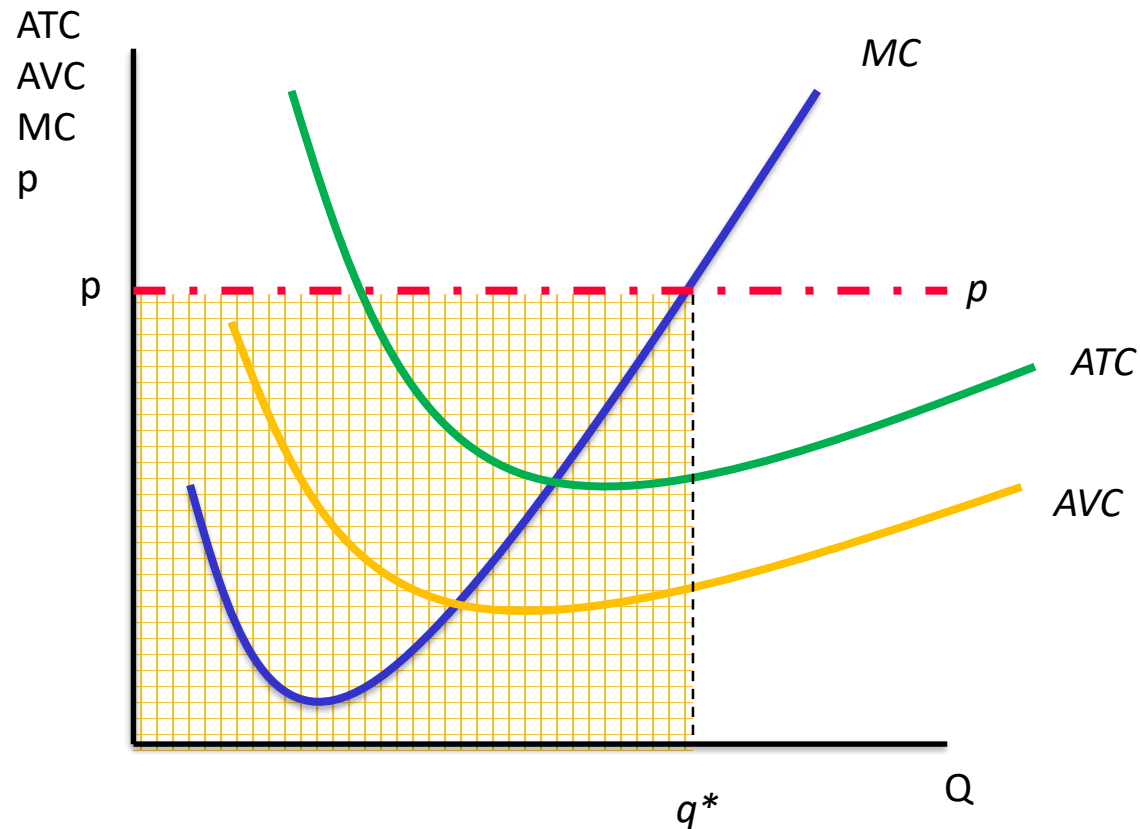
LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

In the following graph:



LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

In the following graph:

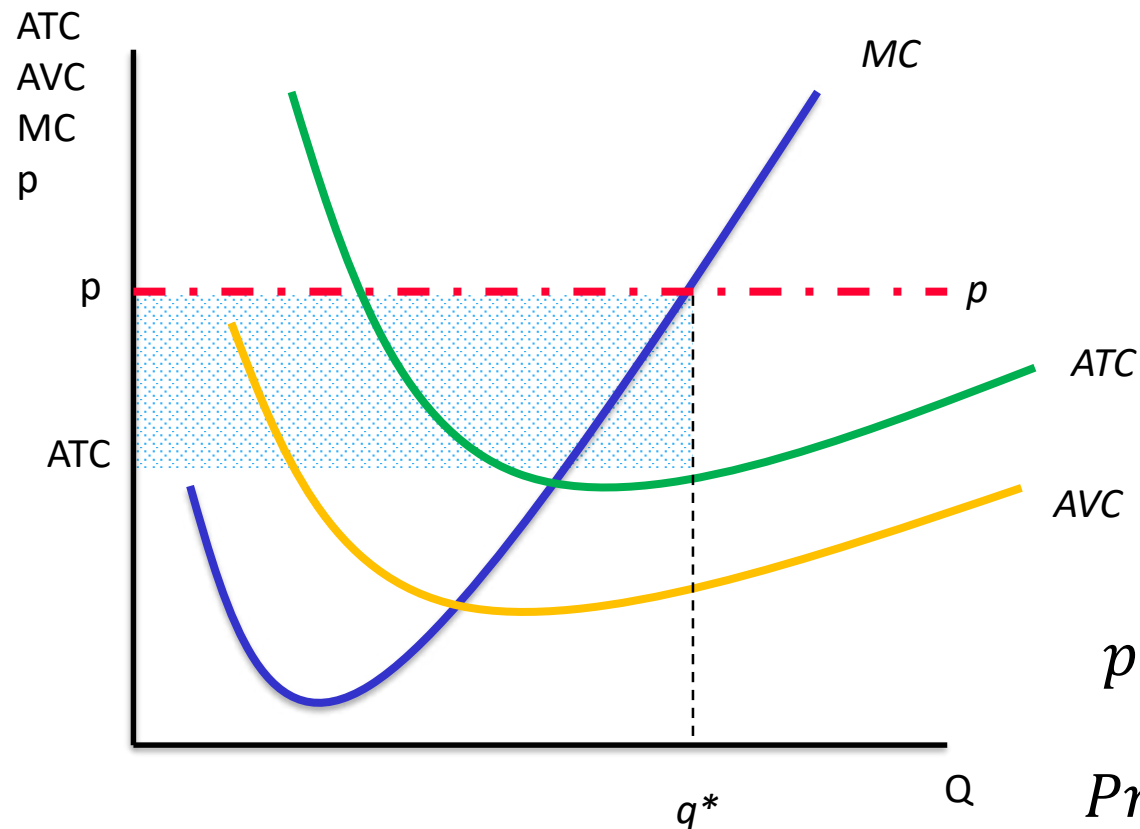


Total Revenue

$$TR = p \cdot q^*$$

LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

In the following graph:



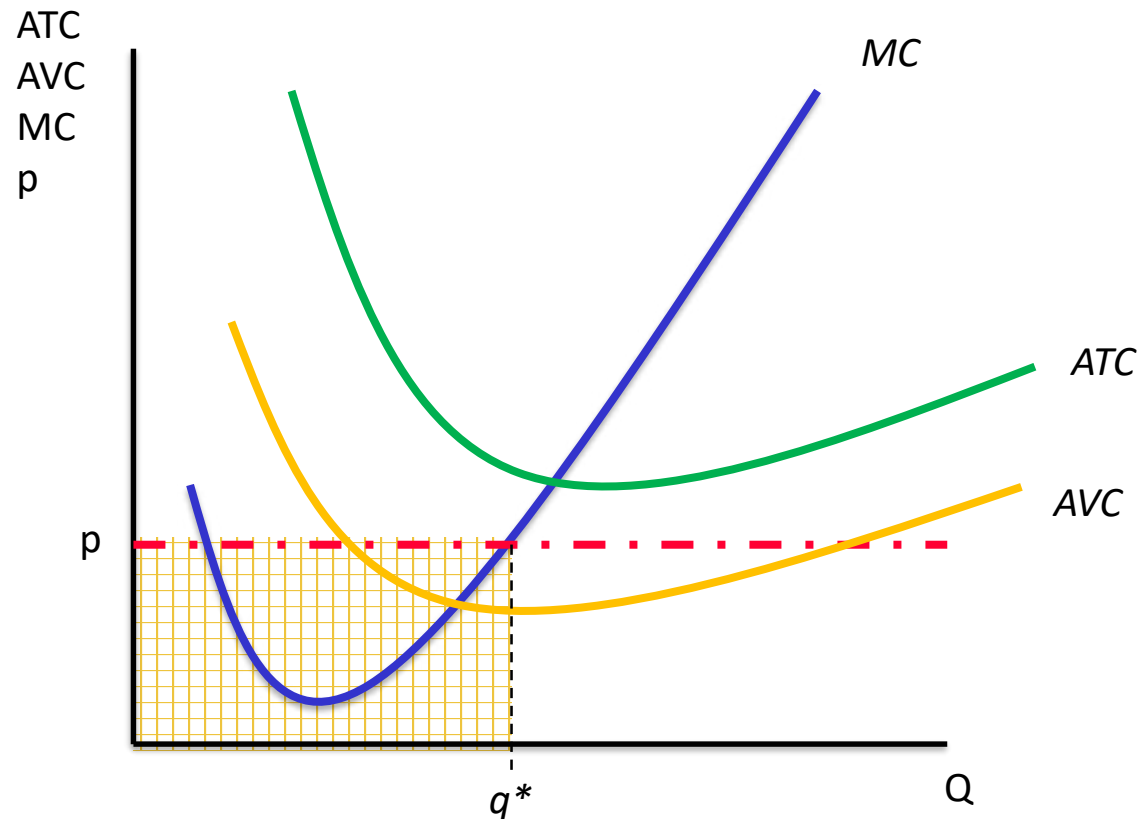
Profit

$$p > ATC \rightarrow Profit > 0$$

$$Profit = (p - ATC) \cdot q^*$$

LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

If the firm has losses:

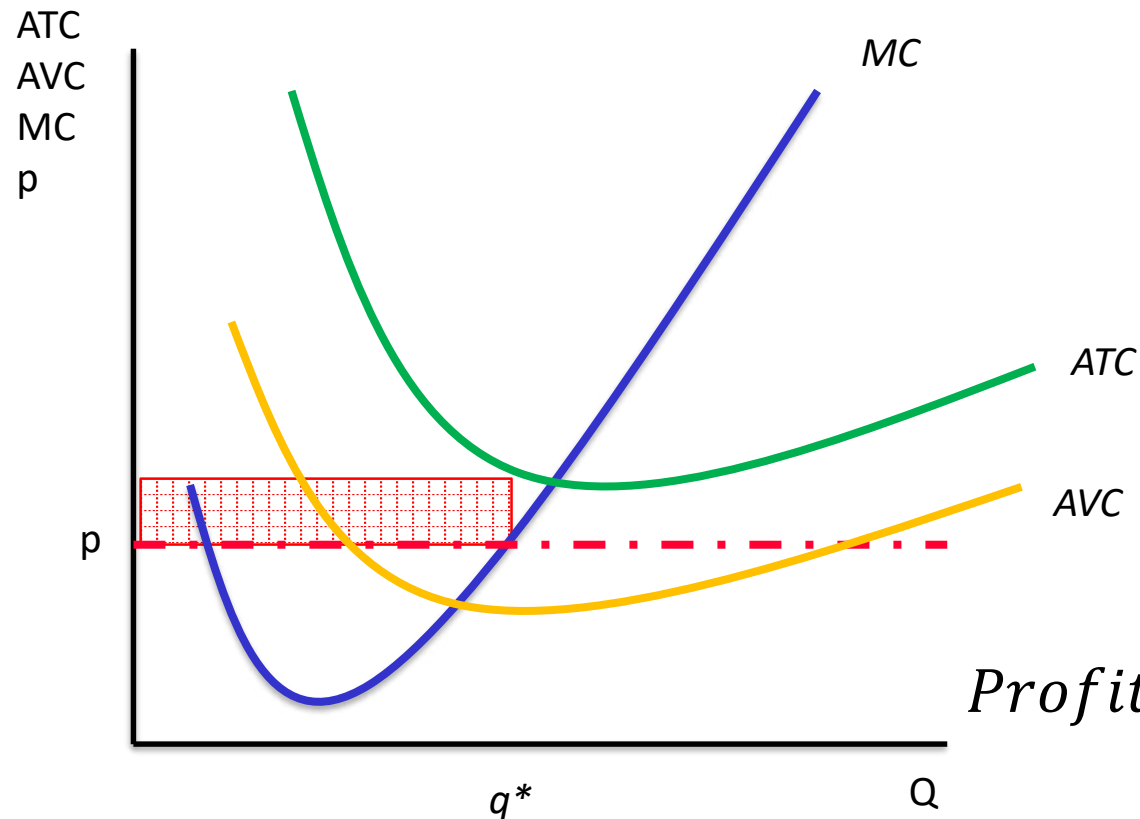


Total Revenue

$$TR = p \cdot q^*$$

LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

If the firm has losses:



Losses

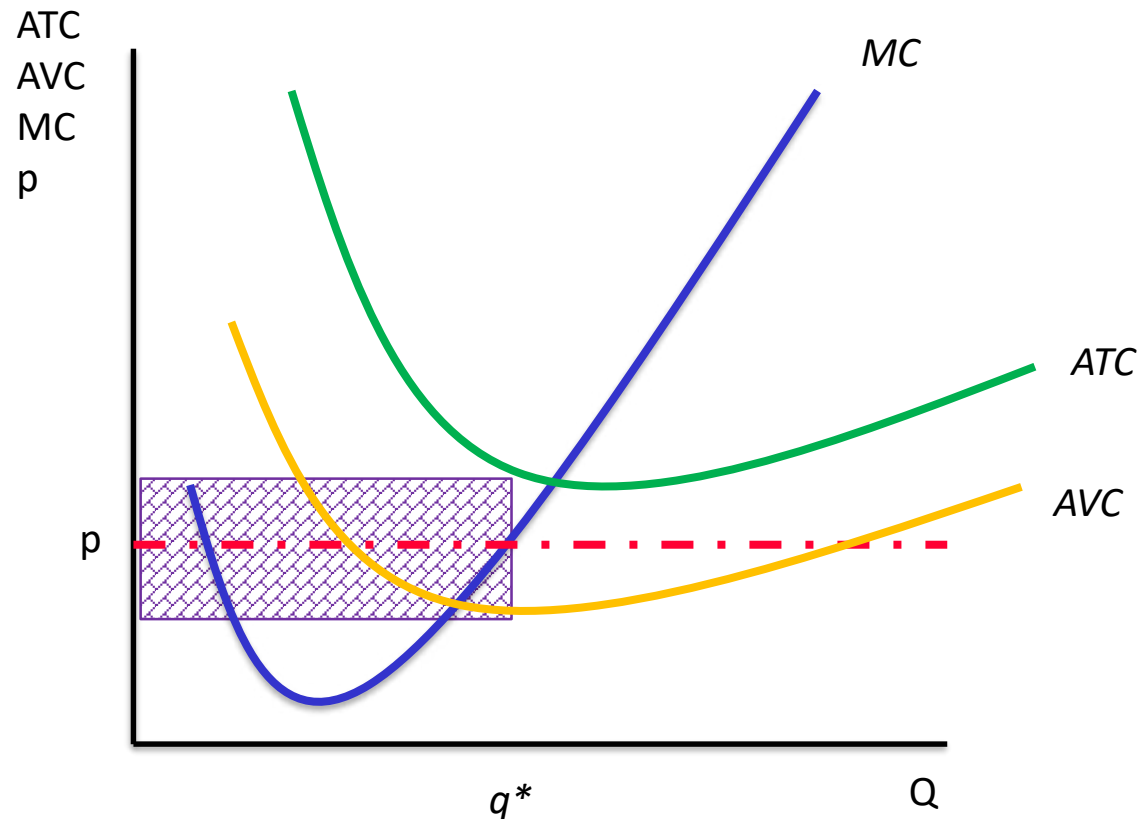
$$ATC > p > AVC$$

$$Profit = (p - ATC) \cdot q^* < 0$$

$$|Pr| < |-CF|$$

LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

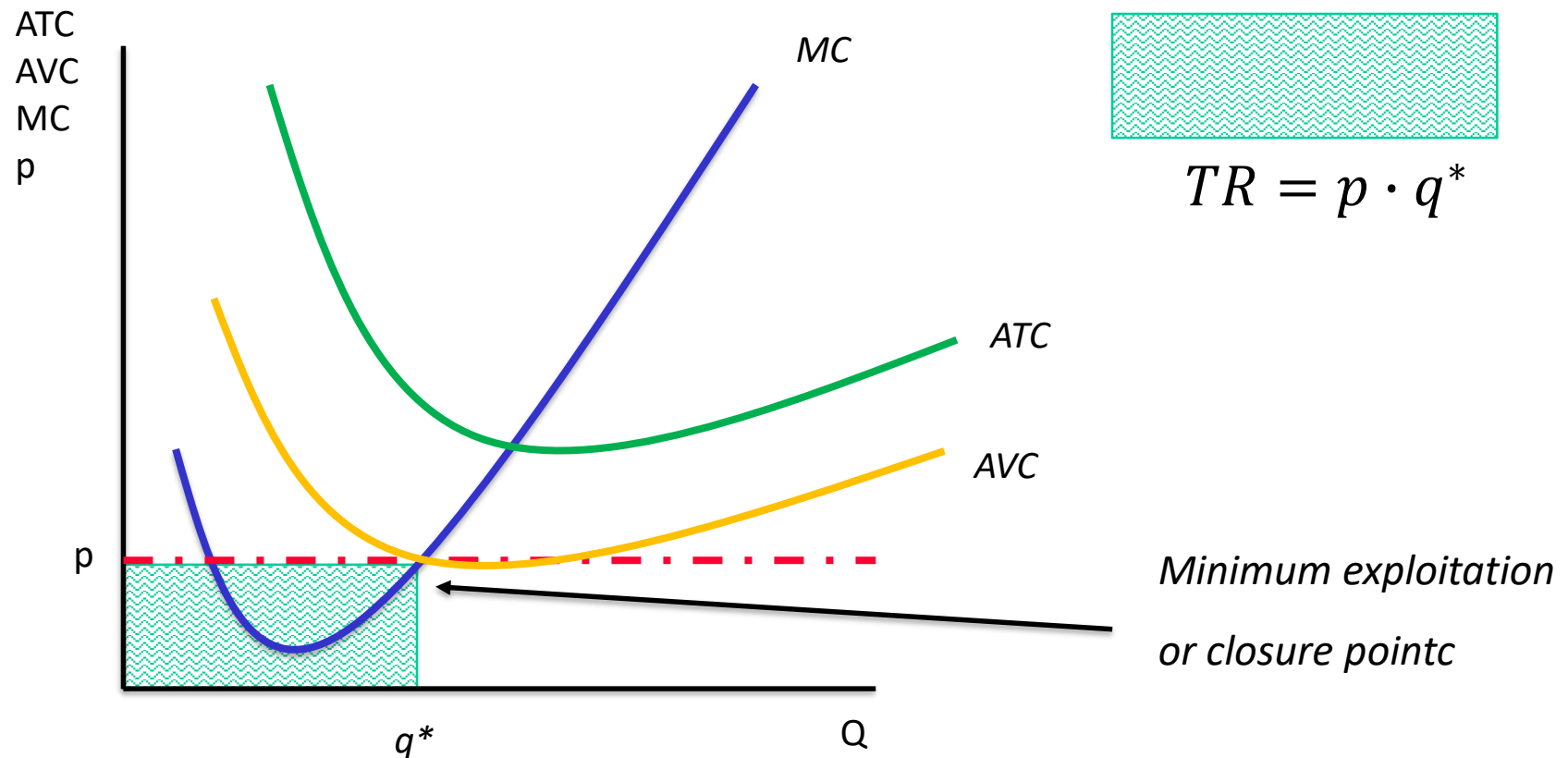
If the firm has losses:



In this case the company will continue producing even if it has losses because if it decided to shut down its losses would be even greater

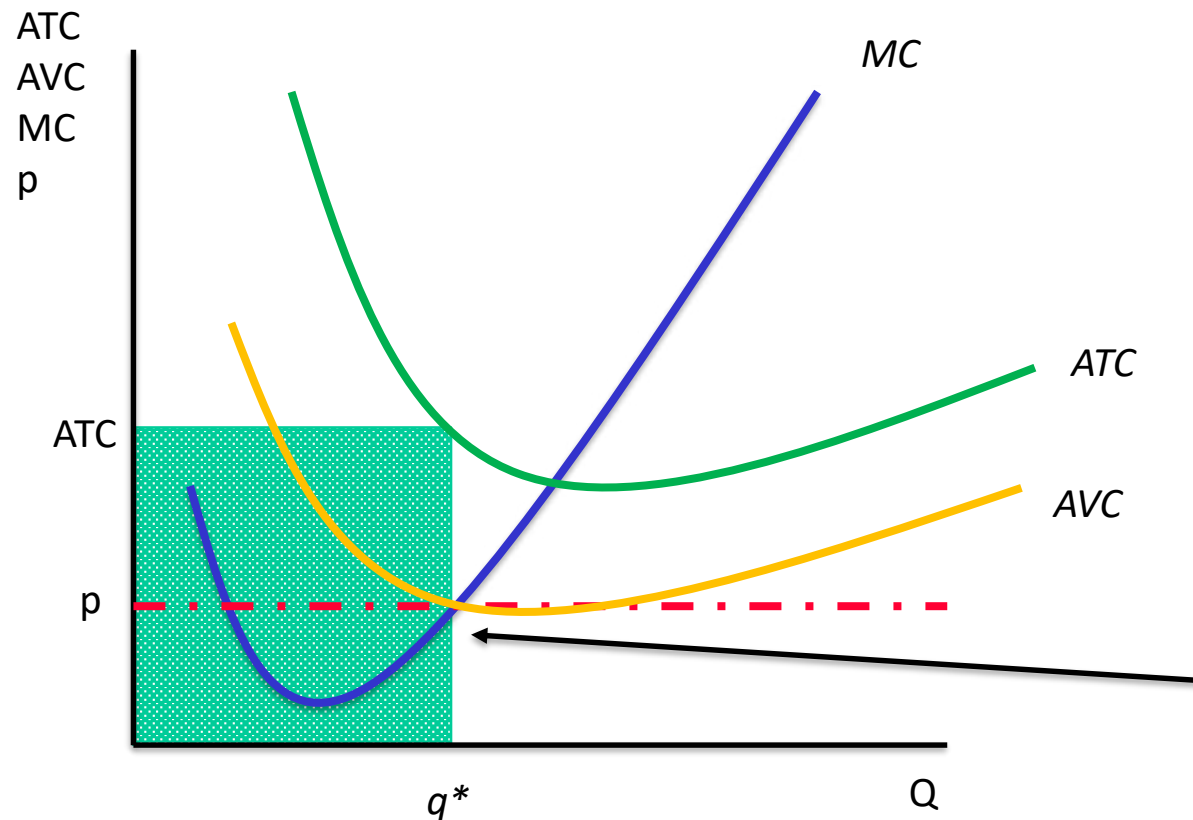
LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

When is the firm indifferent to producing or not?

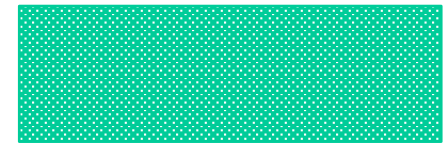


LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

When is the firm indifferent to producing or not?



Total Costs

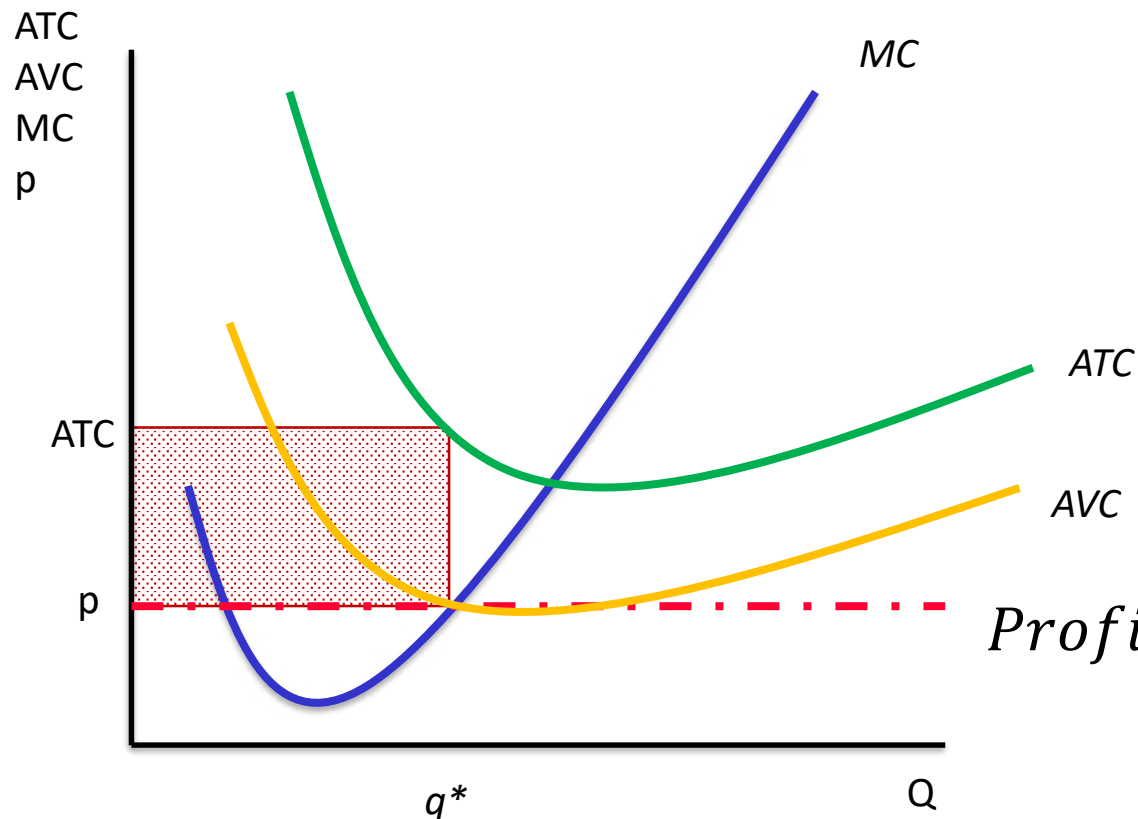


$$TC = ATC \cdot q^*$$

*Minimum exploitation
or closure pointc*

LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

When is the firm indifferent to producing or not?



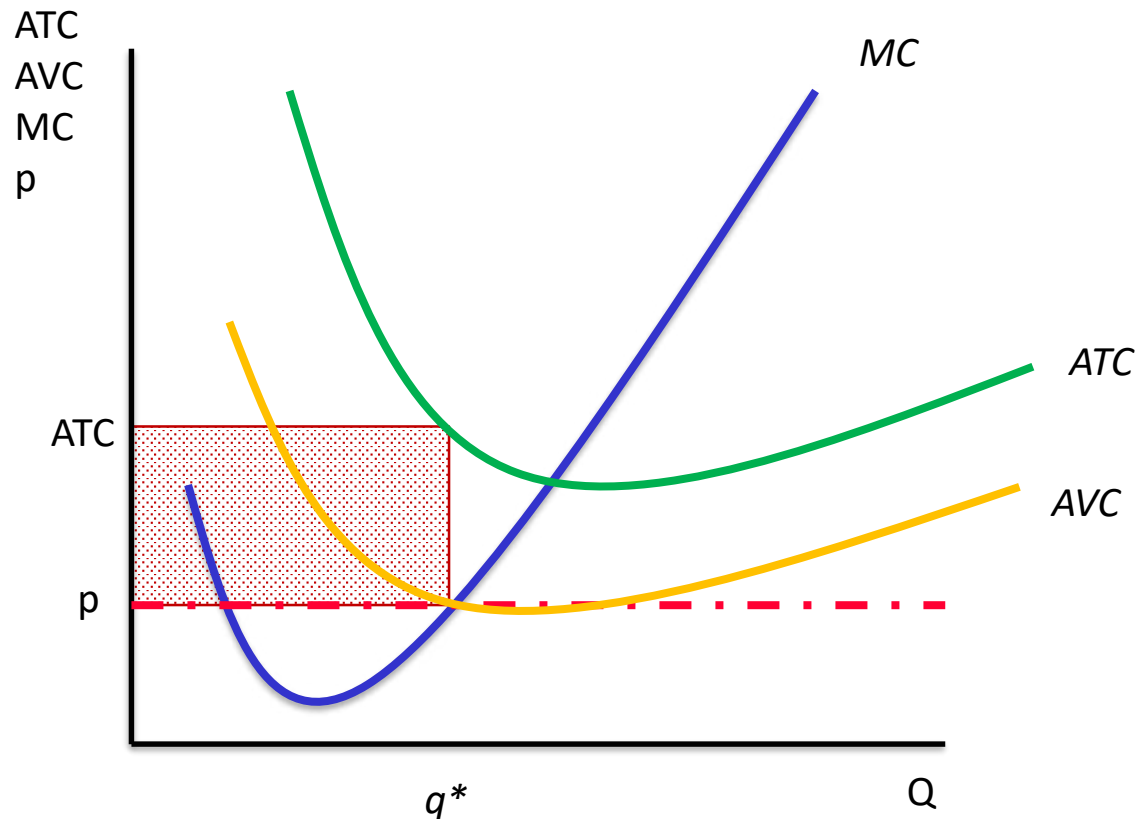
$$p < ATC$$
$$Profit < 0$$

$$Profit = (p - ATC) \cdot q^* < 0$$

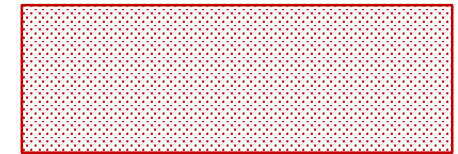
$$Profit = -FC$$

LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

When is the firm indifferent to producing or not?



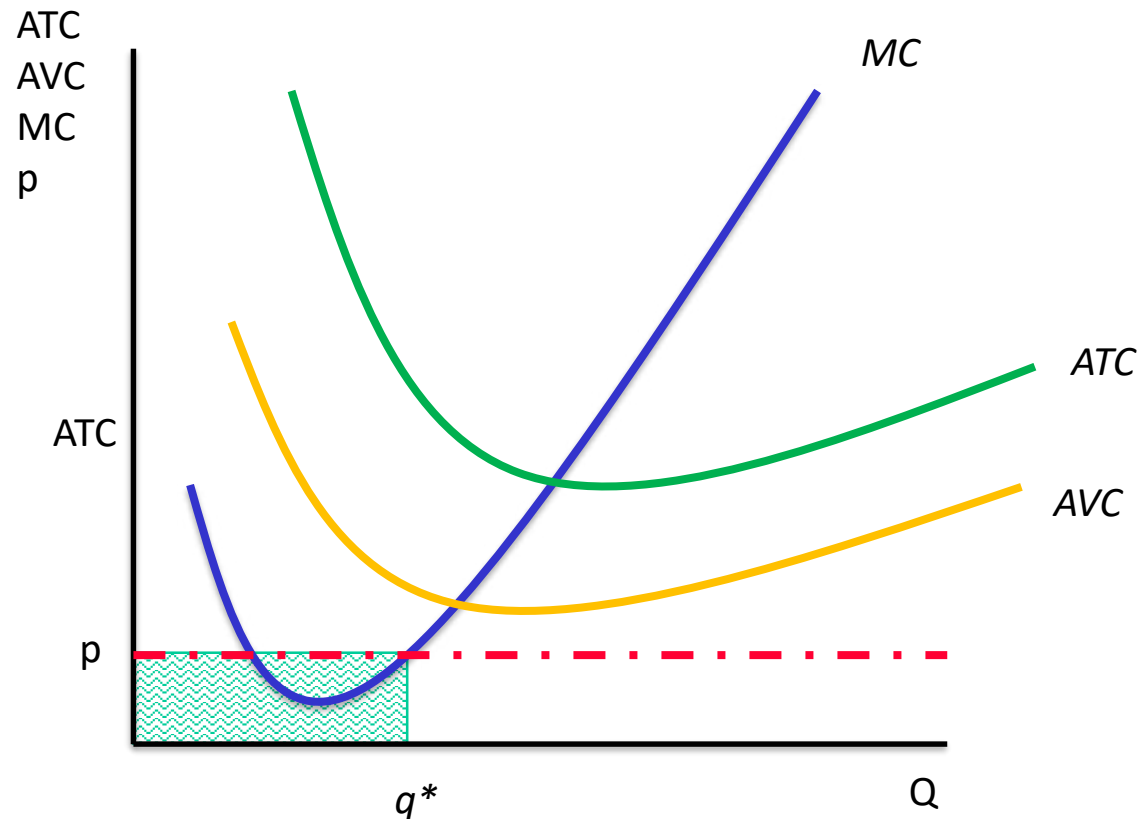
Losses



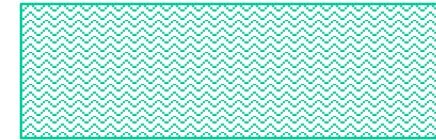
In this case, the company is indifferent between continuing producing or shutting down since in both scenarios its losses are equal to the fixed costs.

LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

When will the firm shut down?



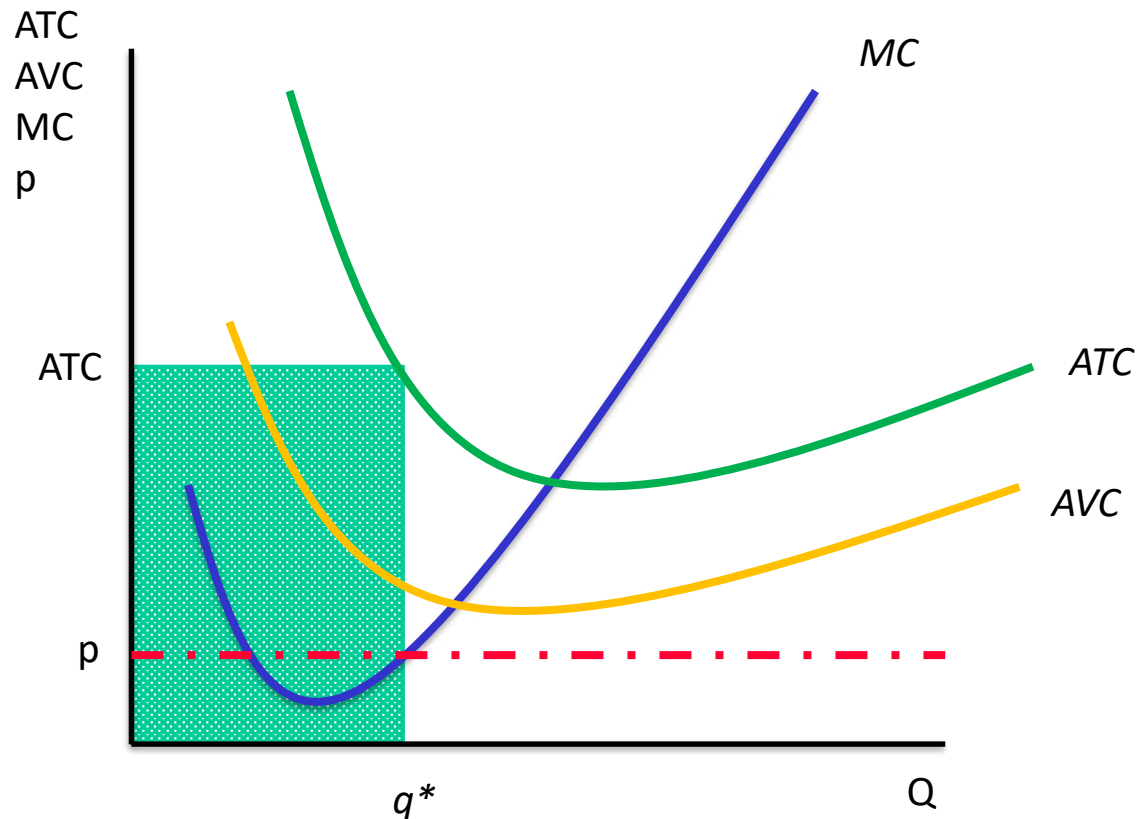
Total Revenue



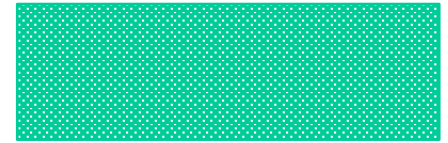
$$TR = p \cdot q^*$$

LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

When will the firm shut down?



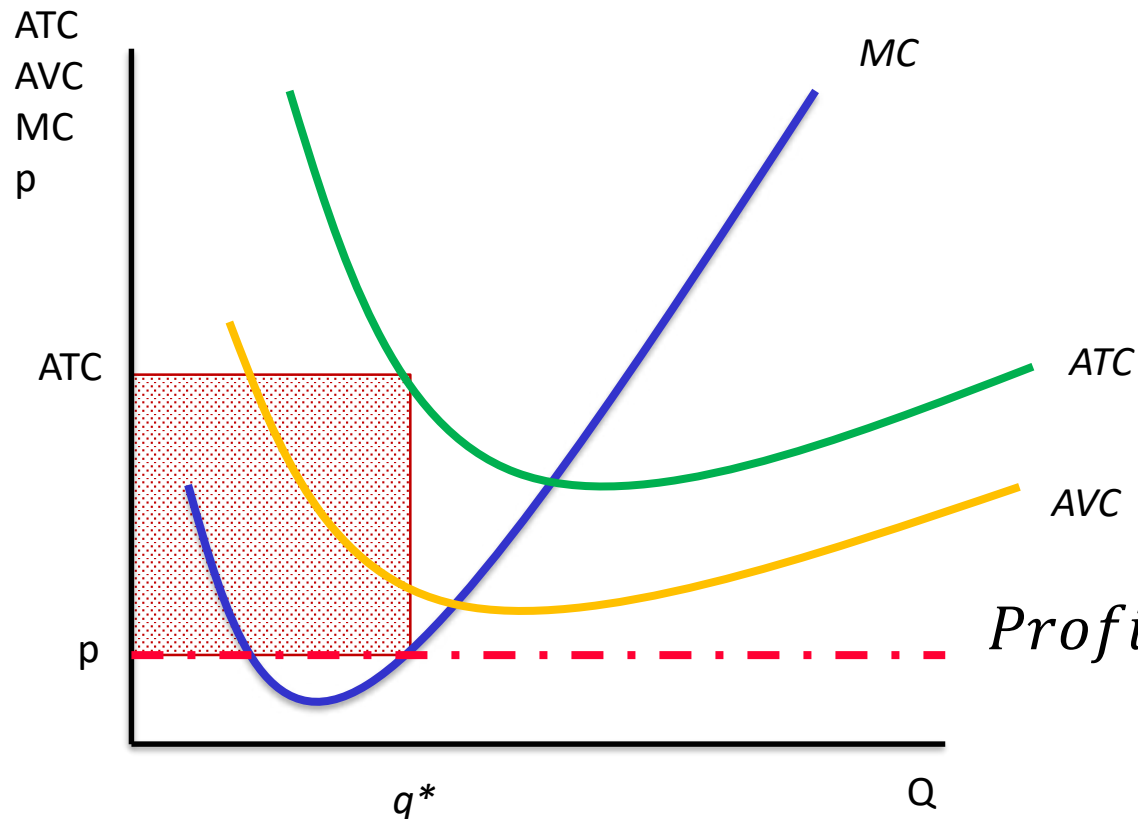
Total Costs



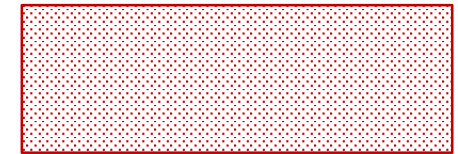
$$TC = ATC \cdot q^*$$

LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

When will the firm shut down?



Losses



$$p < ATC$$

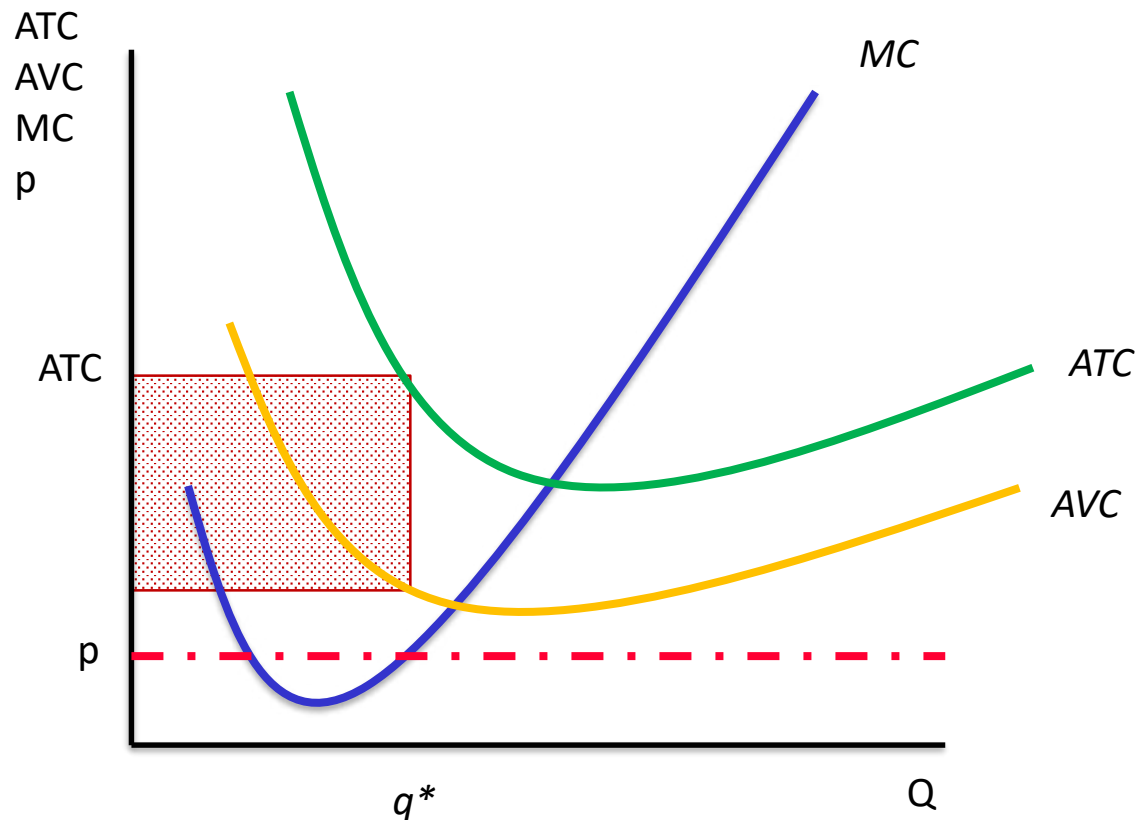
$$Profit < 0$$

$$Profit = (p - ATC) \cdot q^* < 0$$

$$|Profit| > |-FC|$$

LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

When will the firm shut down?



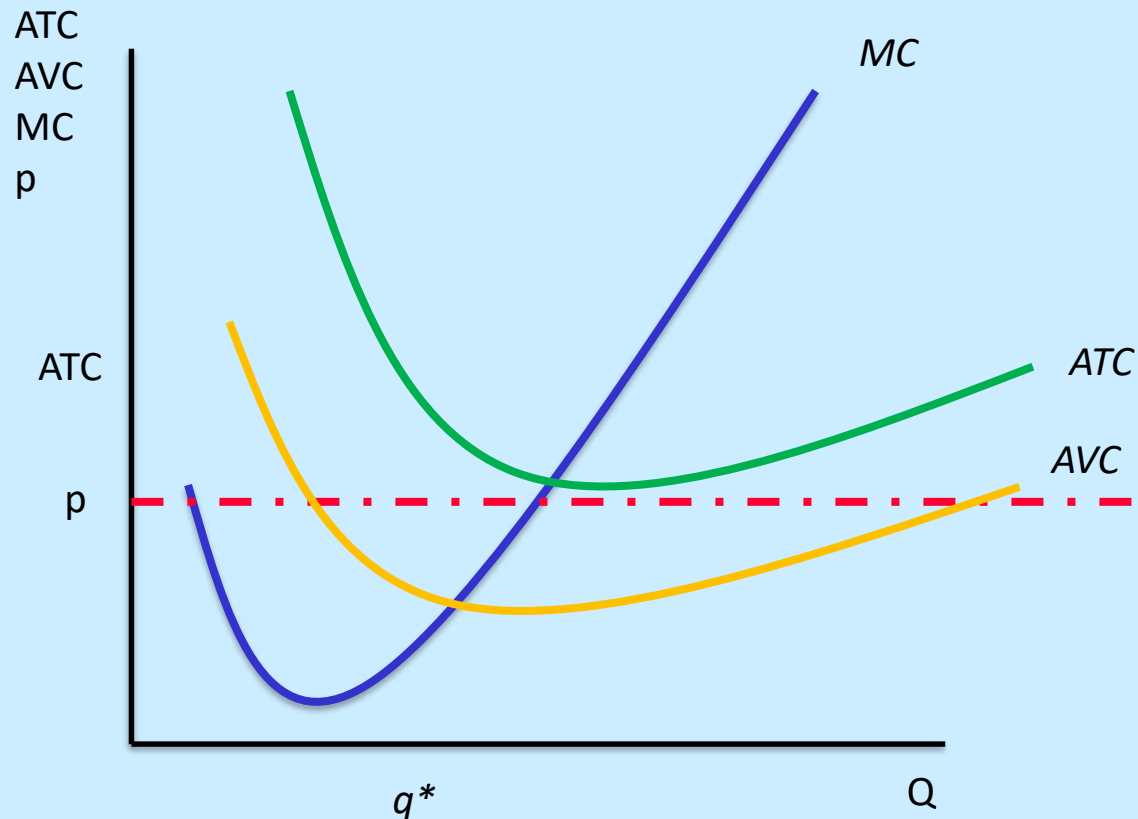
Losses

In this case the company will shut down limiting its losses to the fixed costs.

$$Profit = -FC$$

LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

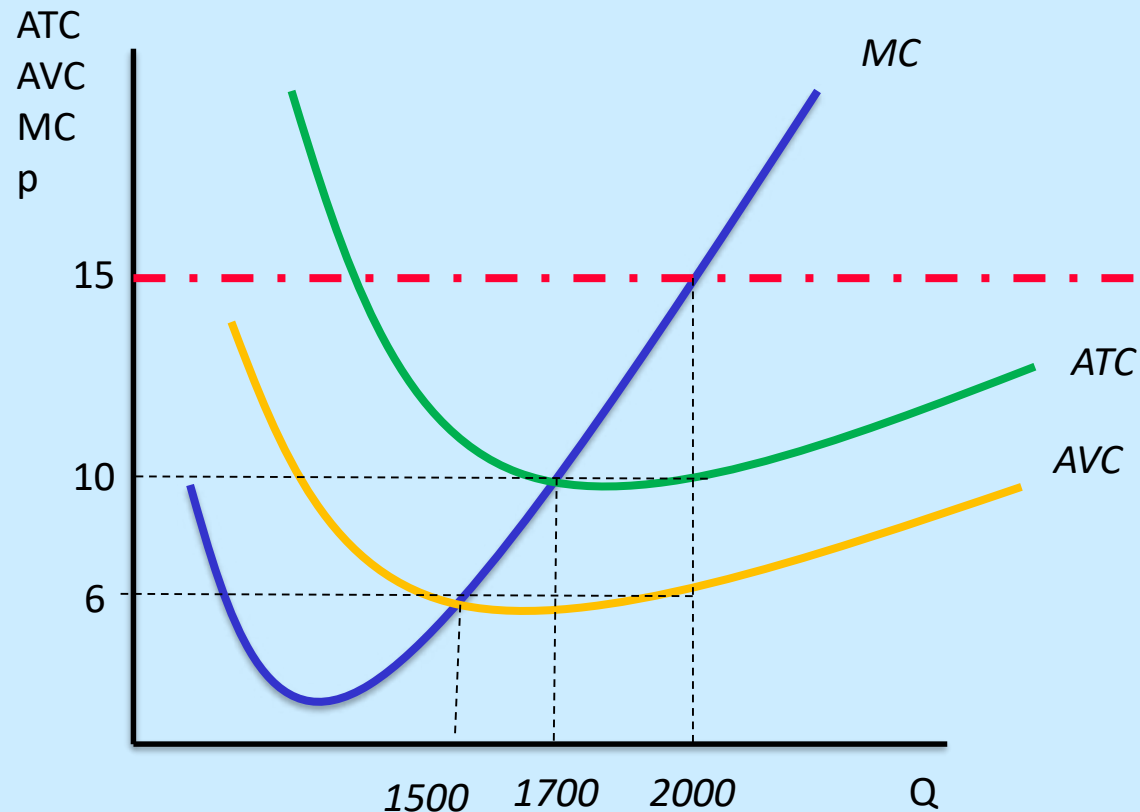
Question



*Will this firm
still produce in
the short-run?*

LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

Question

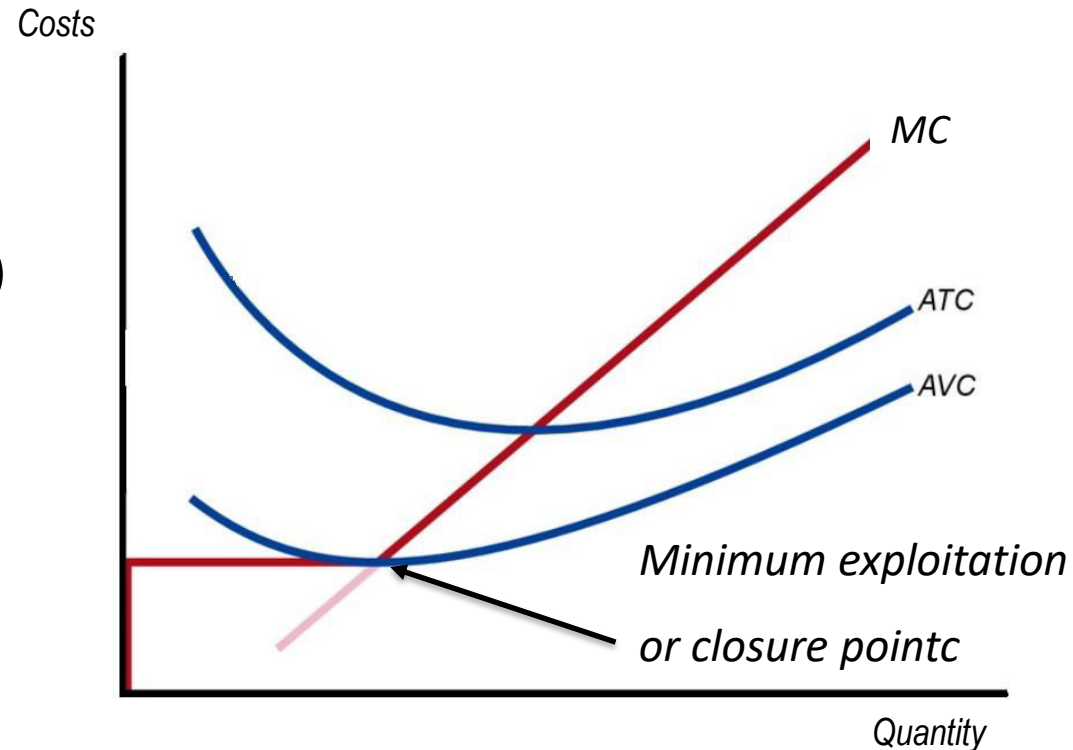


*Calculate the
TR, TC, VC, FC
and profit for
this firm*

LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

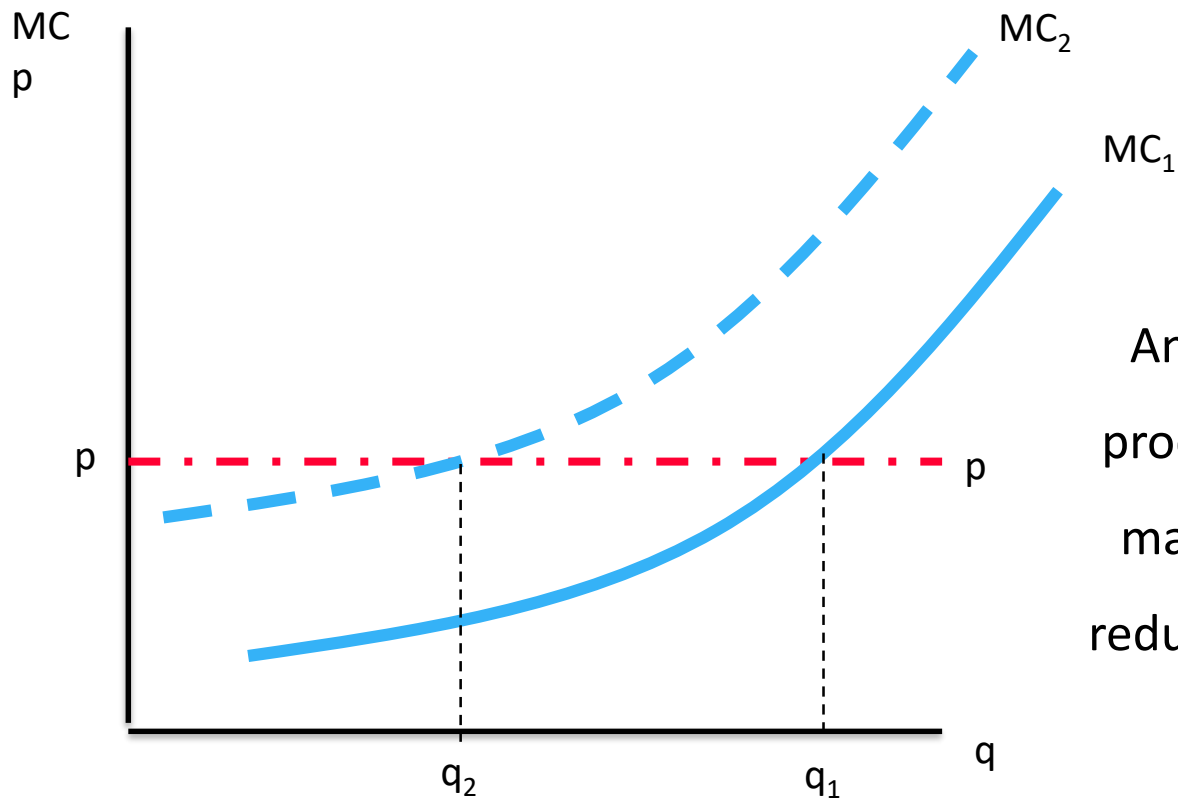
FIRM'S SUPPLY CURVE

In the short-run the firm's supply curve is its marginal cost curve (MC) above the average variable cost (AVC), that is, above the minimum exploitation or closure point.



LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

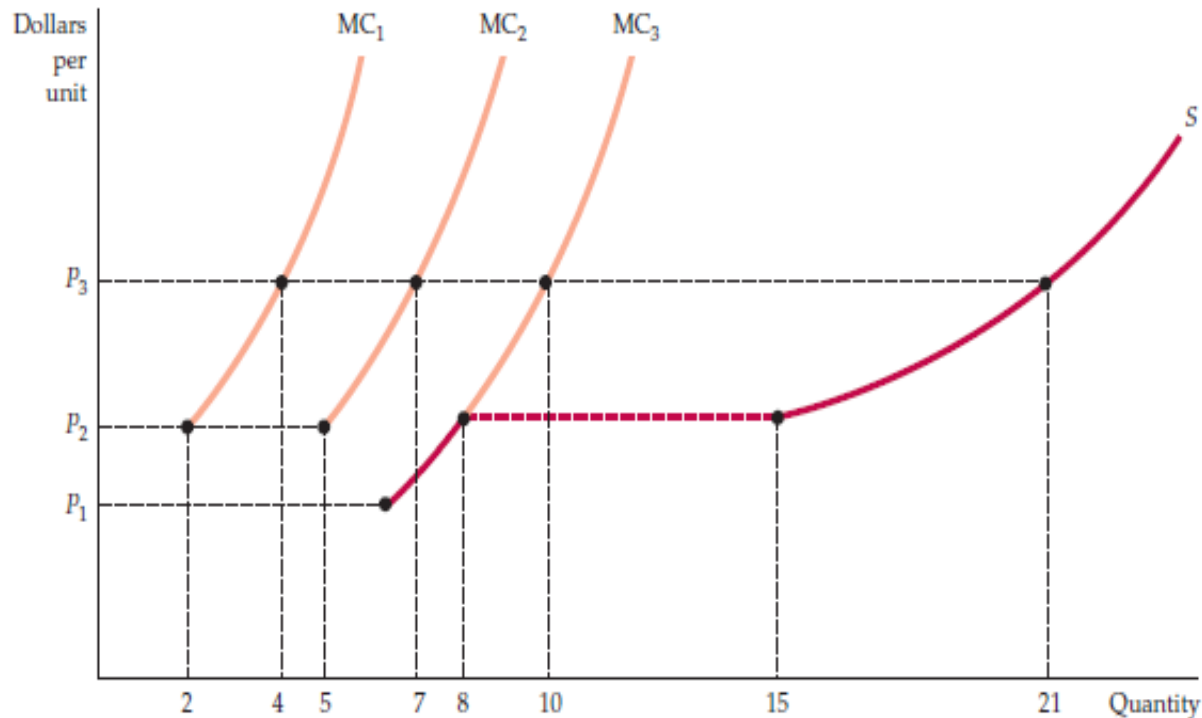
Effects of an increase in the cost of production factors



An increase in the price of a production factor shifts up the marginal cost curve (supply), reducing the quantity produced by the firm.

LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

MARKET'S SUPPLY CURVE



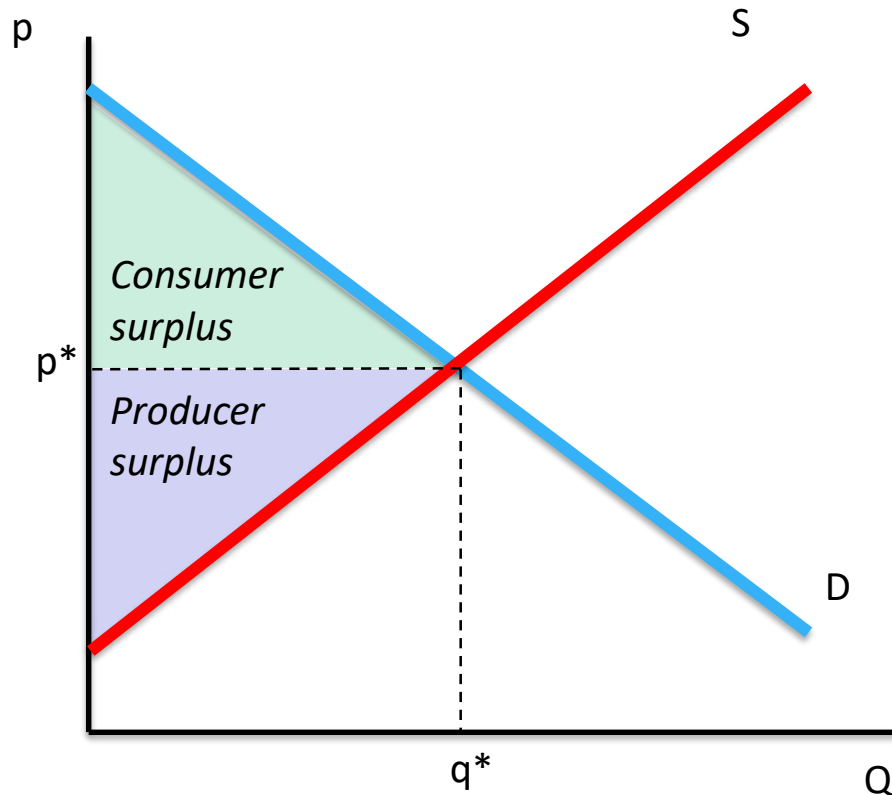
The market supply curve is the horizontal sum of the supplies of the firms that make up the industry.

Source: Pindyck & Rubinfeld

LESSON 5: ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

- 5.1. Market structure
- 5.2 Main characteristics of perfect competition
- 5.3 Profit maximization in firms
- 5.4 The decision to produce and the short-run market supply curve.
- 5.5 *The efficiency of the competitive equilibrium in the short-run.*
- 5.6 The long-run supply curve of the firm.
- 5.7 Competitive equilibrium. The functioning of competitive markets

LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.



A perfectly competitive market is economically efficient, by achieving the maximization of the aggregate surplus of consumers and producers

$$\begin{aligned} & \textit{Max Total Surplus} \\ &= \textit{Consumer Surplus} \\ &+ \textit{Producer Surplus} \end{aligned}$$

Limitations of competition

The presence of market failures implies that an unregulated competitive market does not reach economic efficiency.

In these cases, prices do not send the correct signals to consumers and producers... and public sector intervention would be necessary.

LESSON 5: ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

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LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

In the long run, firms enter and leave the market when the price changes; therefore, it is not possible to calculate the supply curve of the industry as the horizontal sum of the supply curves of all the firms.

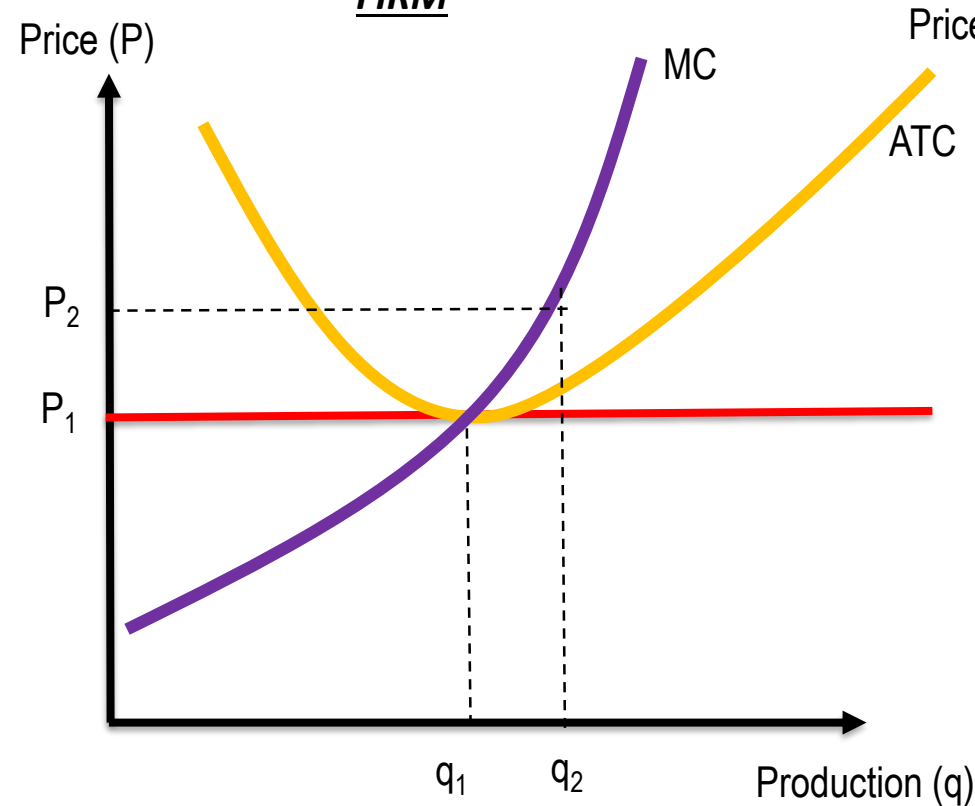
According to the long-run supply curve we distinguish three types of industries:

- Of constant costs
- Of increasing costs
- Of decreasing costs

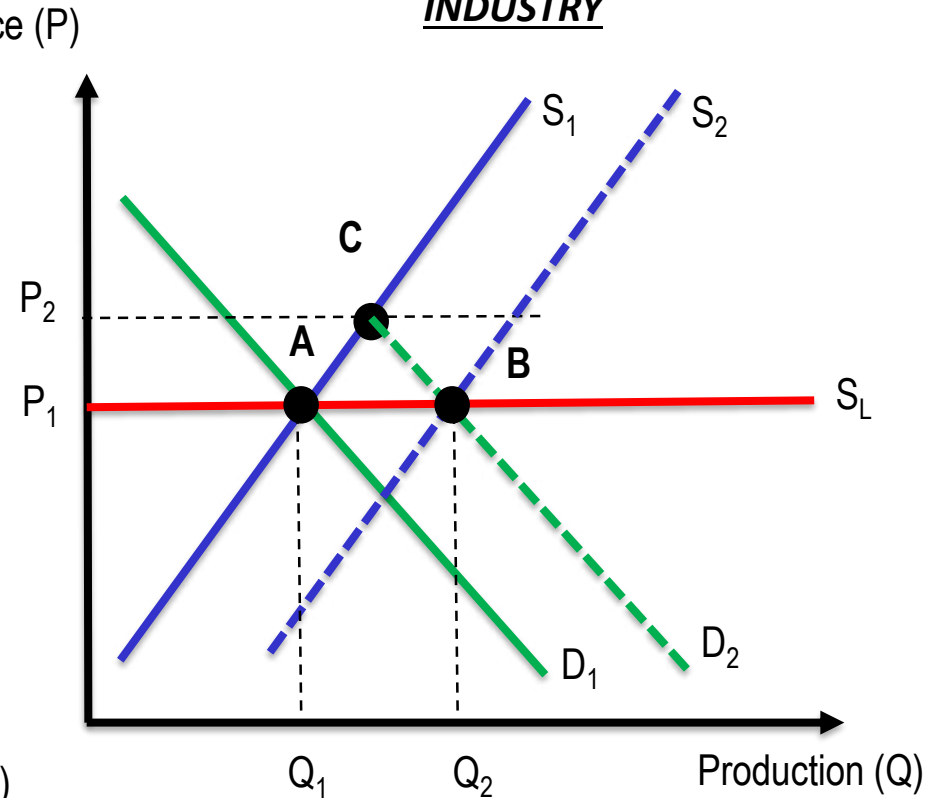
LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

Constant costs

FIRM



INDUSTRY



LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

Constant costs

When the demand increases it initially leads to an increase in the price (from A to C). The firm increases its production from q_1 to q_2 but the economic profits attract new companies.

Supply increases to S_2 and the market returns to long-run equilibrium.

Q_1 increases to Q_2 .

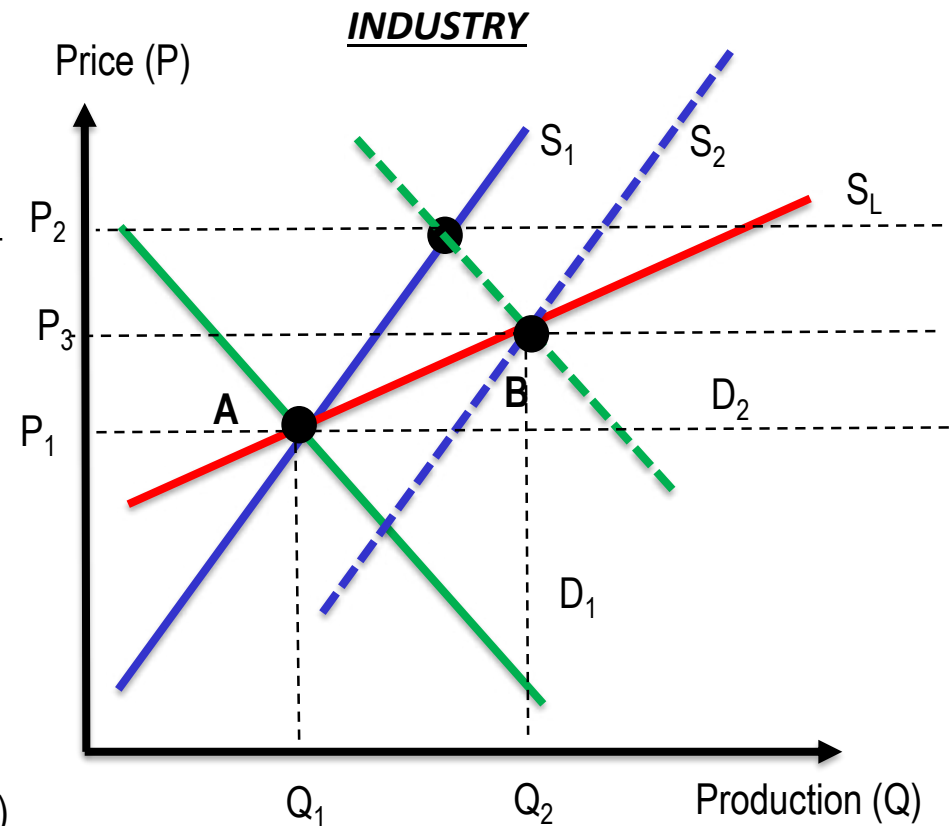
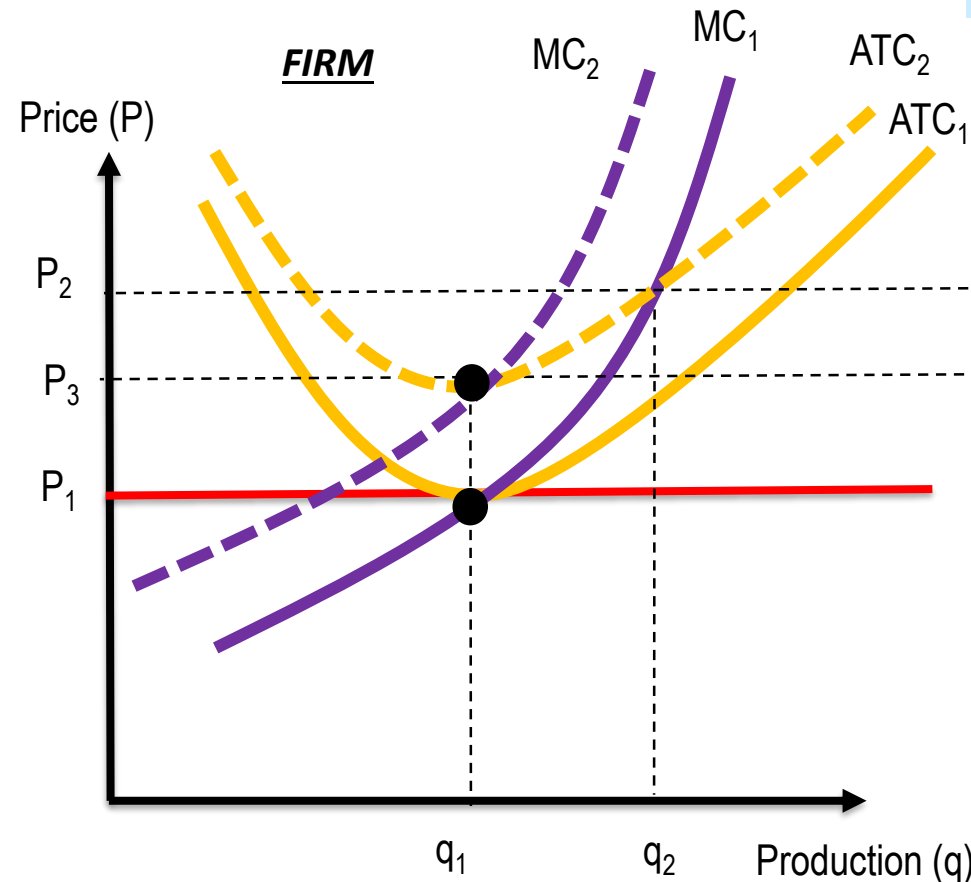
Long-run supply = $S_L = ATC_{lr}$

The variation in production does not affect production factors cost.

LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

Increasing costs

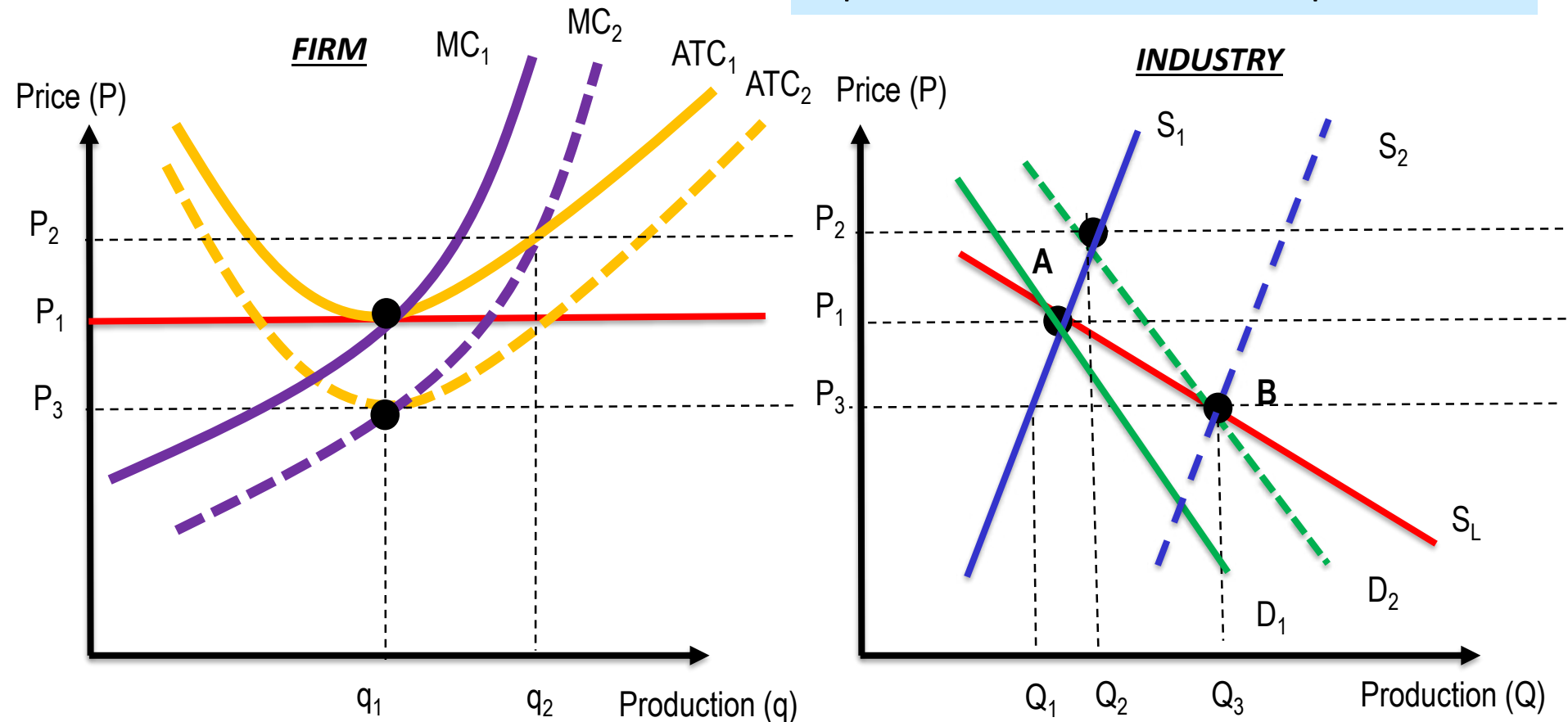
Due to the increase in the prices of the production factors, the long-run equilibrium reaches a higher price.



LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

Decreasing costs

Due to the fall in the prices of the production factors, the long-run equilibrium reaches a lower price.

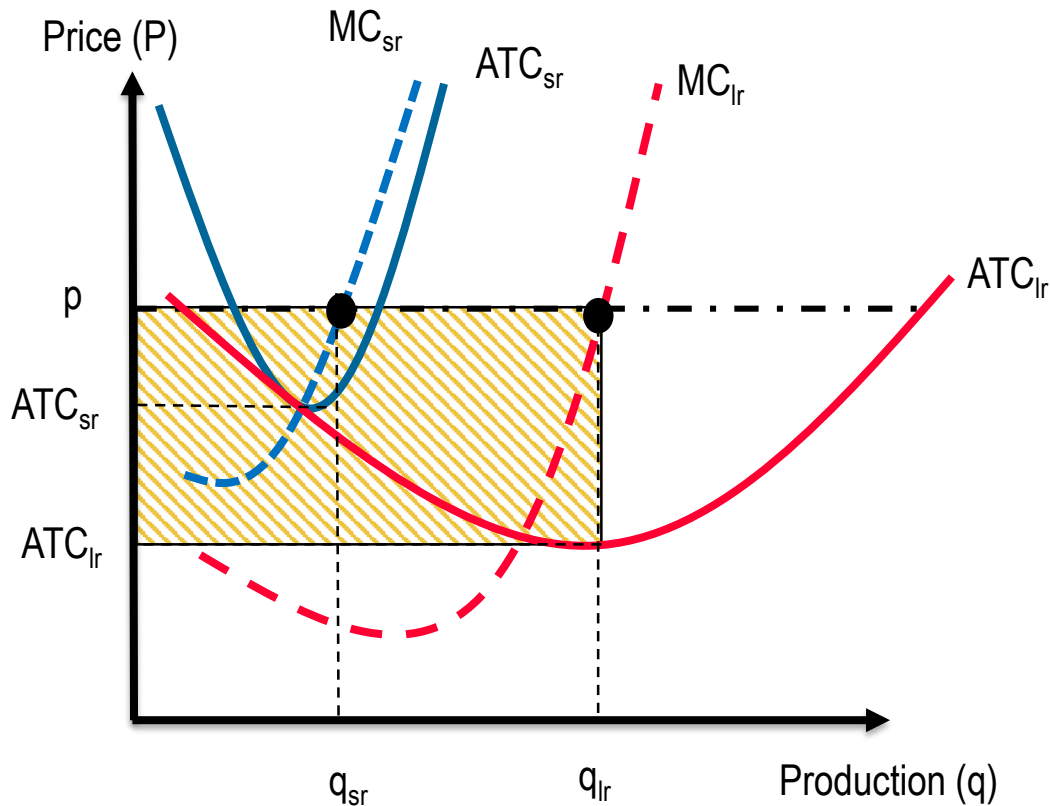


LESSON 5: ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

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LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

LONG-RUN PROFIT MAXIMIZATION

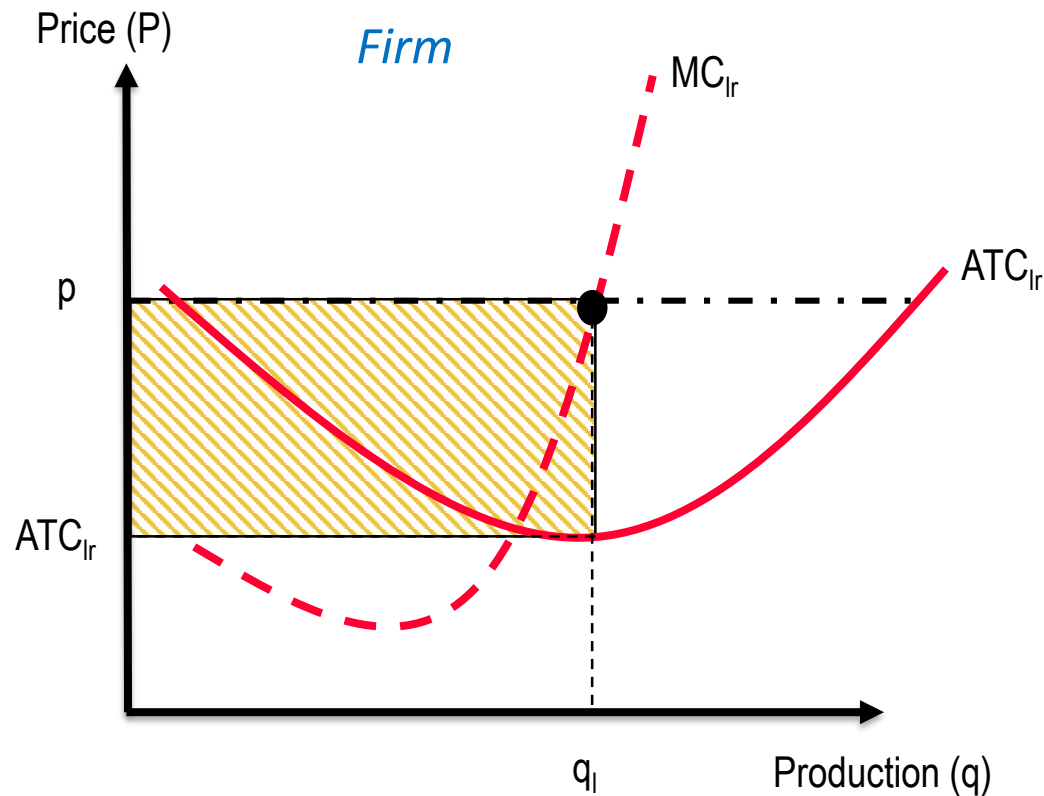


In the short run, the company maximizes profit with the production level q_{sr} .

In the long run, the company increases its production maximizing profit in q_{lr} , increasing its profits.

LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

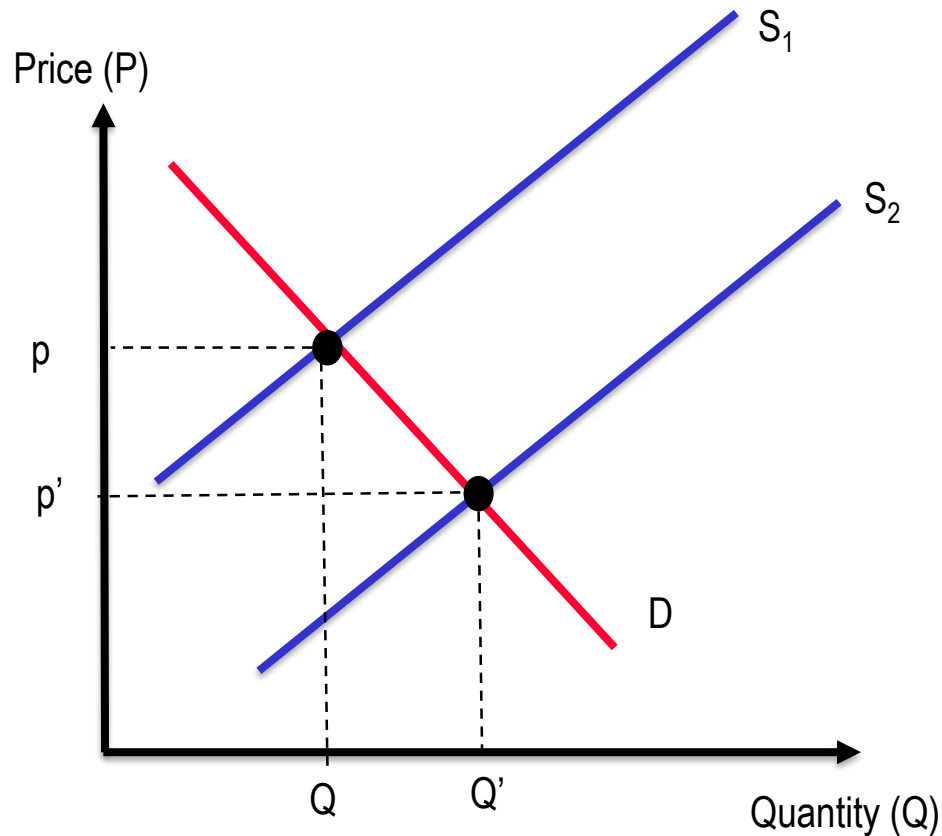
COMPETITIVE EQUILIBRIUM IN THE LONG-RUN



The existence of extraordinary profits will attract new firms to the industry..

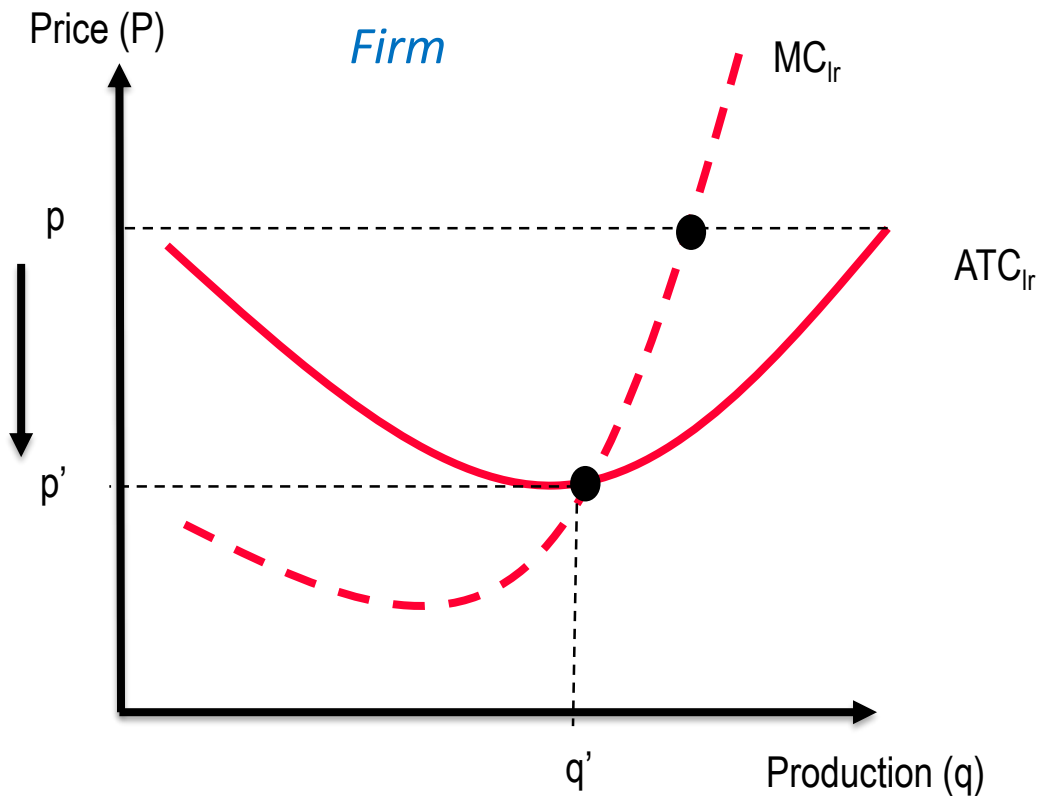
LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.

Industry



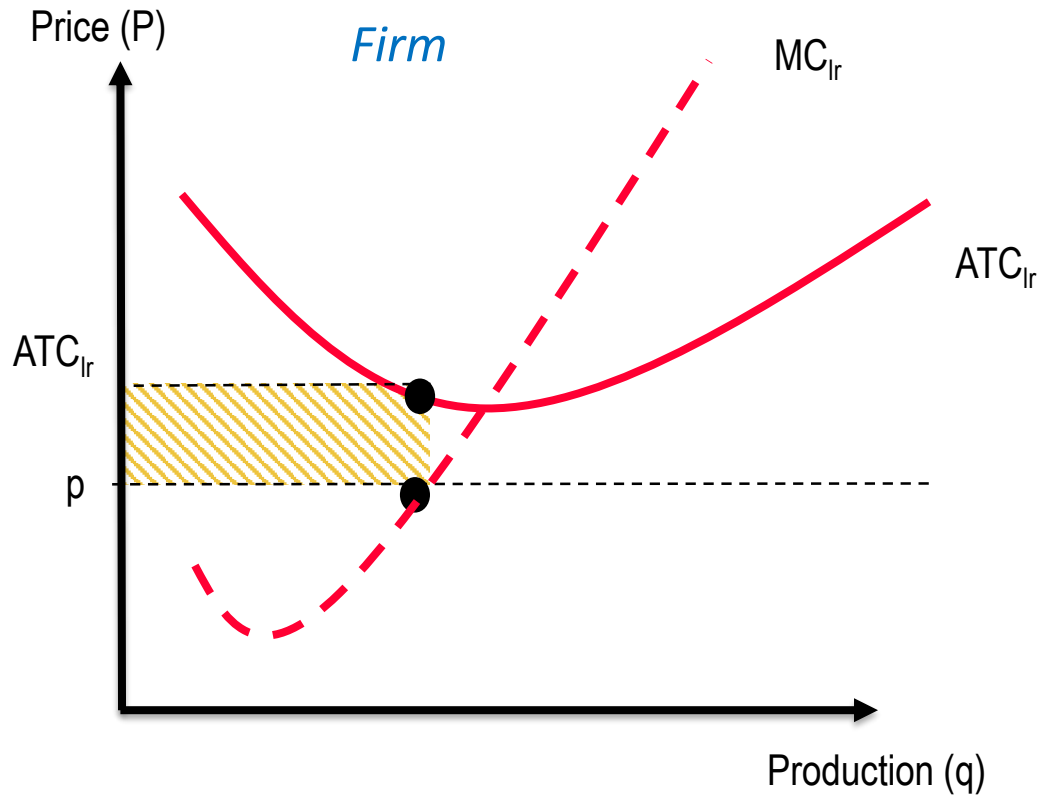
New firms will enter the market so the the supply curve shifts to the right, reducing the market equilibrium price.

LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.



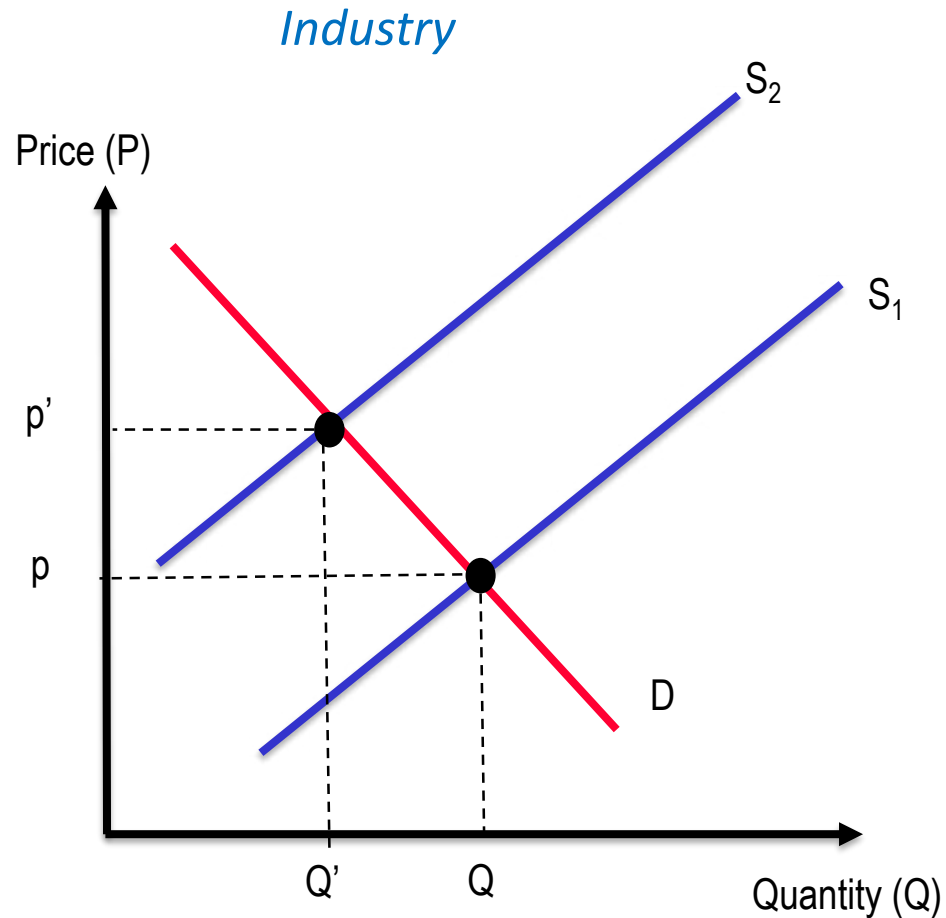
The decrease in the market price will reduce the extraordinary profits of the firm until they do not exist. When that moment comes, new firms stop entering the market.

LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.



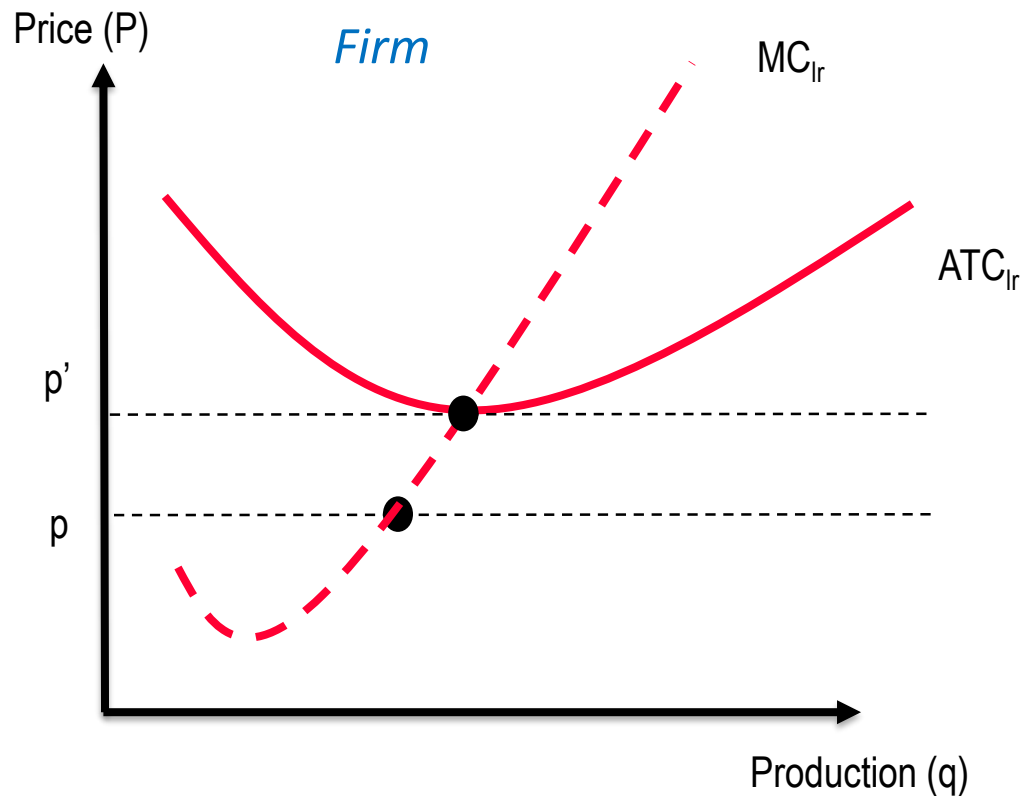
The existence of losses will cause firms to leave the market.

LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.



The exit of companies from the industry will shift the supply curve to the left, increasing the market equilibrium price.

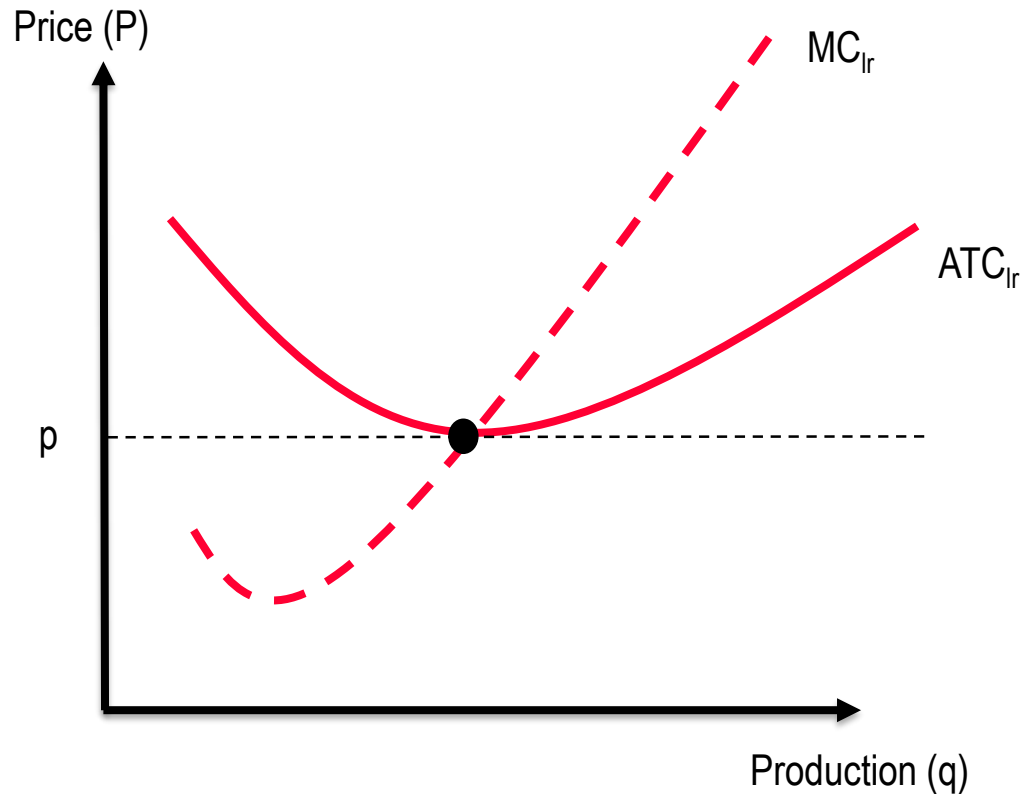
LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.



The increase in the market price will reduce the losses of the companies until they are zero.

At that moment, companies in the industry stop leaving the market.

LESSON 5. ANALYSIS OF THE STRUCTURE OF THE MARKETS. PERFECT COMPETITION.



Therefore, the equilibrium of the company in the long run will occur for that level of production when:

$$\min ATC_{lr} = p$$

LESSON 6: MARKET POWER. MONOPOLY AND MONOPOLISTIC COMPETITION.

6.1 *Market failures.*

6.2 The monopoly: Concept and origin.

6.3 Pricing in monopoly.

6.4 Monopolistic competition versus perfect competition.

6.5 Product differentiation: the role of advertising.

6.6 The equilibrium in the short and long run in monopolistic competition.

6.7 Economic inefficiency of monopoly and monopolistic competition.

6.1 Market failures

- The results provided by the allocation of resources through the market are not always satisfactory for society.
- The market does not allocate resources efficiently, it “fails”.
- Where do market failures occur? Why?
 - Lack of public goods (collective consumption).
 - Externalities.
 - Imperfect competition and market power.
 - The lack of equity in the distribution of income.

6.1 Market failures

Market power and imperfect competition

Open markets tend to favour situations of imperfect competition. In some sectors, companies have significant market power (monopoly power).



Inefficiency in the allocation of resources



Government role: improve efficiency



Market regulation
Protection of competition

LESSON 6: MARKET POWER. MONOPOLY AND MONOPOLISTIC COMPETITION.

- 6.1 Market failures.
- 6.2 *The monopoly: Concept and origin*
- 6.3 Pricing in monopoly.
- 6.4 Monopolistic competition versus perfect competition.
- 6.5 Product differentiation: the role of advertising.
- 6.6 The equilibrium in the short and long run in monopolistic competition.
- 6.7 Economic inefficiency of monopoly and monopolistic competition.

6.2 The monopoly: Concept and origin.

A firm is considered a monopoly if:

- ☐ It is the only seller of a product.
- ☐ The product does not have close substitutes.

The main reason for the existence of monopolies is the existence of entry barriers.

6.2 The monopoly: Concept and origin

MONOPOLY ORIGIN

- Exclusive control of strategic production factors.
- Existence of economies of scale in the production process:
natural monopoly.

Legal monopolies



- Patents - The government grants the right to exploit an invention to a single company. That way, research and development are encouraged.
- State concessions or licenses - The government grants the right to exploit a certain product or service to a company for a certain period of time.

6.2 The monopoly: Concept and origin

MONOPOLY ORIGIN: Some examples

- A legal concession: Postal and Telegraph Service (Correos y Telégrafos)
- A patent: New medicine.
- The sole ownership of a resource: a motorway toll.
- The formation of a cartel: Organization of the Petroleum Exporting Countries (OPEC)
- The existence of significant economies of scale: a gas or electricity company.

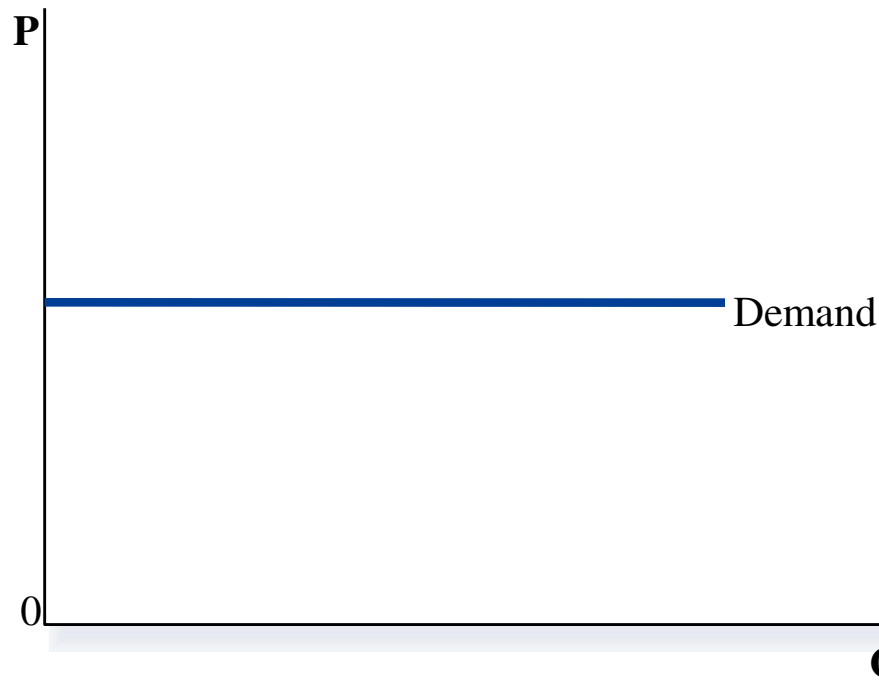
6.2 The monopoly: Concept and origin

Monopoly: Market supplied by a **single supplier**, in which the product sold has no close substitutes.

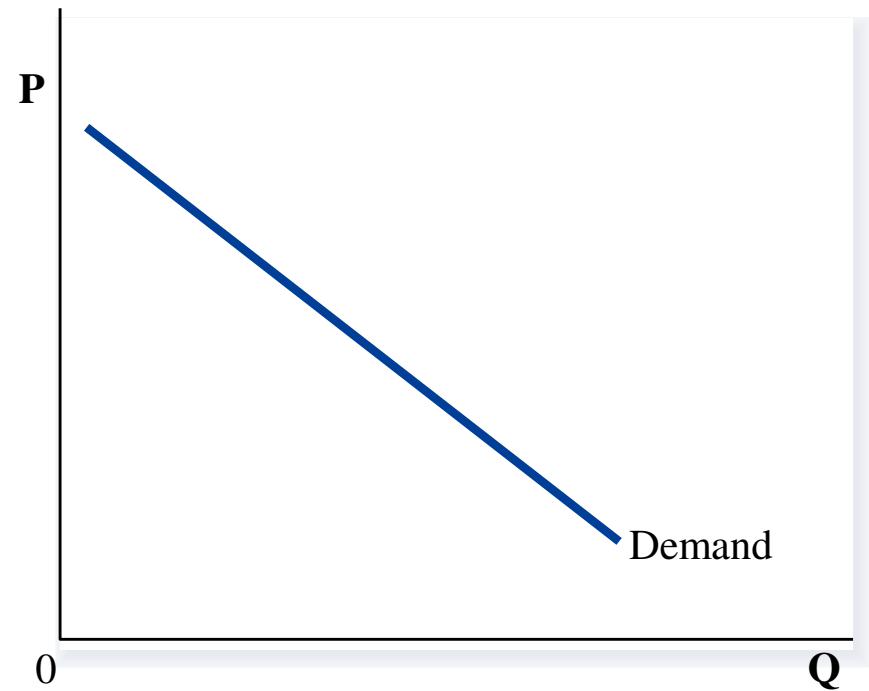
<i>Monopoly</i>	<i>Perfect Competition</i>
Price-bidder firm	Price-taker firms
<i>Differentiated?</i> product	Homogeneous product (perfect substitutes)
No long-run perfect mobility of productive factors (there are barriers)	Long-run perfect mobility of productive factors
There is perfect information	There is perfect information

6.2 The monopoly: Concept and origin

Demand curve of a competitive company



Demand curve of the monopoly

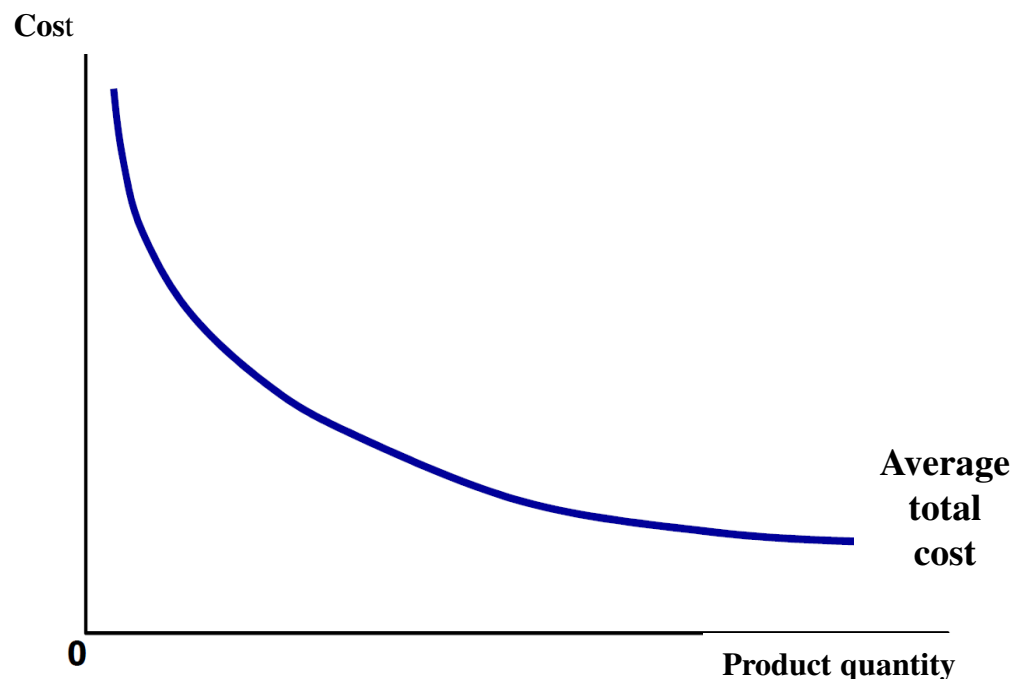


6.2 The monopoly: Concept and origin

NATURAL MONOPOLY

An industry is a natural monopoly when a single firm can offer a good or a service to an entire market with fewer costs than two or more companies.

A natural monopoly appears when there are economies of scale in a relevant interval of production.



6.2 The monopoly: Concept and origin

Monopsony

- ✓ A monopsony is a monopoly of demand.
- ✓ Monopsony happens when a single person / institution, demands the product or service and, therefore, can set the market price at its will. So, the demand can get part of the producer surplus.
- ✓ It generally occurs:
 - ☐ With certain production factors. For example, the demand for a certain type of specialized work.
 - ☐ With raw materials and goods in process;
 - ☐ With special consumer goods: the army of a country may be the only buyer of a certain class of equipment or weapons, or certain distributors or wholesalers may be the only demanders of a certain type of good that they will then offer to the public.

LESSON 6: MARKET POWER. MONOPOLY AND MONOPOLISTIC COMPETITION.

- 6.1 Market failures.
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6.3 Pricing in monopoly

- The monopolist is the supply side of the market and has absolute control over the amount of production that it puts up for sale.
- **Profits will be maximized, as usual, at the level of production when marginal revenue equals marginal cost.**
- Remember that a monopoly is a situation of imperfect competition where market power is held by a single company. **This company will be able to fix the quantity to produce OR the price, but not both simultaneously.**

6.3 Pricing in the monopoly

- The market demand will determine the quantity demanded at the prices given by the monopolist or viceversa

How many units will a monopoly produce?

- Like any firm, the monopolist will try to maximize profits or economic utilities, so that;

$$Max Profit = TR - TC$$

$$MgR = MgC$$

- The MgC depends on the monopolist's cost curve (its cost curve will like the one analysed in lesson 4)
- The MgR depends on the market demand faced by the monopolist (nearly always a demand with a negative slope).
- How do we get the monopolist's optimum?

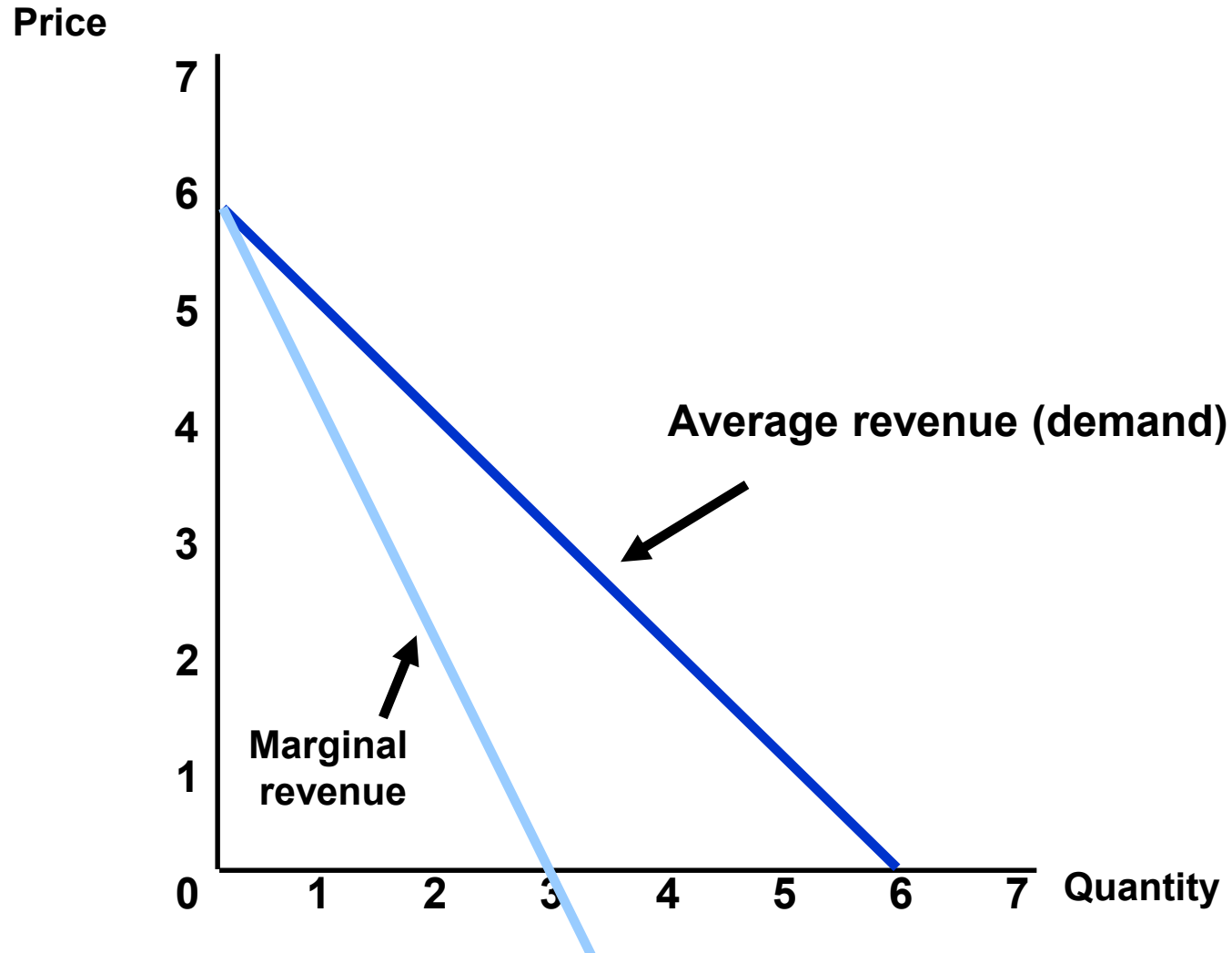
6.3 Pricing in the monopoly

- *Calculation of marginal revenue*

- As the only producer, the monopolist depends on the market demand to determine the quantity of output and the price.
- If, for example, a firm had the following demand curve:

- $P = 6 - Q$

6.3 Pricing in the monopoly



6.3 Pricing in the monopoly

- For a monopolist, the additional revenue received from selling one more unit has two components. On the one hand, it obtains the additional revenue for the sale of that extra unit (p). On the other hand, to sell one more unit, the monopolist has to lower the price it charges for all the units that were previously sold at a higher price.
- The sum of both effects gives us the expression of the monopolist's MgR .

$$MgR = \frac{\Delta TR}{\Delta Q} = \frac{\Delta(P \cdot Q)}{\Delta Q}$$

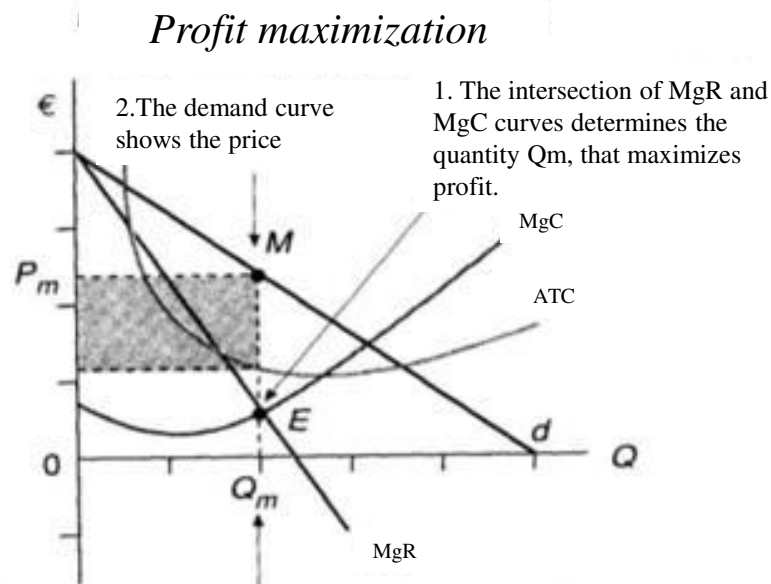
6.3 Pricing in the monopoly

Given that the MgR is the increase in revenue when a marginal change in quantity happens, the slope of the total revenue curve is, therefore, the marginal revenue

That MgR is NO longer equal to the price (p) because now there is a negative relationship between P and Q (given by the demand function)

The point where profit is maximized can be graphically found in the point where the slope of TR and TC are equal ($MgC = MgR$)

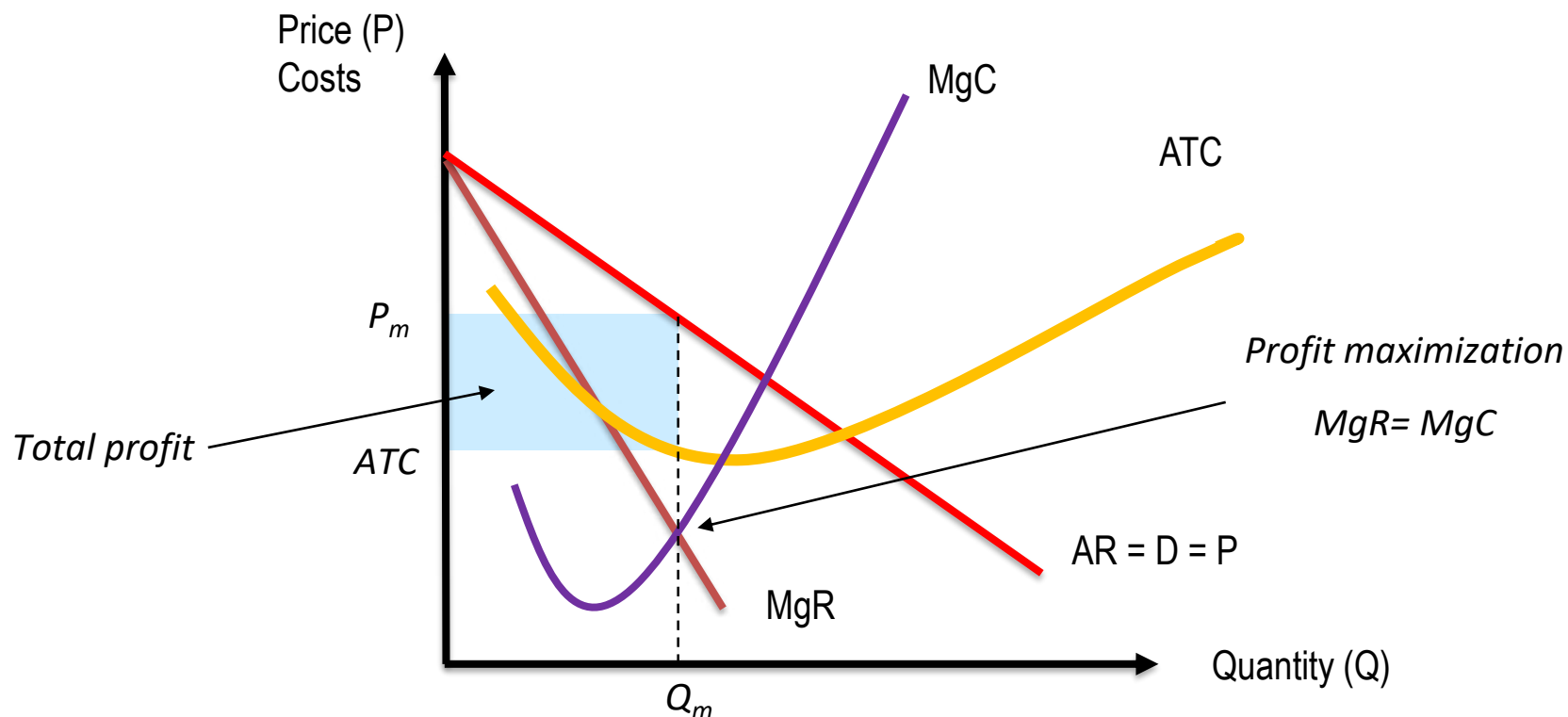
This situation is also known as **the optimal condition for the monopolist**.



6.3 Pricing in the monopoly

1. Cases in which the monopolist makes *extraordinary profits*

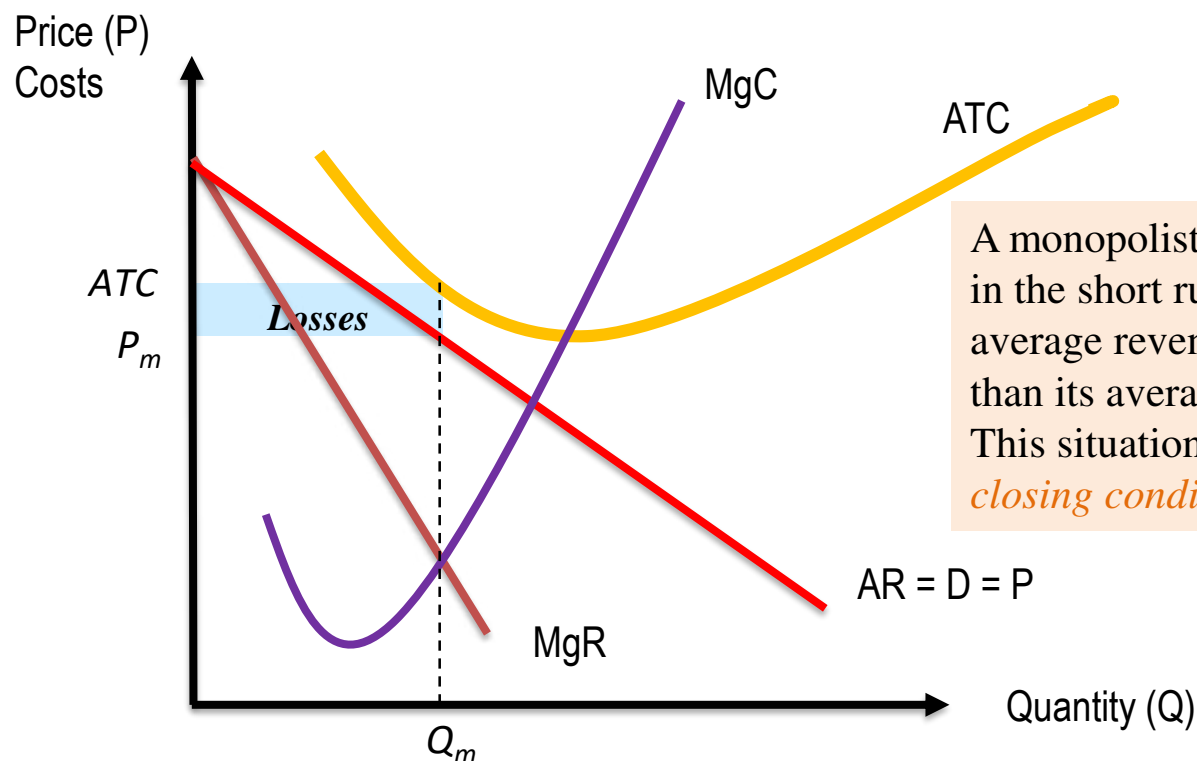
- The ATC is below the AR (demand)
- The revenue obtained ($p \cdot q$) is higher than the cost of producing ($ATC \cdot q$)



6.3 Pricing in the monopoly

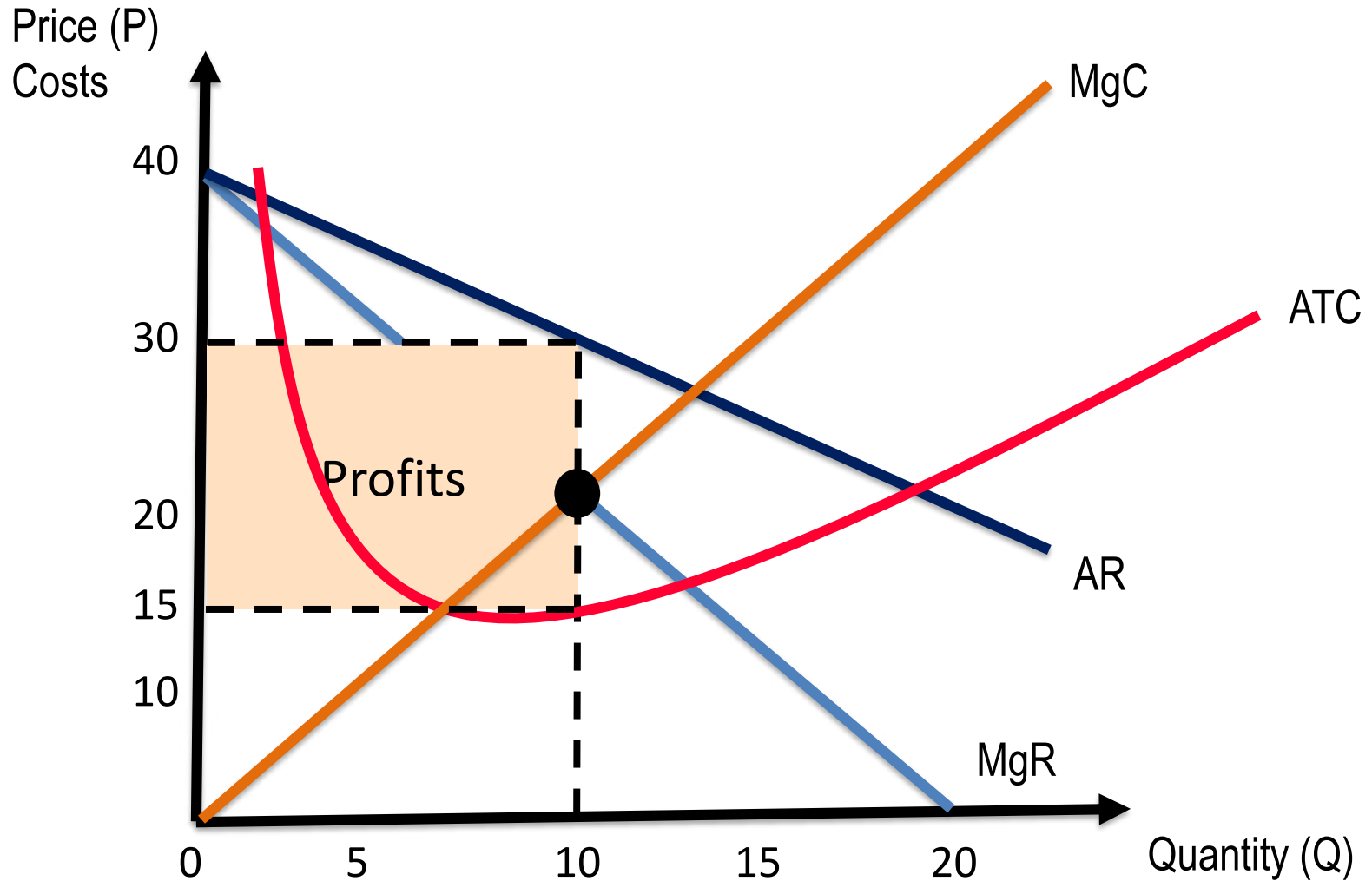
2. Cases in which the monopolist has *losses*

- The ATC is above the AR (demand)
- The revenue obtained ($p \cdot q$) is lower than the cost of producing ($ATC \cdot q$)



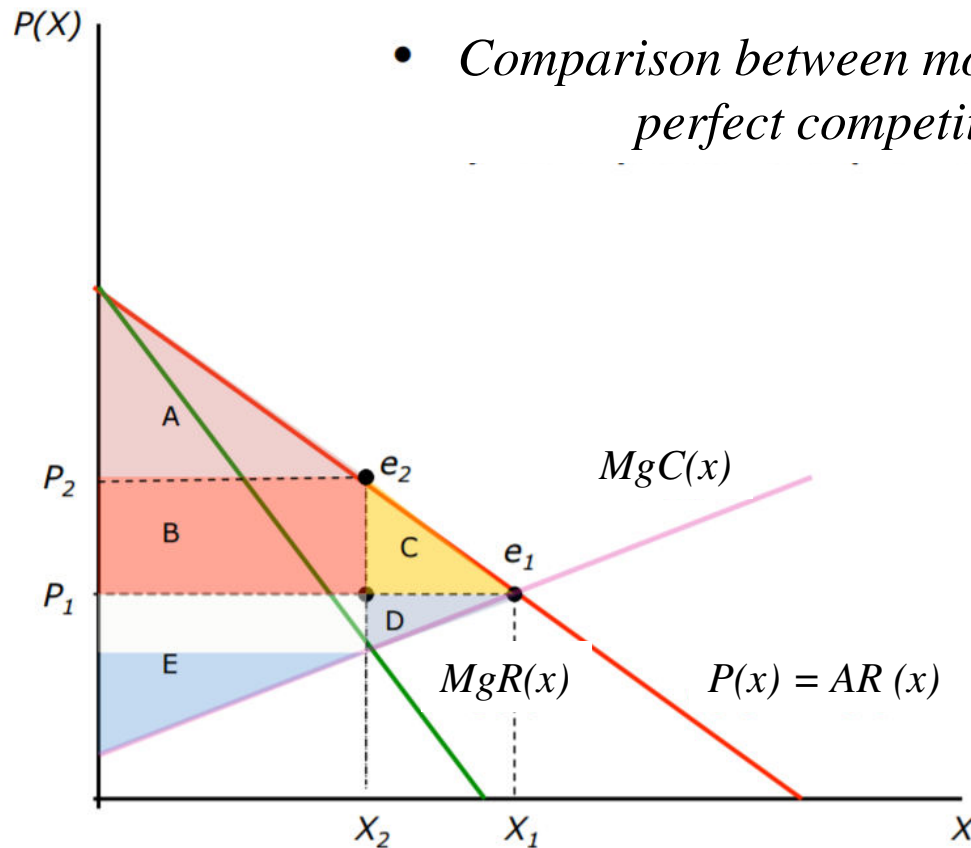
A monopolist must shut down in the short run when the average revenue (price) is lower than its average variable cost. This situation is known as *closing condition*.

6.3 Pricing in the monopoly



6.3 Pricing in the monopoly

- *Comparison between monopoly and perfect competition*



6.3 Pricing in the monopoly

- Monopoly effects on social welfare.

	Perfect competition	Monopoly	Variation (€)
Consumer Surplus (CS)	$A + B + C$	A	$-B - C = \Delta CS$
Producer Surplus (PS)	$D + E$	$B + E$	$-D + B = \Delta PS$
Social welfare	$A + B + C + D + E$	$A + B + E$	$-C - D = \Delta SW$

- Direct and indirect effects
- The optimal amount of production is reduced.
 - The market price of the good increases.
 - The company does not produce at the minimum of average costs.

6.3 Pricing in the monopoly

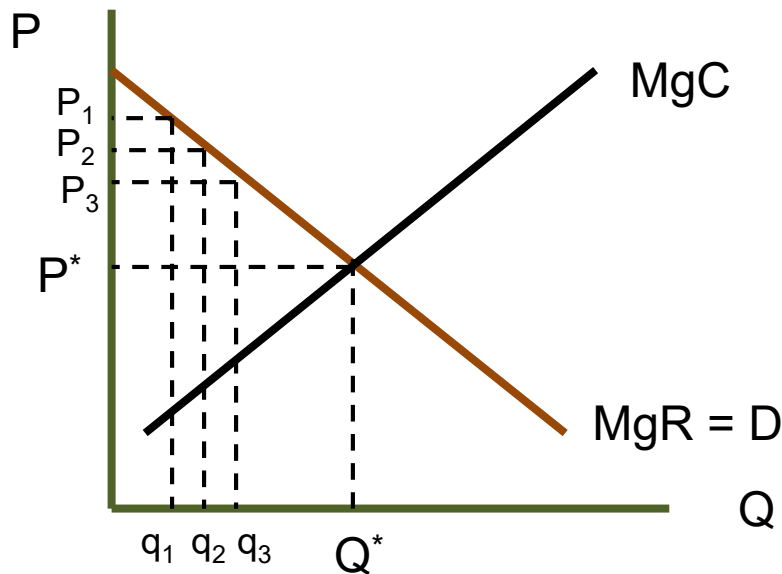
PRICE DISCRIMINATION TO MAXIMIZE PROFIT

- **First degree** → A different price is set for each *consumer*.
- **Second degree** → A different price is set depending on the sales volume (*segment*) → Price table (for ex. wholesale / retail)
- **Third degree** → Different prices are set for each *market*.

6.3 Pricing in the monopoly

FIRST DEGREE: PERFECT DISCRIMINATION

Each unit has a different price, and is equal to the maximum that consumers are willing to pay (it is given by the demand curve). We call this price **reservation price**.



$$TR(Q^*) = P_1 + P_2 + P_3 + \dots + P^*$$

MgR = Marginal price = Inverse function
of demand

$$\text{Max Profit} = TR(Q) - TC(Q)$$

$$MgR(Q^*) = MgC(Q^*)$$

$$\text{Marginal price} = MgC(Q^*)$$

6.3 Pricing in the monopoly

SECOND DEGREE: PRICE TABLE

The price that the consumer pays is a function of the sales volume

Consumers are split based on how much they want to buy (**block pricing**).

Examples:

- In a large container, the price per unit is lower: mineral water, detergent...;
- 3x2 offer;
- Wholesale / Retail;
- Two part tariff: Entry fee (or fixed fee) and usage fee (or marginal price). Like telephone lines, city gas, some subways...

6.3 Pricing in the monopoly

SECOND DEGREE: PRICE TABLE

Two part tariff

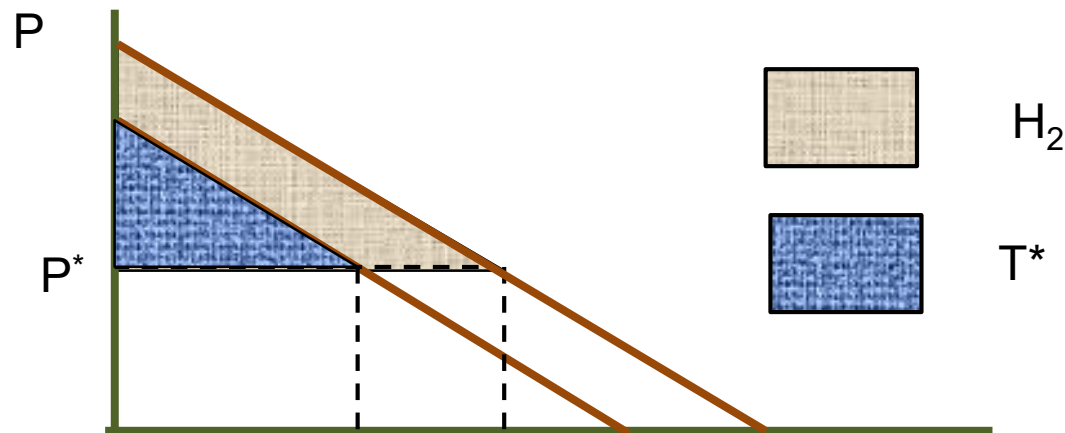
- Consumer 'j' pays to consume the quantity ' q_j ': $T + P \cdot q_j$
- The entrance fee (T): amount that must be paid for the availability of the service, whether the buyer does finally consume or not.
- The usage fee (P): amount that must be paid for each unit consumed.
- The case of two consumers will be studied:

6.3 Pricing in the monopoly

SECOND DEGREE: PRICE TABLE

Two part tariff

Each consumer values the product differently. If the entry fee were the one that cancels the surplus of the consumer who values the product the most (denoted by H_2), the company would lose the other consumer.



Consequently, the entry fee will equal the surplus of the consumer who values least the product.

6.3 Pricing in the monopoly

THIRD DEGREE: SEPARATE MARKETS

The price paid by the consumer is a function of the consumer's characteristics: the company separates consumers into groups.

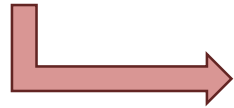
Examples:

- Circus, Aquarium, ...: child –adults



(depending on the age of the consumer)

- Cinema: normal –student or retired



(depending on the status of the consumer)

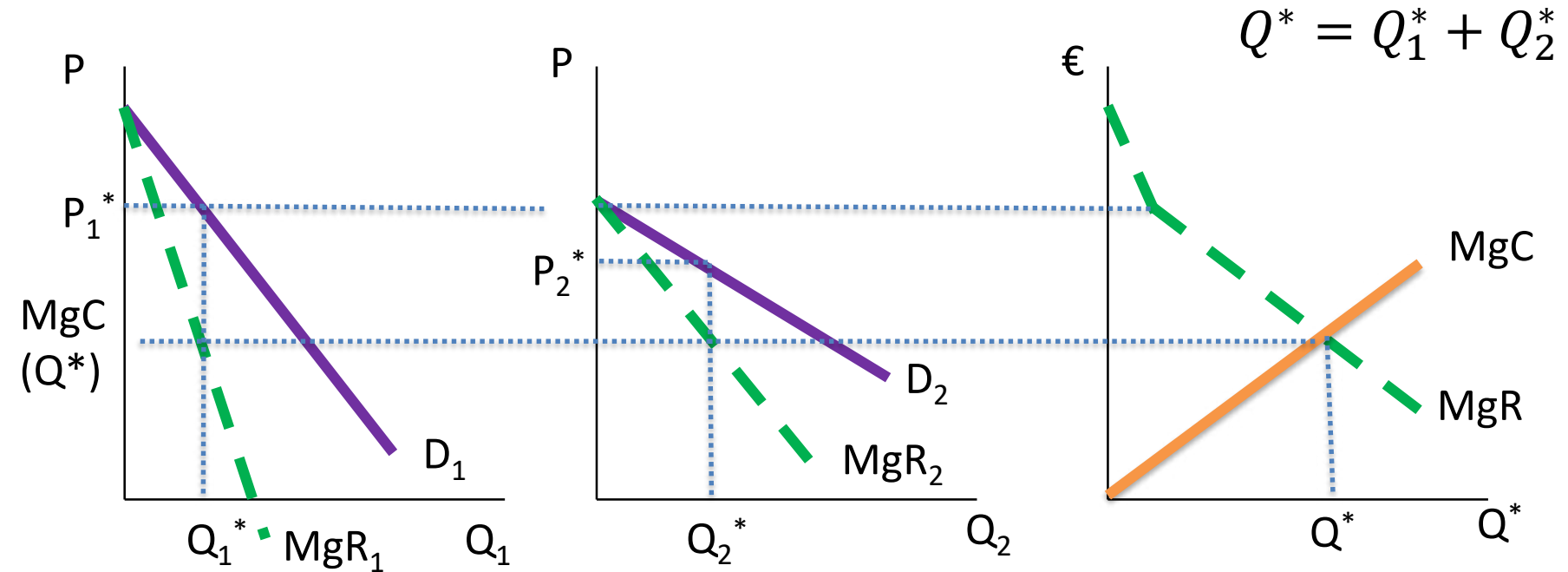
- Metro: normal –student –retired



(depending on the status of the consumer)

6.3 Pricing in the monopoly

THIRD DEGREE: SEPARATE MARKETS



- The MgR curve is the result of the horizontal sum of MgR_1 and MgR_2 .
- The price to be set for each group is determined by their respective demand curves.

LESSON 6: MARKET POWER. MONOPOLY AND MONOPOLISTIC COMPETITION.

- 6.1 Market failures.
- 6.2 The monopoly: Concept and origin
- 6.3 Pricing in monopoly.
- 6.4 *Monopolistic competition versus perfect competition.*
- 6.5 Product differentiation: the role of advertising.
- 6.6 The equilibrium in the short and long run in monopolistic competition.
- 6.7 Economic inefficiency of monopoly and monopolistic competition.

6.4 Monopolistic competition versus perfect competition

- The term monopolistic competition was coined by the economist Edward Chamberlain (Harvard University) in 1930.
- According to Gregory Mankiw: “monopolistic competition is a market structure in which many companies sell similar, but not identical, products”.
- Samuelson and Nordhaus: “Many sellers offer goods that are close substitutes, but not perfect substitutes”.

6.4 Monopolistic competition versus perfect competition

Characteristics

- *Differentiated product:* Firms compete by selling differentiated products that are easily substituted by the other varieties.
- *Many firms:* Each firm is a monopolist of its variety and determines the level of production that maximizes its profits ($MgR(q^*) = MgC(q^*)$). The firm has monopoly power since it can influence the price of its own variety.
- *No barriers:* There are no entry or exit barriers for firms.

6.4 Monopolistic competition versus perfect competition

A monopolistic competition market **is similar** to a perfect competition market in two key aspects:

- ✓ There are many firms
- ✓ The entry of new firms is not limited.

But it **is different** in:

- ✓ The product is differentiated.

Each firm looks for a brand or version of the product that differs by its quality, its appearance or its reputation, and each firm is the only producer of its own brand / variety.

The degree of monopoly power that the firms has depends on its success in differentiating its product from the other companies.

LESSON 6: MARKET POWER. MONOPOLY AND MONOPOLISTIC COMPETITION.

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6.5 Product differentiation: the role of advertising

➤ Incentives to advertise in order to attract customers:

- ✓ The product is differentiated
- ✓ Prices are above marginal cost

➤ The 'amount' of advertising varies by product:

- Companies that sell **differentiated products**

↳ perfumes, cereals, cars... 15-20% TR

- Companies that sell **industrial products**

↳ screws, communication satellites... very little

- Companies that sell **homogeneous products**

↳ wheat, barley... No amount spent.

6.5 Product differentiation: the role of advertising

CRITICISM OF ADVERTISING

1. Manipulates consumer preferences.
2. Non-informative psychological advertising.
3. Creates wishes artificially.
4. Prevents competition.
5. Tries to convince of nonexistent differences.
6. Encourages brand loyalty.
7. Consumers end up paying different prices for similar goods.
8. Lead to less elastic demand curves.
9. Firms charge a higher margin on marginal cost.

6.5 Product differentiation: the role of advertising

DEFENSE OF ADVERTISING

1. **Transmits information** to consumers about:
Prices, existence of new products, place of purchase, etc.
Better choice for consumers and more efficient allocation of resources.
2. **Encourages competition**, as customers are more informed and can take advantage of price differences.
3. Each company has **less market power**.
4. Public authorities have accepted advertising as a **mean of increasing competition**.
5. The willingness of the firm to spend a large amount of money on **advertising** is a **signal** to consumers **about the quality of the product**: The information is not in the content of the advertisement but simply in its existence and expense.

The consumer thinks that “if the firm is willing to spend so much money to advertise this product, it must be really good”.

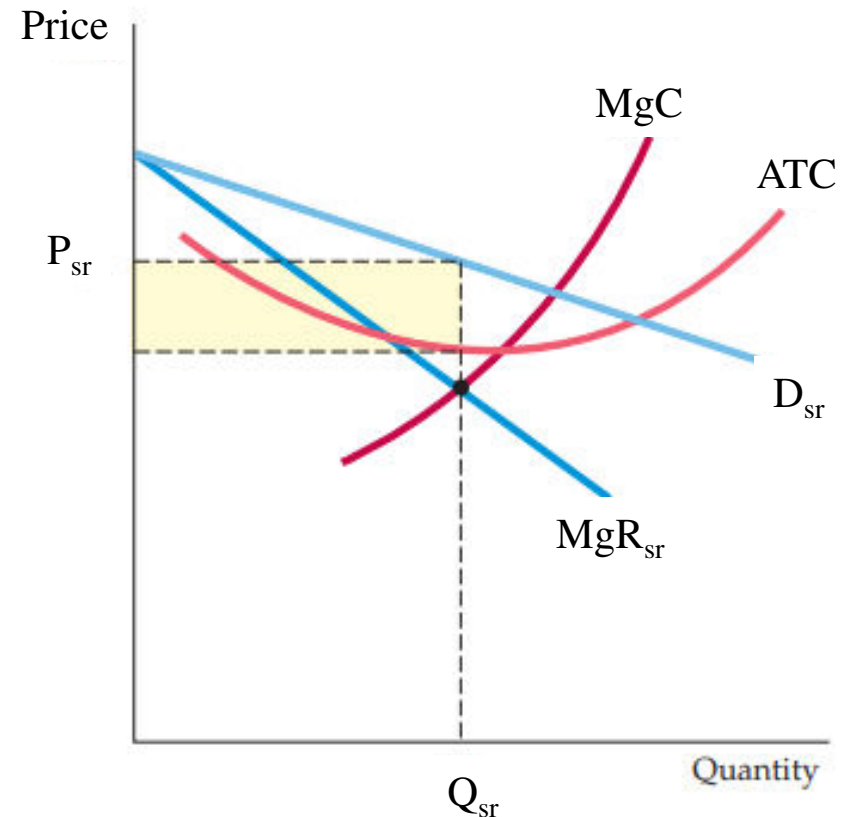
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6.6 The equilibrium in the short and long run in monopolistic competition.

EQUILIBRIUM IN THE SHORT RUN

- ❑ Because the firm's product differs from its competitors', its demand curve D_{SR} is downward sloping (FIRM'S demand, not the market demand)
- ❑ The profit maximizing quantity Q_{SR} is found where $MgR = MgC$
- ❑ Since $P_{sr} > ATC$ the firm has extraordinary profits.

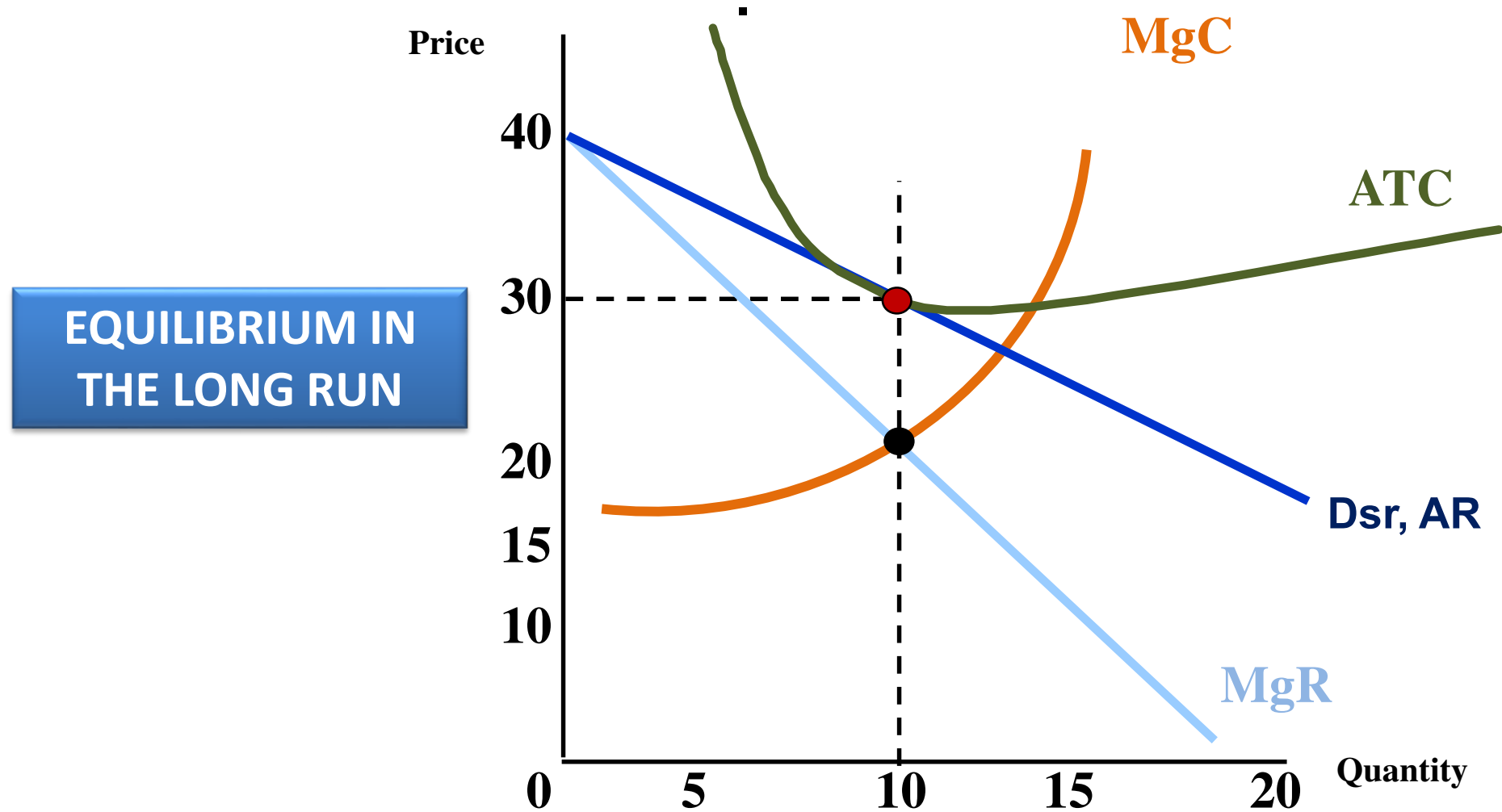


6.6 The equilibrium in the short and long run in monopolistic competition.

EQUILIBRIUM IN THE LONG RUN

- ❑ Extraordinary profits **attract new businesses.**
- ❑ The demand for the firm decreases, it shifts to the left until **the profits disappear.**
- ❑ The higher degree of competition causes **the production and the price of each firm to decrease.**
- ❑ This mechanism operates until $P = ATC$ (zero profits), despite which, in equilibrium, $P > MgC$ (firms continue to have monopoly power).

6.6 The equilibrium in the short and long run in monopolistic competition.



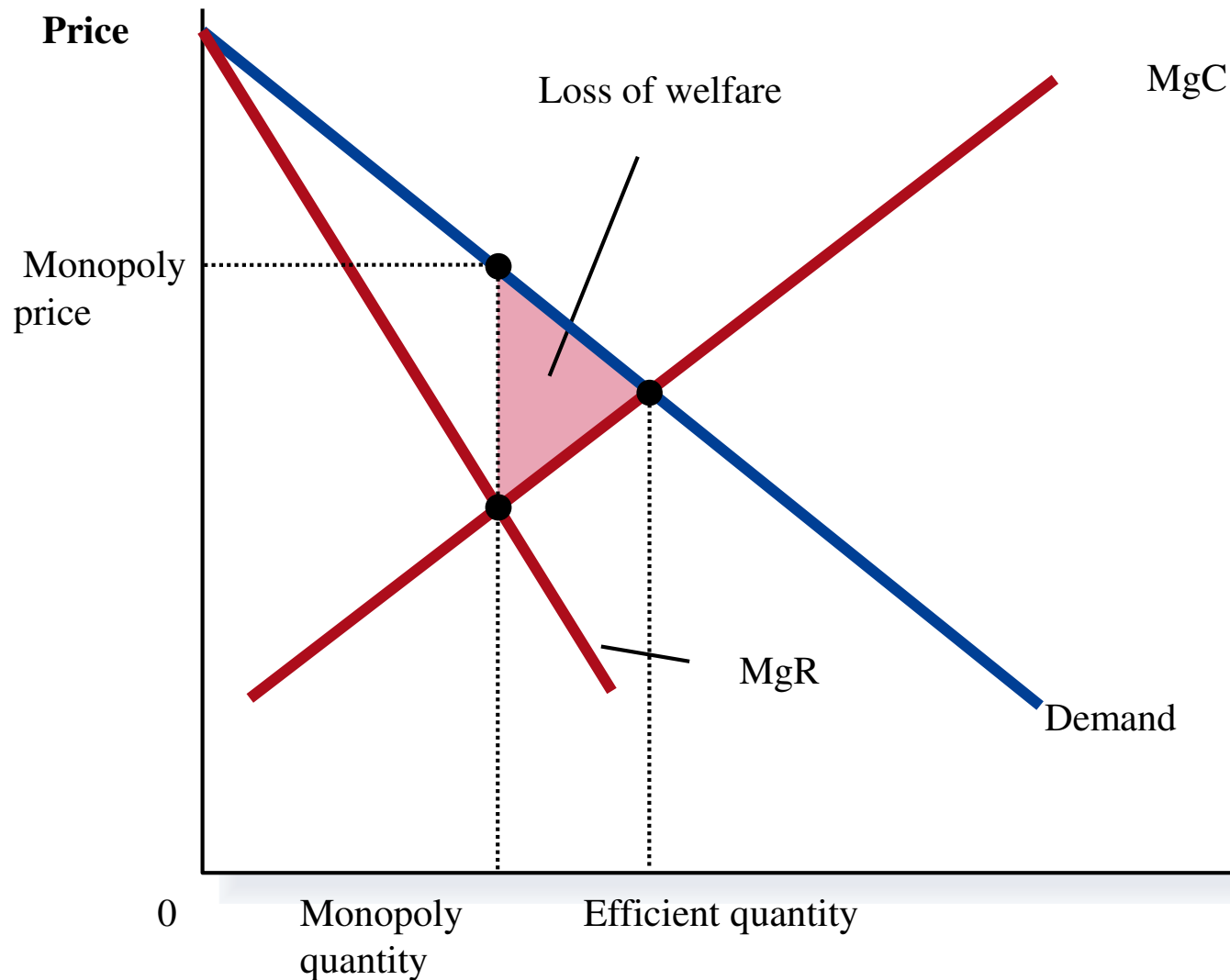
LESSON 6: MARKET POWER. MONOPOLY AND MONOPOLISTIC COMPETITION.

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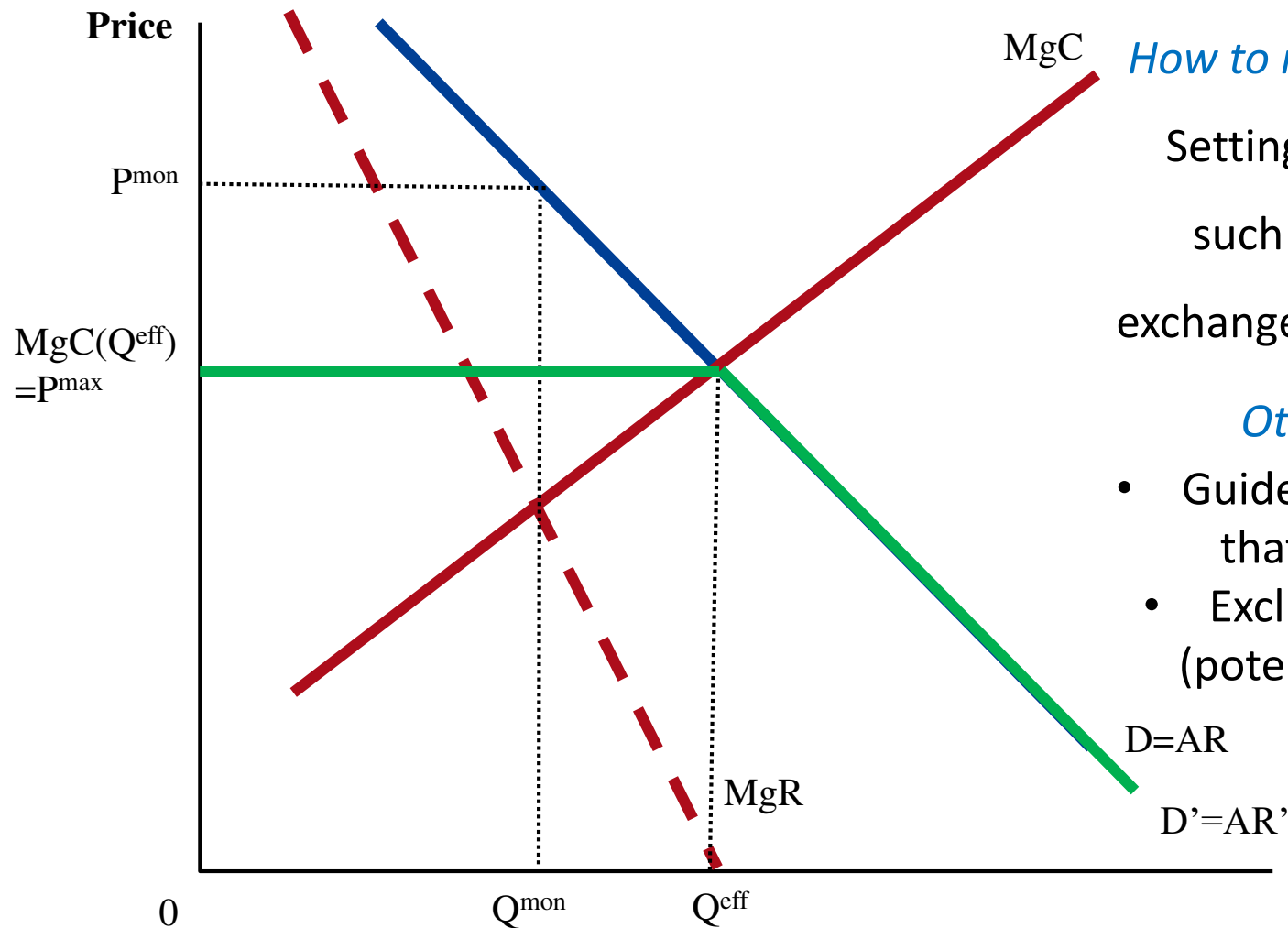
6.7 Economic inefficiency of monopoly and monopolistic competition.

- A market is efficient if social welfare (SW) is maximized.
- The SW reaches its maximum when the value of the last unit consumed (which is given by the demand curve) equals the MgC of producing it.
- The difference between the maximum social welfare and the effective social welfare is known as the irrecoverable loss of efficiency.
- A monopoly can generate irrecoverable loss of efficiency as long as it exchanges less than the efficient quantity.

6.7 Economic inefficiency of monopoly and monopolistic competition.



6.7 Economic inefficiency of monopoly and monopolistic competition.



How to regulate the market?

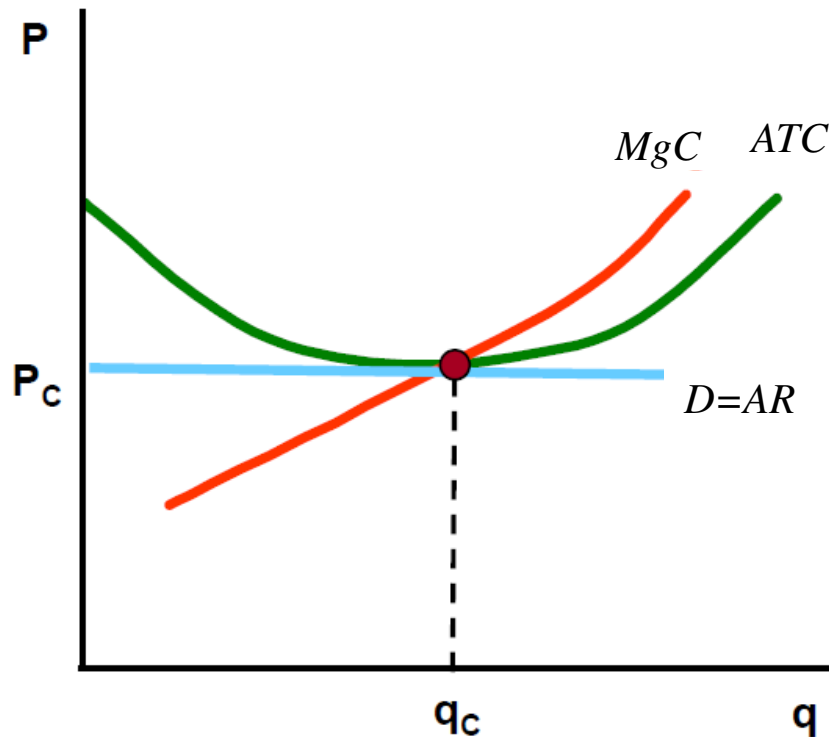
Setting a maximum price
such that the quantity
exchanged is the efficient one

Other options:

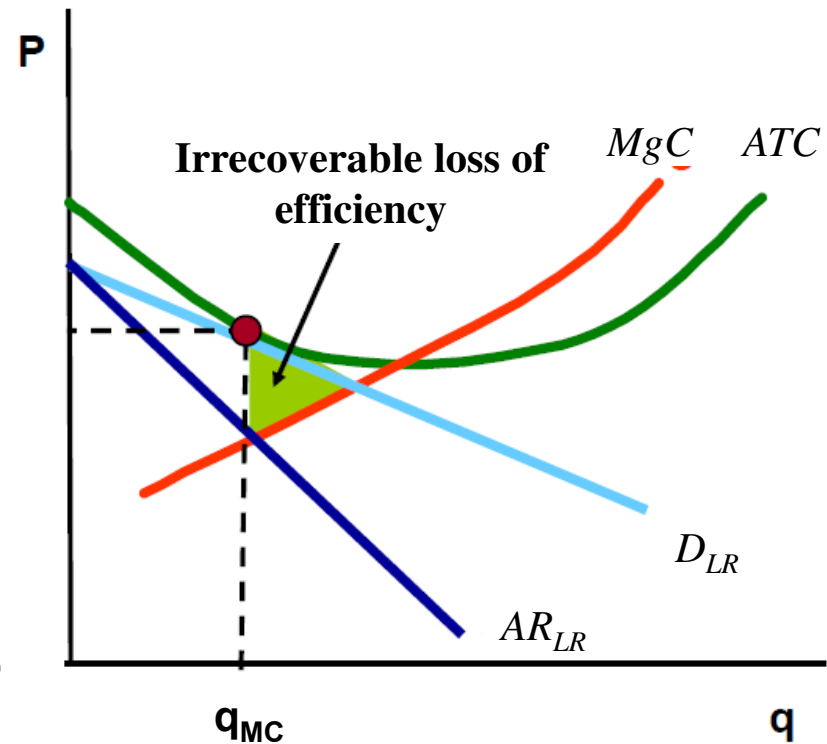
- Guidelines or standards that control pricing.
- Exclusive contracting (potential competition).

6.7 Economic inefficiency of monopoly and monopolistic competition.

Perfect competition



Monopolistic competition



6.7 Economic inefficiency of monopoly and monopolistic competition.

Two sources of economic inefficiency:

- ✓ Monopoly power generates $P > MgC$ and therefore, an irrecoverable loss of efficiency.
- ✓ Companies have excess of capacity: their production is lower than that which minimizes the average cost.

Is it a socially negative market structure?

- ✓ If there is a high elasticity of demand, the monopoly power is small, as well as the irrecoverable loss of efficiency and excess of capacity
- ✓ The diversity of products allows the consumer to choose from a wide variety of products

Variety is valuable, but pricey

EXERCISES

LESSON 2

EXERCISE 1

A consumer has the following utility function:

$$U = (x + 1) \cdot (y + 2)$$

Find combinations of goods that belong to the same indifference curve as the point (2, 3).

EXERCISE 2

A consumer has the following utility function:

$$U = (x + 1) \cdot (y + 4)$$

Calculate the slope at that point (2,3)

EXERCISE 3

A consumer has the following utility function:

$$U = (2 + x) \cdot (1 + y)$$

- A. Find combinations of goods that belong to the same indifference curve as the point (3,2).**
- B. Calculate the slope at that point**

EXERCISE 4

A consumer has the following utility function:

$$U = (x + 3) \cdot (y + 2)$$

- a) Calculate the slope at the point $x=2, y=4$**
- b) Find two combinations of goods that belong to the same indifference curve as the point $(3, 2)$.**
- c) Calculate the MRS at the point $(1,3)$**

EXERCISE 5

An individual shows his preferences for a market basket using the following utility function:

$$U = x^2 + y^2$$

- a) What will be the shape of the indifference curves?**
- b) Find the marginal rate of substitution**

EXERCISE 6

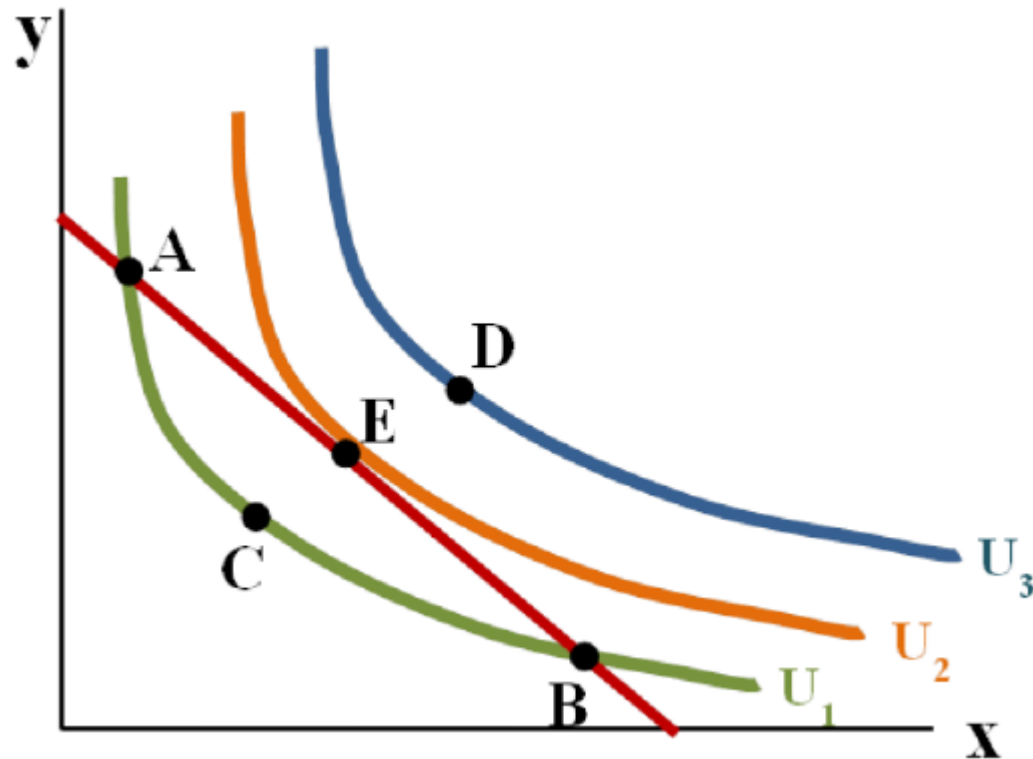
An individual shows his preferences for a market basket using the following utility function:

$$U = x^2 y^2$$

- a) Calculate the slope of the indifference curve at the point $x=2$, $y=4$**
- b) Find two combinations of goods that belong to the same indifference curve as the point $(3, 2)$.**
- c) Calculate the MRS at the point $(1,3)$**

EXERCISE 7

What would be the point or points at which the consumer would maximize utility?



EXERCISE 8

A worker has a monthly income of 1500€ and buys two types of goods: food (x) and clothes (y) on the market. His utility function is given by the following formula, where $P_x = 10€$ and $P_y = 20€$.

$$U = 5xy$$

- a) Calculate the slope of the budget constraint**
- b) Draw in a graph the budget line**
- c) If his income decreases by 200€, determine and draw the new budget constraint**
- d) If the price of clothes increases up to 30€, how would the budget constraint change? Calculate its slope and draw the new budget constraint.**
- e) What would happen if the price of food drops by 20%? Show it graphically**

EXERCISE 9

Please indicate whether the following statements are true or false. Justify your answer

- a) For the study of economic theory, one of the most important assumptions is that decision-makers are emotional, that is, their decisions are influenced by their emotions and feelings.
- b) The indifference curves show the scarcity of resources that the consumer is facing.
- c) The different points on the same indifference curve represent different combinations of goods, and each combination gives a different utility to the consumer.
- d) A basic rule in microeconomics is that the consumers will always choose the combination of goods that maximizes their utility.
- e) Transitivity means that if A is preferred to B, and C is preferred to B, then A is preferred to C.

EXERCISE 9

- f) The curves furthest from the origin will be the ones that give the consumer the greatest utility
- g) The indifference curves are convex because the MRS increases as the consumption of one good increases with respect to the other.
- h) The MRS shows the amount that we are willing to give up of one good to get one more unit of another good.
- i) The MRS is the subjective valuation of the good because it shows us the valuation that the decision-maker makes of the importance of each good.
- j) The consumer will always choose the basket of goods in the indifference curve furthest from the origin

EXERCISE 10

It is known that the preferences of a consumer are given by the following utility function:

$$U = (x + 1) \cdot (y + 2)$$

- a) If the consumer has a monthly income of 500€ and the prices are $P_x = 8€$ and $P_y = 5€$, graphically and analytically represent his budget constraint.**
- b) What would the budget constraint look like if the price of good x were 10€?**
- c) What would happen if his income increased by 100€ and the price of the good y increased up to 8€? Draw it graphically**

EXERCISE 11

Javier has a fixed income of 100€. and spends it on two goods, X and Y. The price of good X is 6€ and the price for good Y is 8€. If Javier's utility function is:

$$U = X^{1/2}Y^{1/3}$$

- a) What would be the optimal consumption for Javier?**
- b) Express graphically and numerically what would happen to Javier's optimum if:**
 - The price of Y doubles**
 - Income doubles**
 - Both double**

LESSON 3: INDIVIDUAL AND MARKET DEMAND

3.1 Consumer demand

3.2 Changes in the budget constraint line: Engel curve

3.3 Market demand

3.4 Income effect and substitution effect

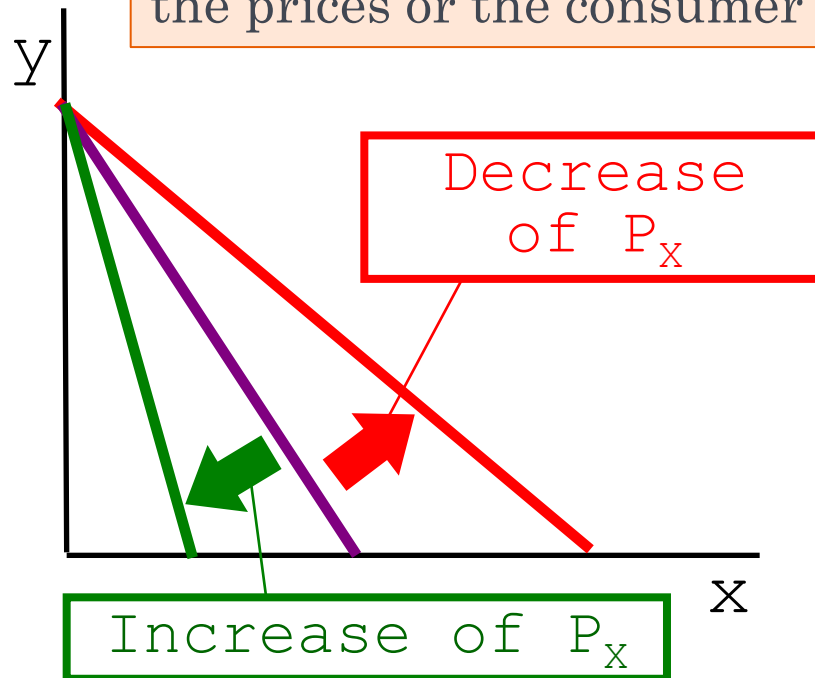
3.5 Elasticity of demand.

3.1 CONSUMER DEMAND

CHANGES IN THE BUDGET CONSTRAINT LINE

Variations in the price of a good lead to variations in the budget constraint line.

The price of a good changes without modifying the rest of the prices or the consumer's income.

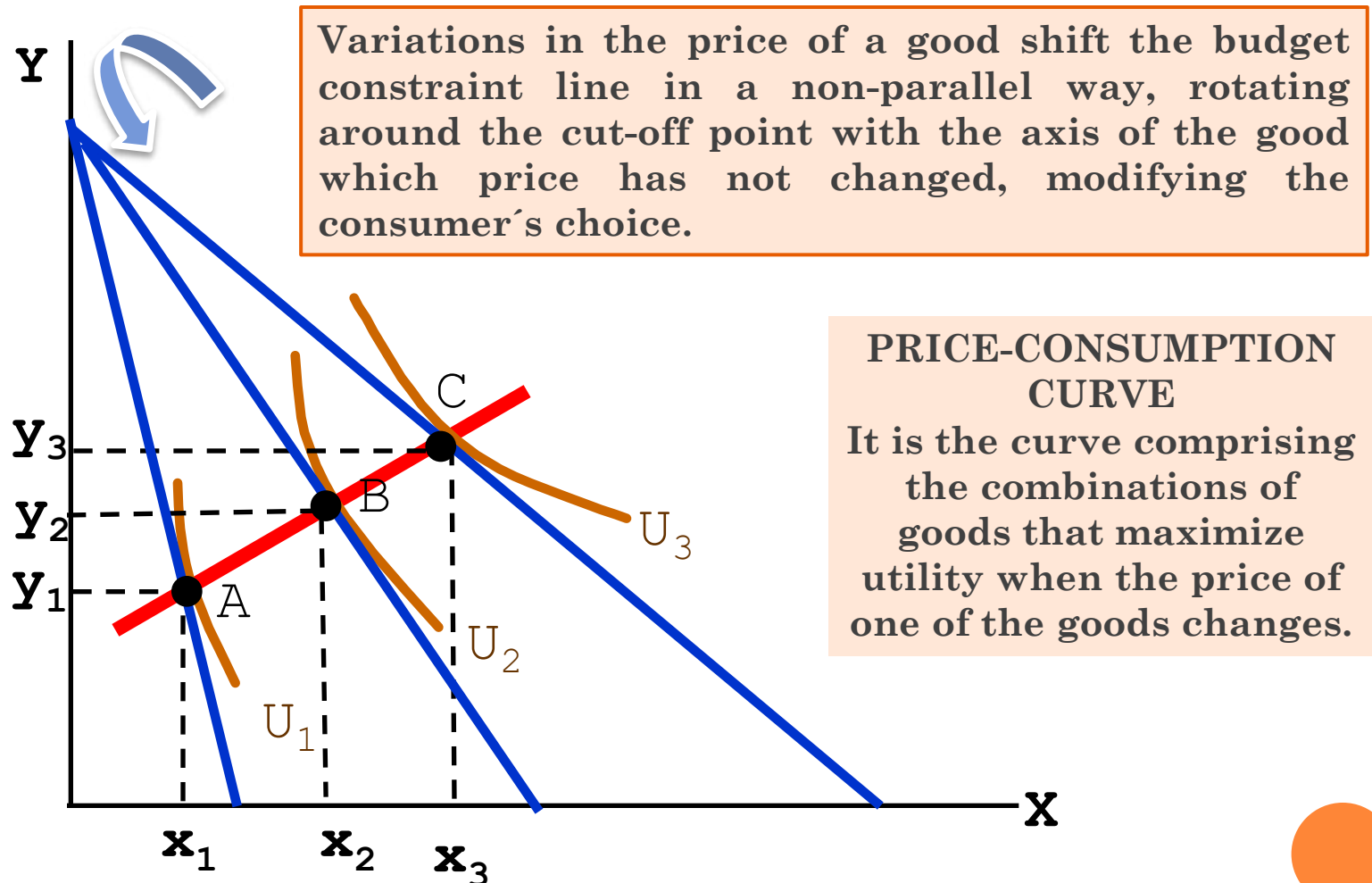


The new budget constraint line will not be parallel to the initial constraint line.

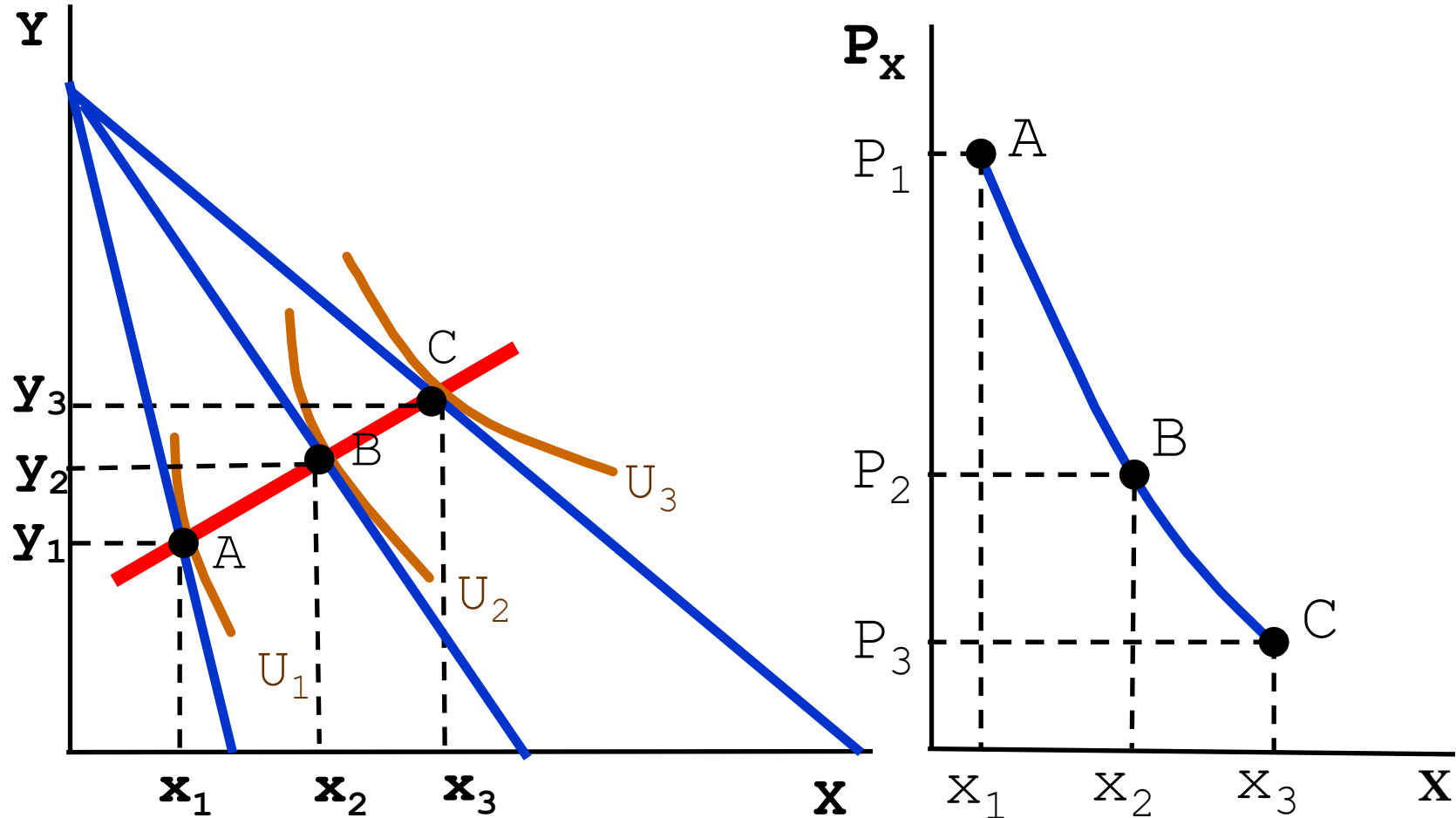


3.1 CONSUMER DEMAND

Price-consumption curve

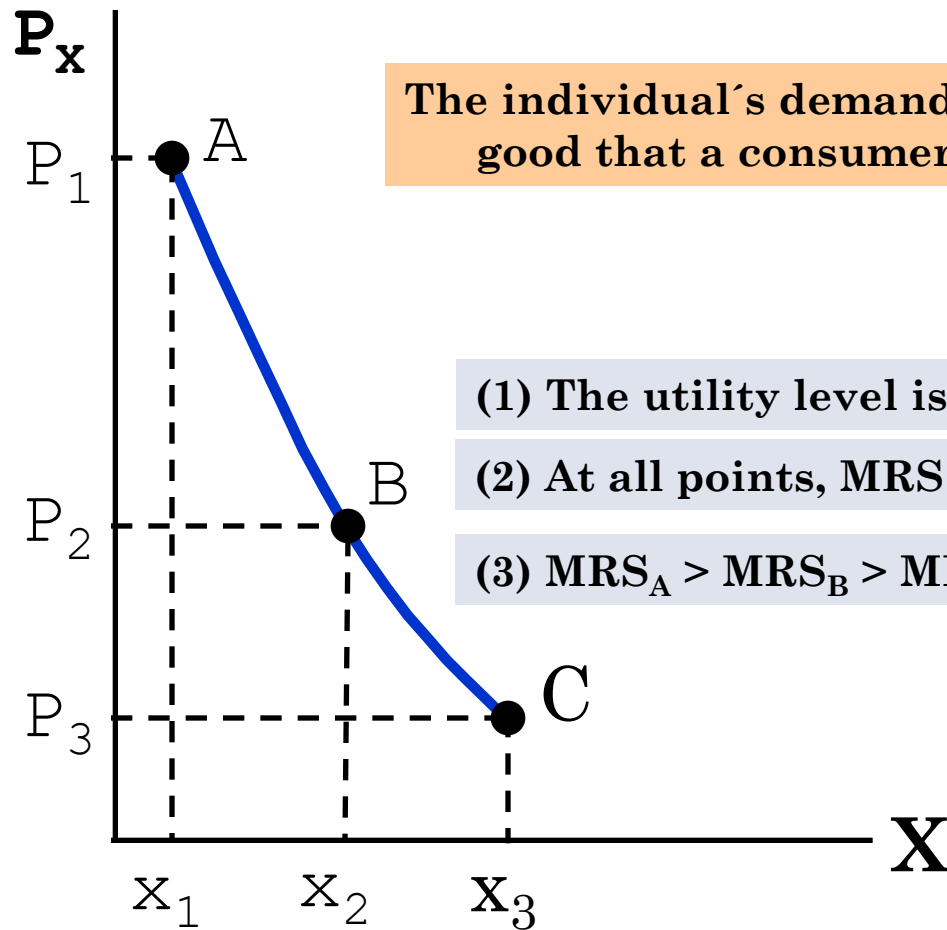


PRICE-CONSUMPTION CURVE AND DEMAND CURVE



3.1 CONSUMER DEMAND

Individual demand curve



The individual's demand curve links the amount of good that a consumer will buy with its price.

(1) The utility level is different at each point.

(2) At all points, $MRS = RP$.

(3) $MRS_A > MRS_B > MRS_C$



3.1 CONSUMER DEMAND

Two properties of the individual demand curve:

- The **utility level** that is achieved varies as we move along the demand curve: the **lower the price** of the product, **the higher the utility level**.
- At all points on the demand curve, the individual's **utility is being maximized**, so that as **we move down the demand curve**, the relative value of the good decreases as its consumption increases.



3.1 CONSUMER DEMAND

COMPLEMENTARY, SUBSTITUTE OR INDEPENDENT GOODS

What happens to the consumption of a good when the price of another good changes?

When the price of a good decreases, the consumer's purchasing power increases, then the quantity demanded of other goods may increase. We can talk about 3 different types of relationship between goods:

- ❑ Two goods are **complementary** if the variation in the price of one of them causes a variation in the opposite direction of the quantity demanded of the other.
- ❑ Two goods are **substitutes** if the variation in the price of one of them causes a variation in the same direction of the quantity demanded of the other.
- ❑ Two goods are **independent** if the variation in the price of one of them does not affect the quantity demanded of the other.

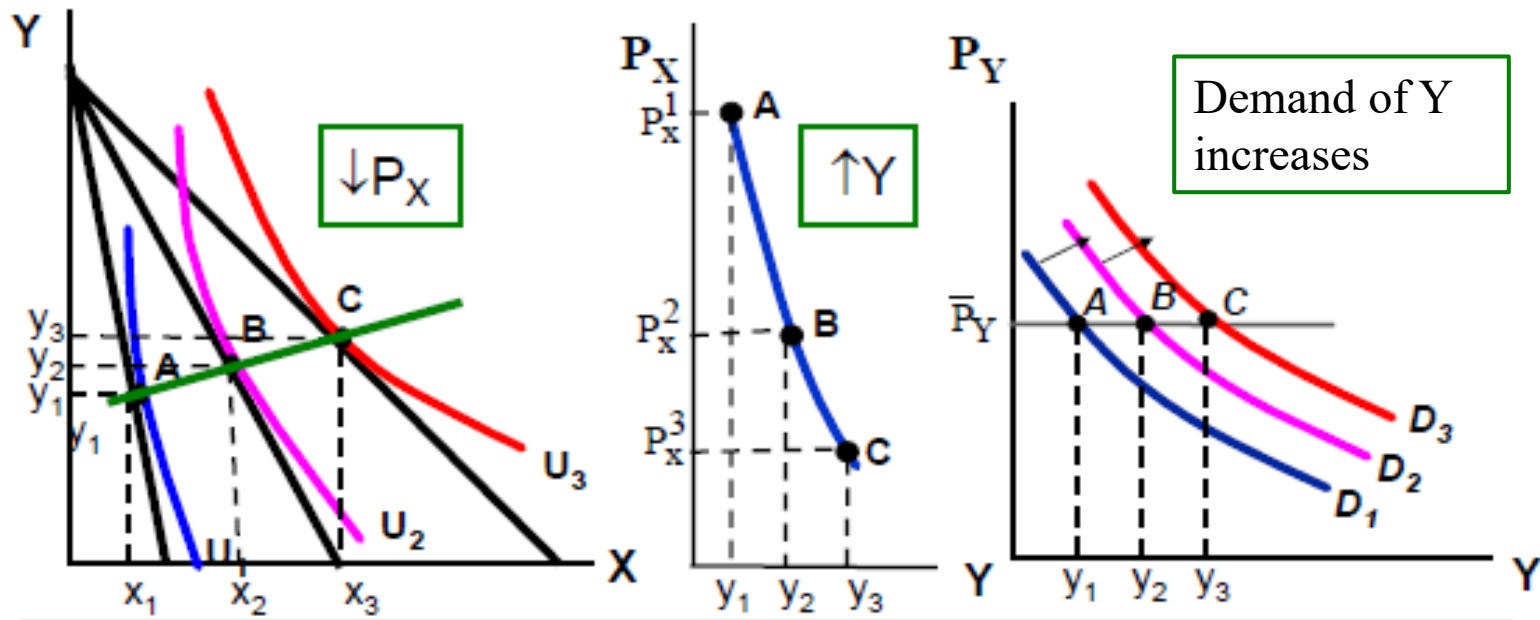


3.1 CONSUMER DEMAND

Quantity demanded
of good Y

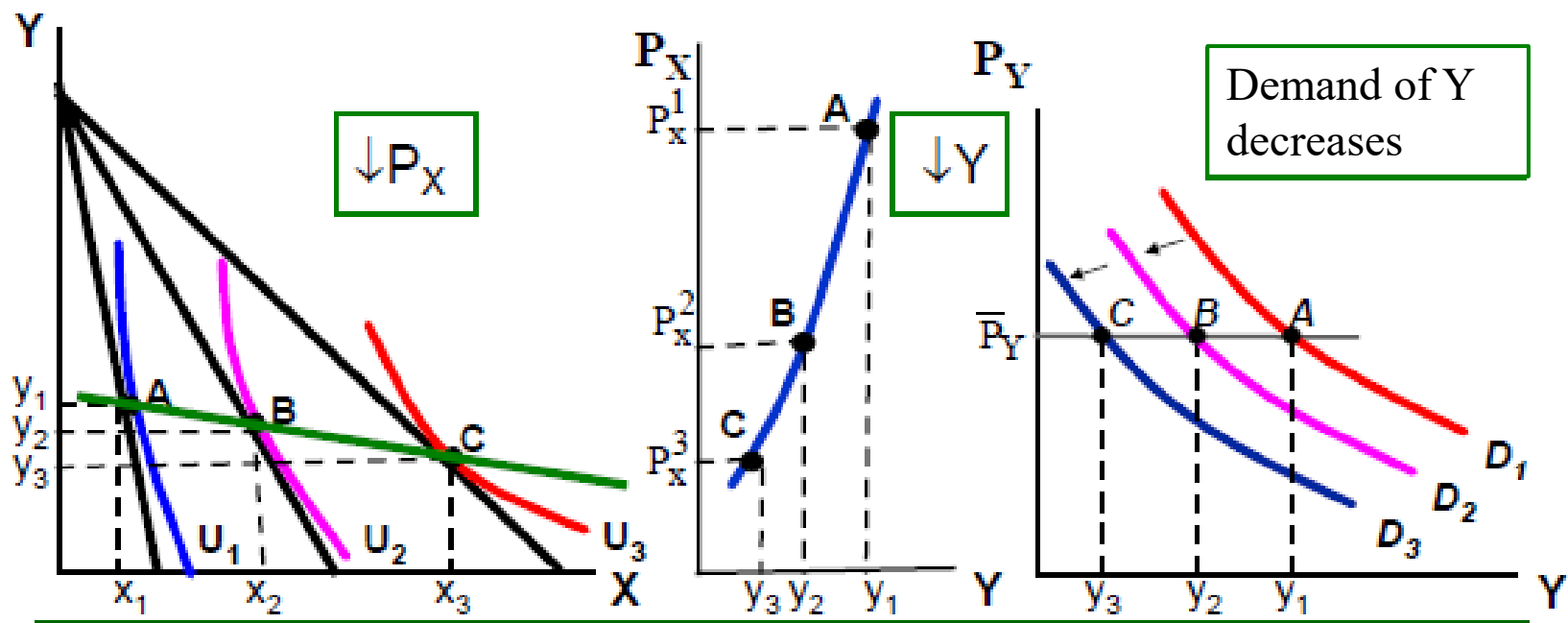
Complementary goods $\frac{\partial Y}{\partial P_x} < 0$

If the price of good X increases, the demand for good y decreases



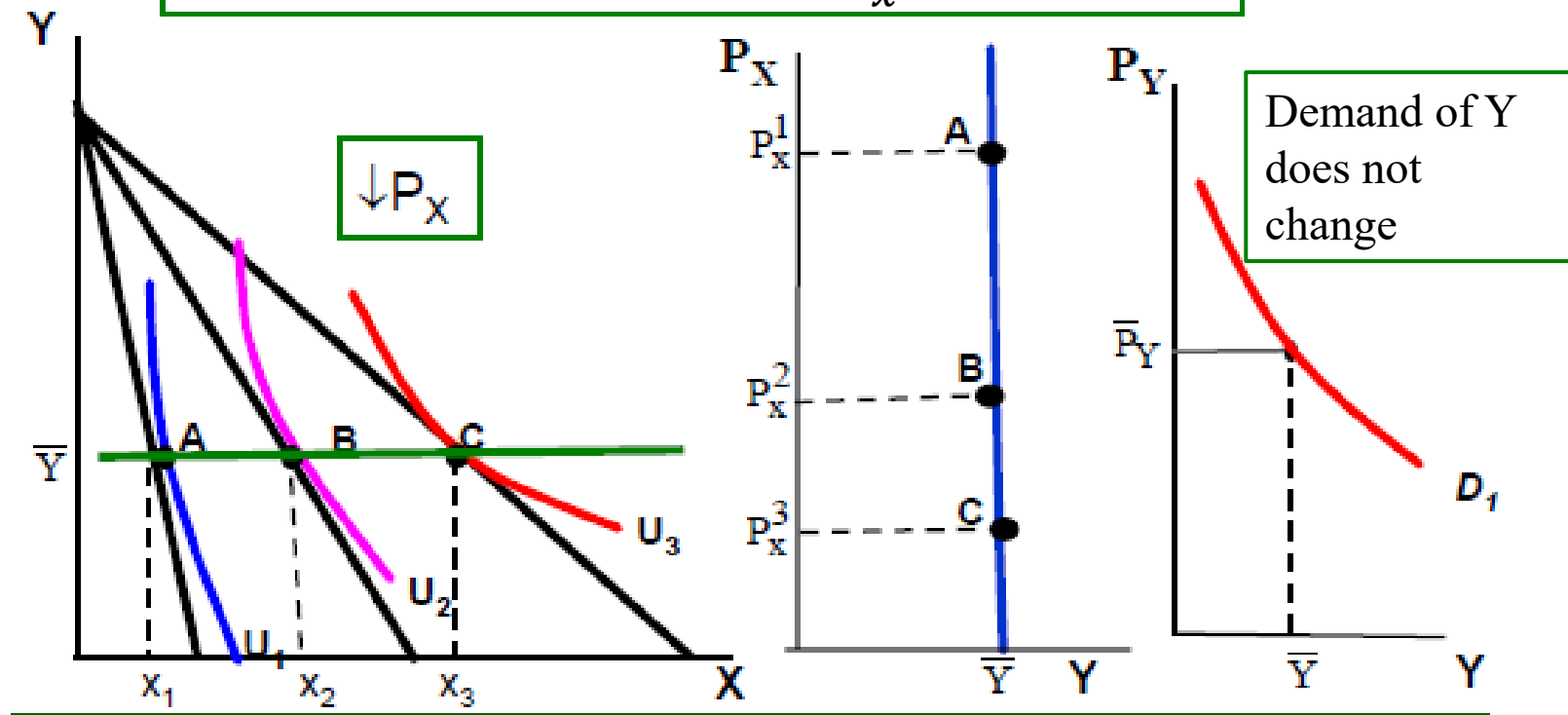
3.1 CONSUMER DEMAND

Substitute goods $\frac{\partial Y}{\partial P_x} > 0$



3.1 CONSUMER DEMAND

Independent goods $\frac{\partial Y}{\partial P_x} = 0$



LESSON 3: INDIVIDUAL AND MARKET DEMAND

3.1 Consumer demand

3.2 Changes in the budget constraint line: Engel curve

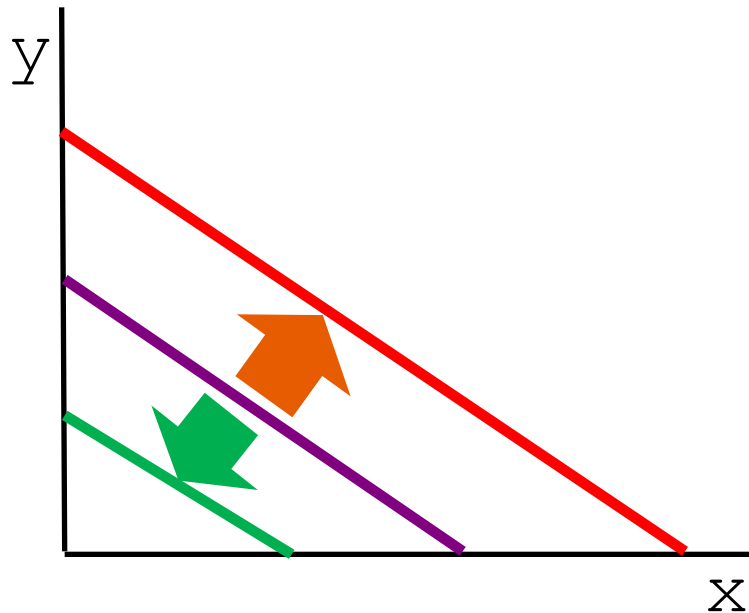
3.3 Market demand

3.4 Income effect and substitution effect

3.5 Elasticity of demand.

3.2 CHANGES IN THE BUDGET CONSTRAINT LINE

- ❖ Variations in income imply variations in the budget constraint line, affecting the maximum amount of goods that the consumer can afford.
- ❖ Let's suppose that the consumer's income changes but the prices remain the same:



- ❖ The new budget constraint line will be **parallel** to the initial line.

Increase in
income

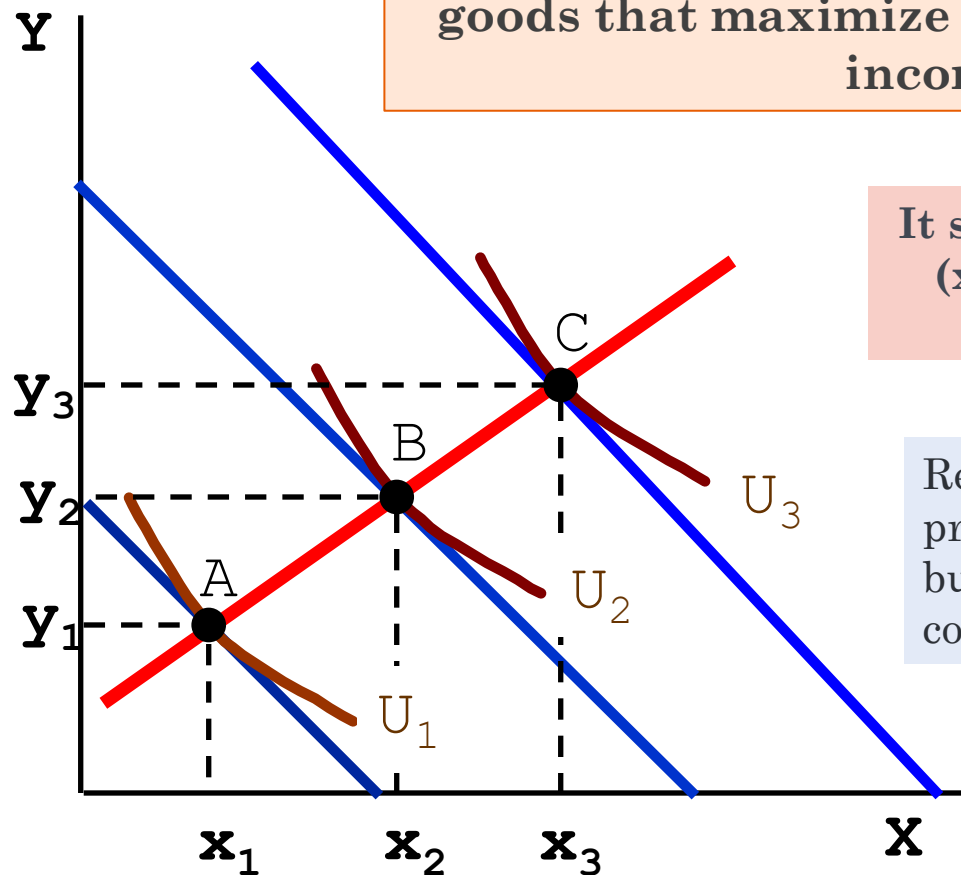
Decrease in
income



3.2 CHANGES IN THE BUDGET CONSTRAINT LINE

Income-consumption curve:

It is the curve comprising the combinations of goods that maximize utility when the consumer's income changes.



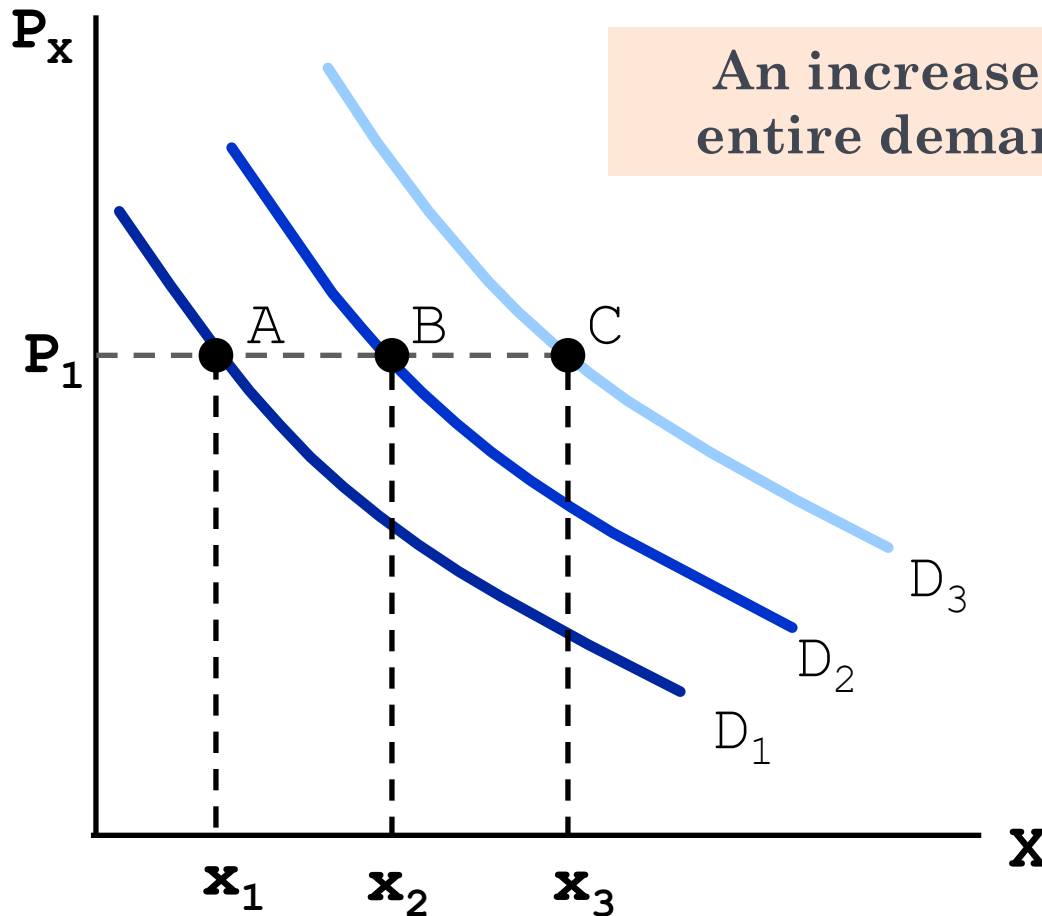
It shows the combinations of goods (x,y) that are utility maximizers for different income levels.

Remember that income variations, if prices do not change, shift the budgetary line in parallel, altering consumer choice.



3.2 CHANGES IN THE BUDGET CONSTRAINT LINE

Effects on the demand curve



An increase in income shifts the entire demand curve to the right.

Each demand curve corresponds to a certain level of income, so a change in income will cause a shift in the demand curve. The greater the shifts in the demand, the greater the income elasticity.



3.2 CHANGES IN THE BUDGET CONSTRAINT LINE

NORMAL AND INFERIOR GOODS

We will classify **goods** as **normal** or **inferior** depending on the shape of the income-consumption curve.

❑ **Normal goods:**

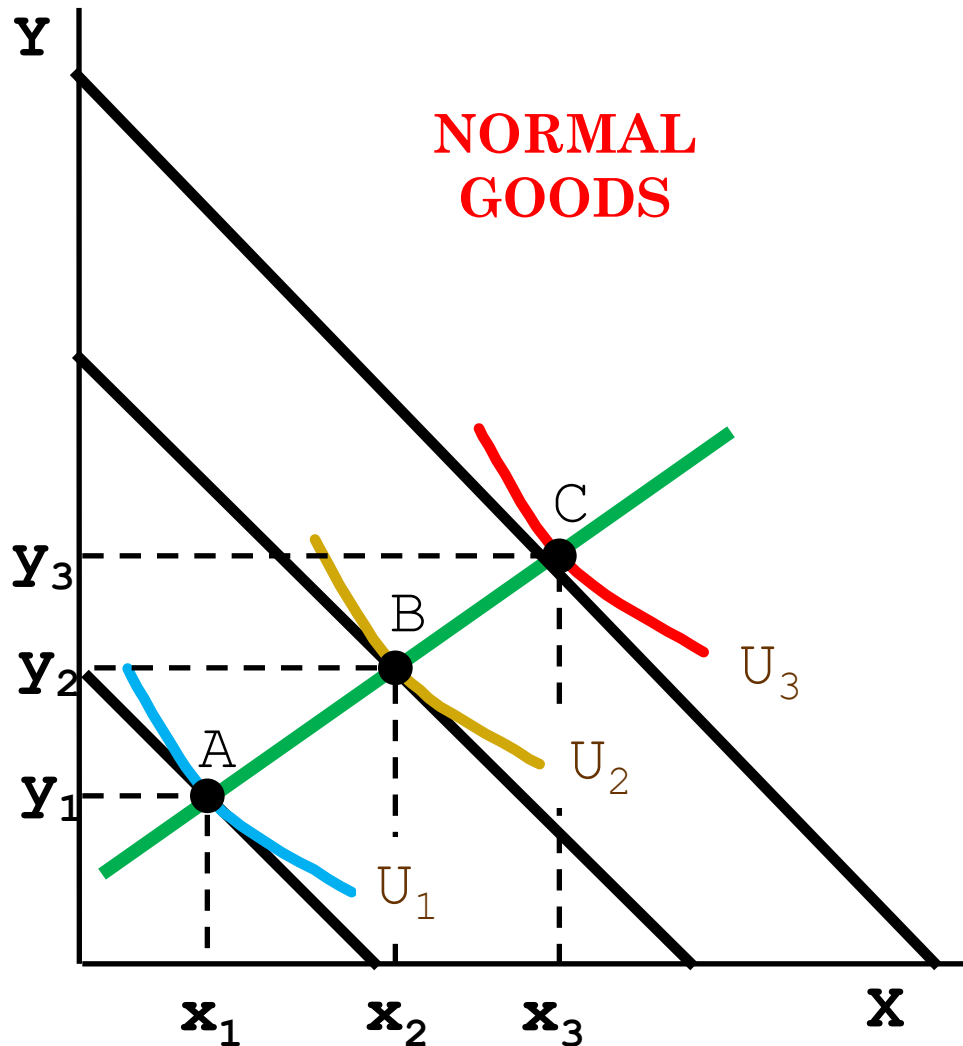
Normal goods are goods which quantity demanded **increases** with income. The income-consumption curve has a **positive slope**.

❑ **Inferior goods:**

Inferior goods are goods which quantity demanded **decreases** with income. The income-consumption curve has a **negative slope**.



3.2 CHANGES IN THE BUDGET CONSTRAINT LINE

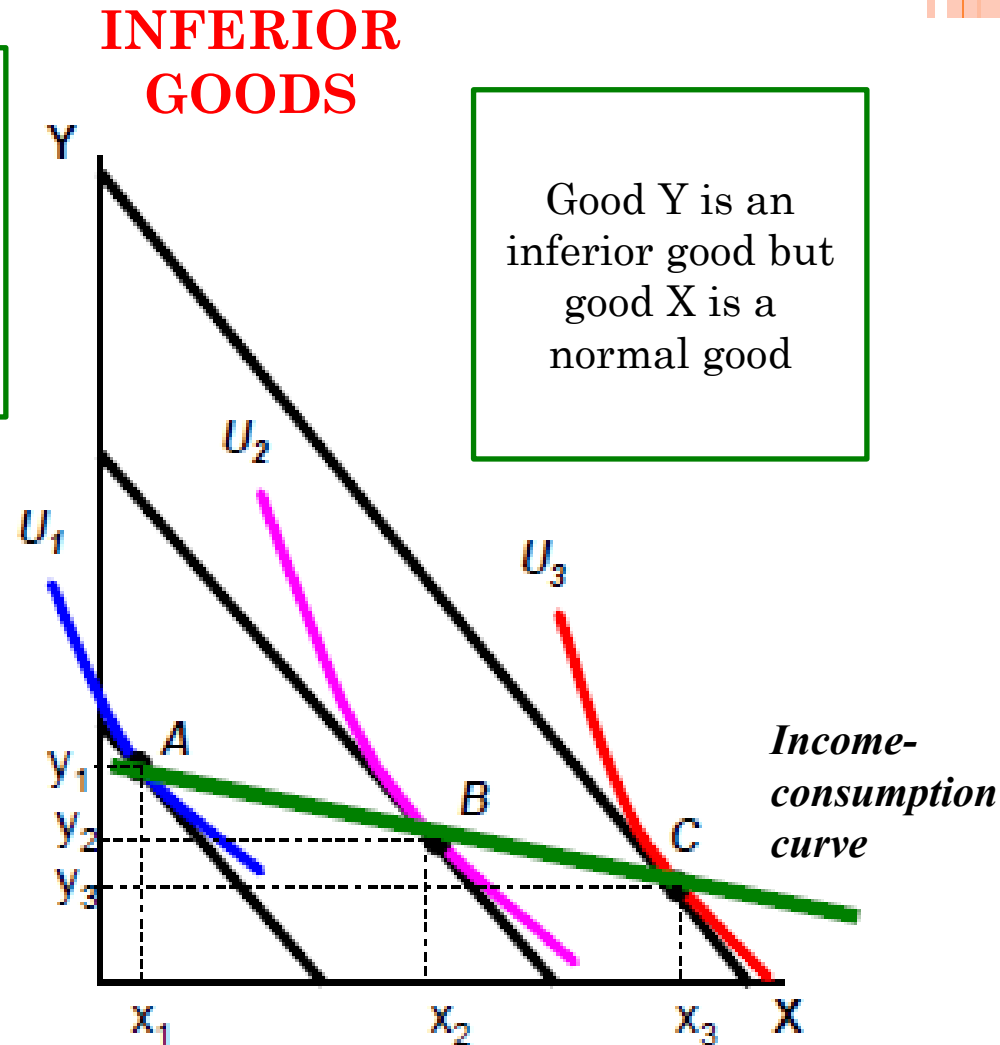
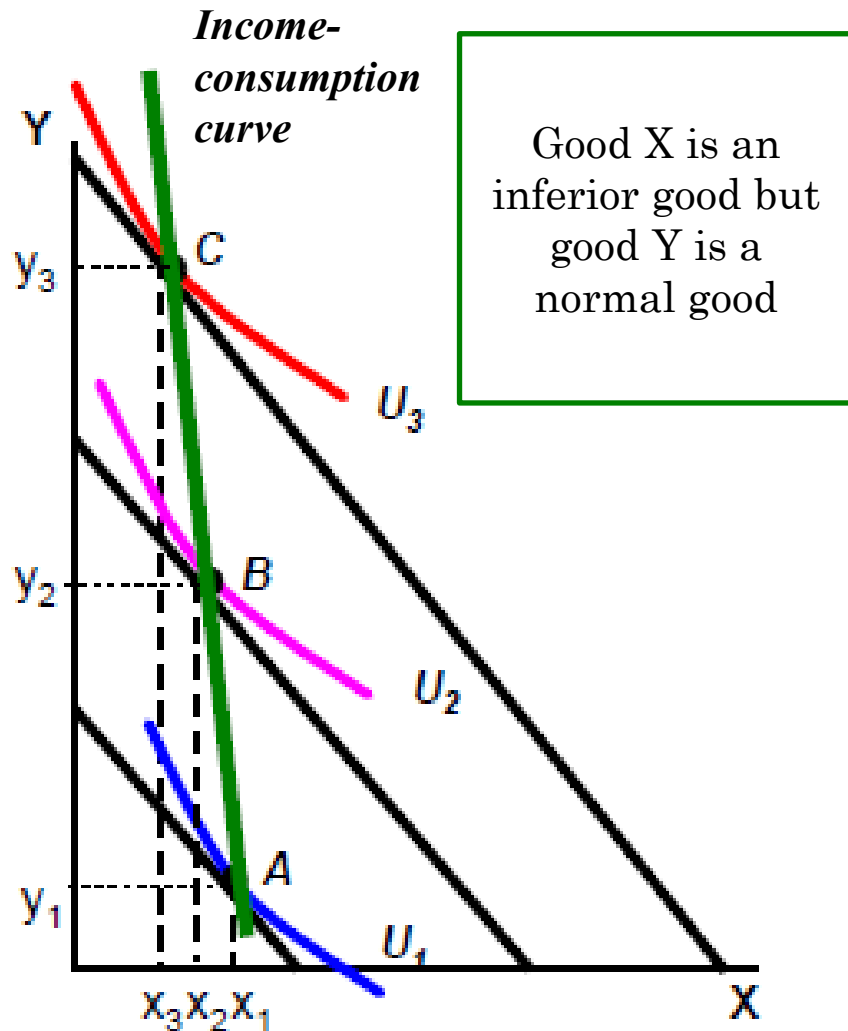


**Income-consumption
curve with positive
slope:**

As income changes, the new consumer choice implies that the consumption of both goods experiences a variation in the same direction (normal goods). The income elasticity of demand is positive.



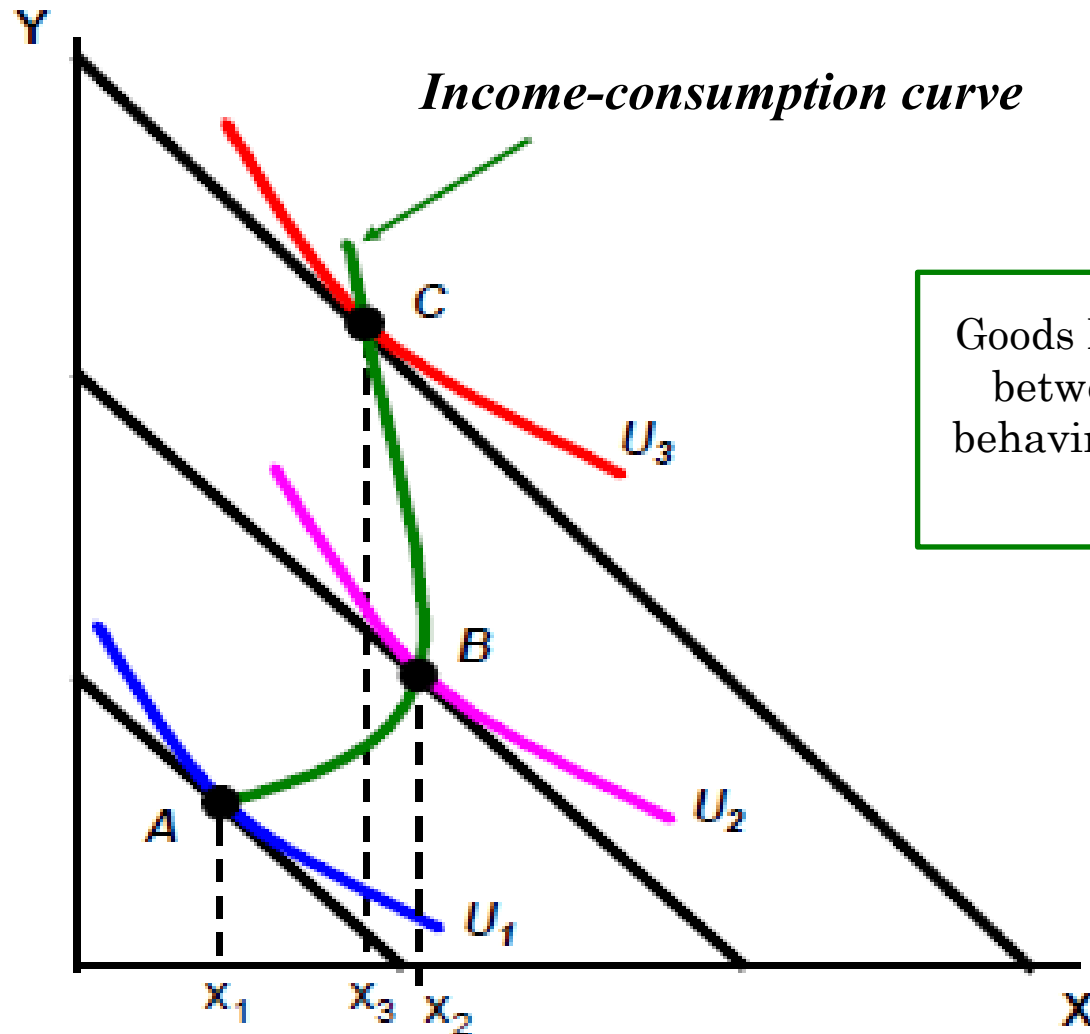
3.2 CHANGES IN THE BUDGET CONSTRAINT LINE



The income elasticity of the demand of the inferior good is negative

3.2 CHANGES IN THE BUDGET CONSTRAINT LINE

NORMAL AND INFERIOR GOODS



Goods X and Y behave as normal goods between A and B , but good X starts behaving as an inferior good between B and C

3.2 CHANGES IN THE BUDGET CONSTRAINT LINE

ENGEL CURVE

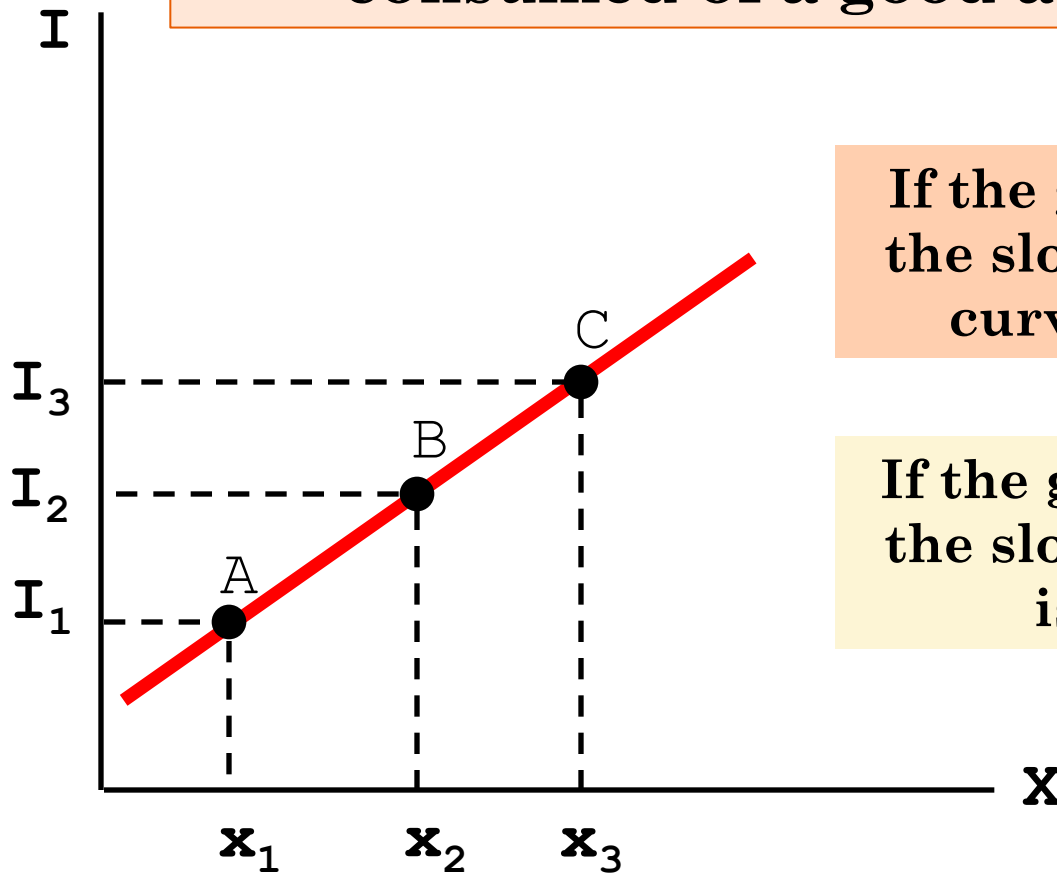
The Engel curve links the quantity of the good consumed and income.

- **Engel curves and types of goods:**
 - ✓ When the good is **normal**, its Engel curve has **positive slope**.
 - ✓ When the good is **inferior**, its Engel curve has **negative slope**.



3.2 CHANGES IN THE BUDGET CONSTRAINT LINE

The Engel curve links the quantity consumed of a good and income

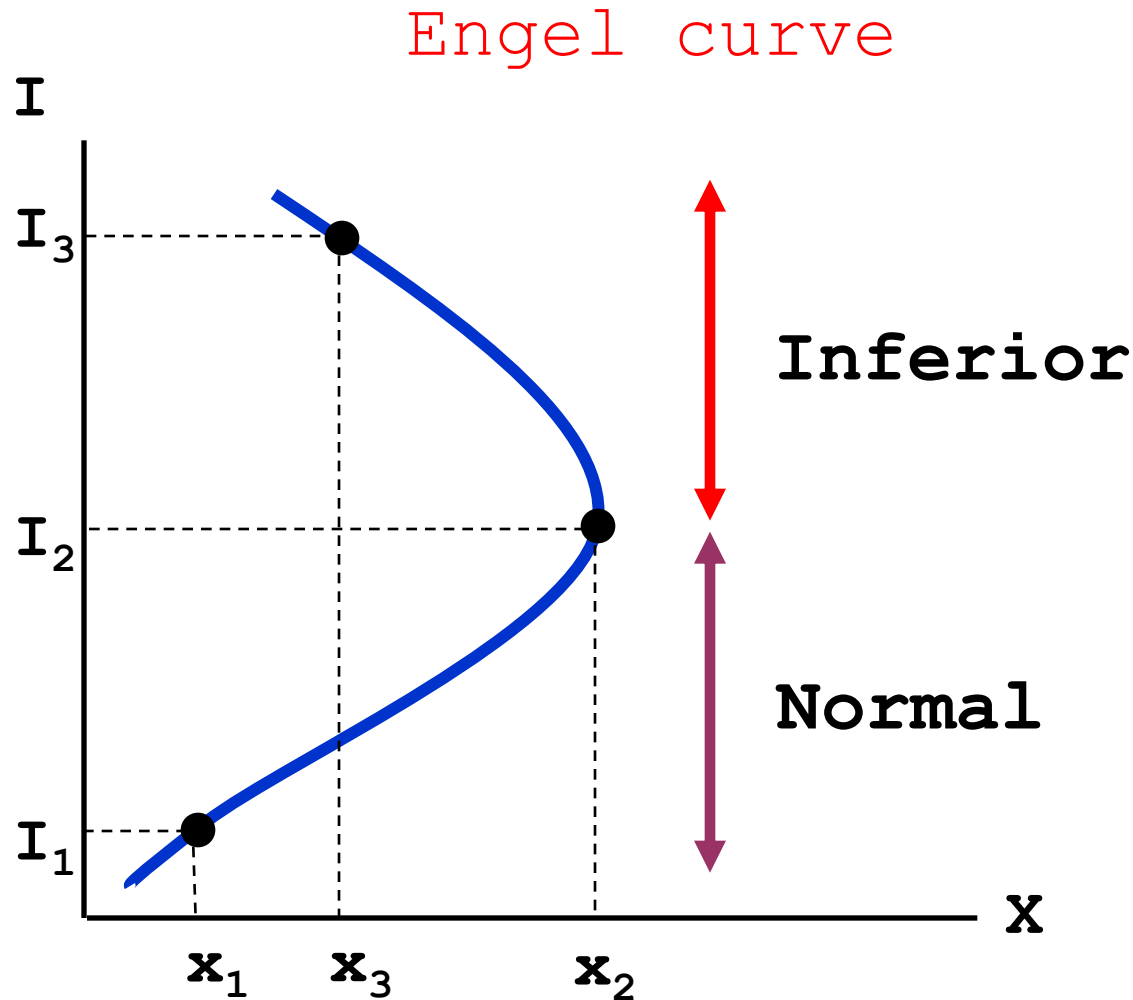


If the good is normal, the slope of the Engel curve is positive.

If the good is inferior, the slope of the Engel curve is negative



3.2 CHANGES IN THE BUDGET CONSTRAINT LINE



LESSON 3: INDIVIDUAL AND MARKET DEMAND

3.1 Consumer demand

3.2 Changes in the budget constraint line: Engel curve

3.3 Market demand

3.4 Income effect and substitution effect

3.5 Elasticity of demand.

3.3 MARKET DEMAND

Market demand curve

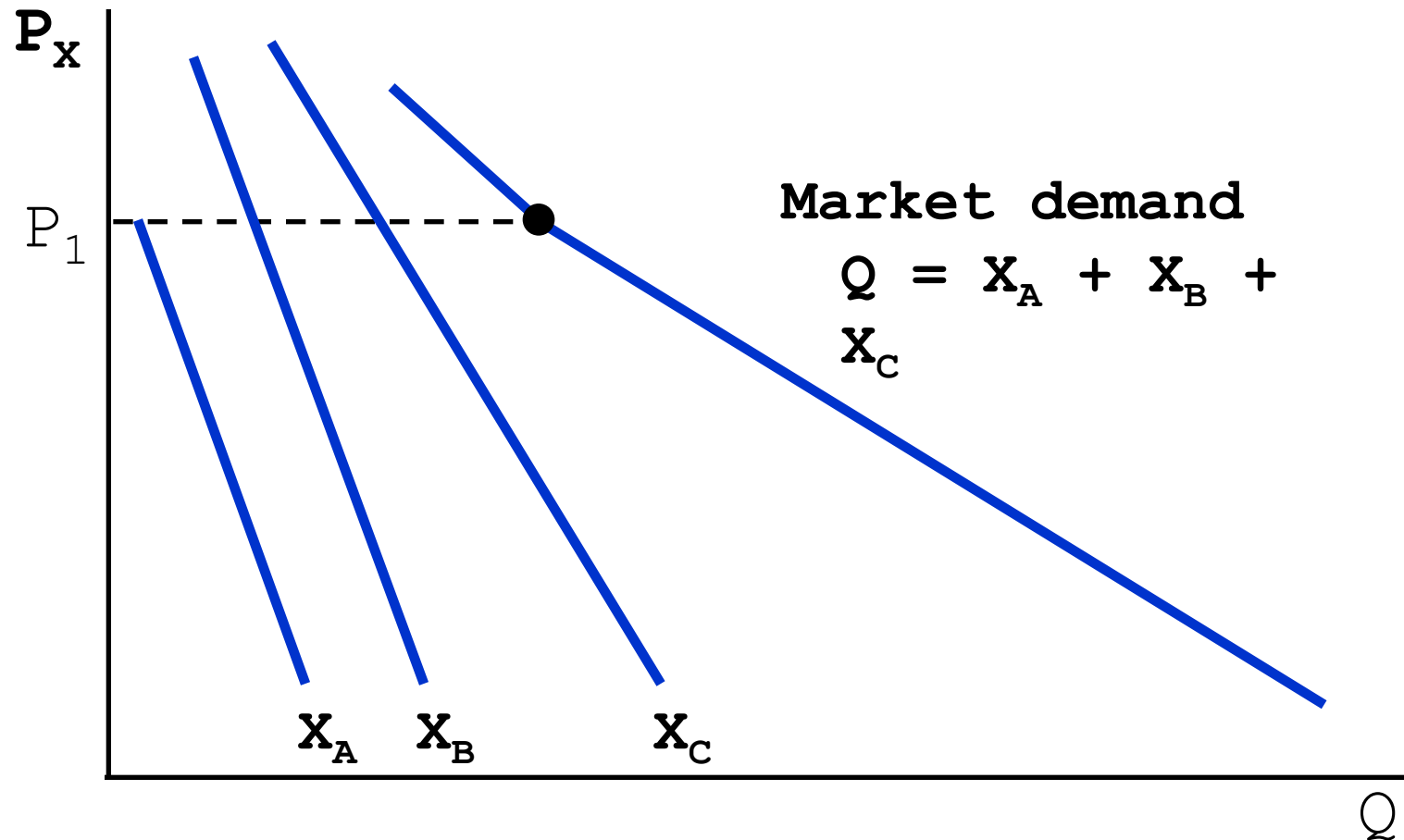
It is the sum of the individual demand curves for all consumers who purchase good X.

Graphically the market demand curve is the horizontal sum of the individual demand curves.



3.3 MARKET DEMAND

Market demand curve



LESSON 3: INDIVIDUAL AND MARKET DEMAND

3.1 Consumer demand

3.2 Changes in the budget constraint line: Engel curve

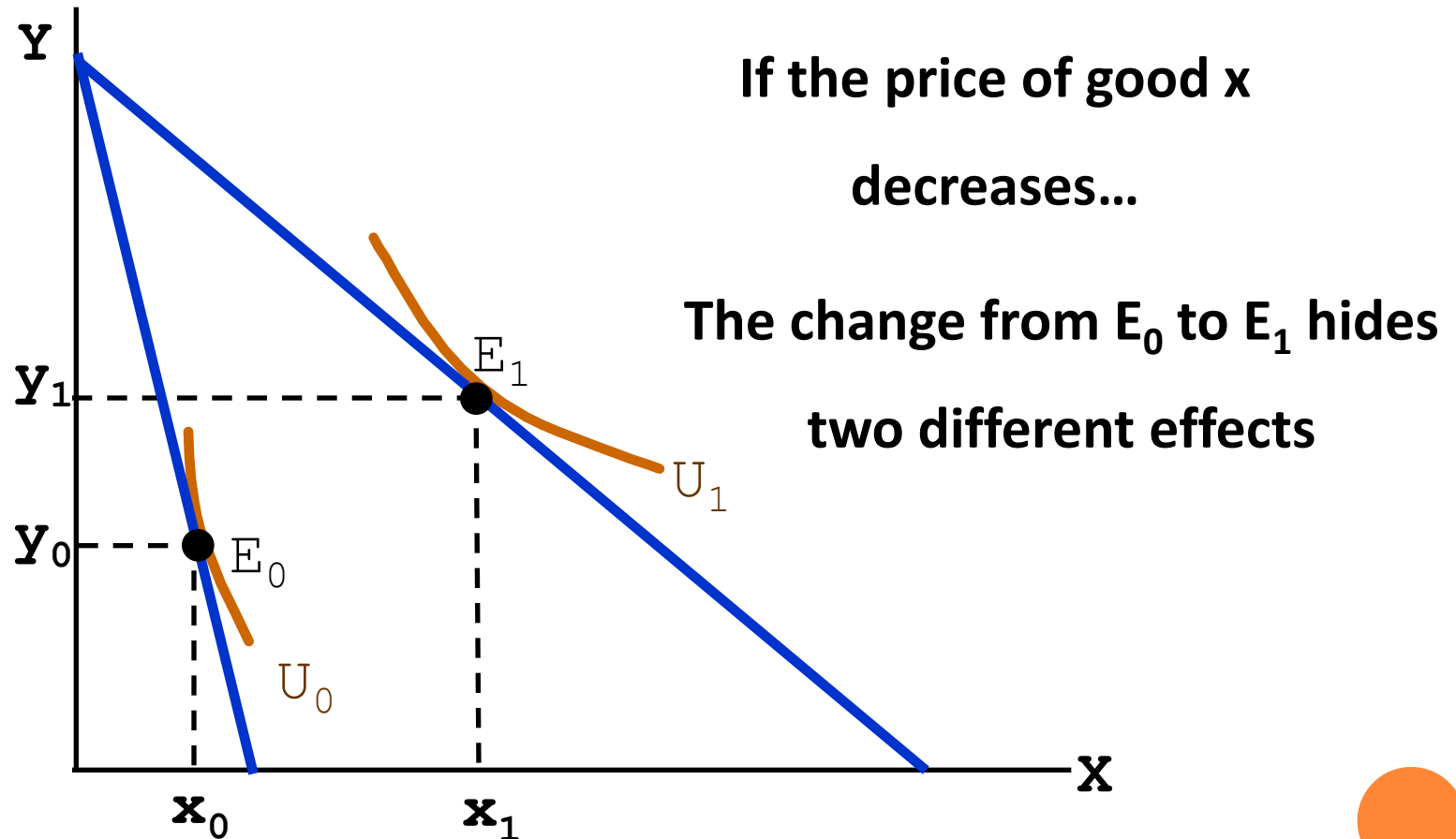
3.3 Market demand

3.4 Income effect and substitution effect

3.5 Elasticity of demand.

3.4 INCOME EFFECT AND SUBSTITUTION EFFECT

What are the income effect and the substitution effect?



3.4 INCOME EFFECT AND SUBSTITUTION EFFECT

What are the income effect and the substitution effect?

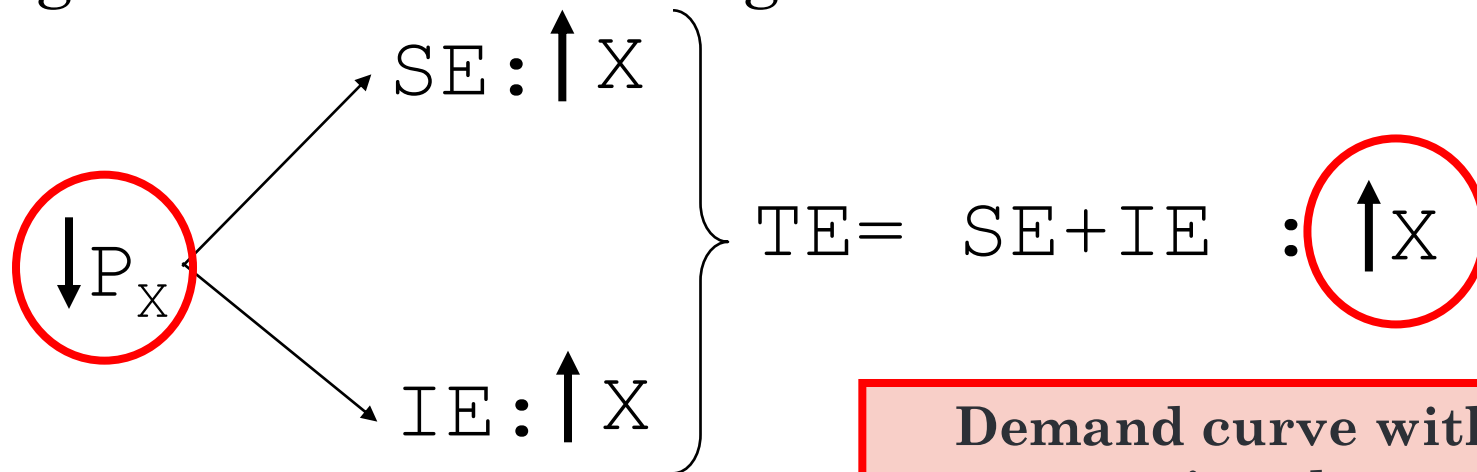
- (1) Good “x” **has become cheaper** in relative terms with respect to “y”. The consumer **will buy more** of the good that has become cheaper and less of the good that has become more expensive (**SUBSTITUTION EFFECT**).
- (2) As the price of good “x” falls, the purchasing power (real income / range of consumption possibilities) **has increased**. The consumer will buy **more of all the goods** that are normal (**INCOME EFFECT**).



3.4 INCOME EFFECT AND SUBSTITUTION EFFECT

What are the income effect and the substitution effect?

If good “x” is a normal good:

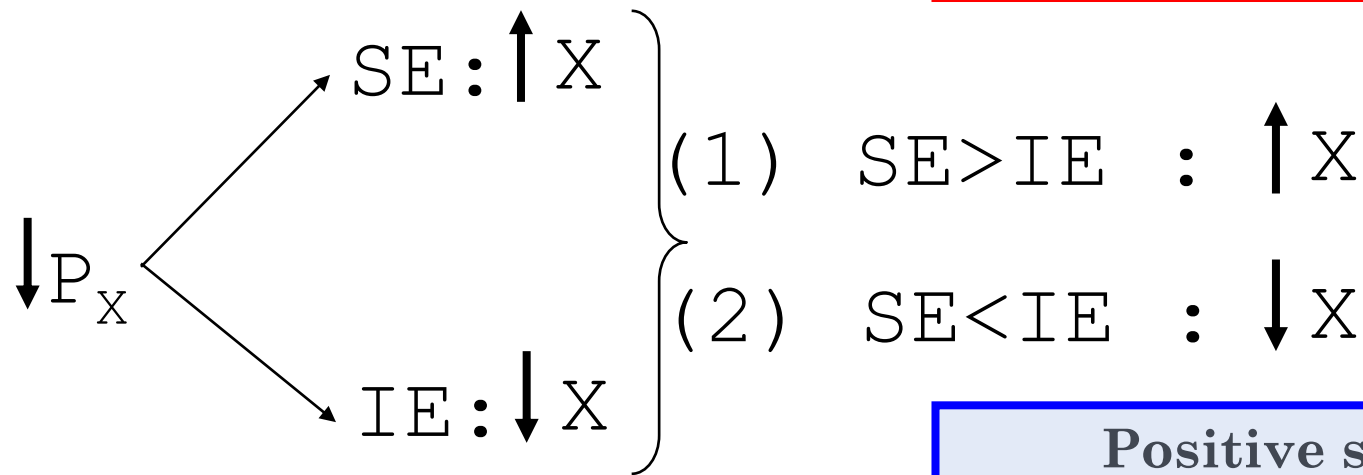


**Demand curve with
negative slope**



3.4 INCOME EFFECT AND SUBSTITUTION EFFECT

If good “x” is an inferior good:



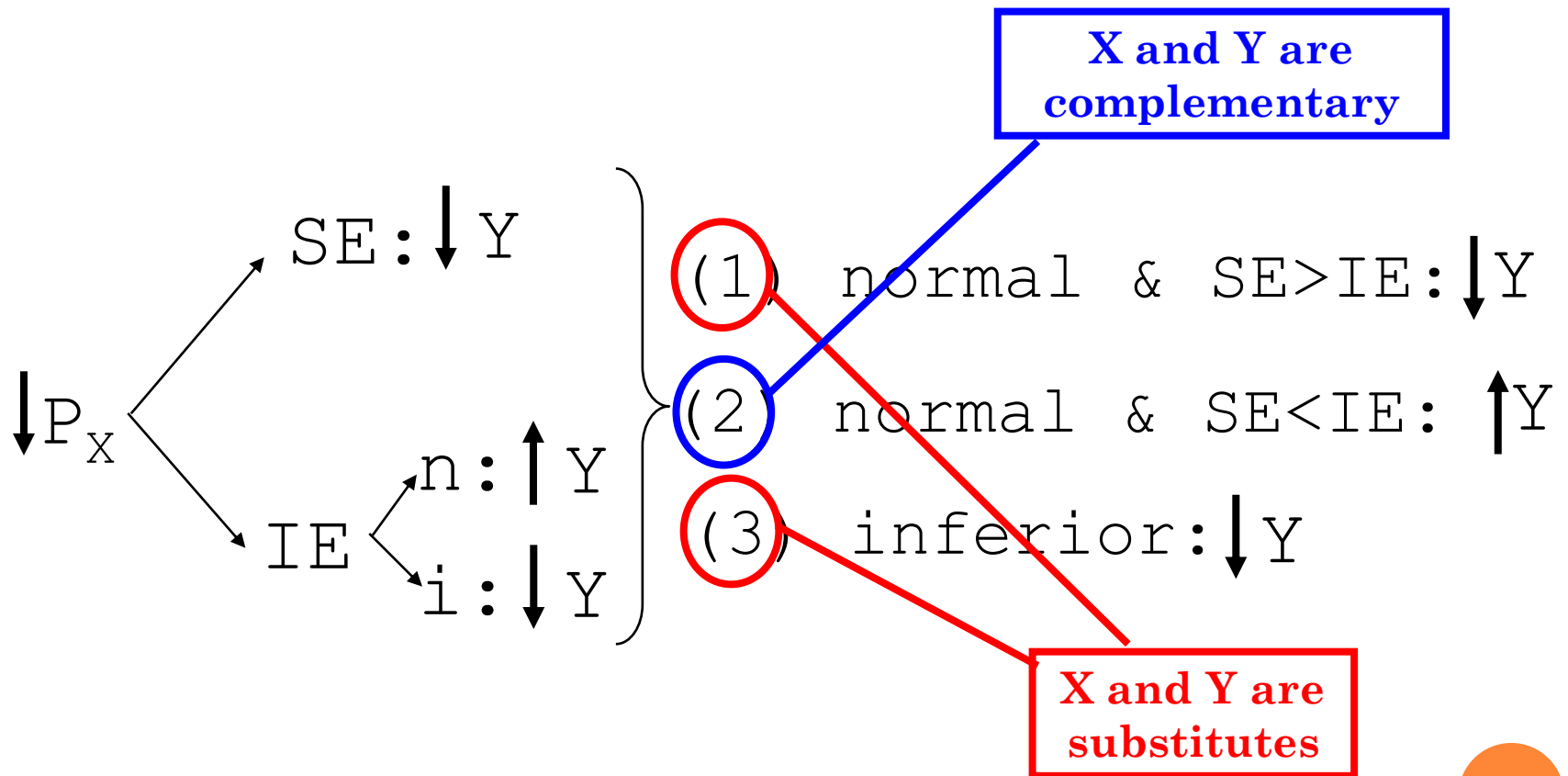
Negative slope

Positive slope:
GIFFEN goods



3.4 INCOME EFFECT AND SUBSTITUTION EFFECT

The effects on good Y can also be analysed:



3.4 INCOME EFFECT AND SUBSTITUTION EFFECT

How can we quantify these effects?

There are two criteria for finding the final equilibrium:

(1) **SLUTSKY method**: I' is fixed so, in that way, the initial combination of goods can be purchased with the final prices.

$$I' = X_0 \cdot P^1_X + Y_0 \cdot P^0_Y.$$

(2) **HICKS method**: I' is fixed so, in that way, the initial indifference curve can be reached.

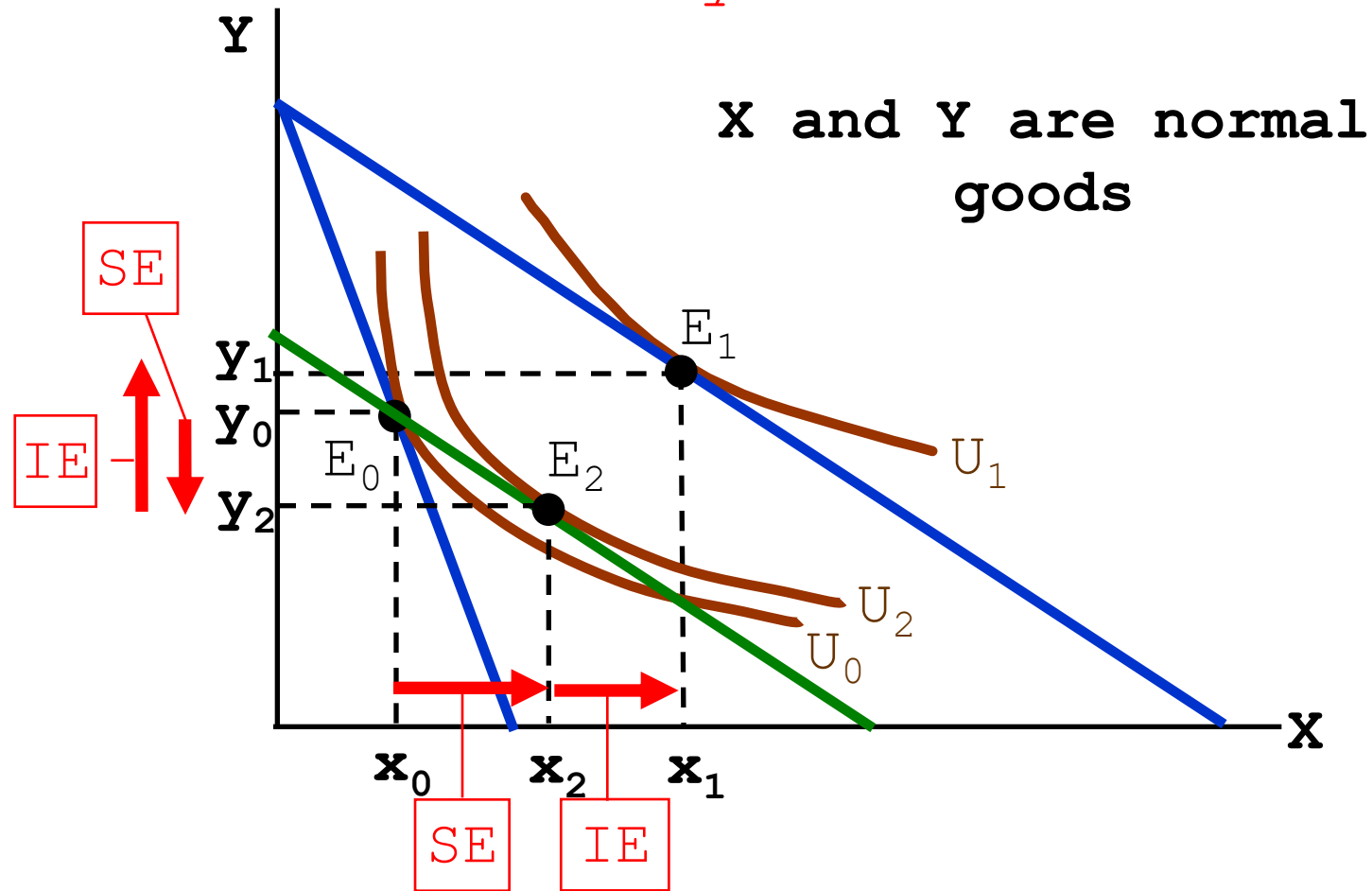
3.4 INCOME EFFECT AND SUBSTITUTION EFFECT

(1) **Substitution effect**: it is the variation experienced by the consumption of a good when the relative price changes and the purchasing power (Slutsky) or the utility level (Hicks) remains constant.

(2) **Income effect**: it is the variation in the consumption of a good caused by an increase in real income, remaining the relative price constant.

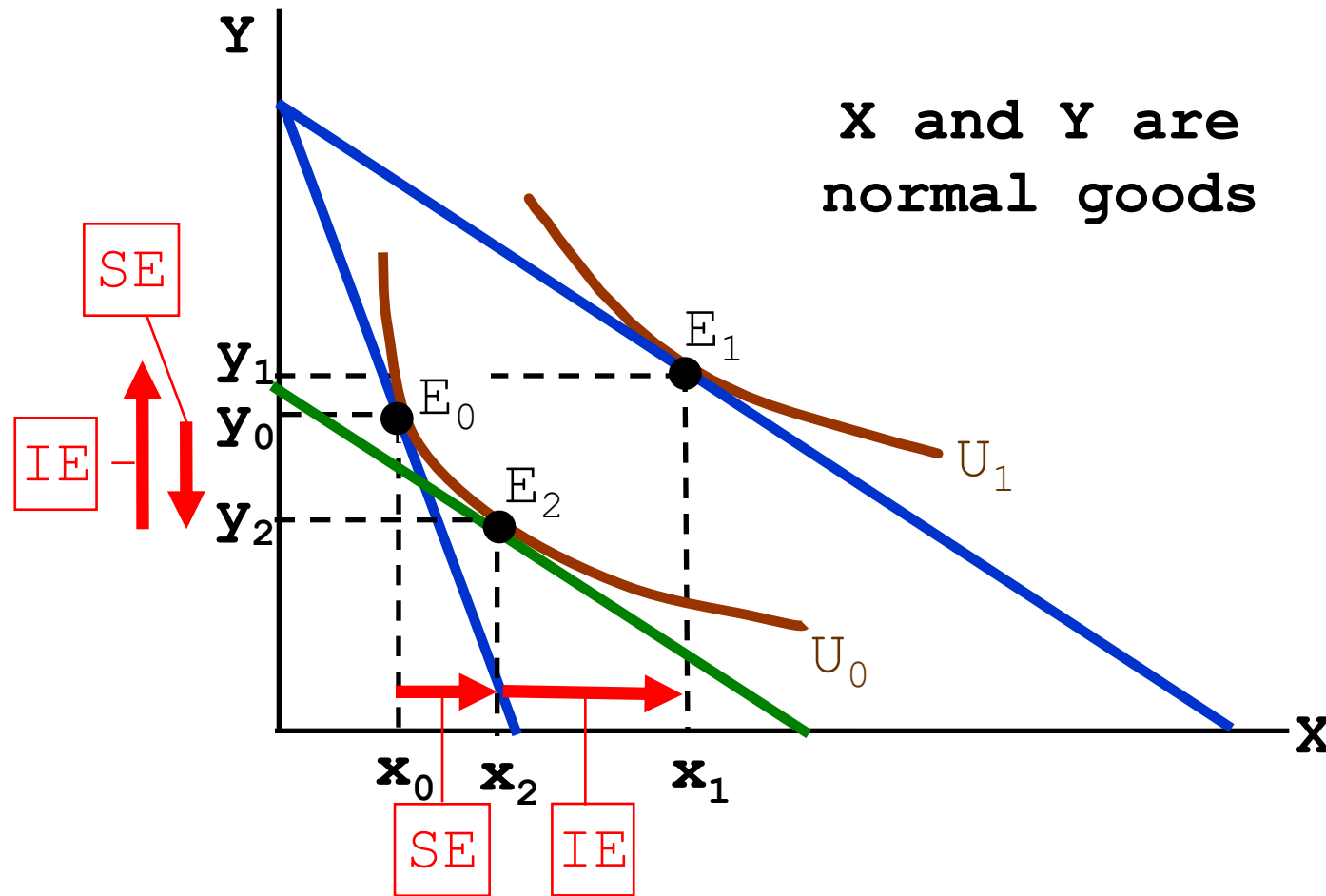
3.4 INCOME EFFECT AND SUBSTITUTION EFFECT

Slutsky Method



3.4 INCOME EFFECT AND SUBSTITUTION EFFECT

Hicks Method



LESSON 3: INDIVIDUAL AND MARKET DEMAND

3.1 Consumer demand

3.2 Changes in the budget constraint line: Engel curve

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3.5 Elasticity of demand.

3.5 ELASTICITY OF DEMAND

Elasticity is an instrument that determines the degree of sensitivity of the quantity demanded (or supplied) to changes in any of the variables on which it depends.

$$Q^d_i = f(p_i, p_1, \dots, p_n, I, P, \dots)$$

The price elasticity of demand measures the percentage change in the quantity demanded of a good when there is a percentage change in price of 1%.

$$\varepsilon = \frac{\Delta Q/Q}{\Delta P/P} = \frac{P}{Q} \frac{\Delta Q}{\Delta P} = \frac{P}{Q} \frac{1}{\text{slope}}$$



3.5 ELASTICITY OF DEMAND

Income elasticity of demand

$$e_{id} = \frac{\% \text{ change in quantity demanded}}{\% \text{ change in income}} = \frac{\frac{Q_{d1} - Q_{d0}}{Q_{d0}} \cdot 100}{\frac{I_1 - I_0}{I_0} \cdot 100}$$

If the good is **normal**, an increase in income will lead to an increase in consumption. In this case, the income elasticity will be positive. $e_{id} > 0$.

- Basic goods $1 > e_{id} > 0$
- Luxury goods $e_{id} > 1$

If it is an **inferior good**, increases in consumer income will cause decreases in the quantity demanded, in this case the income elasticity will be negative ($e_{id} < 0$)

$$e_{id} = \frac{\frac{\Delta Q}{Q}}{\frac{\Delta I}{I}} = \frac{I}{Q} \cdot \frac{\Delta Q}{\Delta I}$$



3.5 PRICE ELASTICITY OF DEMAND

Cross price elasticity of demand

The cross price elasticity between two goods X and Y is defined as the percentage change in the quantity demanded of good X when there is a percentage change (of 1%) in the price of good Y.

$$e_{cd} = \frac{\% \text{ change in quantity demanded of good X}}{\% \text{ change in the price of good Y}} = \frac{\frac{Q_{d1} - Q_{d0}}{Q_{d0}} \cdot 100}{\frac{p_1 - p_0}{p_0} \cdot 100}$$

- Complementary goods $e_{cd} < 0$
- Substitutes $e_{cd} > 0$
- Independent goods $e_{cd} = 0$



EXERCISES LESSON 3

EXERCISE 1

A consumer has the following utility function:

$$U(x, y) = x \cdot y$$

- A. If both goods cost 100€ ($P_x=100€$ and $P_y=100€$), calculate and represent graphically the income-consumption curve.
- B. Get the Engel curve. What kind of goods are they and why?
- C. If the income of this consumer is 2000€ determine the quantity of both goods that would maximize the utility.

EXERCISE 2

The utility function of a consumer is:

$$U = (x + 5) \cdot (y + 6)$$

He has a monthly income of 1000€ and the prices are 5€ for good Y and 4€ for good x.

- A. Calculate the slope of the indifference curve at the point (2,2)
- B. Find other bundles of goods that would provide the same utility
- C. Represent graphically the indifference curve for a utility of 2000.
- D. Calculate the slope of the budget constraint and include it in your graph.
- E. Calculate the consumer's choice for $P_x=4€$, $P_x=8€$ and $P_x=10€$. Draw the demand for x according to the consumer's choice.
- F. Calculate the budget constraint for an income of 500€ and 1500€
- G. From the results obtained represent Engel's curve and explain which type of good are we studying.

EXERCISE 3

Indicate whether the following statements are true or false. Justify your answer.

- A. A fall in the price of one good causes that, in relative terms, the price of the other good increases.
- B. In the price-consumption curve, the utility level of the different goods demanded is the same because they belong to the same demand curve.
- C. At each point of the price-consumption curve, utility maximization is fulfilled because the slope of the budget constraint is equal to the relative prices.
- D. The lower the price of the good, the lower the utility level because you will pay more for the goods.

EXERCISE 3

- E. A change in income shifts the budget line, but the same level of utility remains
- F. Whenever there is an increase in income, the demand for the good increases
- G. According to the Engel curve, which links the price and quantity demanded, when the slope is negative, the good is an inferior good

EXERCISE 4

Using the data from Lesson 2, Exercise 10:

- A. If the student had a monthly income of 1250€, represent graphically and analyse his budget constraint.
- B. Calculate and show consumer choice in the graph.
- C. Calculate the optimum when the price of good y changes to 7€.
- D. Draw the price-consumption curve
- E. Draw the demand curve for good y.

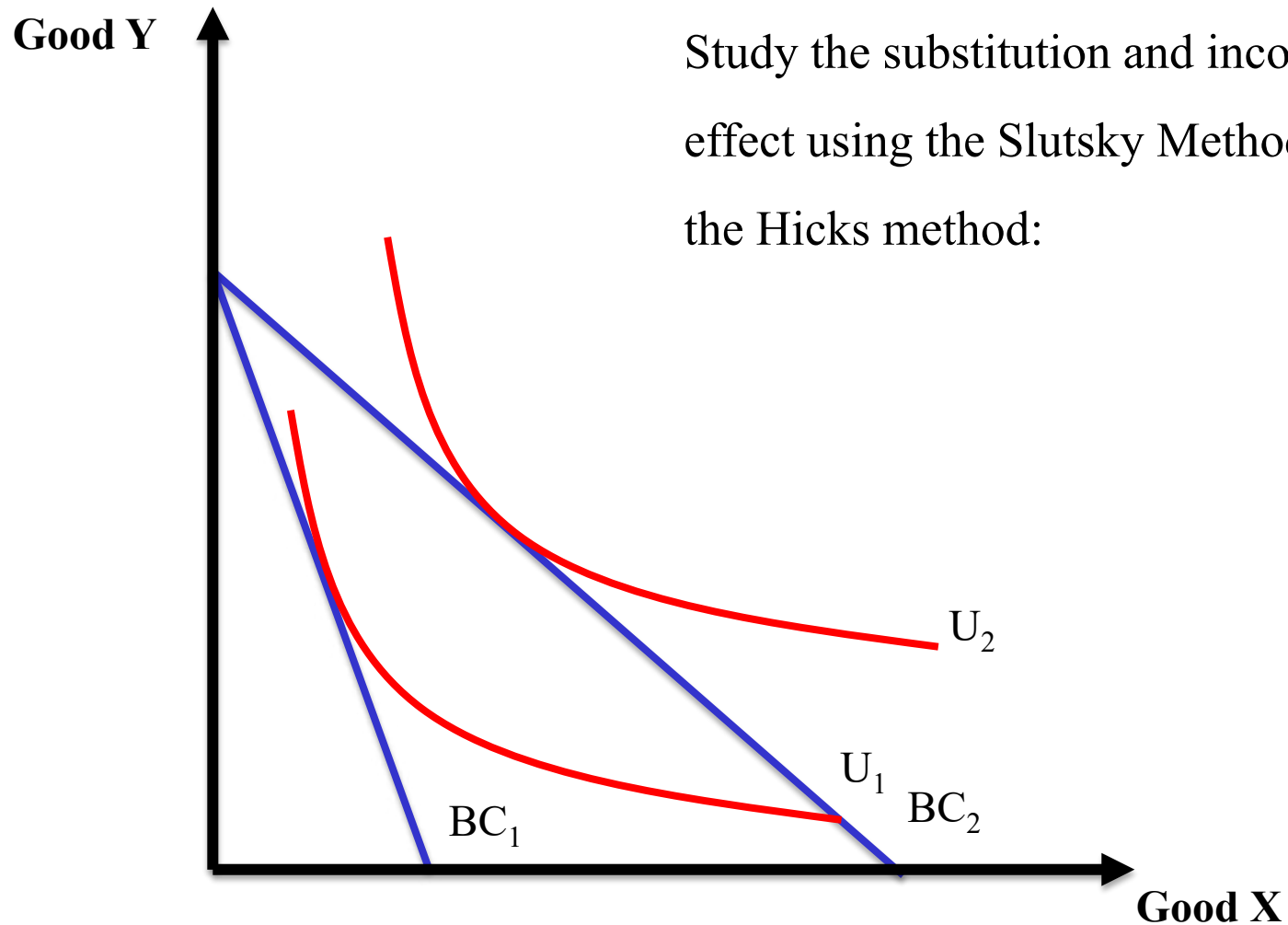
EXERCISE 5

We know that the preferences of a typical university student are represented by the following utility function:

$$U = 2xy$$

- A. If the student has a weekly income of 100 and the prices are $P_x = P_y = 2\text{€}$.
Represent graphically and analyse his budget constraint.
- B. Calculate and show consumer choice in the graph.
- C. Calculate the optimum when the price of good x changes and is now 1€
- D. Draw the demand curve for good x
- E. If his weekly income increases by 200€ (same prices as in question a),
determine and draw the new Budget constraint as well as the new equilibrium.
- F. Draw the income-consumption curve and the Engel curve for goods x and y.

EXERCISE 6



EXERCISE 7

Indicate whether the following statements are true or false. Justify your answer.

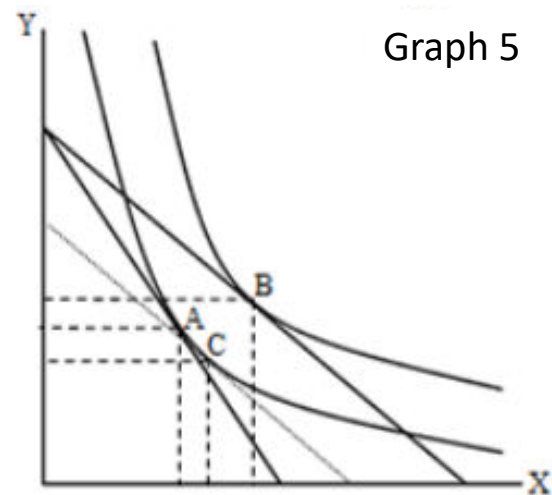
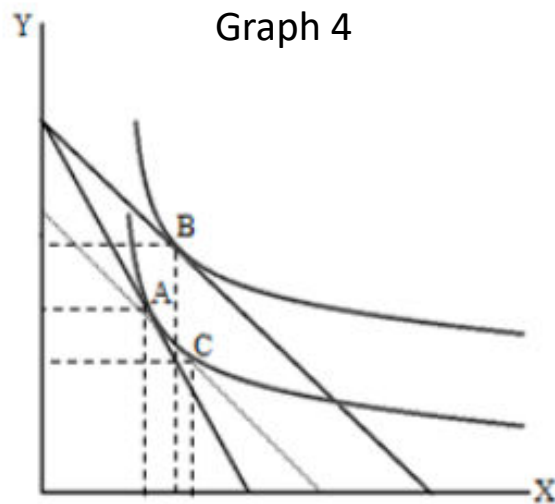
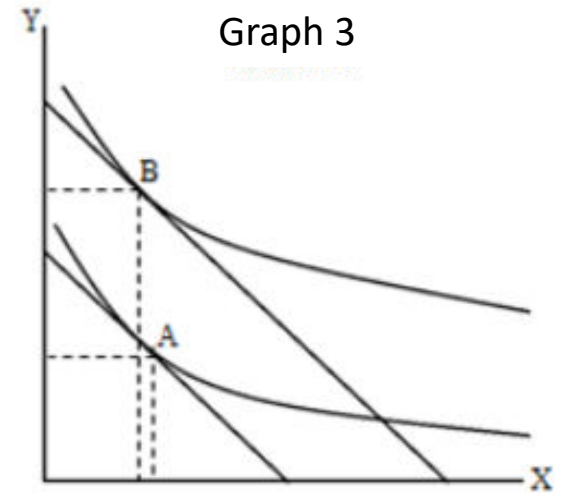
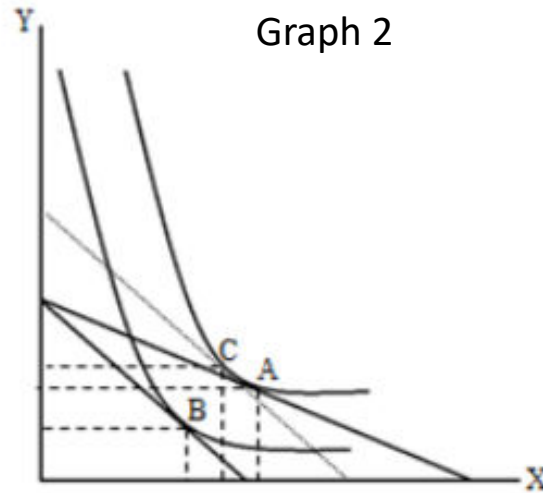
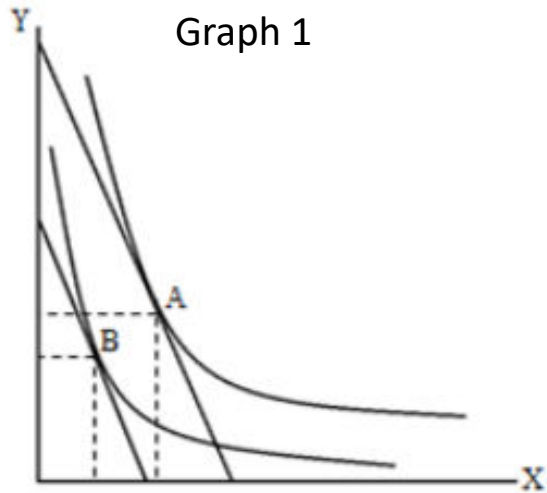
- A. To know the effect that a change in the price of good X has on the demand for X, it is necessary to know previously if good X is a normal good or an inferior good.
- B. To know the effect that a change in the price of good X has on the demand for good Y, it is necessary to know if they are substitutes or complements.
- C. The substitution effect will always cause an increase in demand for the good itself, since in relative terms, it is cheaper

EXERCISE 7

- D. When the good is an inferior good, a fall in the price of X increases the demand for X and we call that income effect.
- E. If the good is a normal good, a reduction in its price will lead to an increase in demand
- F. A giffen good perfectly reflects the behavior of the consumer according to the law of market demand (higher price, lower demand)

EXERCISE 8

Check the following graphs:



EXERCISE 8

In the graphs A is the original bundle of goods X and Y.

B is the new bundle after a change and C is the bundle of goods ‘in-between’.

Complete the sentences:

- a) Graph _____ shows an increase in the price of good X and the graph _____ shows an increase in income.
- b) In the graph/s _____ X is a normal good.
- c) In the graph/s _____ X is an inferior good.

EXERCISE 9

Indicate whether the following statements are true or false. Justify your answer.

- A. The price elasticity of demand gives us information about the direction and magnitude of the change in demand.
- B. An example of elastic demand is when a price change of 1% causes the quantity demanded to increase by 25%.
- C. With substitute goods, there will be less competition and less elasticity in those goods / services
- D. When the goods are luxury goods, the price elasticity of demand will be lower than 1.
- E. If the price elasticity of the demand for apples is unitary. If the price increase by 40% the quantity demanded will increase by 1%.

EXERCISE 10

According to the following information, indicate whether we are referring to normal, inferior, basic, luxury, substitute or complementary goods.

- a) The income elasticity of demand of the tomato is 0,3
- b) The cross price elasticity of demand between bread and honey is -1
- c) The income elasticity of demand of Audi C7 cars is 1,8
- d) The cross price elasticity of demand between sugar and honey is 0,9
- e) The income elasticity of demand of beef is -0,9
- f) The cross price elasticity of demand between a movie ticket and a sports event ticket is 0,3
- g) The cross price elasticity of demand between water and gasoline is 0
- h) The income elasticity of demand of branded lingerie is 1,2
- i) The cross price elasticity of demand between oranges and pineapples is 1,7
- j) The income elasticity of demand of low-cost cellphones is -0,9

EXERCISE 11

- A. The population income increases by 20%, leading to an increase in the quantity demanded of books from 4682 units to 5100 units. Calculate the income elasticity of demand.
- B. The price of good A increased from 10€ to 12€. That lead to a decrease in the quantity demanded of good A from 250 to 220 units and to an increase in the quantity demanded of good B from 325 to 350 units. Calculate the cross price elasticity of demand between good A and good B.
- C. Calculate the consumer surplus if the market is in equilibrium and the demand function is $Q_d = 240 - 2P$ and the price is one-third of the quantity demanded.